# Praxia Update

Daily orthopaedic literature digest · Wednesday, 26 August 2026 · 487 new articles across 10 journals

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AJSM

## American Journal of Sports Medicine

Volume 54, Issue 10 · August 2026 · 28 new articles

### The Rise of Remplissage.

*Owens BD*

PubMed · DOI

### Risk of Revision and Patient-Reported Outcomes After ACL Reconstruction: Influence of Concomitant MCL Injury and Graft Choice: Analysis of 35,139 Reconstructions From the Norwegian Knee Ligament Register.

*Birkenes T, Chahla J, Lygre SHL, Williams A, Inderhaug E*

BACKGROUND: Concomitant medial collateral ligament (MCL) injury is common in anterior cruciate ligament (ACL) tears, but optimal management of these injury components remains controversial.

PURPOSE: To compare patient-reported outcome measures and the risk of ACL revision surgery in patients with concomitant MCL injuries (treated operatively and nonoperatively) versus isolated ACL injuries and to assess whether MCL treatment strategy or ACL graft choice influenced outcomes.

STUDY DESIGN: Cohort study; Level of evidence 3.

METHODS: Data were obtained from the Norwegian Knee Ligament Register. Patients undergoing primary ACL reconstruction (ACLR) between 2004 and 2024 were included for graft-survival analyses, and those operated by December 31, 2022, were eligible for 2-year Knee injury and Osteoarthritis Outcome Score (KOOS) outcomes. Patients with concomitant ligament injuries other than to the MCL, ACL repairs, or those with unknown ACL graft type were excluded. Patients were stratified into isolated ACL and ACL + MCL injury groups. Kaplan-Meier survival analyses (unadjusted), multivariable Cox regression, and linear/logistic regression models were used, adjusted for descriptive data, body mass index, meniscal/cartilage injury, time to surgery, smoking, pivoting sports, graft choice, and baseline KOOS.

RESULTS: A total of 35,139 ACLRs were included (2410 ACL + MCL and 32,7 […]

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### Short-term Patient-Reported Outcomes After Primary Anterior Cruciate Ligament Reconstruction With and Without Lateral Extra-articular Tenodesis: A Matched-Cohort Analysis From the Swedish Knee Ligament Registry.

*Pruneski JA, Zsidai B, Öttl F, Heder Ternell K, Cristiani R, Musahl V, Senorski EH, Samuelsson K et al.*

BACKGROUND: Despite reduced graft failure rates reported in randomized controlled trials comparing anterior cruciate ligament reconstruction (ACL-R) with and without lateral extra-articular tenodesis (LET), the impact of LET on postoperative patient-reported outcomes (PROs) remains poorly defined.

PURPOSE: To compare short-term PROs, as well as anterior cruciate ligament (ACL) revision rates and clinical failure rates (defined as Knee injury and Osteoarthritis Outcome Score [KOOS] Quality of Life [QoL] value <44) between patients undergoing primary ACL-R with and without concomitant LET.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Data were extracted from the Swedish Knee Ligament Registry for all patients undergoing primary ACL-R between January 1, 2005, and June 25, 2025. Patients who underwent ACL-R + LET were matched to patients who underwent ACL-R without LET using 1:4 matching based on age, sex, body mass index, graft type, time from injury to reconstruction, and concomitant meniscal and cartilage injury status. The primary outcomes were KOOS values at the 1- and 2-year follow-ups.Secondary outcomes included revision ACL-R and clinical failure within 1 and 2 years. Categorical variables were compared using Fisher exact or chi-square tests, continuous variables were compared using Fisher nonparametric permutation tests, and within-group KOOS changes were […]

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## ACL Reconstruction With Attachment-Sparing Hamstring Autograft Results in Earlier Graft Maturation and Better Short-Term Clinical Outcome in Comparison to Free Graft: A Randomized Controlled Trial.

*Sinha S, Arumugham KC, Singh DK, Kumar N, Bera AP, Uppal H, Vakharia KS*

BACKGROUND: Attachment-sparing anterior cruciate ligament reconstruction (ACLR) is gaining acceptance over the use of free grafts because of reported early graft ligamentization in both animals and humans. However, the literature regarding clinical outcomes and the correlation of early graft healing is unclear.

HYPOTHESIS: Tibial attachment-sparing hamstring graft (AS) results in earlier graft maturation, which correlates with a better clinical outcome than a free hamstring graft (FG).

STUDY DESIGN: Randomized controlled trial; Level of evidence, 1.

METHODS: A total of 56 patients were randomly assigned to 2 groups at the commencement of the study. Of these patients, 25 (AS group) underwent ACLR using the attachment-sparing technique, while 25 patients (FG group) underwent ACLR using the free hamstring graft technique. Prospective sequential magnetic resonance imaging (MRI) evaluations were performed at 3, 6, and 9 months to document graft maturation, characteristics, orientation, and tunnel dilation. Clinical evaluation was done at 3, 6, 9, and 12 months to document Lysholm, International Knee Documentation Committee (IKDC), and Tegner scores and anterior tibial translation (ATT) by KT-2000 arthrometer, by independent observers.

RESULTS: The overall follow-up rate was 78% (19 cases and 20 controls of 25 in each group). MRI Figueroa scores demonstrated significant differenc […]

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## MRI-Based Volumetric Analysis of the Quadriceps Muscles Correlation With Knee Extension Strength in Patients Undergoing Anterior Cruciate Ligament Reconstruction.

*DeFoor MT, Knurr KA, Heiderscheit B, Riem L, Cognetti DJ, Shaw J, Blaylock M, Antosh IJ et al.*

BACKGROUND: Preoperative quadriceps and hamstring muscle weakness may predispose patients to strength deficits and prolonged recovery in the setting of anterior cruciate ligament (ACL) injury. Three-dimensional (3D) magnetic resonance imaging (MRI) analysis has been developed to provide a comprehensive assessment of the thigh musculature.

PURPOSE: To (1) describe the magnitude of quadriceps and hamstring muscle weakness relative to the uninjured limb preoperatively in patients with known ACL tears, and (2) determine if preoperative quadriceps and hamstring muscle volume correlate with quadriceps knee extension strength and hamstring knee flexion strength.

STUDY DESIGN: Cross-sectional study; Level of evidence, 3.

METHODS: Patients undergoing primary ACL reconstruction (ACLR) underwent MRI scans of the bilateral lower extremities and strength testing preoperatively within 7 days of ACLR. Three-dimensional MRI-based autosegmentation was used to measure the volume of individual quadriceps and hamstring muscles. Preoperative knee extension and flexion strength were assessed isometrically at 90° and isokinetically. Linear mixed-effects models were used to assess muscle volume by limb interaction effect on strength. Pearson correlation coefficients (r) were used to assess the association between muscle volume and strength in the injured and uninjured limbs.

RESULTS: In total, 72 […]

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## Knee Synovial Fluid Biomarker Type and Concentration With Long-Term Outcomes After ACL Reconstruction With Concomitant Meniscal Injury.

*Ehlers M, Estes LC, Derry K, Kaplan DJ, Strauss EJ*

BACKGROUND: Anterior cruciate ligament (ACL) rupture frequently occurs alongside meniscal injury, together altering the intra-articular environment. While ACL injury triggers an inflammatory cascade within synovial fluid, meniscal injury may intensify these responses, contributing to joint degeneration. However, little is known about how combined ACL and meniscal injuries shape the synovial inflammatory milieu or whether these profiles influence long-term outcomes.

PURPOSE/HYPOTHESIS: The purpose was to characterize synovial fluid inflammatory phenotypes at the time of ACL reconstruction with concomitant meniscal injury and assess their association with patient-reported outcome measures (PROMs) and clinically meaningful improvement. It was hypothesized that (1) distinct inflammatory phenotypes would exist, (2) these phenotypes would correlate with symptom severity and recovery, and (3) individual biomarkers' effects on outcomes would vary by phenotype.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Patients undergoing arthroscopic ACL reconstruction with concomitant meniscal injury were prospectively enrolled between July 2011 and January 2024. Synovial fluid was aspirated before incision, and the concentrations of 10 biomarkers were quantified. Unsupervised k-means clustering identified high- and low-inflammation phenotypes. PROMs were collected preoperatively o […]

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## Quantitative Investigation of Tibiofemoral Cartilage Early Degeneration After ACL Reconstruction: An Integrated Biomechanical and Longitudinal MRI Analysis.

*Lin J, Cheng R, Yan Y, Zeng X, Huang W, Tsai TY, Deng C, Wang S et al.*

BACKGROUND: Anterior cruciate ligament reconstruction (ACLR) restores stability but is often followed by early cartilage degeneration. The contribution of altered dynamic contact pressure during gait to this degeneration remains poorly understood.

PURPOSE: To investigate the biomechanical mechanisms underlying early cartilage degeneration after ACLR, with a focus on dynamic contact pressure distribution during gait.

STUDY DESIGN: Controlled laboratory study.

METHODS: In a 3-year longitudinal magnetic resonance imaging (MRI) study of patients with ACLR ($n = 30$), cartilage thickness of the tibial plateau was quantitatively assessed using deep-learning-based 3-dimensional segmentation techniques. In parallel, cadaveric knees ($n = 8$) were tested under intact, ACL-deficient, and ACL-reconstructed conditions using a 6 degree-of-freedom robotic simulator replicating gait cycles. Tibiofemoral contact pressure and pressure center trajectories were recorded using pressure-sensitive film.

RESULTS: Quantitative MRI analyses revealed significant cartilage thinning in the posterior tibial subregion 3 years after ACLR. On the medial plateau, the central medial tibia, internal medial tibia, and posterior medial tibia subregions exhibited mean reductions of 10% ($P = .027$), 21% ($P = .019$), and 13% ($P = .041$), respectively. On the lateral plateau, significant decreases were observed in centra […]

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## Comparable Midterm Survival After Medial Meniscus Root Repair for High and Low Intraoperative Outerbridge Classification Grades.

*Grandberg C, Boduch AN, Bilodeau RE, Cohen D, Herman ZJ, Ahrendt GM, Drain NP, Setliff JC et al.*

BACKGROUND: While medial meniscus posterior root repairs (MMPRRs) have been shown to reverse harmful biomechanical effects of medial meniscus posterior root tears, it is questionable whether meaningful improvement occurs in patients with advanced chondral loss.

PURPOSE/HYPOTHESIS: The objectives of this study were to compare midterm survival of MMPRR and to determine patient-reported outcomes (PROs) after MMPRR based on intraoperative Outerbridge grades (0-2 vs 3 and 4). It was hypothesized that patients undergoing MMPRR with Outerbridge grades 3 and 4 would have a decreased rate of MMPRR survival and inferior PROs compared to patients with Outerbridge grades 0 to 2.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Consecutive patients who underwent primary isolated MMPRR performed at a single institution between July 2003 and November 2022 with a minimum follow-up of 1 year were included and categorized according to medial compartment Outerbridge grade (0-2 or 3 and 4) based on diagnostic arthroscopy. Failure was defined as reoperation of the index knee. PROs collected included International Knee Documentation Committee (IKDC), visual analog scale (VAS) for pain, and Marx activity scale scores, which were compared to the previously established minimal clinically important difference (MCID) and/or clinically meaningful change. A minimum follow-up of 2 years was use […]

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### The Effect of Smoking History on Inflammatory Biomarkers in Synovial Fluid of Patients Undergoing Arthroscopic Knee Surgery for Meniscal Injury.

*Kurtz JL, Ehlers M, Montgomery SR, Kaplan DJ, Strauss EJ*

BACKGROUND: Smoking has been linked to alterations in cellular inflammatory pathways and adverse outcomes in orthopaedic procedures. It is unclear how patient smoking affects the intra-articular microenvironment in the setting of symptomatic meniscal tears.

PURPOSE: To investigate the association between tobacco use and variation in cytokine concentrations in knee synovial fluid in patients undergoing arthroscopic knee surgery for symptomatic meniscal injury.

STUDY DESIGN: Cohort study; Level of evidence, 3Methods:Patients who underwent knee arthroscopy for meniscal injury were prospectively enrolled between July 2011 and June 2019. Synovial fluid was aspirated and the concentrations of 10 biomarkers were measured by immunoassay. Smoking status, pack-years, and time since smoking cessation were collected. Log-normalized biomarker concentrations and smoking status were analyzed using analysis of variance (ANOVA) and linear regression.

RESULTS: A total of 297 patients (mean age, 45.7 ± 12.5 years; mean body mass index, 28.3 ± 5.3 kg/m2; 55.6% male) were included. Patients were divided into current smokers (n = 27), former smokers (n = 54), and nonsmokers (n = 216). No descriptive differences between groups were found. ANOVA showed significant differences (P < .01) in regulated on activation, normal T cell expressed and secreted (RANTES), monocyte chemotactic protein-1 (MCP-1), […]

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### A Prospective Clinical and Magnetic Resonance Imaging-Based Follow-up Beyond 10 Years After Arthroscopic Matrix-Induced Autologous Chondrocyte Implantation.

*Ebert JR, Edwards PK, Klinken S, Wood DJ, Janes GC*

BACKGROUND: Long-term studies are emerging reporting encouraging outcomes after matrix-induced autologous chondrocyte implantation (MACI). While MACI has historically been undertaken via open arthrotomy, arthroscopic techniques have been reported. However, longer term clinical outcomes and/or graft survivorship after arthroscopically performed MACI are yet to be reported.

PURPOSE: To present longer term clinical and radiological outcomes in patients after MACI performed arthroscopically on the femoral condyles.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: A total of 45 patients were prospectively recruited and underwent arthroscopically performed MACI to the medial femoral condyle (MFC; n = 31; defect size, 1.0-7.7 cm2) or lateral femoral condyle (LFC; n = 14; defect size, 1.5-5.0 cm2). Outcomes were collected before surgery and at 1, 2, and 5 years as well as at a mean of 13.6 years (range, 11-17 years) after surgery. These included scores for a variety of patient-reported outcome measures (PROMs) including the Knee Injury and Osteoarthritis Outcome Score, Lysholm Knee Score, Tegner Activity Scale, and patient satisfaction. Active knee range of motion as well as peak isokinetic knee extensor and flexor strength were assessed, while magnetic resonance imaging (MRI) was undertaken. The Magnetic Resonance Observation of Cartilage Repair Tissue (MOCART) scoring too […]

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### A Novel Autologous Osteoperiosteal Composite for Osteochondral Regeneration.

*Yao L, Zhu S, Wu Y, Ye G, Wang G, Niu Z, Ding S, Xiang Z et al.*

BACKGROUND: Osteochondral defects result in persistent knee pain and functional impairment, and remain a clinical challenge to repair effectively.

PURPOSE: To evaluate the efficacy of an autologous inverted subchondral bone-periosteum composite graft for osteochondral reconstruction.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 30 mature New Zealand White rabbits were randomly assigned to 3 groups: microfracture (MF) (n = 10), osteoperiosteal composite graft (OPC) (n = 10), and untreated control (n = 10). A 2 mm-diameter osteochondral defect was created in the control and MF groups and left untreated in the control group. In the MF group, microfracture was then performed by creating four 4 mm-deep perforations in the defect bed using Kirschner wires. A cylindrical osteochondral autograft (2 mm diameter × 4 mm height) was aseptically harvested from the intercondylar fossa using a calibrated coring drill only in the OPC group. Group-specific treatments included defect creation without reparative intervention (control), defect creation followed by subchondral perforation (MF), and inverted autograft implantation with periosteal wrapping (OPC). Animals were euthanized at 6 and 12 weeks postoperatively (n = 5 per group per time point) for gross and histological evaluation.

RESULTS: At 6 weeks, fibrocartilaginous tissue partially filled the defects in all groups […]

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## Weakening of the Active Muscle Restraint Against Valgus Loading During Repeated Baseball Pitching: Direct Evidence of Fatigue in Dynamic Valgus Stabilizers of the Elbow Joint in Collegiate Baseball Pitchers.

*Yanai T, Onuma K*

BACKGROUND: Although research has shown the effect of repeated pitching on throwing arm strength, no studies have examined the weakening in the overall ability of dynamic stabilizers to restrain the medial elbow against valgus loading. Fatigue in the dynamic stabilizers increases reliance on the ulnar collateral ligament (UCL) to withstand the valgus loads during pitching.

HYPOTHESIS: It was hypothesized that (1) the muscular varus strength weakens as the pitch count accumulates and (2) the muscular varus strength becomes insufficient to overcome the valgus loads in later innings.

STUDY DESIGN: Descriptive laboratory study.

METHODS: A total of 28 collegiate baseball pitchers threw 7 innings of 15 pitches in a bullpen. Seven genlocked cameras were used to record pitching performances, and the elbow valgus load was determined. Muscular varus strength was measured before the first inning and after completing the fourth and seventh innings using an isokinetic dynamometer system and an ultrasound device. A mixed-effect model was used to examine changes over the innings in the valgus load and varus strength, and t tests were used to compare the varus strengths and the largest valgus load in corresponding innings.

RESULTS: Neither the largest valgus load of the inning nor the mean valgus load of the inning significantly changed across the innings, whereas the varus strength was re […]

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## Changes in Rate and Indications for Remplissage When Performing Arthroscopic Bankart Repair for Anterior Shoulder Instability.

*Brej BL, Rauck RC, Huntley KS, David T, Denard PJ, Lin A, Bishop JY*

BACKGROUND: Remplissage is used as an adjunct to arthroscopic Bankart repair (ABR) to reduce recurrent instability. Despite growing evidence supporting efficacy and broader application of the procedure, the absence of a specific procedural code limits knowledge of its utilization, and current surgical indications are not well defined.

HYPOTHESIS: It was hypothesized that remplissage utilization would increase over the study period and that indications would expand beyond traditionally defined off-track Hill-Sachs lesions.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Patients undergoing arthroscopic anterior shoulder stabilization from 2012 to 2024 were retrospectively reviewed across multiple institutions. Patients were grouped by ABR alone (ABR) or with remplissage (REMP). Temporal trends were evaluated by comparing early (2012-2020) and late (2021-2024) cohorts. Preoperative magnetic resonance imaging was used to assess bipolar bone loss using established measurement techniques, including glenoid track (GT), glenoid bone loss (GBL), and distance to dislocation (DTD).

RESULTS: The cohort included 244 patients who underwent remplissage and 1100 patients who underwent ABR. The proportion of REMP increased 4.5-fold over the study period (7.7% in 2012 vs 34.7% in 2024). Later year of surgery was significantly associated with increased remplissage utilization (odd […]

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## Morphological Characteristics After Single Versus Recurrent Traumatic Posterior Shoulder Dislocation Events.

*Gibbs HP, McQuade AM, Ramirez V, Feeley SM, McCarthy CF, Hoyt BW, Dickens JF*

BACKGROUND: Posterior shoulder dislocations are rare but particularly challenging in high-demand populations. While anterior instability has been well studied, limited data exist on the morphological consequences of traumatic first-time and recurrent posterior dislocations.

PURPOSE: To quantify posterior glenoid bone loss (pGBL), glenoid version, reverse Hill-Sachs lesions, and acromial morphology in patients with traumatic posterior shoulder dislocations and compare these features between the single and recurrent dislocation groups.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: A retrospective review was conducted using the Military Health System Data Repository to identify patients who sustained traumatic posterior shoulder dislocations between January 2017 and December 2024. These were defined as events requiring manual reduction with subsequent advanced imaging demonstrating a posterior labral tear or posterior bony Bankart lesion, with or without a reverse Hill-Sachs lesion. Morphological features were independently assessed by 2 raters. Measured factors included pGBL (measured by the perfect circle technique (PCT), two-thirds glenoid height methods, and one-third glenoid height method), glenoid version, reverse Hill-Sachs Lesion parameters, and acromial tilt and height. Patients were stratified into single and recurrent dislocation groups. Statistical compa […]

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## Comparison of Long-term Supraspinatus Tear Progression After Arthroscopic Isolated Subscapularis Repair With and Without Comma Tissue Preservation: A Minimum 10-Year MRI Follow-up Study.

*Do WS, Chun YM, Kim SJ, Sim W, Yoon TH*

BACKGROUND: Long-term outcomes after arthroscopic isolated subscapularis repair are limited, and magnetic resonance imaging (MRI)-based assessments extending to 10 years are particularly scarce. Therefore, it remains unclear whether preserving the comma tissue (in-continuity technique) provides long-term advantages over disrupting the upper subscapularis margin (margin disruption technique).

HYPOTHESIS: The in-continuity and margin disruption techniques would demonstrate comparable 10-year radiological and clinical outcomes in patients undergoing arthroscopic isolated subscapularis repair.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Among patients who underwent arthroscopic isolated subscapularis repair between 2008 and 2015, those with an intact repair on 6-month postoperative MRI and at least 2 years of clinical follow-up were identified. From this cohort, 24 patients treated with the in-continuity technique and 24 treated with the margin disruption technique were included. Ten-year MRI findings (subscapularis Goutallier grade, tendon integrity, and supraspinatus tear progression) and clinical outcomes (active range of motion [ROM], visual analog scale [VAS] pain score, Subjective Shoulder Value [SSV], American Shoulder and Elbow Surgeons [ASES] score, and University of California, Los Angeles [UCLA] score) were compared. A subgroup analysis was performed to […]

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## Intraoperative Zoledronic Acid for Arthroscopic Rotator Cuff Repair Results in Distinct Circulating Micro-RNA Profiles Indicating Improved Tendon-to-Bone Healing.

*Schanda JE, Hackl M, Bischl R, Ullrich R, Boesmueller S, Mittermayr R, Redl H, Grillari J et al.*

BACKGROUND: A prospective, randomized, placebo-controlled trial showed reduced retears after arthroscopic rotator cuff (RC) repair in patients without osteoporosis by a systemic single-dose of zoledronic acid. Distinct micro-ribonucleic acids (miRNAs) related to inflammation, fibrosis, and tendon-to-bone healing are associated with RC injuries.

PURPOSE: To investigate the longitudinal effects of a single-dose of zoledronic acid in patients without osteoporosis undergoing arthroscopic RC repair on circulating miRNAs in order to explore the molecular mechanism of this treatment.

STUDY DESIGN: Controlled laboratory study.

METHODS: Data were collected in the course of a single-center, prospective, randomized, placebo-controlled, triple-blinded (investigator, surgeon, patient) phase II trial. A total of 80 patients underwent arthroscopic RC repair and were intraoperatively randomized to the zoledronic acid group (n = 40) or the control group (n = 40). Circulating plasma miRNAs were assessed preoperatively, 2 days postoperatively, and 3 months postoperatively using small RNA sequencing and reverse transcription quantitative polymerase chain reaction (RT-qPCR).

RESULTS: Six miRNAs were selected for validation by RT-qPCR based on small RNA-sequencing analysis. No statistical differences in miRNA plasma levels were observed preoperatively between the 2 study groups. Two days after s […]

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## Association of Central Acetabular Osteophytes With Microinstability and Increased Combined Anteversion in Borderline Dysplasia Hips.

*Zhu JB, Ding R, Huang Y, Chen YZ, Zheng GY, Uchida S, Shen C*

BACKGROUND: Central acetabular osteophytes (CAOs) have been implicated in abnormal hip biomechanics. However, their relationship with microinstability in borderline developmental dysplasia of the hip (BDDH) is unclear.

PURPOSE/HYPOTHESIS: The purpose of this study is to compare radiographic parameters between BDDH hips with and without CAOs and to determine whether CAOs are associated with microinstability. It was hypothesized that CAOs would correlate with increased version abnormalities and microinstability.

STUDY DESIGN: Case-control study; Level of evidence, 3.

METHODS: We retrospectively reviewed patients with BDDH, defined as a lateral center-edge angle of 18° to 25°, who underwent hip arthroscopy from 2020 to 2024. Patients were grouped as BDDH with CAO (cBDDH) or without CAO (nBDDH). Radiographic parameters, including anterior/posterior acetabular coverage, femoral neck version, acetabular version, and combined anteversion, were compared. Intraoperative findings of microinstability and ligamentum teres (LT) pathology were analyzed. Binary logistic regression identified predictors of CAO.

RESULTS: A total of 157 patients were included (cBDDH, 76; nBDDH, 81). The cBDDH group demonstrated higher rates of microinstability and LT tears, as well as increased femoral neck, acetabular, and combined anteversion. Logistic regression showed that microinstability ($\beta$ = .027; P < […]

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## Minimum 5-Year Outcomes After Arthroscopic Subspine Decompression in Patients With Femoroacetabular Impingement: A Propensity-Matched Study.

*Kahana-Rojkind AH, Rana K, Barda SR, Quesada-Jimenez R, Flynn ME, Sugarman EP, Domb BG*

BACKGROUND: Subspine impingement (SSI) is a frequently coexisting pathology in patients undergoing hip arthroscopy for femoroacetabular impingement (FAI). While short-term outcomes after arthroscopic subspine decompression (SSD) are promising, midterm outcomes remain unclear.

PURPOSE: To evaluate minimum 5-year patient-reported outcomes (PROs), survivorship, and rates of clinically meaningful improvement after arthroscopic SSD in patients undergoing hip arthroscopy for FAI; and to compare these results to a matched cohort without SSD.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: The data of patients undergoing primary hip arthroscopy for FAI between 2009 and 2019 were retrospectively reviewed. Inclusion criteria included a confirmed diagnosis of FAI and SSI and complete PROs (modified Harris Hip Score, Nonarthritic Hip Score, Hip Outcome Score-Sports-Specific Subscale, International Hip Outcome Tool-12 score, and visual analog scale score) at a minimum of 5 years. Exclusion criteria included prior ipsilateral hip surgery, workers' compensation claims, and Tönnis grade >1. Patients with SSD were matched 1:2 to a control group without SSI. Matching variables included age, sex, body mass index, follow-up duration, lateral center-edge angle, alpha angle, Tönnis grade, acetabular labrum articular disruption grade, and labral treatment. Outcomes were compared between […]

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### Revision Hip Arthroscopy: Propensity Score-Matched Comparison of Labral Reconstruction Versus Repair With Minimum 5-Year Outcomes.

*Quesada-Jimenez R, Cohen MF, Rana K, Kahana-Rojkind AH, Domb BG*

BACKGROUND: There is a paucity of literature comparing the midterm outcomes of arthroscopic labral reconstruction and arthroscopic labral repair in the revision setting.

PURPOSE: To evaluate the minimum 5-year patient-reported outcomes (PROs) of patients who underwent revision hip arthroscopy, comparing labral reconstruction with labral repair.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective analysis was performed for patients who underwent revision hip arthroscopy, either labral repair or labral reconstruction, to treat femoroacetabular impingement and labral tears between 2008 and 2019. Patients who underwent reconstruction were matched 1:1 to patients who underwent repair. Included patients had complete preoperative and minimum 5-year follow-up scores for at least one of the following PRO measures: modified Harris Hip Score, Non-Arthritic Hip Score, Hip Outcome Score-Sports-Specific Subscale, patient satisfaction, and visual analog scale for pain. Rates of achieving the minimal clinically important difference and patient acceptable symptom state as well as secondary surgery rates were compared between groups.

RESULTS: A total of 68 patients were included, with 34 patients who underwent labral reconstruction successfully matched to 34 patients who underwent labral repair, controlling for age, body mass index, sex, acetabular Outerbridge grade, a […]

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### Has the Protective Effect of Hormonal Contraception Been Overestimated? A Systematic Review and Meta-analysis of Oral Contraceptive Pills and ACL Injury.

*Paska J, Watson SL, Larrabee GA, Ndjonko LCM, Tjong VK*

BACKGROUND: Hormonal contraception (HC), including oral contraceptive pills (OCPs) and long-acting reversible contraceptives, is widely used by reproductive-aged women for various medical indications. Emerging evidence suggests a possible association between HC use and reduced anterior cruciate ligament (ACL) injury risk; however, no meta-analysis has systematically evaluated this relationship.

PURPOSE: To determine whether HC use reduces the risk of ACL injury in females through a systematic review and meta-analysis.

STUDY DESIGN: Meta-analysis; Level of evidence, 3.

METHODS: A systematic search of the PubMed, OVID (MEDLINE), and Embase databases was conducted from inception through March 2025 following PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Studies were included if they reported ACL injury incidence among both HC users and nonusers. Data were synthesized using a random-effects meta-analysis to calculate a pooled risk ratio. Publication bias was evaluated using funnel plots and Egger regression.

RESULTS: Eight studies consisting of 18,103,890 participants met inclusion criteria. All studies evaluated OCPs, and 3 included other forms of HC. The pooled risk ratio for ACL injury among HC users versus nonusers was 0.88 (95% CI, 0.77-1.01). Heterogeneity was high ($I2$ = 97.3%), and Egger regression showed no significant publicatio […]

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## Surgical Timing and Outcomes in Medial Meniscus Posterior Root Tear Repair: A Systematic Review.

*Moews LD, Thamrongskulsiri N, Morgan JT, Vega TF, Perez Lloveras GO, Casanova F, Verma NN, Chahla J*

BACKGROUND: Medial meniscus posterior root tears (MMPRTs) compromise hoop stress transmission and tibiofemoral load distribution, accelerating degenerative changes comparable with total meniscectomy. While early repair has been recommended, many patients present late because of a delayed diagnosis or an insidious onset of symptoms. The effect of delayed surgical management on outcomes remains unclear.

PURPOSE: To systematically review the literature comparing clinical outcomes, radiological progression, and complication rates of MMPRT repair performed at <3 months, 3 to 6 months, and >6 months after the injury.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: A systematic search of PubMed, Embase, and Scopus was conducted in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines and registered with PROSPERO. A total of 11 studies (366 patients) reporting isolated transtibial pullout repair with a documented time from injury to surgery were included. Extracted data included patient-reported outcome scores (International Knee Documentation Committee [IKDC], Lysholm), osteoarthritis progression by the Kellgren-Lawrence grade, meniscal extrusion, healing status, and surgical failures or complications.

RESULTS: Across all surgical timing groups, MMPRT repair resulted in significant functional improvements. Patients t […]

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## Hip Arthroscopy in Patients With Acetabular Dysplasia: A Systematic Review of Clinical Outcomes at Long-term Follow-up.

*Sparks CA, Monty TL, Brinkman JC, Barton KI, Nho SJ*

BACKGROUND: Hip arthroscopy has shown satisfactory outcomes for the treatment of intra-articular pathology and femoroacetabular impingement syndrome (FAIS) in the setting of acetabular dysplasia at short- and mid-term follow-up. However, the long-term outcomes remain less understood.

PURPOSE: To systematically review the literature to determine patient-reported outcomes (PROs), clinically significant outcome (CSO) achievement, reoperation rates, and predictors of failure in patients with acetabular dysplasia undergoing hip arthroscopy at long-term follow-up.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: Three databases were searched around the following terms: hip arthroscopy, dysplasia, and long-term follow-up. Articles available in English, presenting original data, and reporting on a mean follow-up of ≥10 years after primary hip arthroscopy using either PROs, CSOs, or conversion to total hip arthroplasty (THA) and/or revision surgery were included. Quality assessment was completed using Methodological Index for Non-Randomized Studies (MINORS) assessment.

RESULTS: Five studies (555 hips) were included in the systematic review: patients had a mean age of 30 to 41 years, 13% to 88% of patients were female, and follow-up ranged from 9.6 to 12 years. MINORS assessment ranged from 11 to 21. All studies included PROs and reported statistically significant impr […]

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## Psychiatric Diagnoses and Outcomes After Hip Arthroscopy: A Systematic Review and Meta-analysis.

*Villegas Meza AD, Nocek M, Morales Lugo B, Mitchell BC, Philippon MJ*

BACKGROUND: Psychiatric comorbidity-most commonly depression and anxiety-is frequently reported among patients undergoing hip arthroscopy and may influence postoperative trajectories, but existing analyses vary in scope and methods, limiting clinical interpretation.

PURPOSE: To determine whether recorded psychiatric diagnoses are associated with worse surgical, patient-reported, and health care utilization outcomes after hip arthroscopy.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 3.

METHODS: The authors conducted a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)-compliant search (International Prospective Register of Systematic Reviews [PROSPERO] CRD420251101816) of the MEDLINE, Embase, Web of Science, and PsycINFO databases to identify studies comparing postoperative outcomes in patients undergoing hip arthroscopy with and without recorded psychiatric diagnoses. Two reviewers independently screened, extracted data, and assessed the risk of bias using the Risk of Bias in Non-randomized Studies of Interventions tool. Primary endpoints were arthroscopy failure, revision, conversion to total hip arthroplasty (THA), and readmission. Secondary endpoints included the modified Harris Hip Score (mHHS), International Hip Outcome Tool-12 (iHOT-12) score, visual analog scale (VAS) pain score, and costs. Random-effects meta-analyses […]

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## Platelet Concentration Does Not Influence Clinical Efficacy and Retear Rates of Rotator Cuff Repair With Platelet-Rich Plasma: A Systematic Review and Meta-analysis With Meta-Regression.

*Lim JJ, Stepaniuk EA, Belk JW, Keeter C, Oeding JF, Dragoo JL, McCarty EC, Frank RM*

BACKGROUND: Platelet-rich plasma (PRP) is emerging as a popular augmentation technique in the context of rotator cuff repair (RCR) to strengthen the repair construct and promote healing. Previous studies examining the efficacy of PRP augmentation during RCR have described heterogeneous results.

PURPOSE: To determine whether platelet concentration of a PRP injection as an adjunct of RCR impacts clinical outcomes and retear rates.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 2.

METHODS: A systematic review was performed by searching the PubMed, Cochrane Library, and Embase databases to identify level 1 and 2 studies that evaluated the clinical efficacy of RCR augmented with PRP. The search phrase used was "rotator cuff repair AND (PRP OR platelet rich plasma OR platelet-rich plasma)." Included randomized controlled trials were classified as utilizing high-dose or low-dose PRP based on their platelet concentration factor as a >4-fold increase over whole blood. The outcomes evaluated were the standardized mean differences of visual analog scale scores, Constant-Murley scores, American Shoulder and Elbow Surgeons scores, and University of California-Los Angeles scores, and pooled odds ratios of retear rates. Mixed-effects and random-effects meta-analyses were performed along with meta-regression to evaluate relationships in patient outcomes associated wit […]

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## Episode Count Should Not Substitute for Anatomy in Arthroscopic Bankart Repair: Response.

*Lin RT, Gilbert R, Lin A*

PubMed · DOI

## Time After ACL Reconstruction: Degenerative Clock or Vulnerable Knee Exposure? Response.

*Agostinone P, Romandini I, Rotzius P, Zaffagnini S, Forssblad M, Sandon A*

PubMed · DOI

## Episode Count Should Not Substitute for Anatomy in Arthroscopic Bankart Repair: Letter to the Editor.

*Qu X, Li D, Tang X, Chen B*

PubMed · DOI

## Time After ACL Reconstruction: Degenerative Clock or Vulnerable Knee Exposure? Letter to the Editor.

*Tang X, Zhang J, Yu B*

PubMed · DOI

# BJJ The Bone & Joint Journal

Volume 108-B, Issue 8 · August 2026 · 13 new articles

## Beyond BMI cut-offs : optimizing risk, not excluding patients.

*Alvand A, Rudran B, Haddad FS*

PubMed · DOI

## Clinical efficacy to climate accountability : the case for embedding environmental outcomes in orthopaedic trials.

*Farrow L, Purdie S, Evans JT, Krause T, Campbell M*

There is increasing global recognition of the importance of sustainability in the design and delivery of healthcare, including trauma and orthopaedics. There has, however, to date been little focus on providing high-quality research which includes environmental outcomes to guide clinical practice. This annotation explores the importance of including sustainability outcomes using three key carbon-related domains, and highlights the urgent need for the incorporation of sustainability outcomes into future clinical trials.

PubMed · DOI

## Periprosthetic joint infection : lessons from the INFORM research programme.

*Moore AJ, Beswick AD, Whitehouse MR, Blom AW*

Infection is one of the most devastating complications following arthroplasty. The National Institute for Health and Care Research (NIHR) funded the multinational INFection ORthopaedic Management (INFORM) research programme to study this problem comprehensively, and then funded INFORM: Evidence into Practice to translate findings into the INFORM guidelines. This annotation outlines the key findings with particular emphasis on hip and knee arthroplasty, including risk factors, impact, diagnosis, and treatment, and suggests guidelines underpinned by high-quality evidence from the programme. Finally, future directions for research are discussed.

PubMed · DOI

## Glucagon-like peptide-1 receptor agonists are associated with improved 90-day outcomes after total hip and knee arthroplasty : a systematic review and meta-analysis.

*Rudran B, Little CDJ, Palmer AJR, Judge A, Carli A, Price AJ, Alvand A*

AIMS: Obesity and diabetes are risk factors for developing osteoarthritis and associated with worse outcomes following total hip (THA) and total knee arthroplasty (TKA). Glucagon-like peptide-1 receptor agonists (GLP-1as) improve glycaemic control and promote weight loss. The aim of this study was to assess the impact of GLP-1a treatment on 90-day surgical complications following THA and TKA.

METHODS: A systematic review was undertaken according to PRISMA guidelines. Studies were included if they reported at least one outcome of interest following primary THA or TKA. In all the studies which were included, a GLP-1a was prescribed for the management of type 2 diabetes mellitus or for weight loss in obese patients. The primary outcome was the 90-day surgical complication rate in the combined THA and TKA cohort. The 90-day surgical complications were defined as: periprosthetic joint infection (PJI), wound dehiscence, periprosthetic fracture (PPF), haematoma, nerve injury, or surgical site infection (SSI). Secondary outcomes included 90-day medical complications and readmission rates, all-cause revision, healthcare costs, and length of stay.

RESULTS: A total of 78 studies were reviewed, and ten matched cohort studies with a total of 96,356 patients (30,350 THAs and 66,606 TKAs) were included. The mean age of the patients was 61.9 years; 60.1% female (n =

57,939). Two studies had […]

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### Autologous impaction bone grafting of a cemented acetabular component in primary total hip arthroplasty in young patients: a randomized controlled trial.

*de Boer DR, Pasman P, Munnik-Hagewoud R, Steinweg MJQ, Edens MA, van Driel PBAA, Ettema HB*

AIMS: This randomized controlled trial (RCT) aimed to compare clinical and radiological outcomes of cemented acetabular component in primary total hip arthroplasty (THA) in patients aged under 60 years with and without autologous impaction bone grafting (IBG).

METHODS: A single-centre, triple-blinded RCT was conducted in patients aged 18 to 59 years undergoing primary THA without acetabular defects. Patients were randomized to receive either IBG or no IBG. Noninferiority of IBG was assessed for the primary endpoint, the Hip disability and Osteoarthritis Outcome Score (HOOS) sub-scale for activities of daily living (ADL) at one-year follow-up. Secondary outcomes included other patient-reported outcome measures (PROMs), perioperative outcomes, complications, and radiological outcomes. Radiographs assessed acetabular component and centre of rotation positioning, offset changes, and the presence of radiolucent lines in the bone-cement interface.

RESULTS: A total of 131 patients were randomized. No significant group differences were found for the HOOS-ADL sub-scale at one year (mean difference 1.9 points (95% CI -4.85 to 8.7); $p = 0.578$), meeting the criteria for noninferiority. Furthermore, there was no effect of group on the scores for other PROMs. Surgery duration (+ 7.5 minutes; $p < 0.001$) and blood loss (+ 100 ml; $p = 0.001$) were significantly higher in the IBG group. Complic […]

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### Increased risk of early failure of multiply revised total hip arthroplasty.

*D'Apollonio TM, Hopman MLT, Sullivan T, Abrahams JM, Sheth NP, Solomon LB, Callary SA*

AIMS: As the demand for revision total hip arthroplasty (rTHA) grows, understanding how successive revision procedures influence implant survivorship and patient outcomes becomes increasingly important. This study aimed to evaluate the association between revision number and implant survivorship following aseptic rTHA.

METHODS: We analyzed 845 aseptic rTHAs performed between January 2003 and December 2023. Index primary THA dates, rTHA sequence, and reoperations were sourced and verified using hospital records, national registry data, and radiological records. Cases were stratified by revision number: first (R1), second (R2), or third and beyond (R3+). Re-revision risk was evaluated using cumulative incidence functions and Fine-Gray competing-risk regression, with death treated as a competing event.

RESULTS: Of 845 rTHAs (717 patients), 838 (710 patients) with confirmed index THA were included: 598 R1, 155 R2, and 85 R3+. At two years, cumulative incidence of re-revision was 10.2% (95% CI 8.0 to 12.9) for R1, 14.2% (95% CI 9.6 to 20.8) for R2, and 17.7% (95% CI 11.0 to 27.7) for R3+. Unadjusted competing-risk regression demonstrated a higher cumulative incidence of re-revision in R2s compared with R1s. However, after multivariable adjustment, revision number was not independently associated with a statistically significant increase in re-revision risk. Across all revision gro […]

PubMed · DOI

### Preoperative predictors of patient-reported outcomes after joint-preserving forefoot surgery for rheumatoid arthritis : a case series.

*Etani Y, Noguchi T, Hirao M, Okamura G, Sugimoto A, Okada S, Nakata K, Ebina K*

AIMS: Forefoot deformities are common in patients with rheumatoid arthritis (RA) and often impair quality of life. Patient-reported outcome measures (PROMs), including the Self-Administered Foot Evaluation Questionnaire (SAFE-Q), are increasingly used to assess surgical outcomes; however, the preoperative conditions needed to achieve favourable postoperative PROMs remain unclear. This study aimed to identify factors associated with postoperative SAFE-Q scores (mean score across five sub-scales, maximum 100) and determined preoperative cutoff values predictive of achieving predefined postoperative SAFE-Q thresholds (≥ 70 or ≥ 90).

METHODS: In this multicentre, retrospective case series, 101 feet in 80 patients (including 21 bilateral cases) with RA who underwent forefoot surgery between July 2015 and July 2023 were analyzed (mean age at surgery 65.8 years (SD 10.7); 79 female patients). Inclusion criteria were symptomatic forefoot deformities with available SAFE-Q data pre-surgery and at ≥ one year post-surgery. Procedures included modified scarf osteotomy and offset shortening osteotomies for the metatarsals. SAFE-Q scores, Japanese Society for Surgery of the Foot scores, radiological parameters, and disease activity were evaluated pre- and postoperatively. Multiple regression identified preoperative factors associated with postoperative SAFE-Q scores, and significant variable […]

PubMed · DOI

### Non-medical surgical delay increases one-year mortality following hip fracture.

*Busby C, Nightingale J, Deacon C, Deakin D, Valdes A, Ollivere BJ, Moran C*

AIMS: This study aimed to assess the association between surgical delay due to non-medical reasons and one-year mortality in patients with hip fracture, and to explore whether a dose-response relationship exists between increasing time to surgery and mortality risk.

METHODS: A retrospective cohort analysis was performed using prospectively collected data from the Nottingham Hip Fracture Database. A total of 5,414 patients aged ≥ 60 years admitted with a fragility femoral fracture to a major trauma centre between April 2016 and March 2023 were reviewed. Time to theatre was measured from emergency department arrival to anaesthetic start. Delays beyond 36 hours were categorized as either medical or logistical. The primary outcome was all-cause one-year mortality with a secondary outcome being length of stay (LOS). Multivariable Cox proportional hazards models were used to assess associations, adjusting for age, sex, comorbidity, and cognition.

RESULTS: Of the cohort, 38.5% (n = 2,083) experienced surgical delay beyond 36 hours; 27.5% (n = 1,490) due to limited surgical capacity and 11.0% (n = 593) due to medical unfitness. After adjusting for age, sex, comorbidity, and cognition, delay due to logistical factors was associated with a 20% increase in one-year mortality (hazard ratio (HR) 1.20 (95% CI 1.06 to 1.36); p = 0.004). A secondary analysis excluding medically unfit patient […]

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### Use of neurolytic nerve blocks and mortality in nonoperative management of hip fracture patients : a nationwide population study in the Netherlands.

*van Bremen HE, Lommerse MI, Hegeman JH, Schuijt HJ, Kallewaard JW, Wijnen HH, Seppala LJ, van der Velde N et al.*

AIMS: Nonoperative management may be suitable for very frail hip fracture patients; in these instances, neurolytic nerve blocks with phenol could act as a long-term analgesic, with potentially fewer disadvantages compared with systemic analgesia. While neurolytic nerve blocks are being used more frequently in end-of-life care, there are limited data on how often they are used, and what long-term outcomes they achieve in patients with hip fractures. This study aims to evaluate the association between neurolytic nerve block use and mortality in patients with hip fractures managed nonoperatively, while also exploring the characteristics of patients who receive these blocks.

METHODS: A retrospective population-based cohort study was conducted including patients with a hip fracture aged 70 years and older, who were treated nonoperatively between 1 January 2024 and 31 December 2024. The primary outcome was mortality, comparing patients who received a neurolytic block with those who did not.

RESULTS: Of 1,415 nonoperatively managed patients (mean age 86 years (SD 6.75)), 633 (45%) received neurolytic nerve blocks. These patients were more frail, had greater levels of dependence, malnourishment, and immobility, and were diagnosed with a displaced fracture more often than those who did not receive a block. Nerve block patients also showed significantly lower mortality at seven days (2 […]

PubMed · DOI

### A matter of definition : variation in treatment success of fracture-related infections.

*Reinert N, Poggioli A, Metsemakers WJ, Clauss M, Kuehl R, FRI-Basel Research Group, Morgenstern M, FRI-Basel research Group et al.*

AIMS: The establishment of the fracture-related infection (FRI) consensus definition and the FRI classification system provide a foundation for research and the development of clinical recommendations. To date, however, the evaluation of treatment success is still impeded by a lack of clear and standardized outcome definitions. We hypothesized that applying different outcome measurements to the same cohort of FRI patients results in significantly different treatment success rates.

METHODS: Adult patients undergoing curative surgery for FRI at the Center of Musculoskeletal Infections, University Hospital Basel, between 1 January 2020 and 31 December 2023 were included in this prospective cohort study. Treatment success was evaluated by review of patient charts; after one year of follow-up according to seven predefined outcome definitions (S1 to S7): no infection persistence at one year (S1); no recurrence within one year (with the same (S2) or a new (S3) pathogen); bony union (S4); no infection recurrence, or persistence and bony union (S5); no unplanned revision surgery (S6); and bony union and no infection recurrence, or persistence and no unplanned revision surgery (S7).

RESULTS: In total, 106 patients with 112 confirmed FRIs were included. At one year, treatment success varied significantly ($p < 0.001$) between the different outcome definitions: for S1 the success rate was […]

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## Reducing complications after resection of pelvic bone sarcomas : an aggregation of marginal gains.

*Jeys LM, Kurisunkal V, Parry MC, Morris G, Stevenson JD, Laitinen MK*

AIMS: Primary pelvic bone sarcomas are rare and their treatment remains complex, with reconstruction after resection remaining controversial due to high complication and revision rates. Both prosthetic and non-prosthetic reconstructions carry risks including infection, implant failure, wound problems, and reoperation. The aim of this study was to review the changes in the management of these patients over the 20-year period between 2003 and 2023.

METHODS: This was a retrospective cohort study including 479 patients who underwent surgery for a primary pelvic sarcoma during this time, divided into a historical (2003 to 2012) and a modern (2013 to 2023) group. Postoperative complications and the rates of return to theatre (RTT) were compared, and factors associated with improved outcomes were analyzed.

RESULTS: Early RTT within six weeks decreased significantly from 22% in the historical cohort ($n = 38$) to 14% in the modern cohort ($n = 42$; $p = 0.021$). Historically, the ilioinguinal incision was associated with a 44% early rate of RTT ($n = 8$), compared with 10% when using the question mark incision in the modern cohort ($n = 10$). In the modern cohort, the rate of early RTT was similar between those with prosthetic (17%; $n = 13$) and non-prosthetic reconstructions (13%; $n = 29$) ($p = 0.344$), although the long-term rate of RTT remained significantly higher in the prosthetic group (p = […]

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## Natural history of solitary osteochondroma : a retrospective analysis of 236 patients in South Korea.

*Sung MJ, Kim CH, Jung ST, Kim S*

AIMS: The aim of this study was to analyze the natural history of solitary osteochondroma and determine how skeletal maturity influences tumour behaviour and treatment outcomes.

METHODS: A retrospective analysis of 236 patients with solitary osteochondroma (151 skeletally immature, 85 skeletally mature) was undertaken. Skeletal maturity was defined as bone age ≥ 18 years in males and ≥ 17 years in females. Tumour characteristics, disease progression, indications for surgery and recurrence rates were evaluated. Interobserver reliability for bone age assessment was excellent (intraclass correlation coefficient (ICC) 0.92 (95% CI 0.88 to 0.95)).

RESULTS: Spontaneous regression occurred exclusively in skeletally immature patients (17.2% ($n = 26$) vs 0%; $p < 0.001$). Skeletally immature patients predominantly presented with sessile lesions (76.2%; $n = 115$), had lower surgical needs (31.8%; $n = 48$), but higher progression rates (9.9% ($n = 15$) vs 1.2% ($n = 1$); $p = 0.013$). Skeletally mature patients showed more stable disease but required surgery more often (54.1%; $n = 46$). Among skeletally immature patients with regression, independent predictors were younger age at diagnosis (odds ratio (OR) 0.82 (95% CI 0.73 to 0.92); $p < 0.001$), proximal humeral location (OR 10.59 (95% CI 2.01 to 55.85); $p = 0.005$), and humeral shaft location (OR 38.93 (95% CI 4.81 to 315.10); $p = 0.001$). The optim […]

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### Cost-effectiveness of screening for developmental dysplasia of the hip in children born in England and Wales.

*Plumpton CO, Perry DC*

AIMS: The aim of this study was to evaluate the cost-effectiveness of screening programmes for identifying developmental hip dysplasia (DDH) in children born in England and Wales.

METHODS: The costs and quality-adjusted life-years (QALYs) associated with DDH screening programmes were estimated using a decision-analytic model. Six programmes were modelled from an NHS perspective: 1) Current practice - 'selective' screening: clinical examination with later ultrasound screening (USS) for those with an abnormal examination or specific risk factors (breech birth, first degree family history of DDH); 2) No hip screening: no clinical examination or risk factor assessment; 3) Clinical examination: treatment decisions based solely upon a positive Barlow or Ortolani manoeuvre; 4) Universal USS at birth: USS in the neonatal period in addition to a postnatal check; 5) Universal USS at six weeks: USS in addition to the six-week postnatal check; 6) Sex-stratified screening: clinical examination with selective USS for males and universal USS in the neonatal period for females. The incremental net monetary benefit (INMB) was estimated at a cost-effectiveness threshold of £20,000 per QALY gained. Whether the result was robust to assumptions was assessed using analyses of uncertainty. Secondary outcomes included an estimate of the value of reducing the uncertainty of the decisions.

RESULTS: Un […]

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## Clinical Orthopaedics & Related Research

Volume 484, Issue 8 · August 2026 · 38 new articles

### A Conversation With ... Malcolm Willett, Cartoonist Who Looks at Us With a Wry Smile.

*Leopold SS*

PubMed · DOI

### Editor's Spotlight/Take 5: How Does Traumatic Hand Injury Impact Patients in a Safety-net Healthcare System? A Mixed Methods Study.

*Wongworawat MD*

PubMed · DOI

### How Does Traumatic Hand Injury Impact Patients in a Safety-net Healthcare System? A Mixed Methods Study.

*Higashi RT, Lee J, Hernandez Y, Pontón M, Liao JM, Cross KC, Billig JI*

BACKGROUND: Traumatic hand injuries can result in substantial harms to patients' functional abilities and financial health, disproportionately affecting individuals who are of working age, low income, and uninsured. However, little is known about how patients and their families cope with the functional limitations and economic consequences that follow, and how these injuries affect patients' mental and emotional health. Exploring clinician and staff perceptions about the impact of traumatic hand injuries on patients may provide insight into how to help address the various burdens patients face.

QUESTIONS/PURPOSES: In this paper, we aimed to explore the following questions: (1) What is the impact of a traumatic hand injury on patients' financial health? (2) How does a traumatic hand injury affect patients' physical health and disability? (3) What kind of toll does a traumatic hand injury take on patients' emotional and mental health? (4) What health system challenges function as additional stressors after a traumatic hand injury?

METHODS: This descriptive study consisted of surveys and semistructured interviews. We recruited patients from an outpatient hand surgery clinic at a safety-net hospital in a southern US city. Eligible patients were English- or Spanish-speaking, age 18 years or older, who presented to the clinic after a traumatic hand injury. Between January and April […]

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## Clinical Faceoff: Social Media for Orthopaedic Surgeons-Tool or Vice?

*Fourman MS, Daniels AH, Herron PD*

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## Behind the Mask: Setting Expectations.

*Zhang SE*

PubMed · DOI

## Value-based Healthcare: Moving From PROMs as Population Benchmarks to Clinical Tools for Patient Goal Setting and Shared Decision-making.

*Ahlfinger Z, Bozic KJ*

PubMed · DOI

## Virtue Ethics in a Value-driven World: Soliciting Donations From Grateful Patients.

*Humbyrd CJ*

PubMed · DOI

## Not the Last Word: Seers, Soothsayers, and Surgeons.

*Bernstein J*

PubMed · DOI

## Editorial Comment: Selected Proceedings From the 2025 Musculoskeletal Infection Society Meeting.

*Zalavras CG*

PubMed · DOI

## Can We Validate the Musculoskeletal Infection Society Criteria in Patients Suspected of Infection After Transfemoral Osseointegration Surgery?

*Bozzay AB, Rivera JA, Rabin SE, Souza JM, Potter BK, Forsberg JA*

BACKGROUND: Major complications that may result in the removal of osseointegrated implants in patients with limb loss include peri-implant fracture, severe bone resorption, and infection. It remains challenging to diagnose and manage peri-implant infection in patients who have undergone osseointegration surgery. The Musculoskeletal Infection Society (MSIS) developed criteria to identify periprosthetic joint infection (PJI) after total joint arthroplasty during the 2018 MSIS/International Consensus Meeting; however, there is no such validated system for patients who have received osseointegrated implants, and some of the measures on the MSIS/International Consensus Meeting criteria are not applicable to osseointegrated implants. Nevertheless, establishing an adapted version of the MSIS criteria, or a comparable standardized system, would be valuable for enabling consistent diagnosis of osseointegration-associated peri-implant infections.

QUESTIONS/PURPOSES: (1) Is there a difference in the serum parameters of the 2018 MSIS criteria between a cohort of patients without infection and a cohort with an infection? (2) Can the serum parameters of the 2018 MSIS criteria be validated in a population that has undergone osseointegration? (3) Can we develop a new, valid set of criteria to gauge the likelihood of infection from serum parameters for those who have undergone osseointegration […]

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## CORR Insights®: Can We Validate the Musculoskeletal Infection Society Criteria in Patients Suspected of Infection After Transfemoral Osseointegration Surgery?

*Allison DC*

PubMed · DOI

### Synthesis and Characterization of an Antimicrobial Honey-based Composite Bone Cement.

*Kemp LE, Delgado-Alvarado I, Rossiter TE, Chen AF, Tower RJ, Tatara AM*

BACKGROUND: Orthopaedic infections are a major cause of morbidity and mortality with increasing prevalence. Antibiotics are often added to bone cement or polymethyl methacrylate (PMMA) for local delivery to prevent or treat infection. However, PMMA is not inherently antimicrobial and has poor antibiotic elution kinetics. A biomaterial composite of PMMA and antimicrobial medical-grade honey (PMMA-H) could address some of these shortcomings of current bone cements to better inhibit bacteria.

QUESTIONS/PURPOSES: (1) How does honey affect the architecture and static mechanical properties of bone cement? (2) How does honey affect antibiotic elution? (3) What is the in vitro antimicrobial activity of PMMA-H with and without antibiotics?

METHODS: Three different formulations of bone cements were created by mixing a generic dental bone cement without honey (0 wt%), with a low amount of medical-grade manuka honey (15 wt%), or with a high amount of honey (30 wt%). Vancomycin or gentamicin were added to the three formulations at a typical cement loading dose (3 wt%) or no additional antibiotic for a total of nine formulations. We characterized the subsequent total porosity and percentage of open porosity of constructs synthesized from the nine formulations using microcomputed tomography. Both the compressive and bending yield strength and modulus were measured by mechanical testing. Con […]

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### CORR Insights®: Synthesis and Characterization of an Antimicrobial Honey-based Composite Bone Cement.

*DeBaun MR*

PubMed · DOI

### What Is the Fate of Retained Antibiotic Spacers After First-stage Revision for Periprosthetic Joint Infection?

*Sarfraz A, Khury F, McCormick K, Aziz HH, Koljaka S, Rozell JC, Schwarzkopf R, Aggarwal VK*

BACKGROUND: Prolonged retention of antibiotic-loaded articulating spacers after the first stage of a two-stage revision for periprosthetic joint infection (PJI) can occur because of patient preference, surgeon preference, or medical reasons that prevent the planned second stage. Little is known about the frequency of persistent infections, mechanical complications, and functional results in patients with retained spacers.

QUESTIONS/PURPOSES: At a minimum follow-up of 2 years after spacer placement, among patients who do not undergo the second stage revision (replacement of the spacer with a definitive prosthesis of the hip or knee): (1) What was the survival of the spacer free from unplanned reoperation or removal, the cumulative incidence of symptomatic infection, and the overall (Kaplan-Meier) survivorship of the patients? (2) What is the cumulative incidence of mechanical complications (spacer fracture or dislocation)? (3) What is the ambulatory status of patients who have retained their spacers?

METHODS: Between March 2011 and July 2023, a total of 111 and 152 patients underwent first-stage revision with an articulating spacer placement as part of a planned two-stage procedure for chronic PJI after THA and TKA at our institution, respectively. Of these, 21% (23 of 111) in the THA group and 24% (37 of 152) in the TKA group did not undergo the anticipated second-stage reimp […]

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### CORR Insights®: What Is the Fate of Retained Antibiotic Spacers After First-stage Revision for Periprosthetic Joint Infection?

*Hartman CW*

PubMed · DOI

### Can a Targeted Dissemination Strategy of the PREPARE Trial Results Change Surgical Antiseptic Practice?

*Nesbit SE, Hong I, O'Hara NN, Sprague S, Slobogean GP, Levack AE*

BACKGROUND: The PREPARE trial (Pragmatic Randomized trial Evaluating Pre-operative Alcohol skin solutions in Fractured Extremities), published in February 2024, demonstrated that 0.7% iodine povacrylex in 74% isopropyl alcohol reduced surgical site infections in patients with closed fractures compared with chlorhexidine gluconate in 70% isopropyl alcohol. Prior studies have shown that the translation of clinical trial findings into routine clinical practice and surgical procedures frequently takes longer than a decade, and strategies to accelerate implementation in orthopaedic surgery remain poorly defined. Engaging in a multifaceted implementation strategy with minimal barriers to entry for providers may be able to reduce this time to widespread adoption.

QUESTIONS/PURPOSES: (1) Can the low-cost single-center dissemination efforts of the PREPARE trial results change antiseptic usage? (2) If a true practice change occurs, can the change be sustained over time without additional dissemination efforts? (3) Does the adoption of new evidence vary between fellowship-trained orthopaedic trauma surgeons and orthopaedic trauma surgeons who are not fellowship-trained? (4) What is the economic impact of a practice change to iodine povacrylex on hospital purchasing costs?

METHODS: This was a retrospective, historically controlled comparative study of patients who underwent operative tre […]

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### CORR Insights®: Can a Targeted Dissemination Strategy of the PREPARE Trial Results Change Surgical Antiseptic Practice?

*Ring DC*

PubMed · DOI

### Do Published Orthopaedic RCTs Match Their Registered Protocols? A 6-year Analysis of Leading Orthopaedic Journals.

*Poursalehian M, Sahebi M, Tajvidi M, Sabaghian A, Asgari AM, Tabaie SA, Hoveidaei AH*

BACKGROUND: RCTs are a key block in the evidence pyramid, but their quality relies on detailed, consistent reporting, and one best-practice standard is prospective registration. Prospective trial registration was intended to reduce publication bias, and adherence to a prespecified protocol helps limit bias from selective reporting; any protocol or end point changes should be transparently documented and justified. However, the degree to which articles published in the leading journals on orthopaedic surgery comply with this best-practice standard has, to our knowledge, not been evaluated.

QUESTIONS/PURPOSES: (1) Do RCTs published in leading, general-interest orthopaedic surgery journals comply with best practices regarding prospective clinical trial registration? (2) Do major discrepancies exist between registered protocols and published orthopaedic RCTs? (3) Are there specific study types that are more likely to demonstrate discrepancies?

METHODS: A review was performed on RCTs published in the top five general-interest orthopaedic surgery journals, based on the 2022 scientific journal rankings (from SciMago): Journal of Bone and Joint Surgery , Bone and Joint Journal , Clinical Orthopaedics and Related Research ®, Journal of the American Academy of Orthopaedic Surgeons (JAAOS) , and Acta Orthopaedica . During the study period, all journals maintained editorial policies requ […]

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### CORR Insights®: Do Published Orthopaedic RCTs Match Their Registered Protocols? A 6-year Analysis of Leading Orthopaedic Journals.

*Woo EJ*

PubMed · DOI

## Do Personalized Recovery Trajectories Enhance the Utility of Patient-reported Outcome Measures After Orthopaedic Surgery?

*Graber J, Jin X, Juarez-Colunga E, Dibblee P, Woodfield D, Minick KI, Chamberlain AM, Bade M et al.*

BACKGROUND: Patient-reported outcome measures (PROMs) could be useful tools for supporting patient care after orthopaedic surgery, but they can be difficult to interpret and apply in practice. Existing methods for PROM interpretation, such as meaningful change values and population norms, have practical limitations and are unresponsive to patient-specific factors that influence recovery. Personalized recovery trajectories could help overcome these limitations, making PROM scores more clinically meaningful and actionable.

QUESTIONS/PURPOSES: (1) Does a "people-like-me" modeling approach to crafting personalized recovery trajectories provide accurate and precise trajectories of PROM recovery early after total shoulder arthroplasty (TSA)? (2) What advantages do personalized recovery trajectories provide over alternative methods for PROM interpretation (minimum detectable change, minimum clinically important difference, and normative values) when applied to individual patients?

METHODS: We used data from 28 outpatient physical therapy clinics within a single integrated health system from 2017 to 2022. We included patients who underwent unilateral anatomic or reverse TSA and had postoperative data available from the DASH questionnaire. We excluded patients who were missing key modeling variables, who had fewer than three postoperative DASH assessments, or who lacked a DASH assessm […]

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## CORR Insights®: Do Personalized Recovery Trajectories Enhance the Utility of Patient-reported Outcome Measures After Orthopaedic Surgery?

*Navarro RA*

PubMed · DOI

## Does Liposomal Bupivacaine Provide Superior Postoperative Analgesia Compared With Ropivacaine With Dexamethasone in Suprainguinal Fascia Iliaca Compartment Block for THA? A Randomized Controlled Trial.

*Wu W, Hu X, Yang C, Sun Y, Li J, Wang L*

BACKGROUND: Choosing the most effective pain management is important for enhancing functional recovery after surgery. The fascia iliaca compartment block (FICB) is recommended for analgesia after THA, but whether liposomal bupivacaine is more effective for this application remains uncertain.

QUESTIONS/PURPOSES: (1) Did a single-injection suprainguinal FICB using liposomal bupivacaine provide superior pain control by a clinically important margin? (2) What was the effect of liposomal bupivacaine on perioperative opioid consumption? (3) How did liposomal bupivacaine affect the duration of the sensory block and quadriceps motor weakness in the operated limb?

METHODS: This was a prospective, parallel-group, assessor-blinded, single-center, randomized controlled clinical trial. Patients who underwent unilateral primary THA via the posterolateral approach were included in the study. Exclusion criteria included a BMI of > 32.5 kg/m 2 (which is the diagnostic criterion for moderate obesity in China), contraindications to peripheral nerve block, known allergy to study medications, psychiatric or cognitive disorders that could interfere with pain assessment, severe hepatic or renal dysfunction, and chronic pain necessitating long-term opioid therapy. According to these criteria, 76% (60 of 79) of screened patients met eligibility requirements and were subsequently randomized into eithe […]

PubMed · DOI

## CORR Insights®: Does Liposomal Bupivacaine Provide Superior Postoperative Analgesia Compared With Ropivacaine With Dexamethasone in Suprainguinal Fascia Iliaca Compartment Block for THA? A Randomized Controlled Trial.

*Sadeghirad B*

PubMed · DOI

## Is Functional Reconstruction Feasible With Modified Hip Transposition Using a Customized 3D-printed Femoral Prosthesis After Pelvic Tumor Resection? A Preliminary Study.

*Wang H, Shen J, Zuo D, Liu K, Sun W*

BACKGROUND: Functional reconstruction after resection of malignant pelvic tumors involving zones I + II ± IV remains a major challenge in orthopaedic oncology. Conventional hip transposition can reduce prosthesis-related complications, but it is often associated with limb shortening and femoral head malrotation. We propose a modified hip transposition technique-femoral lengthening and retroversion hip transposition with a customized three-dimensionally (3D) printed femoral prosthesis-to address deficiencies in limb length, femoral head positioning, and fixation stability, and we evaluate its effectiveness in reducing complications and improving functional outcomes.

QUESTIONS/PURPOSES: In the context of a small, initial patient series, we asked: (1) What was the postoperative functional outcome, as assessed by Musculoskeletal Tumor Society 1993 (MSTS-93) score, after reconstruction using this technique? (2) What were the frequency and nature of complications associated with the method? (3) How well was limb length restored at a minimum follow-up of 3 years?

METHODS: Between January 2019 and December 2021, a total of 49 patients underwent resection and reconstruction for pelvic zone I + II ± IV tumors. Of these, 18% (9) received the modified hip transposition with a customized 3D-printed femoral lengthening and retroversion prosthesis. This approach was selected for patients in […]

PubMed · DOI

## CORR Insights®: Is Functional Reconstruction Feasible With Modified Hip Transposition Using a Customized 3D-printed Femoral Prosthesis After Pelvic Tumor Resection? A Preliminary Study.

*Potter BK*

PubMed · DOI

## Does Aspiration With Culture Augmented With Percutaneous Core Biopsy Analysis Have Better Diagnostic Value Than Culture Alone for PJI in Large Joints? A Prospective Enhancement and Feasibility Study.

*Borucki RJ, Leutloff D, Labs K*

BACKGROUND: Existing diagnostic approaches in periprosthetic joint infections (PJIs) are primarily based on joint aspiration, with a reported sensitivity of 40% to 86%. "Dry taps," occurring in up to 40% of aspirations, reduce diagnostic accuracy. Core biopsies, which enable minimally invasive tissue sampling, are widely used across various medical specialties and represent a promising tool for application in orthopaedics.

QUESTION/PURPOSE: Can the inclusion of core biopsy analysis, as opposed to a sole joint aspiration, enhance diagnostic accuracy for PJI in large joint (hip, knee, and shoulder) arthroplasties?

METHODS: In this prospective enhancement study, 175 patients with painful large arthroplasties were recruited between September 2020 and September 2023. All patients underwent joint aspiration and core biopsy, leading to 181 primary data sets, of which 111 were excluded from final analysis for not reaching the endpoint or because of unrelated death, protocol breach, or a dry tap, which occurred in 34% (62 of 181) of examined joints. Diagnostic accuracy was assessed for sensitivity and specificity against the 2018 International Consensus Meeting (ICM) criteria, constituting the primary endpoint. Subsequent revision surgery, septic or aseptic, along with the microbiological evaluation of tissue samples and the sonicated prosthesis served as the confirmatory endpoint.

R […]

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## S1P Modulation via S1P Lyase Inhibition Enhances Bone Regeneration in a Murine Critical-size Defect Model of Atrophic Nonunion.

*Reinkemeier F, Korte R, Bakri A, Wallner C, Drysch M, Schmidt SV, Fueth M, Puscz F et al.*

BACKGROUND: Atrophic nonunion is a challenging complication of fracture healing with few noninvasive therapeutic options. Given that sphingosine-1-phosphate (S1P) regulates bone remodeling and angiogenesis, systemic S1P modulation may offer a regenerative strategy; however, its efficacy in nonunion has not been established.

QUESTIONS/PURPOSES: We therefore asked whether S1P lyase inhibition (1) enhances osteogenesis and consolidation in a murine critical-size bone defect model of atrophic nonunion, (2) modulates bone remodeling (osteoblast differentiation and osteoclast activity), and (3) augments cellular proliferation and endothelial activity, and (4) whether quantitative micro-CT corroborates histologically observed improvements.

METHODS: In a validated murine critical-size bone defect model of atrophic nonunion, 12-week-old male and female C57BL/6J mice were treated by continuous systemic administration of the S1P lyase inhibitor 4-deoxypyridoxine (DOP) for 5 weeks, while a reference group remained untreated after nonunion induction. Bone regeneration was assessed by histology and immunohistochemistry to evaluate osteoblast differentiation, osteoclast activity, cellular proliferation, and angiogenesis (n = 7 per group) and by high-resolution micro-CT to quantify newly formed bone within the defect region (n = 6 per group).

RESULTS: DOP treatment resulted in increased min […]

PubMed · DOI

### CORR Insights®: S1P Modulation via S1P Lyase Inhibition Enhances Bone Regeneration in a Murine Critical-size Defect Model of Atrophic Nonunion.

*Tatara AM*

PubMed · DOI

### Letter to the Editor: Is Displacement of the Lesser Trochanter Associated With Functional Outcome in Older Adults With Intertrochanteric Fractures?

*Qin K, Xu Y, Liang W*

PubMed · DOI

### Reply to the Letter to the Editor: Is Displacement of the Lesser Trochanter Associated With Functional Outcome in Older Adults With Intertrochanteric Fractures?

*Ratanasermsub N, Amarase C, Tantavisut S, Sirichuchnin P*

PubMed · DOI

### Letter to the Editor: All-inside ACL Reconstruction Offers No Advantage in Clinical Outcomes, Graft Healing, or Tunnel Widening Compared With the Complete Tibial Tunnel Technique: A Prospective Randomized Trial.

*Lu J, Jiang K, Du X, Ji J*

PubMed · DOI

### Reply to the Letter to the Editor: All-inside ACL Reconstruction Offers No Advantage in Clinical Outcomes, Graft Healing, or Tunnel Widening Compared With the Complete Tibial Tunnel Technique: A Prospective Randomized Trial.

*Okutan AE, Gürün E*

PubMed · DOI

### Letter to the Editor: Patients With Nonanaphylactic Penicillin Allergy Are Not at Increased Risk of Allergic Events After Receiving Prophylactic Preoperative Cefazolin Prior to Closed Fracture Repair.

*Kocak S*

PubMed · DOI

**Letter to the Editor: Does Insertional Reattachment for Acute Sleeve Avulsion Fracture of the Achilles Tendon Provide Sustained Function and Sports Participation?**

*Spiezia F, Oliva F, Maffulli N*

PubMed · DOI

**Reply to the Letter to the Editor: Does Insertional Reattachment for Acute Sleeve Avulsion Fracture of Achilles Tendon Provide Sustained Function and Sports Participation?**

*Xiong S, Xie X, Guo Q*

PubMed · DOI

**Letter to the Editor: Equity360: Gender, Race, and Ethnicity: Sex and Fairness in Sports.**

*Lichtblau E*

PubMed · DOI

**Reply to the Letter to the Editor: Equity360: Gender, Race, and Ethnicity: Sex and Fairness in Sports.**

*Jain R*

PubMed · DOI

**Reply to the Letter to the Editor: Equity360: Gender, Race, and Ethnicity: Sex and Fairness in Sports.**

*O'Connor MI*

PubMed · DOI

JBJS

## JBJS (American)

Volume 108, Issue 16 · August 2026 · 17 new articles

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**Why Intraosseous Medication Application Is Less Beneficial for Pain Management than for Periprosthetic Joint Infection Prophylaxis: Commentary on an article by Devon R. Pekas, MD, MS, et al.: "Effectiveness of Intraosseous Morphine for Pain Control in Total Knee Arthroplasty. A Double-Blinded, Randomized Trial".**

*Boettner F*

PubMed · DOI

**How Bad Is Operating Room Smoke During Orthopaedic Surgery?: Commentary on an article by Ryuto Tsuchiya, MD, PhD, et al.: " Intraoperative Surgical Smoke During Arthroplasty. Early Unhealthy-Level Exposure and Approach-Dependent Risk ".**

*Abidi NA*

PubMed · DOI

**Quantifying Protrusio Risk in the Metastatic Acetabulum: A Step Toward Precision: Commentary on an article by Will Jiang, BS, et al.: "Development of a Radiographic Scoring System to Estimate Acetabular Protrusion Risk in Patients with Osteolytic Periacetabular Metastases".**

*Gonzalez MR*

PubMed · DOI

### Unplanned Excision and the Race to Cure Sarcomas: Commentary on an article by Jacob Jahn, BS, et al.: " Effect of Timing of First Consultation with a Sarcoma Specialist Following Unplanned Excision. Oncologic Outcomes of Patients with Soft-Tissue Sarcomas ".

*Avedian RS*

PubMed · DOI

### To Fuse or Not to Fuse, That Is the Lingering Question: Commentary on article by Gregor Fischer, MD, et al.: " Surgical Treatment of Recurrent Lumbar Disc Herniation: To Fuse or Not to Fuse. A Single-Center Analysis of Clinical and Radiographic Characteristics and Surgical Outcomes of 450 Patients ".

*Lindsey RW*

PubMed · DOI

### Are Indications for Rotating Hinges Too Constraining and Should They Be Loosened?: Commentary on an article by E. Bailey Terhune, MD, et al.: " Surprisingly Low Rates of Aseptic Loosening in 575 Rotating-Hinge Total Knee Arthroplasties ".

*Odgaard A*

PubMed · DOI

### What's New in Limb Lengthening and Deformity Correction.

*Vogt B, Frommer A, Roedl R, Laufer A*

PubMed · DOI

### Surgeon Ownership in the Ambulatory Arthroplasty Era: Preserving Transparency and Clinical Neutrality as Site-of-Service Shifts.

*Siddiqi A, Jacob PB, Yousuf KM*

The rapid migration of total joint arthroplasty from hospitals to ambulatory surgery centers (ASCs) represents one of the most consequential structural changes in contemporary orthopaedics in the United States. In parallel, surgeon equity ownership in ASCs has expanded, aligning clinical decision-making with financial participation. Although surgeon ownership is legal, common, and often associated with efficiency and innovation, it introduces complex questions regarding transparency, case selection, and public trust. As higher-risk patients remain hospital-based while lower-risk patients migrate to ASCs, outcome comparisons may become distorted, and incentives may be less visible. Current disclosure practices and benchmarking frameworks have not evolved to reflect these structural shifts. This article argues that surgeon ownership in the ASC era necessitates updated safeguards, including standardized ownership disclosure, risk-adjusted site-of-service reporting, independent case review mechanisms, and transparent equity life-cycle planning. Proactive governance, rather than reactionary regulation, is essential to preserve professionalism and patient trust as ambulatory arthroplasty continues to expand.

PubMed · DOI

### AI vs AI: Health Care's Spy vs Spy: A Comedy of Optimization.

*DiFazio FA*

PubMed · DOI

### I Had Never Prayed in an Operating Room Before: Reflections on Global Orthopaedics, Faith, and the Illusion of Control in the Operating Room.

*Asturias AM*

PubMed · DOI

## Federated Computing in Orthopaedic Surgery: A Paradigm Shift in Collaborative Multicenter Research.

*Hill BG, Nyul TE, Moschetti WE, Schilling PL*

➢ Multicenter studies remain rare in orthopaedics (9.9% of recent publications) due to regulatory complexity, restrictive data-sharing policies, and administrative delays, despite their proven value in generating robust, generalizable evidence. ➢ Federated computing (FC) brings analytic code to each institution's data and returns only encrypted, aggregated results. This preserves patient privacy, maintains institutional control, and streamlines compliance by reducing or eliminating the need for complex data-use agreements. ➢ Extensive applications in other medical fields show that FC can match or exceed the performance of centralized analyses across diverse data types (imaging, electronic health records, genomics, and surgical video) without centralizing sensitive information. ➢ For orthopaedics, FC enables scalable, privacy-preserving collaboration on rare events, low-frequency outcomes, and practice variation across subspecialties. Broad adoption could expand research scope, improve evidence quality, and accelerate translation of findings into clinical care. ➢ FC enables secure, privacy-preserving multicenter collaboration in orthopaedics, overcoming long-standing barriers to data-sharing and expanding the scope of clinically impactful research.

PubMed · DOI

## Effectiveness of Intraosseous Morphine for Pain Control in Total Knee Arthroplasty: A Double-Blinded, Randomized Trial.

*Pekas DR, Adrados M, Lee MM, Lee Y, Burks WG, Martino JM, Coobs BR, Moskal JT*

BACKGROUND: Effective pain management following total knee arthroplasty (TKA) is crucial to optimizing patient outcomes and experiences. Multimodal pain management protocols vary between institutions, with some recently proposing the addition of an intraosseous (IO) injection of morphine intraoperatively. The purpose of this study was to investigate whether the addition of an intraoperative, IO injection of morphine during elective primary TKA would lead to improved pain control and decreased narcotic consumption during the postoperative period.

METHODS: In this double-blinded, randomized controlled trial, 100 patients undergoing elective primary TKA were prospectively enrolled. All patients received spinal anesthesia and intravenous sedation combined with an intraoperative, surgeon-administered adductor canal block. The experimental group received an intraoperative, IO injection containing 10 mg of morphine and 500 mg of vancomycin in 110 mL of normal saline solution. The control group received the same injection but without morphine. All patients received 6 daily text-message surveys (3 in the morning and 3 in the evening) for 14 days postoperatively to collect pain scores, morphine milligram equivalent (MME) consumption, and nausea and vomiting events. Data on demographics, operative factors, post-anesthesia care unit (PACU) pain scores, PACU MME consumption, and patient-re […]

PubMed · DOI

## Intraoperative Surgical Smoke During Arthroplasty: Early Unhealthy-Level Exposure and Approach-Dependent Risk.

*Tsuchiya R, Suzuki R, Yano S, Ikenoue S, Mimura M, Sameda H, Someya Y, Ohara T et al.*

BACKGROUND: Surgical smoke is a gaseous intraoperative byproduct that contains numerous toxic compounds, including fine particles with a diameter of ≤2.5 µm (PM2.5). However, intraoperative exposure during joint arthroplasty has not been well characterized. In this study, we aimed to quantify PM2.5 exposure during arthroplasty and to determine, by procedure and approach, how often it reached potentially harmful levels.

METHODS: We conducted a single-center prospective study of 83 primary arthroplasties (20 bipolar hemiarthroplasties [BHAs] via a direct lateral approach, 35 total hip arthroplasties [THAs] via a direct anterior approach, and 28 total knee arthroplasties [TKAs]). Intraoperative PM2.5 concentrations were measured using a portable monitor and were mapped to Air Quality Index (AQI) categories. Summary metrics, including maximum and cumulative PM2.5 exposure, were calculated by procedure. Logistic regression was used to estimate odds ratios (ORs) for exceeding AQI thresholds.

RESULTS: In BHA and TKA, PM2.5 concentrations increased sharply during field development, whereas they remained low throughout THA. The median maximum PM2.5 concentrations were 129.50 µg/m 3 (interquartile range [IQR], 85.44 to 275.28 µg/m 3 ) for BHA, 6.14 µg/m 3 (IQR, 2.54 to 23.94 µg/m 3 ) for THA, and 118.80 µg/m

3 (IQR, 56.23 to 208.04 µg/m 3 ) for TKA. The concentrations were significantl […]

PubMed · DOI

### Development of a Radiographic Scoring System to Estimate Acetabular Protrusion Risk in Patients with Osteolytic Periacetabular Metastases.

*Jiang W, Gan D, Tommasini S, Latich IY, Lindskog D, Lee FY*

BACKGROUND: For patients with periacetabular metastases, protrusio acetabuli is a severely painful and mobility-impairing complication that requires subsequent open joint surgery. We aimed to identify specific structural changes that are associated with progression to protrusio acetabuli and to create a scoring system to guide risk stratification.

METHODS: In this single-institution cohort study, we identified all patients who underwent primary surgical stabilization for periacetabular metastases with osteolytic or mixed osteolytic-osteoblastic characteristics from October 2017 through January 2025. Cases of protrusio acetabuli prior to surgical intervention were identified. Pain and ambulatory functional scores and treatment history were recorded. Locations of bone destruction were evaluated using coronal-cut computed tomography (CT) scans obtained within 3 months before clinical presentation (and earlier, as available). Trabecular and subchondral cortical bone mass of the periacetabular weight-bearing portions were indirectly assessed via Hounsfield unit ratio comparisons across scans. Univariable analysis of each feature was performed. The highest-scoring features were used to create a scoring system and analyzed using a receiver operating characteristic (ROC) curve. Finite element analysis was performed for biomechanical validation.

RESULTS: Eighty-seven patients (67 non- […]

PubMed · DOI

### Effect of Timing of First Consultation with a Sarcoma Specialist Following Unplanned Excision: Oncologic Outcomes of Patients with Soft-Tissue Sarcomas.

*Jahn J, Dean KK, Travis LM, Ramos-Pascua LR, García de la Blanca JC, Temple HT, Pretell-Mazzini J*

BACKGROUND: Unplanned excisions (UEs) of soft-tissue sarcoma are resections performed without appropriate preoperative imaging or biopsy confirmation. These procedures represent a large proportion of referrals to sarcoma centers and can negatively influence oncologic outcomes. Limited evidence exists regarding the impact of consultation timing after UE. This study aimed to compare oncologic outcomes of patients evaluated early versus late at a sarcoma center following UE.

METHODS: Of 397 patients treated for soft-tissue sarcoma from 2012 to 2020 at 2 tertiary centers, 117 underwent UE followed by later tumor bed excision and were analyzed. Consultation with a sarcoma specialist was defined as the patient's first visit with a multidisciplinary sarcoma team member, marking entry into the coordinated cancer center. Patients were stratified into early (≤2 months) and late (>2 months) consultation groups. Demographic, clinical, and tumor characteristics were collected. Primary outcomes included local recurrence-free survival (LRFS), metastasis-free survival (MFS), and overall survival (OS). Chi-square and t tests were used for univariate comparisons, and Kaplan-Meier analyses were performed. Multivariable Cox regression and logistic regression analyses were performed, adjusting for patient age, sex, and comorbidities; tumor size, depth, grade, stage, and margin status; and/or follo […]

PubMed · DOI

### Surgical Treatment of Recurrent Lumbar Disc Herniation: To Fuse or Not To Fuse: A Single-Center Analysis of Clinical and Radiographic Characteristics and Surgical Outcomes of 450 Patients.

*Fischer G, Kilian E, Schömig F, Vitale J, Oriordan D, Puhakka J, Reitmeir R, Ropelato M et al.*

BACKGROUND: Optimal surgical treatment for recurrence of lumbar disc herniation (LDH) remains controversial, with options ranging from repeat microdiscectomy (MD) to instrumented fusion (IF). This study aimed to guide surgical decision-making by analyzing reoperation rates, clinical and radiographic risk factors for treatment failure, and functional outcomes following MD versus IF.

METHODS: Prospectively collected data from 450 patients in our outcomes database who underwent surgery for recurrent LDH from 2004 through 2023 were retrospectively analyzed. Clinical assessment included predominant symptoms, neurological deficits, and American Society of Anesthesiologists (ASA) grade. Radiographic assessment included disc height, Pfirrmann grade, facet angle, and Modic changes on magnetic resonance imaging, as well as spinopelvic parameters on standing radiographs. Patient-reported outcomes were assessed using the Core Outcome Measures Index (COMI) and achievement of the minimal clinically important change (MCIC) of ≥2.2 points. Propensity-score matching (PSM) was performed to control for confounding factors. Reoperation rates were analyzed with a minimum 5-year follow-up.

RESULTS: Of 450 patients with recurrent LDH, 316 (70.2%) underwent MD and 134 (29.8%) underwent IF. In 192 patients after PSM, IF showed nonsignificantly higher MCIC achievement (odds ratio [OR] = 1.20, 95% conf […]

PubMed · DOI

## Surprisingly Low Rates of Aseptic Loosening in 575 Rotating-Hinge Total Knee Arthroplasties.

*Terhune EB, Carstens MF, Fruth KM, Hannon CP, Bedard NA, Perry KI, Berry DJ, Abdel MP*

BACKGROUND: Contemporary rotating-hinge total knee arthroplasties (RH-TKAs) have shown reasonable short-term survivorship in smaller series, but concerns remain regarding risks of aseptic and septic failure. The purpose of this study was to assess outcomes of contemporary RH-TKAs in one of the largest series to date.

METHODS: We retrospectively identified 575 RH-TKAs (60% used for aseptic etiologies and 40% used during reimplantation in 2-stage treatment of periprosthetic joint infection [PJI]) from 2002 to 2021 at a single institution. The mean age was 67 years, 58% were female, and the mean body mass index was 33 kg/m 2 . Sixty-five percent had Anderson Orthopaedic Research Institute (AORI) type-2B or 3 bone loss. Kaplan-Meier survivorship analyses were performed. The mean follow-up was 6 years (range, 2 to 19 years).

RESULTS: Survivorship free from any revision was 76% at 5 years and 64% at 10 years. The most common revision indications were PJI (54%) and aseptic loosening (20%). RH-TKA used in the primary setting showed better survivorship compared with RH-TKA used during reimplantation after PJI (79% versus 60% at 10 years). Survivorship free from revision for aseptic loosening was 96% at 5 years and 90% at 10 years. Survivorship free from revision for PJI was 84% at 5 years and 81% at 10 years. Survivorship free from revision for PJI was even lower for RH-TKAs used duri […]

PubMed · DOI

# The Journal of Arthroplasty

Volume 41, Issue 9 · September 2026 · 50 new articles

## Letter to the Editor on "Physiologic and Clinical Sequelae After Pneumatic Tourniquet Release in Frail Patients Undergoing Total Knee Arthroplasty Under Regional Anesthesia: A Prospective Observational Study".

*Sun X, Fei J*

PubMed · DOI

## Reply to the Letter to the Editor on "Physiologic and Clinical Sequelae After Pneumatic Tourniquet Release in Frail Patients Undergoing Total Knee Arthroplasty Under Regional Anesthesia: A Prospective Observational Study".

*Patel G, Kahlon S, Singh A*

PubMed · DOI

## Leadership Begins With the Small Things: Lessons for Early Career Arthroplasty Surgeons.

*Griffin D, Scuderi GR, Mont MA*

PubMed · DOI

### Advocacy 101 for the Early-Career Arthroplasty Surgeon: Payment, Policy, and Professional Sustainability.

*Lum ZC, Herndon CL, Wenzlick TS, Dallas-Orr D, Courtney PM, Rana AJ*

Modern hip and knee arthroplasty surgeons graduate prepared to manage complex deformity, revision surgery, complications, and high-volume clinical practice but often receive little formal preparation for the payment and policy systems that determine whether that care remains accessible and sustainable. For early-career surgeons, this gap between clinical work and reimbursement reality can feel confusing, frustrating, and outside their control. Yet, payment policy directly shapes patient access, practice viability, workforce morale, and professional autonomy. This primer provides an "Advocacy 101" primer for arthroplasty surgeons in their first five years of practice, focusing on the core mechanics of physician payment: relative value units, the Medicare conversion factor, budget neutrality, 90-day global periods, and the distinction between physician professional fees and facility payments. It helps with understanding how declining inflation-adjusted reimbursement, increasing postoperative workload, outpatient migration, consolidation, and alternative payment models are reshaping the arthroplasty practice environment. We also discuss opportunities within bundled payment models, Transforming Episode Accountability Model, and gainsharing, emphasizing that surgeons who understand implant costs, postacute pathways, and episode spending are better positioned to lead value-based care […]

PubMed · DOI

### Balanced Peer Review in the Era of Artificial Intelligence.

*Taunton MJ, Browne JA, Scuderi GR, Smitterberg CW, Mont MA*

PubMed · DOI

### Statistical Choices in Propensity Score Matching Influence the Conclusions in Arthroplasty Outcomes Research.

*Verhey JT, Tarabichi S, Novicoff WM, Mont MA, Browne JA, Bingham JS, Spangehl MJ, Clarke HD*

PubMed · DOI

### Outpatient Arthroplasty: The Site of Service Has Changed, but the Work and Financial Burden Have Not.

*Mont MA, Scuderi GR*

PubMed · DOI

### The Shift to Outpatient Total Hip and Knee Arthroplasty: Perioperative Work Has Not Changed.

*Warwick H, Salehi N, Kugelman DN, Yakkanti R, Krueger CA, Courtney PM*

BACKGROUND: Total hip arthroplasty (THA) and knee arthroplasty (TKA) are increasingly being performed on an outpatient basis. With this shift in site of service, the volume and complexity of work required for these procedures may be affected. The purpose of this study was to quantify the amount of perioperative work performed by the surgeon and advanced practice providers for same-day THA and TKA.

METHODS: We prospectively timed preservice, immediate postservice, and discharge coordination activities for a consecutive series of 96 outpatient cases over a 4-week period. Timing data for activities performed in the operating room were obtained retrospectively from our institutional electronic medical record for 500 patients who underwent THA and 500 who underwent TKA. Results were compared with the current approved values reviewed by the Relative Value Scale Update Committee (RUC) in 2019. Cases were also separated into those performed in tertiary care hospitals, specialty hospitals, and ambulatory surgical centers to compare perioperative work by surgical setting.

RESULTS: The average pre-evaluation time was 43.9 ± 7.3 minutes. The average time from the patient entering the operating room to incision was 39.0 ± 9.8 minutes. Immediate post-service tasks took 26.6 ± 7.2 minutes, and discharge coordination tasks took 122.1 ± 33.9 minutes. Overall, tasks measured took 231.6 minutes […]

PubMed · DOI

## Quantifying Clinical Encounters for Orthopaedic Hip and Knee Surgeries: A Retrospective Analysis of Provider Workload.

*Rana AJ, Sewell JK, Sirisoma AV, Walsh ZA, Faour KNW, McGrory BJ*

BACKGROUND: The time spent providing postoperative care for total hip arthroplasty (THA) and total knee arthroplasty (TKA) has increased provider workload, drawing attention to the Relative Value Scale Update Committee's (RUC) current estimation of work relative value units. Our aim for this study was to quantify the postoperative work performed by the surgeon and their team for THA and TKA during the 90-day global period. We hypothesized that the work intensity and time spent on postoperative communication were higher than estimated by current work relative value units.

METHODS: We retrospectively evaluated all patients undergoing primary total joint arthroplasty at our institution between January 1, 2019, and December 31, 2024. Primary outcomes included the number of postoperative interactions from discharge to 90 days after surgery. These included office visits, telehealth visits, phone calls, and patient portal messages.

RESULTS: From 2019 to 2024, the average number of postoperative total joint arthroplasty interactions per patient increased across all modalities. Telephone encounters spiked during the coronovirus disease 2019 pandemic and remain elevated, while portal messages rose more than fivefold. Office visits averaged 2.3 per patient, exceeding the two currently recognized in RUC valuations. Administrative tasks and telehealth also showed steady annual growth. Mea […]

PubMed · DOI

## There Is No Difference in Outpatient Costs Within One Year Post-Total Hip Arthroplasty when Comparing Surgical Approach.

*Barton KI, Kooner P, Petis SM, Somerville LE, Marsh JD, Howard JL, Lanting BA, Vasarhelyi EM*

BACKGROUND: Total hip arthroplasty (THA) continues to be a substantial financial burden to the Canadian health care system. The purpose of this study was to determine the impact of surgical approach on the associated outpatient cost of THA.

METHODS: Patients who underwent a THA via an anterior, posterior, or lateral approach were recruited (not randomized). All patients prospectively completed a cost diary (self-reported use of allied health services, investigations, clinic visits, patient expenses, and number of employment days lost to THA) to estimate total resource use and patient expenses incurred after discharge. Group comparisons were performed using Pearson's Chi-squares and one-way analyses of variance; significance was accepted at $P \leq 0.05$.

RESULTS: A prospective, microcosting analysis was performed on 104 patients undergoing a THA through either an anterior ($n = 37$), posterior ($n = 32$), or lateral ($n = 35$) approach for one year postoperatively. The overall costs of outpatient resources and expenses were comparable between anterior ($1,321.00 CDN), posterior ($1,094.92 CDN), and lateral ($1,153.18 CDN) approaches ($P = 0.114$). When the overall costs were stratified by specific expenses, the costs of allied health services were significantly higher in the anterior group ($361.32 CDN; 95% confidence interval, 257.58 to 465.07) than the posterior group ($206.50 CDN; 95% […]

PubMed · DOI

## Implant Prices and Physician Reimbursement Have Declined More Than Total Costs and Hospital Payments in Total Joint Arthroplasty.

*Yu JS, Dykhouse GL, Heo KY, Mao YV, Christ AB, Premkumar A*

BACKGROUND: Understanding implant price trends is critical amid growing demand for total joint arthroplasty (TJA) and increasing cost containment pressures. Previous studies have documented trends in costs, reimbursements, and volume for TJA. However, the relationship between implant prices, hospital and physician reimbursement, and patient financial burden remains poorly defined. This study evaluated inflation-adjusted implant pricing trends in TJA and their alignment with physician and hospital reimbursement and patient out-of-pocket (OOP) costs.

METHODS: Implant prices for primary total knee arthroplasty (TKA), total hip arthroplasty (THA), revision TKA (rTKA), and revision THA (rTHA) from 2009 to 2021 were obtained from a large publicly available implant registry. Cost, reimbursement, and patient OOP spending data were sourced from a commercial insurance claims database. There were 629,651 total procedures analyzed. All costs, reimbursements, and prices were adjusted for inflation. Trends were analyzed using linear regressions.

RESULTS: The average price for TKA implants was $5,899, $6,776 for THA, $11,576 for rTKA, and $7,419 for rTHA. Between 2009 and 2021, implant prices declined markedly for TKA (-38%), THA (-37%), and rTHA (-28%) and remained stable for rTKA (+8%). Overall costs and hospital reimbursement remained stable or modestly decreased, whereas physician reimb […]

PubMed · DOI

### Implant Costs Consume a Large Proportion of the Reimbursement in Revision Hip and Knee Arthroplasties.

*SmithOram AL, Mears SC, Barnes CL, Stronach BM, Stambough JB*

BACKGROUND: Revision total hip arthroplasty (rTHA) and revision total knee arthroplasty (rTKA) burdens continue to increase while overall reimbursement decreases. We sought to evaluate the percentage and variability of the diagnosis-related group (DRG) reimbursement spent on implants in revision arthroplasty.

METHODS: A consecutive series of 199 rTHAs and 187 rTKAs performed between June 1, 2019, and June 1, 2021, was reviewed at one academic medical center. Patient characteristics, preoperative diagnosis, implant records, and billing data were recorded for DRG 466 (revision hip or knee arthroplasty with major complication), 467 (revision with complication or comorbidity), and 468 (revision without comorbidity or complication). Data were stratified by DRG and diagnosis. Implant, surgery type, and patient comorbidity factors were analyzed for association with increased costs.

RESULTS: Implant costs comprised an average of 24% of reimbursements for DRG 466 (range, 2.4 to 133), 36.7% for DRG 467 (range, 3.5 to 118), and 35% for DRG 468 (range, 2.5 to 175). When stratified by diagnoses, groups with the largest ranges in implant costs as a percentage of the reimbursement were aseptic loosening (range, 6.6 to 178.2), infection (range, 3.2 to 129.3), and metallosis (range, 2.4 to 108.6%). Implants for rTKA were significantly more of the overall reimbursement than rTHA across all DRG […]

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### Safety Net Hospitals Are at Risk for Major Financial Penalties With New Centers for Medicare & Medicaid Services Patient-Reported Outcome Measure Reporting Requirements for Primary Hip and Knee Arthroplasty.

*Salehi N, DeAnnuntis J, Gorleski M, Warwick H, Kugelman DN, Courtney PM*

BACKGROUND: Starting in 2027, the Centers for Medicare & Medicaid Services (CMS) will penalize hospitals that do not have complete paired patient-reported outcome measures (PROMs) data on at least 50% of patients undergoing inpatient total hip (THA) and knee arthroplasty (TKA). Patients must also meet the minimum substantial clinical benefit (SCB) threshold for Hip Disability and Osteoarthritis Outcome Score Joint Replacement and Knee Injury and Osteoarthritis Outcome Score Joint Replacement scores (22 and 20 points, respectively). The purpose of this study was to identify the risk factors for patients who did not have complete paired PROMs and did not meet the minimum SCB following surgery.

METHODS: We included 5,165 primary THA and TKA patients between 2022 and 2023 at 22 hospitals. We collected data on demographics, medical comorbidities, insurance types, and socioeconomic status. Surveys were collected 90 to zero days preoperatively and 300 to 425 days postoperatively, consistent with CMS rules. A multivariate regression analysis was performed to identify independent risk factors for not meeting new CMS PROMs requirements.

RESULTS: There were 764 patients (14.8%) who met the CMS paired PROMs reporting requirement, whereas 554 (72.5%) met the SCB threshold. Risk factors for failing to meet paired PROMs collections included higher Charlson Comorbidity Index (odds ratio (OR) […]

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### Socioeconomic Deprivation and Outcomes After Total Joint Arthroplasty: An Analysis of the American Joint Replacement Registry Using the Area Deprivation Index.

*Zalikha L, Amanatullah DF, Jahan M, Donnelly P, Cohen EM, El-Othmani MM*

BACKGROUND: There is an increasing interest in socioeconomic deprivation and its influence on outcomes in orthopaedics. This study aimed to examine the relationship between neighborhood-level social deprivation and its impact on 90-day readmission, length of stay, and all-cause revision after primary total knee arthroplasty (TKA) and total hip arthroplasty (THA).

METHODS: A retrospective cohort study was conducted using data from the American Joint Replacement Registry merged with the Centers for Medicare and Medicaid Services claims for patients aged 18 years and over who had undergone primary unilateral TKAs and THAs. Comparisons were performed between patients grouped into Area Deprivation Index (ADI) quartiles by postal code (Quartiles 1 to 4, with Quartile 1 being the least disadvantaged). Subanalyses were performed using multinomial logistic regression and Cox proportional hazard models for outcomes with Quartile 1 serving as a reference. A total of 1,654,680 TKA and 1,051,311 THA procedures met the inclusion criteria.

RESULTS: Unadjusted analysis demonstrated significant differences in all outcomes based on ADI quartile ($P < 0.001$ for all outcomes except TKA all-cause revision, $P = 0.003$). In a subanalysis, Quartile 4 was significantly more likely to undergo readmission within 90 days for TKA (hazard ratio 1.631; 95% CI [confidence interval]: 1.528 to 1.741, $P < 0.0001$ […]

PubMed · DOI

### Social Determinants of Health Disparities Are Associated With Prolonged Opioid Utilization After Total Joint Arthroplasty.

*Hohmann AL, Fellheimer HS, Meacock SS, Lizcano JD, Purtill JJ, Fillingham YA*

BACKGROUND: Social determinants of health disparities (SDHDs) have been associated with an increased risk of poor outcomes after total joint arthroplasty (TJA). The purpose of this study was to assess whether SDHDs are associated with prolonged postoperative opioid use after primary TJA.

METHODS: This study was a single-institution, retrospective cohort study of patients undergoing primary, unilateral TJA. Opioid prescriptions for these patients were queried in a Prescription Drug Monitoring Program. Patients who had an opioid prescription filled within one year preoperatively were excluded. Patients were divided by procedure (total hip arthroplasty [THA] versus total knee arthroplasty [TKA]) and split into five cohorts: no postoperative opioid prescription filled, or the last prescription filled within zero to 30, 31 to 90, 91 to 180, or 181 to 365 days postoperatively. Patients were excluded if they did not fill postoperative opioid prescriptions in consecutive time frames. We identified 7,671 eligible patients who underwent THA and 7,707 who underwent TKA.

RESULTS: The mean overall Social Vulnerability Index (SVI) score increased, indicating greater socioeconomic vulnerability, with a later last fill date, with significant differences across cohorts for THA ($P < 0.001$) and TKA ($P = 0.026$) patients. In multilinear regression analyses, the last fill date in the 31- to 90-day […]

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### Substantial Clinical Benefit Thresholds for Hip Disability and Osteoarthritis Outcome Score Subscales and Associated Risk Factors for Nonattainment After Total Hip Arthroplasty: A Prospective Cohort Study of 9,229 Patients.

*Pasqualini I, Elmenawi KA, Khan ST, Hadad MJ, Cleveland Clinic Adult Reconstruction Research Group (CCARR), Piuzzi NS*

BACKGROUND: The Centers for Medicare & Medicaid Services (CMS) now requires patient-reported outcome measures (PROMs) collection after total hip arthroplasty (THA), emphasizing substantial clinical benefit (SCB) in the Hip Disability and Osteoarthritis Outcome Score - Joint Replacement (HOOS-JR). This study aimed to establish SCB thresholds for HOOS-Pain, Physical Function Short Form (PS), and JR and to identify factors influencing SCB achievement after THA.

METHODS: A prospective analysis included 13,324 primary THAs performed between 2016 and 2022 at a tertiary health care institution. Of these, baseline PROMs were completed by 11,766 patients (88.3%), and 1-year follow-up data were available for 9,229 patients (78.4%). Demographic characteristics, comorbidities, and PROM data were collected, and patients were stratified into eight phenotypes based on pain, function, and mental health status. The SCB thresholds were calculated using anchor-based methods and anchored by the Veterans RAND-12 physical health question at one year. Multivariable logistic regression was conducted to determine factors linked to not achieving SCB.

RESULTS: Established SCB thresholds were 40 for HOOS-pain (83.7% sensitivity, 45.3% specificity), 29.3 for HOOS-PS (75.4% sensitivity, 49.5% specificity), and 32.3 for HOOS-JR (79.3% sensitivity, 50.3% specificity). Overall, 72.8% of patients reached SCB. […]

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### Standard of Care and Its Implications on Joint Arthroplasty: A Primer.

*Hershfeld BE, Tarazi JM, Cohn RM, Scuderi GR, Mont MA, Bitterman AD*

"Standard of care" refers to the level of clinical practice that a reasonably competent physician is expected to provide under similar circumstances, based on evidence, specialty guidelines, and professional consensus. This article reviews current literature on the standard of care in total joint arthroplasty and orthopaedic surgery, clarifying its definition, application, and role in reducing malpractice risk through evidence-based protocols. Specifically, the authors examined (1) the definition and evolution of standard of care; (2) the role of clinical guidelines, including arthroplasty specific; (3) legal implications in malpractice; (4) differences in elective versus emergent arthroplasty; (5) postoperative protocols central to practice; and (6) challenges and open questions in defining and updating standards of care.

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### General, Spinal, and Regional Nerve Blocks: Do Different Anesthesia Practices Affect Same-Day Discharge in Primary Total Hip and Knee Arthroplasty?

*Wing CW, Hatami B, Oseili AT, Ubl DS, Springer BD, Ardon AE, Ledford CK*

BACKGROUND: The purpose of this study was to compare the effect of spinal anesthesia (SA) versus general anesthesia (GA) on acute 90-day outcomes in same-day discharge (SDD) total hip (THA) and total knee arthroplasty (TKA) patients, including subgroup analysis of peripheral nerve block usage.

METHODS: A retrospective review of 1,002 SDD patients (402 THAs, 600 TKAs) was performed, including 868 and 134 undergoing SA and GA, respectively. All patients received periarticular injections and multimodal analgesia; 80% of THAs received a paravertebral block (PVB), and 58% of TKAs received an adductor canal block (ACB). Intraoperative and postoperative narcotic usage (morphine milligram equivalents, MMEs) through SDD, numerical pain scores, first ambulation distance, hospital length of stay, and 90-day complications (readmissions or reoperations) were compared.

RESULTS: The SA performed in SDD THA resulted in decreased intraoperative MME (0.1 versus 8.7, $P < 0.01$), postoperative MME (12.3 versus 16.1, $P < 0.02$), and pain scores (1.6 versus 3.8, $P < 0.01$). A similar result was found in TKA, as SA patients had decreased intraoperative MME (0.1 versus 13.1, respectively, $P < 0.01$), postoperative MME (9.7 versus 15.5, $P < 0.01$), and pain scores (1.5 versus 3.2, $P < 0.01$). There was no difference in hospital length of stay between groups, but THA patients undergoing GA showed an increas […]

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### Uncontrolled Diabetes Mellitus in the Year Prior to Total Hip or Knee Arthroplasty Is Associated with Similar Perioperative Glucose Control, Early Complication Rates, and 5-Year Reoperation Rates after Preoperative Diabetes Mellitus Optimization.

*Umelo J, Jubril A, Burger PU, Tran JN, Balkissoon R, Kaplan NB, Thirukumaran CP, Ricciardi BF*

BACKGROUND: Perioperative glycemic control is considered a modifiable risk factor prior to primary total hip and total knee arthroplasty; however, it is still unknown whether preoperatively optimizing hemoglobin A1C values affects perioperative glucose control or postoperative complications. The purpose of this study was to compare patients who required preoperative diabetes mellitus (DM) optimization in the year prior to surgery (defined as hemoglobin A1C ≥ 8%) versus patients who had DM who did not require preoperative DM optimization to compare (1) postoperative glycemic control and (2) 90-day complications and incidence of revision arthroplasty.

METHODS: This was a retrospective single-center analysis. Patients undergoing primary total joint arthroplasty who had a diagnosis of DM (hemoglobin A1C greater than 6.5%) were eligible for inclusion (N = 585). Patients who had previously uncontrolled DM requiring optimization (hemoglobin A1C ≥ 8.0%) in the year prior to surgery (N = 164) were compared to patients who had controlled DM (N = 421). All patients had hemoglobin A1C less than 8% preoperatively. Outcomes included median and peak perioperative glycemic control (postoperative day zero to two), 90-day complications, and incidence of revision arthroplasty at the final follow-up (mean 64 months). Multivariable logistic regressions evaluated associations of underlying preopera […]

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### Associations Between Preoperative Seasonal Vaccination and Complications After Total Joint Arthroplasty.

*Bcharah G, Wininger AE, Elsabbagh Z, Van Schuyver PR, Moore LL, Bingham JS*

BACKGROUND: Vaccinations against common respiratory pathogens are recommended for the general population. However, associations between respiratory vaccinations and outcomes following total joint arthroplasty (TJA) have not been investigated.

METHODS: Using a national registry, 595,339 adults who underwent TJA were identified. Patients were stratified by influenza vaccination within three months before surgery and COVID-19 vaccination within three months of surgery or any time prior. Propensity-score matching controlled for demographics, comorbidities, and recent infections. Primary outcomes included 90-day and 2-, 3-, and 4-year rates of all-cause revision, periprosthetic joint infection (PJI), aseptic loosening, and periprosthetic fracture. The secondary outcomes included 90-day medical complications and mortality.

RESULTS: Influenza vaccination within three months of TJA was associated with higher 90-day medical complications, including deep vein thrombosis (odds ratio (OR) 1.38, 95% confidence interval (CI) 1.13 to 1.69) and pneumonia (OR 1.51, 95% CI 1.18 to 1.95), as well as PJI (OR 1.24, 95% CI 1.03 to 1.48) at four years. In contrast, COVID-19 vaccination within three months of TJA was associated with lower odds of pneumonia (OR 0.62, 95% CI 0.39 to 0.97) at 90 days compared to unvaccinated controls. At two years postoperatively, COVID-19 vaccination any time before T […]

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### Caring for the Caregiver: Caregiver Preparation and Stress Following Total Joint Arthroplasty.

*Sontag-Milobsky IL, Selph TJ, Madhan A, Pagadala MS, Adelani MA, Edelstein AI, Schwarzkopf R, Suleiman LI*

BACKGROUND: Social support improves outcomes after total hip and knee arthroplasties (THA/TKA), but the demands on informal caregivers, especially as surgeries transition to outpatient care, are understudied. This study strived to assess caregiver burden, predictors, and implications following joint arthroplasty.

METHODS: This prospective cohort study enrolled 185 patient-caregiver dyads undergoing primary total hip arthroplasty or total knee arthroplasty for osteoarthritis at a tertiary academic center. Caregivers completed assessments at four weeks postoperatively, including the Caregiver Strain Index (CSI) and Appraisal of Caregiving Scale (ACS), which measure perceived benefit, threat, and stress. Demographic, socioeconomic, and caregiving-related variables were collected. Multivariate linear regression identified factors associated with caregiver strain and experiences. Caregivers had a mean age of 64 years (range, 52.3 to 76.3), and 60% were women. Most (72.4%) were spouses, and 46.5% were retired.

RESULTS: The CSI scores showed considerable strain, especially among women caregivers ($\beta$ = 1.29, P = 0.001), those who had higher daily time commitment postoperatively, and those who had lower preoperative preparedness. Regarding employment status, 7% worked part-time, and 3.2% were homemakers. Among ACS subscales, non-White race ($\beta$ = 0.31, P = 0.035) and homemaker status ($\beta$ […]

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### National Trends in Unicompartmental Knee Arthroplasty During a Period of Major Survivorship Improvements: Insights From the Dutch Arthroplasty Register.

*Burger JA, Vossen RJM, Bayoumi T, Sierevelt IN, Ten Noever de Brauw GV, Spekenbrink-Spooren A, Zuiderbaan HA*

BACKGROUND: The rising use of unicompartmental knee arthroplasty (UKA) among various Western countries raises questions regarding the multifaceted determinants driving this trend. Therefore, this study aimed to examine trends and their associations with 3-year survivorship, focusing on patient, implant, hospital, and revision-related factors.

METHODS: All primary medial UKAs for osteoarthritis registered in the Dutch Arthroplasty Register from 2014 to 2021 were included. Age, sex, BMI, prior ipsilateral knee surgery, implant type (cemented and uncemented mobile-bearing; cemented fixed-bearing), hospital practice (annual volume and usage), reasons for revision, and revision type were analyzed across four consecutive 2-year intervals. The 3-year survivorship was evaluated, and factors associated with revision risk were identified across these intervals.

RESULTS: A total of 27,234 medial UKAs were analyzed, revealing an increased patient age, a higher proportion of men, and fewer prior ipsilateral knee surgeries over time ($P < 0.001$). A profound increase occurred in uncemented mobile-bearing use and high-volume and usage hospitals ($P < 0.001$). The 3-year survivorship increased from 93.9 to 96.2% ($P < 0.001$). Higher hospital volume and usage, older patient age, and fewer prior ipsilateral knee surgeries were associated with reduced 3-year revision risk ($P < 0.001$) across various […]

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## Factor XI and XIa Inhibitors for Venous Thromboembolism Prophylaxis After Total Knee Arthroplasty: A Review From Mechanism to Phase III Trials.

*Smitterberg CW, Ando W, Chen AF, Garvin KL, Hamilton WG, Lee GC, Lieberman JR, Lombardi AV et al.*

BACKGROUND: Venous thromboembolism (VTE) prophylaxis after total knee arthroplasty (TKA) has improved substantially, yet the field continues to balance two competing priorities: reduction of symptomatic deep vein thrombosis and pulmonary embolism while minimizing bleeding, wound complications, and downstream infection risk. Contemporary practice has shifted toward aspirin as prophylaxis for many low-risk patients after TKA, with direct oral anticoagulants (DOAC) and low-molecular-weight heparin (LMWH) reserved for higher-risk patients; however, these agents still affect common-pathway thrombin generation and can increase bleeding or compromise wound healing. Aspirin is not an anticoagulant and, despite its widespread use, does not directly address coagulation-driven thrombosis while still increasing bleeding risk, particularly gastrointestinal bleeding.

METHODS: This review summarizes the biologic rationale for targeting factor XI (FXI) and activated FXI (FXIa), synthesizes completed orthopaedic clinical evidence across various FXI/XIa modalities, and details the design features of ongoing Phase III trials evaluating anti-FXI molecules for post-TKA prophylaxis.

RESULTS: The FXI and FXIa sit upstream in the intrinsic pathway and may contribute disproportionately to thrombosis relative to hemostasis. In orthopaedic surgery, multiple phase II programs, including antisense oligon […]

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## Recognizing Sarcopenia in Total Knee Arthroplasty Patients: A Call for Greater Awareness.

*Ta■c■ M, Özer M, Çamur S*

BACKGROUND: Suspected sarcopenia, marked by early muscle weakness and functional decline, is increasingly recognized in elderly patients and may negatively affect total knee arthroplasty (TKA) outcomes. Although formal diagnosis may not always be practical, early identification through simple screening tools such as the Strength, Assistance with walking, Rising from a chair, Climbing stairs, and Falls (SARC-F) questionnaire and basic functional tests can help detect at-risk patients and improve postoperative management and prognosis.

METHODS: This retrospective study included 97 patients who underwent unilateral TKA. Sarcopenia was assessed using the European Working Group on Sarcopenia in Older People 2 (EWGSOP2) algorithm. All patients completed the SARC-F and Sarcopenia Quality■of■Life (SarQoL), Oxford, and Knee Society Score (KSS), as well as pre- and postoperative Visual Analog Scale (VAS) scores.

RESULTS: Suspected sarcopenia was present in 34 patients (35.1%), while 63 patients (64.9%) had no suspicion. Probable sarcopenia based on handgrip strength was identified in eight patients, whereas no patient met the muscle mass cutoff for confirmed sarcopenia. Patients who had suspected sarcopenia were older ($P= 0.05$), had higher postoperative VAS scores ($P = 0.05$), and more frequently fell below handgrip cutoff values ($P = 0.03$). Gait speed was significantly slower in this g […]

PubMed · DOI

### Intraosseous Sustained-Release Anesthetics Promote Early Ambulation and Functional Recovery After Total Knee Arthroplasty.

*Huang J, Abdullah Ezzi SH, Wu S, Ali Alshabi MM, Zhang W, Malik MA, Cao X*

BACKGROUND: Postoperative pain is a primary impediment to functional recovery in patients who undergo total knee arthroplasty (TKA).This work aimed to evaluate the efficacy of intraosseous sustained-release anesthetics, administered into the subchondral trabecular space before the insertion of a prosthesis during TKA, in promoting early ambulation and functional recovery.

METHODS: This randomized controlled open-label study was carried out on 110 patients who had a mean age of 75 years (range, 65 to 90). Patients were subdivided into two groups: local infiltration anesthesia (LIA) group: 10 mL of sustained-release anesthetic injection solution was taken, and infiltration anesthesia was applied in the joint and around the incision (two mL was injected into the posterior joint capsule and lateral collateral ligaments, two mL was injected into the knee-extension device and the parapatellar, and six mL was injected into the periarticular soft tissues); and LIA + intraosseous infiltration anesthesia (TIA) or group LIA plus TIA: a sustained-release anesthetic injection solution of 20 mL, 10 mL as described before plus intraosseous infiltration anesthesia on the osteotomy surface of the tibia (10 mL).

RESULTS: Visual analog scale, at rest and during walking, was significantly lower at days one and three in the group LIA plus TIA than in the group LIA ($P < 0.05$). Patient-controlled e […]

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### A Comparison of the Efficacy and Complications Between Intraosseous and Periarticular Multimodal Analgesic Cocktail Injections After Primary Total Knee Arthroplasty: A Randomized Controlled Trial.

*Suttikadsanee W, Pinsornsak P, Pinsornsak P*

BACKGROUND: Early postoperative pain control after total knee arthroplasty (TKA) plays a crucial role in patient satisfaction following surgery. Despite current multimodal pain management strategies, postoperative pain remains a concern for patients and can hinder the decision to undergo TKA. With the advent of intraosseous (IO) analgesic injections, the efficacy of IO multimodal analgesic cocktail administration compared to standard periarticular (PA) injections remains controversial.

METHODS: A prospective randomized controlled trial was conducted comparing 45 patients who received an IO multimodal cocktail injection (five mg of morphine, 30 mg of ketorolac, 20 mL of 0.5% bupivacaine, and 0.6 mL of 1:1,000 epinephrine) with 45 patients who received a standard PA cocktail injection. A total of 43 patients in each group completed the study. Postoperative visual analog scale for pain at rest and during motion, morphine consumption, functional outcomes, and postoperative complications were recorded.

RESULTS: In the IO group, visual analog scale for pain was significantly lower both at rest and during motion at four, six, and 12 hours, as well as two weeks postoperatively. The Timed Up and Go test was faster in the IO group at 48 hours, but not at two weeks. However, morphine consumption, time to start walking, time to discharge, postoperative range of motion, and complications […]

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### Physiologic and Clinical Sequelae After Pneumatic Tourniquet Release in Frail Patients Undergoing Total Knee Arthroplasty Under Regional Anesthesia: A Prospective Observational Study.

*Patel G, Kahlon S, Singh A, Chipre S, Anand V*

BACKGROUND: Tourniquet deflation during total knee arthroplasty (TKA) produces abrupt reperfusion physiology, including hyperkalemia, acidosis, and hemodynamic instability. Frail patients are particularly vulnerable; however, this population has not been prospectively studied.

METHODS: We conducted a prospective, observational study of frail adults (Clinical Frailty Scale ≥ 4) undergoing elective, unilateral, cemented TKA under spinal anesthesia with a pneumatic thigh tourniquet. There were 40 frail patients (mean age, 76 years [range, 65 to 88]; 65% women) included. Tourniquet duration averaged 84 minutes (range, 70 to 96). Patients with baseline hyperkalemia, severe renal dysfunction, significant arrhythmias, or planned revision or bilateral arthroplasty were excluded. Hemodynamics and arterial blood gases (pH, $PaCO_2$, lactate, potassium, and electrolytes) were recorded at baseline, predeflation, and 1- to 30-minute postdeflation. The primary endpoint was the change in serum potassium after tourniquet release, and hemodynamic instability was analyzed as a key clinical outcome. The secondary outcomes included the incidence of hyperkalemia (≥ 5.5 mmol/L), acidosis (pH less than 7.30), hypotension, arrhythmias, postanesthesia care unit (PACU) interventions, acute kidney injury, and hospital lengths of stay.

RESULTS: After deflation, serum potassium rose from 4.3 ± 0.3 to 4.8 ± […]

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## Osteonecrosis of the Femoral Head: A Dysbiotic Condition?

*Ferrini A, He M, Mont MA, Goodman SB, Parvizi J*

Osteonecrosis of the femoral head (ONFH) is a progressive and disabling condition of the hip joint that primarily affects young and active individuals, leading to progressive collapse of subchondral bone and often secondary arthritis. Despite extensive investigation, the precise etiology often remains unclear. While high-dose corticosteroids, chronic alcohol ingestion, and smoking are known associated risk factors, approximately 20 to 30% of ONFH cases are classified as idiopathic. Recently, the concept of gut dysbiosis, i.e., disruption of the normal intestinal microbial balance, has gained increasing attention due to its systemic immunologic and metabolic implications. Dysbiosis is associated with an increase in gut permeability, leading to the translocation of bacteria and their metabolic products, including lipopolysaccharides and short-chain fatty acids, into the systemic circulation. This may stimulate proinflammatory cascades throughout the body, including the joints, initiating a bone remodeling process. Emerging evidence from preclinical and human research suggests that specific gut microbiota taxa may influence key mechanisms involved in the pathogenesis of ONFH. In addition, early findings support the therapeutic potential of microbiota-targeted therapies such as probiotics, short-chain fatty acids-enriched diets, and fecal microbiota transplantation. Although a grow […]

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## Cemented Femoral Stem Design Is Not Associated With Risk of Revision After Total Hip Arthroplasty in Patients Aged ≥ 65 Years: An Analysis of the American Joint Replacement Registry.

*Kelly M, Hegde V, Zaniletti I, Anderson L, Gililland JM, De A, Kagan R*

BACKGROUND: Cemented femoral fixation for total hip arthroplasty (THA) in those aged ≥ 65 years has potential benefits. However, few resources exist to assist in selecting cemented femoral implant designs. We examined the associated risk for all-cause revision, risk of periprosthetic femoral fracture, and aseptic loosening based on a modern classification of cemented femoral stem designs.

METHODS: An analysis of primary THA cases in patients aged ≥ 65 years was performed with American Joint Replacement Registry (AJRR) data linked to Centers for Medicare and Medicaid Services data from 2012 to 2023. Patient demographics, revision or open reduction and internal fixation (ORIF) for periprosthetic femoral fracture, and revision for aseptic loosening were recorded. We identified 20,484 primary THAs with cemented stems that were classified via a modern classification. There were 5,437 (26.5%) type I (taper-slip) and 14,796 (72.2%) type II (composite beam) stem designs. Cause-specific Cox proportional hazard models with competing risk of death were used to evaluate the association of cemented stem design with all-cause revision, revision or ORIF due to periprosthetic femoral fracture, and revision for aseptic loosening while adjusting for potential confounders.

RESULTS: Compared to those who had type I (taper-slip) designs, those with type II (composite beam) designs were not associ […]

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## Low Follow-Up Rates After a Total Hip Arthroplasty Prosthesis Recall.

*Cooper LH, Wahl SD, Stambough JB, Siegel ER, Mears SC, Barnes CL, Stronach BM*

BACKGROUND: A modern hip arthroplasty liner was recalled for premature polyethylene wear with the potential for catastrophic osteolysis and failure. Physicians and hospitals are responsible for communicating recall information to patients. Our center instituted a protocol to reach patients who had recalled implants for follow-up. Our study looked at the effectiveness of this effort and risk factors for nonresponse.

METHODS: A retrospective study was performed to measure follow-up after total hip arthroplasty device recall. A standardized letter was used to reach out to all patients to notify them of the implant recall. Demographic information was collected, and an analysis was performed to determine risk factors for nonresponse.

RESULTS: Of the 763 patients, 58.8% (n = 449) followed up after recall letters were sent. There were significant differences in the follow-up rate found among the four surgeons who performed the total hip arthroplasties. Patient proximity to the performing institution (n = 264), performing surgeon, and adequate health literacy (n = 677) were significant predictors of follow-up ($P < 0.05$). Age, sex, body mass index, and race were not associated with follow-up.

CONCLUSION: The overall rate of follow-up was less than 60% despite an organized process for contacting patients to return for evaluation after implant recall. Our findings raise concerns for th […]

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## Can External Rotator Preservation Reduce Dislocation Risk Following Posterior Approach Total Hip Arthroplasty in High-Risk Patients? A Propensity-Matched Comparative Study.

*Won SH, Joh Y, Lim YW, Kwon SY, Kim YS, Kim SC*

BACKGROUND: Although multiple studies have examined the efficacy of muscle-sparing techniques in reducing dislocation after posterior approach THA, their findings are limited by small sample sizes and short follow-up durations, and to our knowledge, none have specifically evaluated outcomes in high-risk patients. This study compared the overall and 90-day dislocation risks of the previously described external rotator preservation (ERP) and standard posterior approach THA in high-risk patients.

METHODS: We retrospectively reviewed patients who underwent THA between January 2008 and December 2022 at three tertiary centers, including those who had osteonecrosis of the femoral head (ONFH), inflammatory arthritis (IA), or a history of lumbar spinal fusion. Patients who underwent the ERP approach were identified from medical records and matched 1:1 to controls (standard posterior approach) by age, sex, body mass index, American Society of Anesthesiologists score, year of surgery, and surgical indication, yielding 501 patients per group. Binary logistic regressions were used to compare overall and 90-day dislocation risk. Kaplan-Meier analysis evaluated revision due to dislocation and all-cause revision. Surgical complications, Harris Hip Score (HHS), and postoperative radiographic parameters were compared. The mean follow-up was 10.7 years.

RESULTS: The ERP group showed lower overa […]

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## Primary Total Hip Arthroplasty in Patients Who Have a Body Mass Index > 50: Is the Risk Worth the Reward?

*Righolt CH, Finney C, Turgeon TR, Bohm ER, Sniderman J*

BACKGROUND: The obesity epidemic has given rise to a growing number of patients who have a body mass index (BMI) exceeding 50. As BMI cutoffs are no longer recommended, arthroplasty surgeons continue to push the limits of surgical feasibility in total hip arthroplasty (THA) despite possible risks.

METHODS: A retrospective cohort study of patients who underwent primary THA (N = 7,458) was developed, comparing outcomes between patients who had a BMI ≥ 50 (N = 147) and those of other weight classes. We used Cox proportional hazard models to estimate the association between patient BMI and revision risk using overweight patients (BMI 25 to 30) as the reference group. Patients entered the study cohort on their date of surgery and exited on the earliest of the date of revision, date of death, or the end of the study. Patient-reported outcome measures were compared pre- and postoperatively.

RESULTS: Increased obesity class led to a gradient of elevated revision risk. In the first year after surgery, THA patients who had a BMI ≥ 50 demonstrated an adjusted hazard ratio (for revision of 11.8 (95% confidence interval [CI] 5.3 to 26.2). This risk was also elevated in patients who had a BMI of 40 to 50 (N = 635, hazard ratio = 3.8, 95% CI = 1.9 to 7.7). Although the patients who had a BMI ≥ 50 had a significantly worse preoperative Oxford-12

Hip score than other overweight patients, at f […]

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### Is the Use of New Ceramic Heads With Titanium Sleeves on Retained Femoral Stems in Revision Total Hip Arthroplasty Associated With Femoral Head or Neck Junction Failure at Mean Eight-Year Follow-Up?

*Seah MKT, Howard LC, Horwood NJ, Masri BA, Garbuz DS, Neufeld ME*

BACKGROUND: Ceramic heads used in revision total hip arthroplasty (rTHA) typically utilize a titanium sleeve adapter to prevent damage by the retained stem taper; however, there is sparse mid-to long-term data on this practice, and risk factors for failure remain relatively unknown. This study aimed to determine revision-free survival, specifically for failure of the head and/or neck junction (HNJ) and all-cause re-revision in this patient population, and to identify factors associated with failure.

METHODS: We identified all consecutive aseptic rTHA that used a new ceramic head with a titanium sleeve on a retained stem at our institution between 2011 and 2022. A total of 316 rTHA (295 patients) were included with a mean follow-up of 7.7 years (range, 2.1 to 14.3). The mean age was 65 years, and 57% were women. The main indications for rTHA were adverse local tissue reaction (46%), instability (29%), and acetabular aseptic loosening (12%). Isolated modular component exchange was performed in 38% of revisions. Kaplan-Meier analysis was used to determine survivorship. We aimed to use multivariate logistic regressions to identify risk factors for HNJ failure and all-cause rerevision.

RESULTS: There were no rerevisions for HNJ failure. There were 66 hips (21%) that underwent rerevision at a mean of 1.8 years, most commonly for instability (39 of 66), infection (17 of 66), and ase […]

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### Aseptic Revision Total Hip Arthroplasty in the Elderly: Is There a Higher Incidence of Early Complications?

*Dandamudi S, Jones CM, Majji I, Yadav AS, Dunleavy ML, Levine BR, Behery OA*

BACKGROUND: There is conflicting evidence on whether revision total hip arthroplasty (rTHA) in elderly patients (age ≥ 80 years) predisposes them to increased early complications compared to younger patients. This study compared 90-day complications, readmissions, and reoperations following aseptic rTHA between these two cohorts.

METHODS: We retrospectively identified all patients who underwent aseptic rTHA at a single academic institution from 2002 to 2023. Patients who had a history of periprosthetic joint infection, less than 90-day follow-up, or missing data were excluded. A total of 124 patients aged ≥ 80 years were eligible and matched to 124 rTHA patients aged less than 80 years using 1:1 propensity score-matching based on revision type (modular exchange, single-component, or two-component revision), body mass index (BMI), sex, and American Society of Anesthesiologists (ASA) score. Demographic data and outcomes of interest were analyzed using nonparametric tests and multivariate regression models.

RESULTS: The Charlson comorbidity index (CCI) was significantly higher in the ≥ 80 group ($P < 0.001$). No differences were observed in rTHA indications ($P > 0.132$) or 90-day medical or surgical complications ($P > 0.052$). Patients of ≥ 80 years were more likely to undergo reoperation for dislocation ($P = 0.035$), though the difference was small and likely not clinically relevant […]

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### Subsidence of Modular Fluted Tapered Stems After Femoral Revision Surgery: Risk Factors and a Novel Classification System.

*Mahlouly J, Terrier A, Borens O, Meylan A, Wegrzyn J, Steinmetz S*

BACKGROUND: Modular fluted tapered (MFT) stems are widely used in femoral revision surgery for their ability to achieve diaphyseal fixation in the setting of bone loss. This study aimed to identify factors influencing MFT stem subsidence.

METHODS: We retrospectively analyzed a cohort of 443 femoral revision procedures performed at a single institution, all using the same model of MFT stem. A total of 180 procedures met the inclusion criteria (≥ four radiographs) and were analyzed for subsidence during follow-up using a radiographic analysis. The mean radiological follow-up was 28.5 months (range, one to 120). Subsidence ≥ five mm was defined as excessive. Implant survivorship was assessed by rerevision rate. The relationship between clinical variables and excessive subsidence was examined using logistic regression and Bayesian modeling, guided by a directed acyclic graph.

RESULTS: The mean stem subsidence at 12 months was 1.6 ± 2.3 mm. Among the full cohort, 18 stems (5.0%) underwent rerevision at a mean of 71.4 months. Excessive subsidence (≥ five mm) was observed in 8.3% of cases. Associative analysis identified severe bone defects (Paprosky IIIA to IIIB) as the main factor associated with excessive subsidence (odds ratio = 6.35, P = 0.001). Short MFT stems, extended trochanteric osteotomies, and men showed weaker, nonsignificant associations. Causal modeling confirmed the […]

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## Identifying Risk Factors for Aseptic Loosening and Infection Following Distal Femoral Replacement for Periprosthetic Fracture in the Nononcologic Patient.

*Brady TT, Bera SR, Grammatopoulos G, Adamczyk AP*

BACKGROUND: Distal femoral replacement (DFR) is increasingly used for periprosthetic distal femoral fractures (PDFFs). Although infection and aseptic loosening are known complications, predictors, particularly of loosening, remain poorly defined in nononcologic populations. This study aimed to identify patient risk factors for aseptic loosening and infection following DFR for PDFF in a nononcologic cohort. Additionally, DFR utilization trends for PDFF were assessed over time.

METHODS: A retrospective study was conducted using an insurance claims database (2015 to 2023). Aseptic loosening, periprosthetic joint infection (PJI), and DFR for PDFF were identified using side-specific diagnostic and procedural codes. Oncologic patients were excluded. Kaplan-Meier endpoints were diagnosis of aseptic loosening or PJI. Failure was considered a reoperation. Multivariable logistic regressions identified independent risk factors. A total of 1,002 DFRs were included; 85% of patients were women, and 82% were over 65 years of age.

RESULTS: After excluding cases with ipsilateral PJI, aseptic loosening occurred in 6.6% of patients (n = 57), of which 39 (4.6%) underwent reoperation. Men (odds ratio [OR] 4.11) and obesity (OR 2.67) were independently associated with increased odds of loosening. A PJI occurred in 159 patients (15.9%), of whom 103 (10.3%) underwent reoperation. Men (OR 3.50), obes […]

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## Draining Sinus Tracts and Periprosthetic Joint Infections: Traditional Synovial Fluid Counts May Be Misleading.

*Alder KD, Dugdale EM, Osmon DR, Bedard NA, Berry DJ, Abdel MP*

BACKGROUND: Serologic and synovial laboratory values guiding the diagnosis of periprosthetic joint infection (PJI) are well-established. We hypothesized that these values would be lower in patients who had a draining sinus tract. The purpose of this study was to determine if patients who had sinus tracts had lower values for serum erythrocyte sedimentation rate (ESR), serum C-reactive protein (CRP), synovial white blood cell (WBC) count, and synovial percent neutrophils.

METHODS: We retrospectively reviewed 611 primary total joint arthroplasties treated for PJI between 2000 and 2020 at a high-volume academic center. Only 191 patients (94 hips, 97 knees) who met the 2011 Musculoskeletal Infection Society (MSIS) major criteria for PJI and did not receive antibiotics within two weeks before laboratory evaluation were included. Patients were divided into those who had sinus tracts (n = 58) and those who did not (n = 133). Laboratory values for early (less than 90 days) and late (≥ 90 days) postoperative PJI cases were analyzed separately. False negative rates for detecting PJI were compared based on the 2018 International Consensus Meeting (ICM) cutoffs for each diagnostic test.

RESULTS: Median synovial WBC count was significantly lower in early (9,978 versus 76,068 cells/μL; P = 0.01) and late (20,365 versus 63,356 cells/μL; P = 0.03) PJI in those who had a sinus tract. False ne […]

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### Multiple Primary Joint Arthroplasties and the Risk of Periprosthetic Joint Infection: Evidence From a Large Retrospective Cohort.

*Schaffler BC, Prinos A, Kennedy MF, Ehlers M, Rozell JC, Schwarzkopf R*

BACKGROUND: There is a growing number of patients who undergo multiple primary hip and knee joint arthroplasties during their lifetime. Whether patients who have multiple replaced joints are at an increased long-term risk of periprosthetic joint infection (PJI) is not known. The purpose of this study was to compare rates of PJI in patients who have more than one primary arthroplasty.

METHODS: We reviewed 36,125 patients who underwent primary total joint arthroplasty at a single institution from 2011 to 2024. Patients were categorized as having one to four primary hip or knee arthroplasties. The PJI incidence was compared using Chi-square testing and binary logistic regressions, and multivariate models adjusted for sex, body mass index, diabetes, renal disease, smoking status, and Charlson Comorbidity Index. Subanalyses compared patients who had one versus two, three, and four arthroplasties.

RESULTS: When comparing patients who had one, two, three, or four primary joint arthroplasties, there was no significant difference in the rates of PJI between groups ($P = 0.112$). Multivariate analyses showed no statistically significant association between the number of arthroplasties and PJI (adjusted odds ratio for two, three, and four arthroplasties versus one: 1.34, 95% confidence interval (CI) 0.96 to 1.74, $P = 0.083$; 1.98, 95% CI 0.77 to 4.12, $P = 0.105$; 1.57, 95% CI 0.09 to 7.24, […]

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### Artificial Intelligence-Driven Decision-Making for Knee Joint Manipulation Following Primary Total Knee Arthroplasty.

*Ghirardelli S, Chan KCA, Valpiana P, Indelli P, Sculco PK, Lee GC*

BACKGROUND: Manipulation under anesthesia (MUA) is a commonly performed procedure to address postoperative stiffness after total knee arthroplasty (TKA), yet optimal timing, motion thresholds, and outcome expectations remain poorly defined due to limited high-quality evidence and heterogeneous patient presentations.

METHODS: This study evaluated the capacity of an artificial intelligence (AI) platform to synthesize a data-driven decision-making framework for MUA following primary TKA. A systematic literature review on the range of motion (ROM) trajectory and manipulation following primary TKA was performed, and the data were used to train the AI model. Scenarios iterating variables, including time from surgery, knee flexion angle, extension deficit, preoperative ROM, and body mass index, were modeled to define the appropriateness of MUA and expected ROM gains.

RESULTS: According to our AI model, MUA is indicated for patients who have knee flexion < 80 degrees and/or extension deficits greater than 20 degrees at six weeks postoperatively, predicting mean improvements of 26 degrees in flexion and three degrees in extension. Delaying MUA beyond 90 days reduced expected flexion gains to 17 degrees. For persistent stiffness beyond three months, the model advised combining MUA with arthroscopic lysis of adhesions, particularly for flexion less than 90 degrees. A body mass index gre […]

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### Predicting Acetabular Fixation Failure and Bone Loss in Total Hip Arthroplasty: A Combined In Silico and In Vitro Approach.

*Ramos A, Matos M*

BACKGROUND: Total hip arthroplasty continues to rise in prevalence in both the United States and Europe. The leading cause of revision surgery remains acetabular failure, primarily associated with instability and peri-implant bone loss. This study aimed to investigate bone loss around acetabular components following total hip arthroplasty.

METHODS: A combined in silico (finite element method, FEM) and in vitro approach was employed to predict short- to mid-term bone loss and assess acetabular implant stability. The FEM model incorporated anatomically realistic geometries of the femur, ilium, and implants (acetabular shell, polyethylene liner, and stem). The in vitro model utilized composite bones instrumented with strain gauges to measure cortical strains. Bone loss and acetabular loosening were simulated in four progressive steps, guided by FEM strain analysis, and reproduced through sequential computer-controlled manufacturing in the in vitro setup to evaluate implant stability. Loading conditions reproduced hip joint forces during gait.

RESULTS: Cortical strain measurements demonstrated strong agreement between the in silico and experimental models ($R^2 = 0.91$). Acetabular implantation reduced cortical bone strains by approximately 60% posteriorly and 80% anteriorly. Bone loss was most pronounced in the posterior-superior region and at cortical areas adjacent to the implant […]

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## The Fate of the Patient Who Has Early Dislocation After Contemporary Primary Total Hip Arthroplasty.

*Fuqua AA, Hrudka BT, Collins EC, Ross BJ, Premkumar A, Wilson JM*

INTRODUCTION: Recurrent instability remains a leading indication for revision following primary total hip arthroplasty (THA). However, data regarding outcomes following early dislocation are dated and limited in scope. Whether these existing data apply to contemporary practice with the broad adaptation of high-stability bearings and modern surgical techniques is unknown. Therefore, we assessed revision and redislocation rates following early dislocation in a contemporary group of patients following primary THA.

METHODS: A national database was used to identify patients undergoing primary THA from 2009 to 2022. Patients were divided into two cohorts: those with one or more dislocation events within 90 days and those who did not dislocate as the control group. Landmark survivorship analysis was utilized, given the potential for immortal time bias. The 2-year cumulative incidence of aseptic revision, septic revision, and recurrent dislocation after 90 days was determined by competing risk analysis.

RESULTS: At two years, the dislocation rate in all patients undergoing a single primary THA (n = 420,053) was 1.2%. After landmark analysis, the incidence of aseptic revision or dislocation after 90 days in patients with an unrevised early dislocation event (n = 1,942) was 29.0 versus 1.3% in controls (n = 360,493) ($P < 0.001$). The incidence of either dislocation or aseptic revision a […]

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## Concentration, Antibacterial Effect, and Safety of Combining Intra-articular Epsilon-Aminocaproic Acid and Vancomycin in Primary Total Knee Arthroplasty: A Randomized Study.

*Zhang Y, Wei J, Su C, Hu M, Xu H, Xiang S*

BACKGROUND: In this study, we aimed to investigate the intra-articular concentration, antibacterial effect, and safety of a combination of vancomycin and epsilon-aminocaproic acid (EACA) in primary total knee arthroplasty (TKA).

METHODS: There were 94 patients who underwent unilateral primary TKA and were randomized into the vancomycin plus EACA (VE) group and the vancomycin (V) group. There were 47 contemporary TKAs propensity score-matched to form the EACA (E) group. The VE group received vancomycin 30 mL (1,000 mg) plus EACA 20 mL, the V group received vancomycin plus saline, and the E group received saline plus EACA. Postoperative blood and drainage samples were collected for vancomycin analysis using liquid chromatography-tandem mass spectrometry. Perioperative blood loss, renal function, and bacterial tests for common periprosthetic joint infection pathogens were evaluated.

RESULTS: Perioperative blood loss in the VE group was equivalent to that in the E group and significantly lower than that in the V group ($P < 0.01$). The intra-articular vancomycin levels in the VE group were higher than those in the V group at eight and 24 hours, while the serum levels were similar. Antimicrobial effects were comparable in the VE and the V groups, both of which were superior to those in the E group. There were no cases of acute renal injury, ototoxicity, or anaphylaxis.

CONCLUSIONS: […]

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## Moderate-to-Severe Varus Deformity Is Associated With Conversion to Arthroplasty in Patients Who Have Subchondral Insufficiency Fracture of the Knee.

*Park JY, Cho BW, Kim TH, Park KK, Lee WS, Moon J, Kwon HM*

BACKGROUND: Although untreated subchondral insufficiency fractures of the knee may result in progression of osteoarthritis, leading to arthroplasty in approximately 30% of cases, the factors associated with the requirement for arthroplasty remain unknown. We hypothesized that varus deformity would be associated with increased risk of nonoperative treatment failure and conversion to arthroplasty in patients who have subchondral insufficiency fractures with early osteoarthritis.

METHODS: A retrospective cohort study was conducted on 162 patients aged greater than 60 years diagnosed with subchondral insufficiency fracture and early osteoarthritis (Kellgren-Lawrence grades 1 to 2) between March 2015 and December 2021. All patients received initial nonoperative treatment and were followed for a minimum of 36 months. Radiographic evaluation included hip-knee-ankle (HKA) angle measurement and osteoarthritis grading. Magnetic resonance imaging assessment evaluated subchondral insufficiency fracture location, meniscal tears, cartilage defects, and bone marrow edema. Kaplan-Meier survival analyses and Cox proportional hazards regressions were used to identify risk factors for progression to arthroplasty. At a median follow-up of 44 months, 50 patients (30.9%) required conversion to arthroplasty due to osteoarthritis progression.

RESULTS: Patients who had moderate-to-severe varus deform […]

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## Excision Versus Preservation of the Infrapatellar Fat Pad During Total Knee Arthroplasty: A Systematic Review and Meta-Analysis.

*Skaik K, Oulousian S, Lameire DL, Abbas A, Khalik HA, Al-Obaedi O, Ravi B*

BACKGROUND: This study aimed to evaluate whether preservation versus resection of the infrapatellar fat pad (IPFP) during total knee arthroplasty (TKA) affects clinical outcomes.

METHODS: A systematic review and meta-analysis, searching multiple databases up to November 2024 for comparative studies of infrapatellar fat pad-preservation (IPFP-P) versus resection in primary TKA, was conducted. Outcomes assessed included rates of complications, visual analog scale for pain, and the rate of anterior knee pain, Knee Society Score, patellar tendon length, range of motion, and operative time. Data were pooled using random-effects meta-analysis. There were 21 studies (3,573 patients, 4,107 TKAs: 2,298 preserved, and 1,809 resected IPFP) included.

RESULTS: The IPFP-P had a 76% reduction in the rate of complications within six weeks (relative risk = 0.24, 95% confidence interval [CI]: 0.12 to 0.48, $P < 0.01$), but not at later follow-up. The visual analog scale pain scores did not differ significantly between groups at any time point. The IPFP-P reduced the risk of anterior knee pain at three months (relative risk = 0.12, 95% CI: 0.02 to 0.94, $P = 0.04$), but this was not sustained at later follow-up. There were no significant differences found for operative time and Knee Society Score. However, infrapatellar fat pad-resection resulted in greater patellar tendon shortening at six and 12 […]

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## Prevalence, Progression, and Clinical Impact of Stem Notching in Ceramic-on-Ceramic Total Hip Arthroplasty: A Minimum 15-Year Follow-Up Study.

*Kang SY, Loh LL, Jeon YJ, Kim HS, Yoo JJ*

BACKGROUND: Stem notching, resulting from impingement between the femoral stem and ceramic liner in ceramic-on-ceramic (CoC) total hip arthroplasty (THA), has been linked to adverse outcomes such as ceramic-related noise and ceramic component fractures. Despite its potential significance, its true incidence is infrequently documented, and the long-term clinical impact remains uncertain. This study aimed to assess the minimum 15-year outcomes and complications associated with stem notching in CoC THA.

METHODS: We performed a retrospective cohort analysis of patients who received CoC THA between November 1997 and December 2003, with a minimum follow-up period of 15 years. Stem notching was diagnosed using radiographic evaluation and was observed in 21.5% of cases (63 of 293). The lesions were monitored for an average of 12.5 years from detection (range, one to 23.2). The primary endpoints included changes in notch depth and the occurrence of ceramic-related complications such as ceramic component fractures and noise generation. The secondary outcomes involved functional evaluation via the modified Harris Hip Score.

RESULTS: Notch depth did not demonstrate meaningful progression beyond five years after identification. Lower cup inclination and higher anteversion were associated with the presence of notching. The incidence of ceramic component fractures was 6.3% in notched hips c […]

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### Does Traumatic Brain Injury Increase Opioid Utilization After Primary Total Knee Arthroplasty?

*Young B, Shankar D, Sabet CJ, Leopold AP, Fernando ND, Hernandez NM*

BACKGROUND: Traumatic brain injury (TBI) is a common neurological injury with widespread systemic effects, leading to increased pain and opioid utilization. In this study, we investigated how a prior TBI affects perioperative opioid utilization in patients undergoing a total knee arthroplasty (TKA) and the risk of prolonged opioid usage.

METHODS: Using an administrative claims database, we identified patients undergoing a primary TKA from 2010 to 2022. Following inclusion and exclusion criteria, patients were categorized based on a prior diagnosis of a TBI, leading to a final TBI cohort of 127,369 patients and a control cohort of 1,116,605 patients. Our primary outcome was perioperative opioid utilization, defined as any opioid prescription between 30 days before and after surgery. Our secondary outcome was persistent opioid usage or continued opioid prescriptions 90 to 180 days after surgery. Multivariate regression models were used to assess the risk of persistent opioid usage based on TBI history, adjusting for demographics and comorbidities.

RESULTS: Patients who had a prior TBI had greater perioperative opioid utilization compared to patients who did not have a prior TBI, with 987.5 versus 896.7 morphine milligram equivalents ($P < 0.001$). The TBI cohort also showed greater rates of persistent opioid usage compared to the control cohort: 22,258 patients (17.5%) versus 162 […]

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### Minimum 10-Year Outcomes of Total Hip Arthroplasty With Highly Cross-Linked Polyethylene in Patients Under 50 Years: A Systematic Review and Meta-Analysis.

*Randall ZD, Mologne MS, Gaziano DJ, Clohisy JC, Bendich I*

BACKGROUND: Total hip arthroplasty (THA) is increasing in utilization among young patients. While THA markedly improves outcomes in those who have degenerative hip disease, patients under 50 years face an extended lifetime risk for complications and revisions. Highly cross-linked polyethylene (HXLPE) is associated with reduced wear, revision rates, and osteolysis compared to conventional polyethylene, potentially enhancing implant longevity in this younger, active population. This study systematically reviewed and meta-analyzed the outcomes of THA using HXLPE in patients under 50 years of age at a minimum 10-year follow-up.

METHODS: A systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Searches were performed in Embase, Ovid Medline, Scopus, and Cochrane databases. Inclusion criteria targeted studies with patients under 50 years reporting original data on THA using HXLPE with a mean follow-up of ≥ 10 years. Data extraction was performed independently by two reviewers, and study quality was evaluated using the Methodological Index for Non-Randomized Studies (MINORS) criteria. Multiple random-effects meta-analyses were performed to summarize outcomes. There were 18 studies included ($n = 1,587$ hips) with a weighted mean follow-up of 14.1 years.

RESULTS: Overall, revision-free survivorship was 98%, […]

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### The Impact of Major Depressive Disorder on Postoperative Somatic and Psychiatric Complications Following Total Knee Arthroplasty: A Database Study.

*Wahid M, Zaidi Z, Syed S, Nasser E, Meza C, Chen AF*

BACKGROUND: Major depressive disorder (MDD) is a psychiatric condition characterized by persistent low mood, anhedonia, sleep disturbances, guilt, fatigue, impaired concentration, appetite changes, psychomotor abnormalities, and suicidal ideation. It is a prevalent comorbidity among patients undergoing total knee arthroplasty (TKA) and may affect recovery. While prior research has emphasized pain and functional outcomes, the broader impact of MDD on postoperative psychiatric, systemic, and mortality outcomes remains underexplored. This study investigated whether MDD is independently associated with increased risk of newly diagnosed psychiatric, somatic, and mortality complications following primary TKA.

METHODS: In a novel analysis of a large electronic health record network, we identified adult patients undergoing primary, elective TKA from 2015 to 2024, excluding those who had prior knee pathology or psychiatric diagnoses other than MDD. A propensity score matching analysis adjusted for demographics, comorbidities, laboratory values, and medication use was used to compare postoperative outcomes at 90 days and one year using odds ratios (ORs).

RESULTS: Compared to controls, MDD patients experienced significantly higher odds of developing generalized anxiety disorder (OR90day 13.66; OR1year 11.73), adjustment disorder (OR90day 11.11; OR1year 6.91), post-traumatic stress disor […]

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## Delayed Administration of Apixaban, but Not Rivaroxaban, Reduces Transfusion Risk Without Increasing Thromboembolic Rates Following Revision Total Hip Arthroplasty.

*Telang SS, Lim MA, Kumaran P, Culler MW, Palmer R, Telang S, Burdick G, Lieberman JR et al.*

BACKGROUND: The optimal timing for initiating direct oral anticoagulants for venous thromboembolism prophylaxis following revision total hip arthroplasty (rTHA) is unknown. This study compared rates of thromboembolic and bleeding complications between patients beginning prophylactic courses of apixaban or rivaroxaban on postoperative day (POD) zero versus POD one.

METHODS: A national health care database including approximately 25% of all surgeries performed in the United States was used to identify patients who underwent aseptic both-component rTHA between 2016 and 2023. Patients receiving apixaban or rivaroxaban for chemoprophylaxis after rTHA on POD zero were compared to patients who received the same anticoagulant on POD one. Primary outcomes included 90-day rates of postoperative bleeding complications (acute anemia, hematoma, hemorrhage, and transfusion) and thromboembolic complications (deep vein thrombosis, pulmonary embolism, stroke, and myocardial infarction). Multivariable regression analysis was used to assess differences between cohorts. In total, 5,607 patients were identified. Of the 2,990 (53.33%) patients administered apixaban, 646 (21.61%) received anticoagulation beginning POD zero compared to 2,344 (78.39%) on POD one. Of the 2,617 (46.67%) patients given rivaroxaban, 598 (22.85%) received anticoagulation on POD zero compared to 2,019 (77.15%) on POD one. […]

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## Recovery Benchmarks After Total Knee Arthroplasty: The Impact of Motion-Restoring Surgery, Contracture, and Comorbidities on Costs and Utilization.

*Karim M, Beckley SJ, Stinton SK, Branch TP*

BACKGROUND: This study aimed to establish a claims-based timeline for recovery following unilateral total knee arthroplasty (TKA), including the impact of comorbidities and postoperative complications such as motion-restoring surgery (MRS) and joint contracture.

METHODS: Health care claims data were analyzed to determine the costs and recovery timeline after unilateral TKA. Costs related to (1) index surgery, (2) complication surgery, (3) TKA revision, (4) nonoperative hospitalization, (5) MRS, and (6) outpatient surgery were reported. The effect of joint contracture and comorbidities, including diabetes, obesity, peripheral vascular disease, joint infection, and cardiovascular disease, was assessed. The effect of postoperative complications was determined using data from patients rehospitalized with or without additional surgery.

RESULTS: Index surgery median cost was $30,346 (interquartile range: 14,203 to 350,304). Median recovery time (from index surgery to last physical therapy claim) was 126 days (interquartile range: 30 to 975), based on physical therapy billing claims, which may not reflect full functional recovery. While 54% of patients completed their postoperative period in six months, 46% took over six months. Recovery periods for patients who required a complicated surgery were six times longer, and costs were 14 times higher than for those who did not have compl

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# Journal of Orthopaedics & Traumatology

Volume 27, Issue 1 · March 2026 · 45 new articles

## Fractures of the coronoid process: state of the art.

*Marinelli A, Riva M, Coliva F, Minerba M, Carbone G, Guerra E*

Coronoid fractures are rarely isolated and are much more frequently associated with other osseous or ligamentous structures injuries. On the basis of the coronoid fracture patterns, described by the O'Driscoll classification, it is possible to recognize three main patterns of injury that differ on traumatic mechanism and on associated lesions: posterolateral rotatory instability, posteromedial rotatory instability, and axial load injuries. The management of coronoid fractures is challenging and varies according to characteristics of the fracture, associated lesions, and amount of elbow instability. In general, operative treatment is indicated in every case the fracture is at least 50% of the whole coronoid, whether the sublime tubercle is involved, and whether the ulno-humeral joint is not perfectly reduced. In conclusion, the correct management of the coronoid, especially in the setting of complex elbow instability, represents a predictive factor for patient outcomes and functional results. The stability of the elbow, rather than the size of the coronoid fragment, is the main parameter for surgical indication, aimed to fix the coronoid and/or repair the associated lesions.

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## Total hip arthroplasty in Italy: an observational, population-based study on surgical volume growth from 2001 to 2023 and forecasts until 2050 with six different statistical models.

*Ciminello E, Cuccu A, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L et al.*

BACKGROUND: The number of total hip arthroplasty (THA) procedures has been steadily increasing worldwide, driven by aging population, improvements in surgical techniques and implant design. This study aimed to analyze the temporal trends of elective THA in Italy since 2001-2023 and forecast THA volumes up to 2050 to provide insights for healthcare planning.

MATERIALS AND METHODS: International Classification of Diseases, 9th Revision, Clinical Modification (ICD9-CM) coding system was used to extract records of interest (elective THA) from the Italian National Hospital Discharge Record database. Six statistical models were applied to forecast future THA volumes: logistic regression; Poisson regression; logarithmic regression; inverse/power regression; Poisson log-normal regression; and hierarchical Poisson regression with temporal effects (HPTE). Model performances were assessed by using error metrics and internal validation on the basis of a rolling-origin approach. An out-of-sample validation was conducted to ensure a robust assessment of forecasting reliability. THA volume forecasts were provided with 95% prediction intervals.

RESULTS: A total of 1,318,400 records for primary elective THAs performed in Italy since 2001-2023 were analyzed. The number of THAs increased by approximately 80%, rising from 68.270 in 2001 to 122.777 in 2023. Among the tested models, HPTE generally […]

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## Clinical relevance of patient-reported outcome measures in the surgical management of focal chondral defects of the knee: a systematic review.

*Migliorini F, Maffulli N, Memminger MK, Hofmann UK*

INTRODUCTION: To evaluate clinical outcome following surgical management of focal chondral defects in the knee, patient-reported outcome measures (PROMs) are used. To give these measures meaning, parameters such as the minimal clinically important difference (MCID), patient-acceptable symptom state (PASS), minimally detectable change (MDC), clinically important difference (CID) and substantial clinical benefit (SCB) have been introduced. This systematic review investigated the MCID, SCB, CID, PASS and MDC of the most commonly used

PROMs for assessing patients following surgical repair of focal chondral defects of the knee.

METHODS: This systematic review was conducted in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and the recommendations of the Cochrane Handbook for Systematic Reviews of Interventions. All clinical studies investigating tools to assess the clinical relevance of PROMs in the surgical repair of focal chondral defects of the knee were reviewed. In April 2025, the following databases were accessed: PubMed, Web of Science and Embase. The PROMs of interest included: the International Knee Documentation Committee (IKDC) questionnaire, the Knee Injury and Osteoarthritis Outcome Score (KOOS) and its related subscales activities of daily living (ADL), pain, quality of life (QoL), sports and recreational […]

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## Simultaneous versus staged bilateral total hip arthroplasty: a meta-analysis.

*Soler F, Hernández J, Lamo-Espinosa JM, Benlloch M, Mariscal G*

BACKGROUND: There is debate regarding the optimal timing of bilateral total hip arthroplasty (THA), with simultaneous or staged approaches considered. Cost-effectiveness is an important factor that influences resource allocation. The main aim of this study was to assess the cost implications of bilateral simultaneous (sim-THA) versus staged (st-THA) total hip arthroplasty. The secondary objective was to evaluate the efficacy and safety of sim-THA compared those with of st-THA.

METHODS: A systematic review and meta-analysis were conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines. Studies comparing simultaneous and staged bilateral THA in terms of cost, complications, length of hospital stay, and patient outcomes were eligible. Odds ratios (ORs), mean differences (MD), and standard mean differences (SMD) with 95% confidence intervals (CIs) were calculated. The risk of bias was assessed using the Methodological Index for Non-randomised Studies (MINORS). Meta-analyses were performed using Review Manager version 5.4 (Cochrane, Oxford, United Kingdom).

RESULTS: A total of 20 observational studies including 13,984 patients were included. Sim-THA was associated with significantly lower total costs (SMD -0.54, 95% CI -0.92 to -0.16; p = 0.005) and shorter hospital stay (MD -2.90, 95% CI -4.38 to -1.42; p = 0.0001) than tho […]

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## Long-term outcomes of the modified Dunn procedure in moderate and severe slipped capital femoral epiphysis: a prospective case series with 7-year follow-up.

*Fahmy M, Abdelazeem AH, Shawky MA*

INTRODUCTION: Slipped capital femoral epiphysis (SCFE) is the most common hip disorder in adolescents and may lead to femoroacetabular impingement, early osteoarthritis, and long-term functional disability if inadequately treated. While in situ pinning remains the standard treatment for mild slips, it fails to correct the deformity in moderate and severe cases, potentially predisposing to degenerative changes. The modified Dunn procedure (MDP) was developed to restore proximal femoral anatomy through surgical hip dislocation while preserving vascular supply. The aim of the study is to evaluate the long-term radiological and functional outcomes of the MDP in patients with moderate (14 cases) and severe (10 cases) SCFE, and to assess the incidence of avascular necrosis (AVN), osteoarthritis, and other complications.

METHODS: A prospective case series was conducted between August 2015 and January 2019 at a single tertiary institution. A total of 24 hips with moderate-to-severe SCFE and open physis were treated using the MDP via surgical hip dislocation. Mild and acute-only slips were excluded. MDP was used as a primary procedure, performed early in severe slips and in selected moderate slips after clinical assessment. Patients were followed clinically and radiologically for a mean duration of 84 ± 2.6 months (range 80-88 months). Functional outcomes were assessed using the Harris […]

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## The presence of a positive blood culture should be regarded a risk factor for DAIR failure in cases of haematogenous total knee arthroplasty infection.

*Pedemonte-Parramón G, Reynaga E, Molinos S, de Los Ríos JD, Vivero A, García-Oltra E, López-Pérez V, Paredes R et al.*

PURPOSE: Late acute haematogenous total knee arthroplasty (LAH TKA) infections represent approximately 20% of all prosthetic infections. Debridement, antibiotics, and implant retention (DAIR) is a widely accepted treatment; however, it carries a significant risk of failure. The study aims to identify risk factors for DAIR failure in LAH TKA infections, focusing on blood cultures (BC) as a potential contributor.

METHODS: A retrospective cohort study was conducted on 37 cases of LAH TKA infections from 2015 to 2023. Patients were divided into two groups: 20 in the success group (SG) and 17 in the failure group (FG). The study analyzes various factors, including demographics, comorbidities, serum and joint fluid biomarkers, blood and intraoperative cultures, prior infection or antibiotic use, and surgical and post-surgical variables.

RESULTS: Positive BC were more frequent in the FG compared with the SG ($P = 0.03$), and were also more common in patients with a history of prior infection ($P = 0.03$). Logistic regression identified positive BC as the only significant predictor of DAIR failure (odds ratio (OR) 12, 95% confidence interval (CI) 1.1-18, $P = 0.04$), even after adjusting for other variables. Positive intraoperative cultures were more frequent in the FG, particularly in those with a prior infection history ($P = 0.08$), even if they were receiving antibiotics ($P = 0.05$).

CON […]

PubMed · DOI

## Iatrogenic femoral neck fractures or separation of the proximal femoral epiphysis during closed reduction of irreducible femoral head fracture-dislocations in children: a review of 12 cases.

*Lu Y, Xue C, Canavese F, De Salvo S, Yao H, Huang D, Lin R, Zheng J et al.*

OBJECTIVES: This study aimed to analyze the radiologic characteristics of irreducible pediatric femoral head fracture-dislocations to prevent potential iatrogenic femoral neck fractures (FNF) or separation of the proximal femoral epiphysis (SPFE).

METHODS: This is a retrospective review of patients who were skeletally immature and diagnosed with traumatic hip dislocations combined with femoral head fractures. The collected data included patient demographics, fracture classification, fragment ratio, combined injuries, urgent reductions, treatment strategies, complications, and final outcomes.

RESULTS: We treated 12 patients with femoral head fractures and dislocations; 11 out of 12 patients underwent urgent closed reduction (91.7%). Six of the patients failed closed reduction and experienced FNF ($n = 2$, 33.3%) or SPFE ($n = 4$, 66.7%). Five patients presented with avascular necrosis of the femoral head after open reduction with internal fixation via a surgical hip dislocation approach, and two patients required further surgical treatment. Analysis of radiographs and computed tomography (CT) scans of irreducible femoral head fracture-dislocations revealed that the fractured femoral head was perched on the sharp angle of the posterior wall of the acetabulum, with a fragment ratio of 18-27%. After recognizing the irreducibility, one case with a fragment ratio of 26% underwent immed […]

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## Orthopaedic publications receive fewer citations than general medicine articles: a scientometric analysis of temporal trends.

*Migliorini F, Vaishya R, Maffulli N, Eschweiler J, Kobbe P, Betsch M, Schäfer L, Oliva F et al.*

BACKGROUND: Orthopaedics, as a surgical speciality with a structurally narrower citing audience and lower citation density than general medicine, is systematically disadvantaged in any cross-disciplinary comparison relying on absolute citation counts. Whether this disadvantage is stable or has changed over time has not been previously quantified.

METHODS: The 100 most-cited articles indexed in the Web of Science Core Collection were identified for medicine and orthopaedics in 2014 and 2024. Citation data were extracted from a single snapshot taken on 19 January 2026. Total citations and citation velocity, defined as citations per year since publication, were compared between fields and across time points using Mann-Whitney U tests. Ratio-based analyses and distributional characteristics were also assessed.

RESULTS: Median citation counts were substantially higher in medicine than in orthopaedics in both years, reaching 5055 versus 288 in 2014 and 720.5 versus 34.0 in 2024, corresponding to 17.6- and 21.2-fold differences, respectively. Citation velocity followed the same pattern, with medicine maintaining an approximately 17- to 21-fold advantage at both time points. Within-field temporal analysis showed that citation velocity remained broadly stable in medicine between cohorts, whilst orthopaedics demonstrated reduced early citation accrual in the more recent cohort. Citatio […]

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### Sleep quality and nocturnal pain in patients with elbow disorders: a prospective multicenter study.

*Schneider MM, Nietschke R, Migliorini F, Schoch C, Burkhart K, Hollinger B, Kimmeyer M, Lehmann L et al.*

BACKGROUND: Mental health disorders have become a growing global concern. Sleep quality is known to have a crucial impact on mental well-being. Poor sleep contributes to increasing economic costs owing to reduced productivity as well as the development of mental disorders. Conversely, anxiety and depression often lead to sleep disturbances. Pain is a key factor affecting sleep quality. Studies have shown that musculoskeletal disorders, especially in the shoulder, may cause sleep disturbances. Research on the relationship between elbow pain and sleep quality remains limited. This study aims to evaluate the impact of common elbow pathologies on sleep quality.

METHODS: This prospective multicenter cohort study used self-reported outcome measures in patients with elbow disorders, including the Pittsburgh Sleep Quality Index (PSQI), to assess sleep quality. The PSQI was compared with other outcome measures.

RESULTS: A total of 664 patients (284 female, 380 male) were included, of whom 81% reported nocturnal pain. The mean age was 47.9 ± 14.2 years, body mass index (BMI) 26.9 ± 6.7 kg/m2 and symptom duration 15.8 ± 7.3 months. The most common diagnoses were lateral epicondylar pain (66%), medial epicondylar pain, arthritis, ulnar nerve entrapment and biceps/triceps tendinopathies. The overall PSQI was 7.3 ± 3.8, disabilities of arm, shoulder and hand questionnaire (DASH) 35.0 ± 19. […]

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### A novel figure-of-eight polymeric cerclage wiring technique for greater trochanter fixation in revision arthroplasty and conversion of failed proximal femur fracture fixations to total hip arthroplasty.

*Leggieri F, Martín Cocilova FN, Secci G, Stimolo D, Zanna L, Civinini R, Innocenti M*

INTRODUCTION: Periprosthetic femoral fractures occur in 0.8-10% of primary total hip arthroplasties (THA) and up to 17.5% for revision THA (revTHA), with isolated greater trochanter (GT) fractures being the most predominant pattern. This study aimed to evaluate a novel figure-of-eight cerclage wiring technique for greater trochanter fixation in hip replacement cases.

METHODS: These retrospectively collected data included 36 hips (36 patients) undergoing revision THA or conversion of failed intertrochanteric fracture to THA with figure-of-eight cerclage wiring between April 2021 and June 2023. Patients under 18 years, with incomplete records, or lost to follow-up before 24 months were excluded. Primary outcomes included Trendelenburg sign, periprosthetic infections, and hip dislocation rates. Secondary outcomes evaluated THA stem subsidence and trochanteric migration using calibrated digital imaging. Statistical analysis included descriptive statistics, independent t-tests, linear regression, and correlation analyses with significance set at $p < 0.05$.

RESULTS: Mean follow-up was 404.76 ± 22.51 days. Patient demographics showed mean age 76.5 ± 12.3 years, body mass index (BMI) 25.8 ± 4.5 kg/m2, with 73.8% female patients. At final follow-up, one patient (2.8%) experienced post-surgical infection, one patient (2.8%) had dislocation, and three patients (8.3%) exhibited positive T […]

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### Minimal clinically important difference (MCID), patient-acceptable state (PASS), minimally detectable change (MDC), and substantial clinical benefit (SCB) in patients who have

### undergone arthroscopic surgery for femoroacetabular impingement: a systematic review.

*Migliorini F, Maffulli N, Schäfer L, Jeyaraman M, Betsch M, Mahmoud M*

INTRODUCTION: Hip arthroscopy has been established as a successful management for femoroacetabular impingement (FAI) syndrome. Patient-reported outcome measures (PROMs) are frequently used to assess treatment results in patients. However, clinically important outcome values have recently been calculated to better understand and interpret PROMs and define successful treatment. This systematic review aimed to investigate the minimal clinically important difference (MCID), patient-acceptable state (PASS), substantial clinical benefit (SCB), and minimally detectable change (MIC) of the most commonly used PROMs for assessing patients who have undergone arthroscopic surgery for FAI.

MATERIAL AND METHODS: This study was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses: the 2020 PRISMA statement. In October 2025, the following databases were accessed: PubMed, Web of Science, and Embase. All clinical studies investigating tools to assess the clinical relevance of PROMs used to evaluate patients who have undergone arthroscopic surgery for FAI were accessed. Only studies that assessed the minimal MCID, PASS, SCB, and/or MIC were considered.

RESULTS: The methodological quality of the 21 included studies was assessed, and the overall risk of bias was found to be low-to-moderate. The main characteristics of the included studies were analyzed and […]

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### Efficacy and safety of MIPO versus ORIF in distal tibial fractures: a systematic review and meta-analysis.

*Hassaan AM, Hassan AA, Abdelkhalik WS, Saad AM, Hosny AS, Elfeky BAEH, Mandour I, Ragheb MG et al.*

BACKGROUND: Distal tibial fractures are challenging to manage owing to limited soft-tissue coverage and compromised blood supply. Minimally invasive plate osteosynthesis (MIPO) and open reduction and internal fixation (ORIF) are among the widely used surgical options, but the optimal approach remains unclear. MIPO preserves periosteal blood flow with smaller incisions, while ORIF offers direct visualisation but increases soft-tissue damage. With growing comparative evidence, an updated evaluation of these techniques is necessary.

AIM: To compare the efficacy and safety of MIPO versus ORIF in adult patients with distal tibial fractures.

METHODS: Following PRISMA guidelines and the Cochrane Handbook, we searched in PubMed, Embase, Cochrane, Scopus, and Web of Science till November 2025. Randomised trials and cohort studies comparing MIPO and ORIF were included. Data extraction was performed independently by two reviewers. Risk of bias was assessed using RoB 2 for RCTs and the Newcastle-Ottawa Scale for observational studies. Random-effects models were used to estimate pooled mean differences and odds ratios, and heterogeneity was assessed using I2. Sensitivity and subgroup analyses were conducted by study design.

RESULTS: Nine studies involving 530 patients met our inclusion criteria. No significant differences were found between MIPO and ORIF in AOFAS scores, union time, oper […]

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### Comparative efficacy and safety of aspirin, low-molecular-weight heparin, and rivaroxaban for deep venous thrombosis prophylaxis after hip arthroplasty for femoral neck fracture: a single-center randomized controlled trial.

*Miao S, Yue Q, Zhang X, Lu J, Zhang Y, Lu N, Wei L*

OBJECTIVE: To evaluate the efficacy and safety of aspirin, low-molecular-weight heparin (LMWH), and rivaroxaban for deep vein thrombosis (DVT) prophylaxis after total hip arthroplasty (THA) or bipolar hemiarthroplasty (BHA) for femoral neck fracture, with subgroup analysis stratified by surgical type.

METHODS: This single-center randomized controlled trial (RCT) was conducted between June 2024 and June 2025. A total of 217 consecutive patients with femoral neck fracture undergoing hip arthroplasty were initially screened, and 161 eligible patients were finally enrolled and randomly assigned to the aspirin group, LMWH group, or rivaroxaban group at a 1:1:1 ratio. All patients received standardized intermittent pneumatic compression (IPC) and graduated compression stockings for mechanical DVT prophylaxis, as well as a unified postoperative rehabilitation protocol. The primary outcome was the incidence of lower extremity DVT within 12 weeks postoperatively, confirmed by color Doppler ultrasound. The secondary outcome was the incidence of bleeding-related complications graded by the International Society on Thrombosis and Hemostasis (ISTH) criteria. All patients were followed up at 6, 8, and 12 weeks postoperatively, with routine bilateral lower extremity ultrasound performed at 12 weeks regardless of symptoms. Subgroup analysis was performed on the basis of surgical type (THA ver […]

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### Does the surgical sequence of lumbar spine fusion and total hip arthroplasty affect the dislocation and revision risk? A meta■analysis and systematic review.

*Liu F, Jia C, Wu X, Zhou Y*

BACKGROUND: Patients who underwent primary total hip arthroplasty (THA) after lumbar spine fusion (LSF) had a higher incidence of complications compared with those who had not previously undergone LSF. This study aimed to determine whether the timing of THA before or after LSF impacts the incidence of THA complications.

METHODS: Following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed, Web of science, Embase, and Cochrane Library from their inception until 1 October 2025. Data from all eligible published studies meeting the inclusion criteria were extracted and subsequently analyzed using a random-effects model to calculate the rates of all-cause revision, dislocation, periprosthetic joint infections, and aseptic loosening.

RESULTS: Our analysis included 10 studies involving 27,605 patients who underwent THA followed by LSF and 32,002 patients who received LSF prior to THA. There are no significant differences in the rates of dislocation (odds ratio [OR] 1.19, 95% confidence interval (CI) 0.73-1.86, $p = 0.45$), periprosthetic joint infections (OR 0.91, 95% CI 0.52-1.62, $p = 0.76$), and aseptic loosening (OR 0.89, 95% CI 0.74-1.08, $p = 0.23$) between the two surgical sequences, while for all-cause revision, the OR was (OR 1.01, 95% CI 0.70-1.47, $p = 0.94$), which reached statistical significance.

CONCLUSIONS: Compare […]

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### Effects of anterior cruciate ligament resection and total knee arthroplasty on lower limb alignment and tibiofemoral rotation.

*Fan Y, Chen S, Jiang L, Chen Z, Wu M, Wang C*

BACKGROUND: The anterior cruciate ligament (ACL) is a key structure that restricts excessive anterior translation of the tibia and maintains rotational stability. This study aims to dynamically and quantitatively assess the direct effects of ACL resection and total knee arthroplasty (TKA) on lower limb alignment and tibiofemoral joint rotation under passive, unloaded conditions and in the pathological state of osteoarthritis.

METHODS: This retrospective study included 110 patients with varus knee osteoarthritis who underwent Mako for robotic-assisted TKA. Dynamic intraoperative measurements of the lower limb mechanical axis (flexion angle at maximum extension; varus angle) and tibiofemoral rotation at multiple flexion angles (min to max flexion) were recorded at three intervals: pre-ACL resection, post-ACL resection and post-TKA. The variation in rotation amplitude was calculated for successive flexion intervals.

RESULTS: Post-ACL resection, a significant decrease in the flexion angle at the maximum extension and varus angle (both $p < 0.001$) was observed, which was accompanied by increased tibial internal rotation at flexion angles of 60° and above ($p < 0.05$). Varus alignment was successfully corrected following TKA ($p < 0.001$). However, tibiofemoral rotational kinematics were significantly modified: the amplitude of internal rotation decreased in the initial flexion arc (min […]

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### Comparison of Bankart repair with remplissage and Latarjet procedure for anterior shoulder instability: a systematic review and meta-analysis.

*Nong K, Liu G, Jiang S, Li Q, Kwabena BR, Yuan D, Xiao W, Zhang K et al.*

BACKGROUND: Anterior shoulder instability is the most common type of shoulder instability. For cases with subcritical glenoid bone loss, both Bankart repair with remplissage (BR) and the Latarjet procedure are viable surgical options. The aim of this study was to compare the postoperative outcomes in patients undergoing BR and Latarjet procedure.

METHODS: This review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. The methodological index for nonrandomized studies (MINORS) was used to characterize the methodological quality. A literature search was conducted using PubMed, the Cochrane Library, Embase, Web of Science, and Scopus. The analyzed outcomes included postoperative Rowe score, visual analog scale (VAS) score, the Single Assessment Numeric Evaluation (SANE) score, range of motion (ROM), return-to-sport (RTS) rate, RTS time, recurrence rate, revision rate, and complications. Subgroup analysis was performed on the basis of follow-up duration.

RESULTS: Overall, 12 studies involving 1186 patients were included. The meta-analysis showed that BR was associated with a significantly lower complication rate than the Latarjet procedure. No significant differences were observed in functional outcomes, ROM, recurrence, revision, RTS, or RTS time. Subgroup analysis showed a significant interaction between f […]

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## Risk of periprosthetic joint infection within 1 year following robotic-assisted versus conventional primary total knee arthroplasty: a propensity-score-matched cohort study.

*Tian S, Li Y, Mu W, Ji B, Zhang X, Cao L, Zhao J*

BACKGROUND: Robotic-assisted total knee arthroplasty (RA-TKA) is increasingly being adopted for its ability to enhance bone-resection accuracy and component alignment. However, whether these technical gains influence the risk of periprosthetic joint infection (PJI) remains uncertain, especially in the context of prolonged operative duration. This study aimed to compare the 1-year rate of PJI following conventional total knee arthroplasty (cTKA) and RA-TKA in a propensity-score-matched cohort.

METHODS: We retrospectively reviewed 1284 consecutive patients who underwent primary TKAs at a single centre between 2021 and 2023. The patients were stratified according to surgical technique (cTKA versus RA-TKA) and subsequently matched 1:1 using propensity score analysis (age, sex, body mass index [BMI], American Society of Anesthesiologists [ASA] score, Charlson Comorbidity Index [CCI] score, CCI components and smoking), resulting in 522 pairs (1044 patients) for the final comparative analysis. Operative time and 1-year PJI were assessed using multivariable logistic regression. Infections were stratified according to timing: ≤ 90 days and from 90 days to 1 year after surgery.

RESULTS: The 1-year rate of PJI was 0.77% (4/522) after RA-TKA and 0.96% (5/522) after cTKA ($P = 1.000$). All PJIs in patients who underwent RA-TKA occurred within 90 days, whereas PJIs in patients who underwent […]

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## Measured resection, gap balancing, and force-sensor-guided total knee arthroplasty result in different femoral and tibial rotational alignment but similar combined knee rotation and clinical outcomes: a randomized controlled trial.

*Hernández-Hermoso JA, Nescolarde L, Yañez-Siller F, Calle-García J, Pérez-Andrés R, García-Perdomo D*

BACKGROUND: Different alignment philosophies and balancing methods may alter femoral and tibial component rotation in distinct ways within the same knee. This study aimed to (1) determine which of three surgical techniques-measured resection (MR), gap balancing with computer-assisted surgery (GB-CAS), or a force-sensor soft-tissue balancing device (FS-STB)-most closely reproduces native femoral, tibial, and combined rotational alignment in mechanically aligned total knee arthroplasty (TKA), and (2) assess whether differences in rotational alignment affect outcomes at 5 years postoperatively.

MATERIAL AND METHODS: A total of 60 patients undergoing primary mechanical alignment TKA were randomly assigned to one of three surgical approaches ($n = 20$ per group): MR, GB-CAS, or FS-STB. Blinded observers assessed the Knee Society Score (KSS), Western Ontario MacMaster Universities Osteoarthritis Index (WOMAC) score, and hip-knee-ankle angle preoperatively and at the 5-year follow-up visit. Pre- and postoperative two-dimensional (2D)-computed tomography scans were used to measure femoral rotation (BFA), tibial rotation, and combined femur-tibia rotation (TE_PTCA and BC_PTCA). Statistical analyses included paired t-tests, one-way analysis of variance, and effect size calculations.

RESULTS: Femoral rotation remained unchanged in the MR and GB-CAS groups, but decreased slightly (1° exter […]

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### Risser growth plate injury in unstable paediatric pelvic fractures: a multicentre retrospective study.

*Oransky M, Aulisa AG, Roncoroni A, Zoppi AR, Mata M, Rohayem MA, Falciglia F, Toniolo RM*

INTRODUCTION: Unstable pelvic fractures in children are rare but severe injuries and are associated with high-energy trauma. Unlike adults, children have open growth plates, particularly the Risser growth plate (RGP), which is vulnerable to injury. The presence of growth plates, such as the triradiate cartilage and Risser's plate, represents point of vulnerability that determines distinctive fracture patterns, similar to those seen with Salter-Harris growth plate injuries. Lateral compression fracture is the most common mechanism and results in an irreducible crushing of the sacral alae, causing a rotational deformity. Both vertical and lateral compression displaced pelvic fractures require careful evaluation of the posterior iliac apophysis-associated injury. Failure to treat Risser growth plate injury (RGPI) can lead to severe long-term problems such as limb length discrepancy and scoliosis. This study aimed to determine how commonly RGPI occurs in unstable paediatric pelvic fractures, link it to specific fracture patterns and propose a subgroup type to improve treatment.

MATERIALS AND METHODS: This multicentre retrospective study included 40 children aged up to 12 years with unstable pelvic fractures (AO/OTA 61B and 61C). The patients were treated between 1987 and 2022 at trauma centres in Italy, Argentina and Venezuela. We reviewed patient demographics, injury mechanisms a […]

PubMed · DOI

### Total femur replacement: function, quality of life, and gait: a multicentric EMSOS study.

*Valentini M, Svehlik M, Leithner A, Bergovec M, EMSOS TFR Study Group, Smolle MA*

BACKGROUND: Total femur replacements (TFRs) involve the replacement of the whole femoral bone, reconstruction of the hip and knee joints, and soft tissue reconstruction for function and joint stability. The aim of this multicentric European Musculo-Skeletal Oncology Society (EMSOS) study is to report on the functional outcome, quality of life, as well as gait analysis in patients with TFR.

MATERIALS AND METHODS: Retrospective data from 12 international participating centers were collected. In total, 65 patients [median (IQR) age at surgery 40 (18-56) years, 60% male] who received a TFR owing to oncologic or nononcologic indications between 1 January 1990 and 31 March 2024 were included. Patient-related outcome measures (PROMs), range of motion (ROM) of the hip and knee, and gait videos were collected.

RESULTS: At a median (IQR) follow-up of 28 (14-126) months, the median (IQR) percentage Musculoskeletal Tumor Society (MSTS) score was 63% (43-73%), 12-Item Short-Form Health Survey (SF-12) was 70% (55-79%), Harris Hip Score (HHS) was 63% (52-77%), and Oxford Knee Score (OKS) was 63% (52-79%); the median (IQR) extension-to-flexion ROM were 80° (70-97.5°, hip) and 70° (40-90°, knee). Gait videos were available for 24/65 patients (37%). The median (IQR) total Edinburgh Visual Gait Score of the operated side was 8.5 (6-13), with deviations from norm in all patients (24/24). Compens […]

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### Vascularized versus nonvascularized free fibular grafts in reconstruction of post-traumatic critical long bone defects, a comparative study.

*Fouaad AA, Massoud AH, Shaban M, Abdel-Naby M, Abulsoud MI*

BACKGROUND: Reconstruction of critical-sized long bone defects is a complex orthopedic challenge. Free fibular grafts, vascularized (FVFG) and nonvascularized (NVFG), are established reconstructive options, but comparative clinical outcomes remain uncertain.

PURPOSE: To compare clinical and radiological outcomes of FVFG and NVFG in patients with post-traumatic critical bone defects more than 10 cm.

METHODS: A randomized controlled trial was conducted with 50 patients assigned equally to FVFG or NVFG groups. The primary outcome was time to union, while secondary outcomes included graft hypertrophy, functional scores (DASH and LEFS), complication rates, and donor site morbidity.

RESULTS: The mean time to union was 5.78 months in the FVFG group and 6.17 months in the NVFG group, showing no statistically significant difference (p = 0.447). Rates of graft hypertrophy, functional recovery, and complications were comparable between the groups.

CONCLUSIONS: Both FVFG and NVFG provide effective reconstruction for critical bone defects, with nearly similar healing times and functional outcomes. NVFG may represent a less complex alternative in selected cases.

PubMed · DOI

## Five-year outcomes of medial open-wedge high tibial osteotomy with lateral supplemental lag screw in patients with obesity.

*Tian YH, Lee KH, Huang CW, Lin SY, Yang JC*

BACKGROUND: Obesity contributes to the accelerated progression of knee osteoarthritis. Medial open-wedge high tibial osteotomy (MOWHTO) is a joint-preserving surgical intervention for unicompartmental knee osteoarthritis; however, its efficacy in patients with obesity remains controversial. This study aimed to evaluate the 5-year clinical and radiographic outcomes of MOWHTO with lateral supplemental lag screw fixation in patients with obesity.

MATERIALS AND METHODS: This retrospective cohort study included patients with obesity who underwent MOWHTO between 2017 and 2020, with a minimum follow-up of 5 years. All procedures were performed by a single surgeon using three-dimensional printed, patient-specific instrumentation, locking plates, and a lateral supplemental lag screw. Clinical outcomes were assessed using the visual analog scale (VAS) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC). Radiographic parameters, including the weight-bearing line percentage (WBL%) and medial proximal tibial angle (MPTA), were evaluated.

RESULTS: Significant improvements in VAS and WOMAC scores were observed at all postoperative time points ($p < 0.001$) and were accompanied by improved radiographic alignment, with WBL% shifting toward the Fujisawa point and increased MPTA values. At 5 years, mild regression in alignment was noted; however, the overall correction was […]

PubMed · DOI

## Outcomes of surgical fixation of osteochondral fracture: a systematic review.

*Pang AS, Oviya R, Tan JJ, Lim KSA, Hui JHP, Tan SHS*

BACKGROUND: Dealing with knee osteochondral defects presents a substantial clinical challenge due to the variable nature of repair outcomes and intricate biomechanical complexities inherent to the knee joint. Numerous surgical alternatives exist for addressing knee osteochondral lesions, yet there is limited evidence available for comparing their respective outcomes.

METHODS: According to the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guidelines, a systematic literature search was conducted to identify publications from inception to September 2023 on PubMed, Cochrane, and Medline academic search engines with the MeSH terms "osteochondral" AND ("patella" OR "patellar" OR "patellofemoral" OR "femoral condyle lesions" OR "trochlear lesions"). Inclusion criteria were clinical human studies, English language, subjects who underwent surgical management (fixation, reconstruction, or debridement) for knee osteochondral lesions with reported treatment outcomes, and a minimum sample size of ten patients. The methodological index for non-randomized studies (MINORS) was used to evaluate the non-randomized studies' methodological quality. Main outcome measurements Pediatric International Knee Documentation Committee (Pedi IKDC) scale, Tegner-Lysholm Scoring Scale, Tegner Activity Score, Visual Analog Scale, Knee Society Score (KSS) Score, Kujala Score, and F […]

PubMed · DOI

## Do modular tapered long stems differ in stability and function? A prospective comparison of two stems in complex femoral revision THA.

*Fahmy M, Shawky MA*

BACKGROUND: Revision total hip arthroplasty (rTHA) in the setting of substantial proximal femoral bone loss is technically challenging. Modular tapered fluted stems provide predictable diaphyseal fixation while allowing independent adjustment of version, offset, and limb length. Among these, two commonly used systems-modular tapered fluted stem Type A (Revitan™) and Type B (LIMA Modulus™)-have limited direct comparative evidence. This study aimed to prospectively compare radiographic stem subsidence (primary outcome), as well as functional outcomes, complications, survivorship, and secondary outcomes, between Type A and Type B modular long stems in femoral rTHA.

METHODS: In this single-center randomized prospective study, 110 patients undergoing femoral revision rTHA were randomly assigned to receive either Type A (n = 55) or Type B (n = 55) stems. All procedures were performed by experienced revision surgeons under standardized perioperative and rehabilitation protocols. Radiographs were analyzed for stem subsidence, osseointegration, and limb-length restoration. Functional outcomes were assessed using the Harris Hip Score (HHS), Oxford Hip Score (OHS), and European Quality of Life Visual Analogue Scale (EQ-VAS) at baseline and final follow-up (mean 61.4 months). Complications and stem survivorship were recorded prospectively. Statistical analysis included paired and unpaired […]

PubMed · DOI

## Clinical comparison of single posterolateral plate with medial-cannulated-screw fixation and double-plate fixation for extra-articular distal humerus fractures.

*Jung HS, Chu MS, Lee JS*

BACKGROUND: Double-plate fixation is the gold standard for extra-articular distal humerus fractures, but it carries a substantial risk of postoperative ulnar neuropathy. Fixation using a single posterolateral plate with a medial cannulated screw may reduce ulnar neuropathy while maintaining fracture stability. This study aimed to compare the clinical outcomes of single-plate-with-medial-screw fixation versus double-plate fixation for extra-articular distal humerus fractures.

MATERIALS AND METHODS: Fifty-six patients who underwent surgery for extra-articular distal humerus fractures (Arbeitsgemeinschaft für Osteosynthesefragen/Orthopaedic Trauma Association [AO/OTA] classification A2 or A3) between January 2018 and August 2024 were divided into a double-plate group and a single-plate-with-medial-screw group. We conducted a retrospective, nonrandomized comparative study. The double-plate fixation was used in 30 patients from January 2018 to October 2021, while the single-plate fixation with a medial screw was used in 26 patients from November 2021 to August 2024. All surgeries were performed using a posterior paratricipital approach. Bony union, radiographic healing, and loss of reduction were evaluated. Postoperative pain scores (visual analog scale at 2 days after the operation), operative time (minutes), elbow range of motion, elbow function (Mayo Elbow Performance Score [MEP […]

PubMed · DOI

## A study on the association between tibial plateau fractures and intra-articular soft-tissue injuries under valgus injury mechanisms.

*Duan S, Wang S, Zhu T, Li Q, Zhang M*

BACKGROUND: While recent investigations have focused on injury mechanism classifications of tibial plateau fractures (TPFs), the association between valgus TPFs and concomitant soft-tissue damage involving menisci and ligaments remains insufficiently elucidated. This study aimed to characterize intra-articular soft-tissue injuries associated with various valgus TPFs and assess the predictive value of lateral plateau depression (LPD) and widening (LPW). Additionally, the analysis extended to other injury mechanisms.

MATERIALS AND METHODS: This study included adult patients with acute tibial plateau fractures who had complete imaging data, excluding patients with open fractures and multiple fractures throughout the body. Imaging examinations were used to assess the fracture mechanism and intra-articular soft-tissue damage.

RESULTS: The study retrospectively analyzed the clinical data of 138 patients with valgus injury TPFs in our hospital. The study compared the incidence of TPFs with intra-articular soft-tissue damage under different injury mechanisms. The result demonstrated that the incidence of valgus hyperextension TPFs combined with medial collateral ligament injuries was relatively high (61.5%) compared with TPFs with other injury mechanism. Multivariable logistic regression and smooth curve fitting revealed significant dose-response relationships of LPD (odds ratio [OR] […]

PubMed · DOI

### The coronoid as the key fragment of trans-ulnar fracture-dislocations of the elbow: Insights from a retrospective cohort comparison using the coronoid-centric Mayo classification system.

*Henssler L, Kerschbaum M, Zanklmaier M, Wieczorek LA, Alt V, Klute L*

BACKGROUND: Trans-ulnar fracture-dislocations of the elbow are rare injuries with complex fracture patterns and variable outcomes. Traditional classification systems offer limited prognostic value. A recently introduced coronoid-centric Mayo classification distinguishes injury subtypes based on coronoid attachment and identifies trans-ulnar basal coronoid (TUBC) fractures as a particularly challenging entity. This study evaluated outcomes across Mayo fracture types and explored factors associated with inferior results in TUBC injuries.

MATERIALS AND METHODS: In this retrospective cohort study, surgically treated trans-ulnar elbow fracture-dislocations managed at a level I trauma center between 2010 and 2022 were identified and classified according to the Mayo system. Demographic data, injury characteristics, surgical management, radiographic outcomes, and complications were recorded. Functional outcomes were assessed after a minimum follow-up of 12 months using the Mayo Elbow Performance Score (MEPS); Oxford Elbow Score (OES); Quick Disabilities of Arm, Shoulder and Hand Questionnaire (QuickDASH); European Quality of Life Five-Dimension, Five-Level Version (EQ-5D-5L); and range-of-motion measurements. Radiographs were analyzed for union, instability, heterotopic ossification, and post-traumatic osteoarthritis (OA).

RESULTS: A total of 52 patients were included (14 trans-olecr […]

PubMed · DOI

### Feasibility study of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

*Hinz N, Althoff G, Thomaschewski M, Bonke LA, Eberenz A, Münch M, Behrends L, Männel G et al.*

BACKGROUND: Minimally invasive stabilization of acetabular and pelvic ring fractures using endoscopic techniques has become increasingly important. The logical advancement of conventional endoscopic techniques is a robot-assisted approach, which benefits from the advantages of robot-assistance systems (e.g., more degrees of freedom for instruments and improved visualization). The aim was therefore to investigate the feasibility of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

MATERIALS AND METHODS: In this two-part feasibility study, four different plate osteosyntheses were performed endoscopically using the Hugo™ robot-assisted surgery system, first on ten synthetic pelvic models and then on ten human cadavers. During robot-assisted dissection for the preperitoneal approach, identification of ten relevant anatomical landmarks was also assessed. In both parts, the success, number of drilling errors, and time for each plate were described as learning curves and analyzed using linear regression.

RESULTS: The infrapectineal plate could be successfully performed in 100% of synthetic models, the posterior column plate in 100%, the suprapectineal plate in 90%, and the superior pubic ramus plate in 80%. Learning curves could be observed for the number of drilling errors per plate (e.g., 0.67 to 0, $p = 0.009$) and the time r […]

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### Favorable subjective clinical outcomes after revision hip arthroscopy for femoroacetabular impingement syndrome despite a high rate of capsular defects and increased capsular thickness.

*Cao Z, Gao G, Lin W, Zhu Y, Zhou X, Wang J, Xu Y*

BACKGROUND: The capsular healing status and capsular thickness changes in patients with femoroacetabular impingement syndrome (FAIS) following revision hip arthroscopy are poorly documented, and their relationship with subjective clinical outcomes remains unclear. The purpose of this work is to evaluate the incidence of capsular defects and changes in capsular thickness using magnetic resonance imaging (MRI) following hip arthroscopy in patients with FAIS, and to compare the subjective clinical outcomes between patients with and without capsular healing after revision hip arthroscopy.

PATIENTS AND METHODS: Consecutive patients with FAIS who underwent revision hip arthroscopy between 2013 and 2023 were included. Patients were categorized into two groups on the basis of capsular healing status after revision hip arthroscopy. Patient-reported outcomes (PROs) were collected preoperatively and at minimum 2-year follow-up, including the modified Harris Hip Score (mHHS), International Hip Outcome Tool-12 (iHOT-12), Hip Outcome Score-Activity of Daily Living Scale (HOS-ADL), Hip Outcome Score-Sport Specific Subscale (HOS-SSS), and visual analog scale (VAS). Achievements in minimal clinically important difference (MCID) and patient acceptable symptom state (PASS), patient satisfaction, and re-revision rates were evaluated and compared between groups. Capsular thickness was assessed by […]

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### From bench to bedside: a novel suture-augmented prosthesis for greater trochanteric fixation in osteoporotic hip arthroplasty.

*Yang M, Yang J, Wang X, Wu Z, Bian S, Sheng R, Liao F, Qin K et al.*

BACKGROUND: Stable fixation of the greater trochanter during hip arthroplasty for unstable osteoporotic intertrochanteric fractures remains a significant challenge. Conventional techniques depend on securing the bone-implant interface, which is particularly compromised in osteoporotic bone. Here, we propose a novel conceptual approach that establishes a direct mechanical bridge from the abductor tendon to the prosthesis, thereby reducing reliance on the fragile bone-implant interface.

METHODS: In this two-stage translational study, we progressed from biomechanical validation to clinical application. First, using a decalcified caprine femoral model simulating osteoporosis, three fixation constructs were compared: locking plate (LP), suture-augmented locking plate (LPSA), and Kirschner-wire tension band (KWTB). Ultimate load and construct stability were evaluated. Biomechanical testing confirmed the principle of suture-mediated load sharing and highlighted the intrinsic weakness of screw fixation in osteoporotic bone. Guided by these results, we designed a novel femoral prosthesis that eliminates screw fixation, employs sutures as the primary load-bearing element, and incorporates integrated suture anchor tunnels. This prosthesis was then assessed in a retrospective series of 15 consecutive elderly patients with osteoporotic intertrochanteric fractures. Clinical outcomes were ev […]

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### Updating radiographic parameters for the healthy growing hip: are current parameters still valid?

*Hütter K, Smolle MA, Butter S, Schroedter R, Tschauner S, Kraus T*

BACKGROUND: This study aimed to evaluate whether the acetabular angle (AA) by Tönnis and lateral center-edge angle (LCEA) by Wiberg-key radiographic parameters in pediatric hip assessment-have changed in a contemporary pediatric population compared with historical reference values, considering trends in earlier skeletal maturation and body composition.

MATERIALS AND METHODS: This retrospective diagnostic accuracy study assessed changes in AA and center-edge angle (CE) angles in a contemporary cohort. A total of 1774 anteroposterior pelvic radiographs (3548 hips) from patients aged 0-18 years without hip pathology were analyzed. Radiographs were obtained between 2006 and 2018 and measured using the Supervisely digital platform. Standardized anatomical landmarks were applied. Data were stratified by age and sex and compared with historical values using two-sample t-tests ($p < 0.05$). To minimize measurement bias, two independent observers conducted all assessments using standardized digital protocols.

RESULTS: A total of 1774 patients aged 0-18 years (mean age, 8.30 ± 5.20 years; 666 females, 1108 males) were evaluated. AA values did not show statistically significant differences in 11 of 16 age groups compared with historical data ($p > 0.05$). In contrast, LCEAs were significantly higher in the contemporary cohort, especially in the 5-8, 9-12, and 12-16 age groups (all $p < 0.01$) […]

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### In-hospital deep vein thrombosis after tibial plateau fractures: incidence, laterality/anatomy, and risk factors in a multicenter retrospective cohort of 3366 patients.

*Yang S, Li G, Li Y, Long Y, Zhang J, Liu L, Zhao Z, Sun C et al.*

BACKGROUND: We aimed to determine the cumulative in-hospital incidence, anatomic distribution, and predictors of duplex ultrasonography-detected lower-extremity deep vein thrombosis (DVT) in patients hospitalized with tibial plateau fractures under routine thromboprophylaxis and protocolized surveillance.

METHODS: We retrospectively analyzed consecutive adults (≥ 18 years) admitted to two level-I trauma centers (November 2014-January 2024). Bilateral duplex ultrasonography was performed on admission/preoperatively (as soon as feasible after arrival and prior to surgery) and repeated serially during hospitalization (approximately every 5-7 days and when clinically suspected). The primary outcome was cumulative in-hospital DVT from admission to discharge; isolated intermuscular calf-muscle vein thrombosis (soleal/gastrocnemius) was recorded descriptively but excluded from the primary endpoint. Multiple injuries (polytrauma) were defined as ≥ 1 additional acute traumatic lesion beyond the index tibial plateau fracture documented on admission. Pulmonary embolism (PE) was assessed only when clinically suspected and confirmed by computed tomography pulmonary angiography (CTPA). Multivariable logistic regression identified independent predictors; receiver operating characteristic (ROC) analysis evaluated discrimination for key continuous predictors.

RESULTS: Among 3366 patients, 675 […]

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## Retained articulating spacers (1.5 stage) in chronic knee periprosthetic joint infections: an analysis of associate factors and outcomes in a single center cohort of patients.

*Balato G, Ascione T, Festa E, De Mauro D, Di Gennaro D, Rosa D, Mariconda M*

BACKGROUND: Two-stage revision remains the gold standard for chronic periprosthetic joint infection (PJI), but it can significantly impact patients' quality of life and functional recovery. The 1.5-stage revision, involving indefinite spacer retention, has emerged as a potential alternative in selected high-risk patients. Our questions were: (1) What are the early functional outcomes, including their infection-free survival? (2) Which pre- and perioperative factors are associated with spacer retention? and (3) What is the mechanical survivorship of retained spacers?

MATERIALS AND METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI between 2016 and 2020 who were followed prospectively for clinical outcomes. Functional status and quality of life were assessed using the EuroQol-5-Dimension-5-Level (EQ-5D-5L), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and Knee Society Score (KSS) before spacer placement and prior to scheduled reimplantation. Infection-free status was defined as the absence of clinical, radiographic, and laboratory signs of infection recurrence over a 24-month follow-up. Univariate and multivariate logistic regression were used to identify factors associated with spacer retention.

RESULTS: All patients initially underwent articulating spacer implantation as part of a planned two-st […]

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## Moderate leg length discrepancy: a long-term risk factor for hip osteoarthritis?

*Aprato A, Donis A, Via RG, Scandurra A, Tuè F, Audisio A, Fusini F, Massè A*

BACKGROUND: Leg length discrepancy (LLD) has been implicated as a biomechanical factor contributing to hip osteoarthritis (OA), yet the extent of its influence remains unclear. This study examines the correlation between moderate LLD (≥ 10 mm) and hip OA progression, focusing on asymmetrical OA distribution in patients over 65 years old.

MATERIALS AND METHODS: A retrospective analysis was conducted from a database of 1672 full-length standing X-rays. Patients under 65 years, with deformities, unilateral limb issues, or prosthetics, were excluded; therefore, the study group was composed of 220 patients. Tibial and femoral lengths were measured bilaterally, and hip OA severity was assessed using the Tönnis classification. Statistical analyses included Pearson's Chi-squared test and linear regression to explore correlations between LLD and OA distribution.

RESULTS: Among the sample, 18% showed an LLD ≥ 10 mm. A significant correlation was found between LLD and the asymmetrical distribution of hip OA ($p = 0.002$), with higher OA severity observed in the hypometric limb. Linear regression analysis suggested that each millimeter of LLD corresponded to a 0.74-unit change in OA severity difference between hips.

CONCLUSIONS: This study highlights a significant association between moderate LLD and contralateral hip OA in the elderly, emphasizing the biomechanical impact of asymmetrical […]

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### Authorship, titles and open access as drivers of citation performance in orthopaedics: a scientometric analysis.

*Migliorini F, Vaishya R, Rivera F, Eschweiler J, Kobbe P, Betsch M, Oliva F, Maffulli N*

BACKGROUND: Bibliometric analyses are increasingly used to explore how scientific knowledge is created, disseminated, and perceived. In orthopaedics, research output has expanded rapidly over the past decade, yet the factors determining whether an article achieves wide visibility and scholarly impact remain poorly understood. Beyond the inherent quality of a study, elements such as authorship patterns, title construction, and open access (OA) availability may play an essential role in shaping citation performance. However, evidence in this field is still limited and sometimes contradictory, highlighting the need for large-scale, field-specific analyses.

METHODS: Orthopaedic publications from 2010 to 2020 were identified in Scopus using the keyword 'orthopaedic'. After duplicate removal, 97,806 unique articles were included with complete data on authorship, titles, citation counts, study design, and OA status. Citation rates were normalised per year since publication. Associations between bibliographic features and citation performance were assessed using multiple linear regression, while differences across title styles and study designs were evaluated with comparative statistical testing. Exploratory modelling was performed to identify combinations of authorship and title characteristics linked to the highest predicted citation rates.

RESULTS: Larger author teams were associa […]

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### Interpreting nonsignificant findings: Reassessing the safety of self-locking standalone cages in octogenarian patients undergoing ACDF.

*Abudayeh A, Fishchenko I*

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### A novel, simple, and affordable technique for arthroscopic soft-tissue tenodesis of the long head of the biceps tendon.

*Fossati C, Ioppoli A, Ravaglia R, Vaja F, Menon A, Randelli PS*

BACKGROUND: The purpose of this study was to describe a novel, simple, and implant-free arthroscopic soft-tissue tenodesis technique of the long head of the biceps tendon (LHBT) to the rotator cuff using an absorbable monofilament suture and to evaluate its clinical and functional outcomes at a minimum 1-year follow-up.

METHODS: A retrospective case series of 23 patients (mean age 58.0 ± 7.8 years) who underwent arthroscopic rotator cuff repair with concomitant LHBT soft-tissue tenodesis between June 2021 and June 2023 was analyzed. Functional outcomes were assessed using Constant-Murley Score (CMS), American Shoulder and Elbow Surgeons (ASES) score, Single Assessment Numeric Evaluation (SANE), visual analog scale (VAS) for pain, and the Long Head of Biceps (LHB) score. Supination strength was measured with a handheld dynamometer and compared with the contralateral side. Incidence of Popeye deformity and tenderness over the bicipital groove were recorded.

RESULTS: Objective Popeye deformity was observed in 13% of patients, with subjective concern reported by only 4.3%. Supination strength and LHB scores were similar to the contralateral side (means of 104.65 versus 104.64 N; LHB 93.7 versus 94.6 points). The mean CMS and ASES scores were 90.3 ± 12.4 and 89.6 ± 15.9 points, respectively. The SANE score averaged 87.4 ± 20.9, and the VAS for pain was low (1.65 ± 2.59 cm).

CONCL […]

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### ORIF of three- and four-part proximal humeral fractures: a retrospective study of long-term clinical and complication outcomes of two-plate fixation methods.

*Conteduca J, Gasbarra E, Rausa I, Casto A, Valentino L, Solarino G, Meccariello L, Rollo G*

BACKGROUND: The aim of this study was to compare clinical outcomes and complication rates during long-term follow-up (≥ 6 years) after treatment for three- and four-part proximal humeral fractures using two different types of locking plates.

MATERIALS AND METHODS: A total of 113 patients with three- and four-part proximal humeral fractures who underwent surgery between September 2012 and January 2019 were enrolled retrospectively. Data for 49 patients [ 9 males, 40 females; mean age [standard deviation] (SD ) 68.9 (5.8) years} treated with a PGR (intrauma) plate (group A) and 51 patients [10 males, 41 females; mean age (SD): 66.0 (7.5) years] treated with a PHILOS humeral plate (PHP, group B) were available at the last follow-up. The mean follow-up periods in groups A and B were 10.8 and 10.2 years, respectively, with a minimum follow-up of 6 years. At the final follow-up evaluation, functional outcomes were assessed using the Oxford Shoulder Scale (OSS), Simple Shoulder Test (SST), and Disabilities of the Arm, Shoulder and Hand (DASH) questionnaires. X-ray evaluation was also performed, and complications were recorded.

RESULTS: The mean OSS score in the PGR and PHP groups at follow-up was 42.9 and 39.1, respectively ($p > 0.05$). The SST score (PGR: 7.5 ± 2.0; PHP: 7.3 ± 4.0) and DASH score (PGR: 22.1 ± 5.5; PHP: 20.5 ± 4.0) were similar in both groups ($p > 0.05$). In our serie […]

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## Incidence rates and treatment of the transcervical fracture of the neck of femur in Italy: is total hip arthroplasty an increasingly preferred approach? A population study on trends between 2001 and 2023 based on 1,120,770 hospital discharge records.

*Ciminello E, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L, Cuccu A et al.*

INTRODUCTION: Transcervical femoral neck fractures (TFNFs) are among the most devastating fragility fractures in the elderly. TFNF are associated with excess 1-year mortality rates ranging from 15% to 30%. Treatments include conservative methods, internal fixation, and arthroplasty (partial or total hip arthroplasty). This study aims to analyze the changes in incidences of TFNF in the Italian population between 2001 and 2023 and the evolution of the choices of treatment.

MATERIALS AND METHODS: Using hospital discharge record (HDR) data from 2001 to 2023, records with ICD9-CM codes for femoral neck fractures (820.0 and 820.1) among diagnoses were selected and categorized into four treatment groups: total arthroplasty, partial arthroplasty, fixation, and conservative. Time series were analyzed with stratification by sex and age.

RESULTS: The extracted data included 1,120,724 records of TFNFs, with 871,161 cases treated surgically (total or partial arthroplasty or internal fixation) and 249,563 treated conservatively; the average patient age was 79.1 years, with a higher proportion of women (72.8%). Partial hip arthroplasty was the preferred treatment overall. For younger patients, in the age classes < 45 and 45-54 years, fixation was the most chosen treatment. Over time, the use of the conservative treatment decreased from 27.5% in 2001 to 14.6% of cases in 2023. The use of par […]

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## Does latissimus dorsi tendon transfer provide a normal scapular rhythm and external rotation strength in posterosuperior massive irreparable rotator cuff tears? A kinematic analysis in a retrospective cohort.

*Calvo C, Fiumana G, Brigo A, Ruggiero ED, Bonfatti R, Donà A, Micheloni GM, Giorgini A et al.*

BACKGROUND: Posterosuperior massive irreparable rotator cuff tears (PMIRT) are rare and disabling conditions. When conservative treatment fails, latissimus dorsi tendon transfer (LDTT) is a viable surgical option for symptom relief in carefully selected patients. However, its effectiveness in restoring glenohumeral function and its influence on scapulothoracic rhythm remain subjects of ongoing debate.

PURPOSE: The purpose of this study was to evaluate the clinical outcomes of LDTT and assess its impact on scapulothoracic rhythm using kinematic and electromyographic analysis.

MATERIAL AND METHODS: A total of 18 patients with PMIRT underwent LDTT. Functional scores consisted of Constant-Murley score (CMS), America Shoulder and Elbow Surgeons (ASES) score, and Quick Disabilities of the Arm, Shoulder, and Hand (QuickDASH) score. Electromyography (EMG) activity of the latissimus dorsi muscle was evaluated in conjunction with three-dimensional (3D) kinematic tracking system. Differences in external rotation strength were compared with and without active adduction.

RESULTS: The mean age was 55.9 years (range 40-69 years), with a mean follow-up of 20.4 months (range 6-39 months). At final follow-up, the mean CMS was 68.2 (95% CI 64.9-71.5), ASES was 76.9 (95% CI 66.1-87.6), and QuickDASH was 17.6 (95% CI 9.5-25.6). A significant difference in external rotation strength was observed […]

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### Is there a need for exploration in pulseless supracondylar fractures of the humerus? A systematic review and individual patient data meta-analysis.

*Haoyang C, Chen JG, Lai H, Lim AKS, Tan SHS, Hui JHP*

BACKGROUND: Different approaches have been proposed to treat patients with pulseless supracondylar humeral fractures (SHF). We aim to analyze the current outcomes of patients who have undergone different treatment options for pulseless SHF, namely watchful waiting versus urgent surgical exploration.

METHODS: Electronic databases including PubMed, Embase, and Cochrane Library were explored. Our study included patients from all age groups but only included English-language articles. Single case reports and case studies with adequate description of patient population, injury, and outcomes were included. An individual patient data meta-analysis was done to evaluate key outcomes of surgical exploration versus no surgical exploration in pulseless patients with SHF.

RESULTS: Overall, the data for 1070 individual patients from a total of 48 studies were included. Patients with pulseless SHF with open fractures have a higher probability of requiring vascular intervention ($p < 0.001$) as they are more likely to have disrupted arteries ($p = 0.050$) and more likely to require vascular repairs ($p < 0.001$) than patients with closed fractures. Similarly, patients with pulseless SHF with pucker sign ($p = 0.003$) and ecchymosis ($p = 0.002$) were more likely to undergo surgical exploration. However, the neurological status had no relation to them undergoing surgical exploration ($p = 0.382$), and al […]

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### Increasing complexity and plateauing outcomes in AO type C distal humerus fractures: a 50-year institutional case series.

*Hönck K, Wenzelburger P, Eibinger N, Sadoghi P, Puchwein P*

BACKGROUND: Distal humerus fractures in adults are rare but complex injuries, particularly AO type C3 fractures, often involving comminution, instability, and osteoporotic bone. Demographic trends indicate rising incidence, especially among elderly patients, with increasing functional demands challenging modern osteosynthesis. Open reduction and internal fixation (ORIF) with bicolumnar plating remains the preferred reconstructive approach, although complications remain substantial. This study aimed to evaluate trends in fracture complexity, patient demographics, and functional outcomes over five decades, focusing on the most recent cohort (2009-2018, series E) and comparing with historical cohorts (series A-D, 1969-2008).

METHODS: This retrospective analysis included five consecutive 10-year institutional cohorts (series A: n = 43; B: n = 29; C: n = 47; D: n = 58) of 233 patients with 235 supradiacondylar distal humerus fractures (AO type C1-3) treated with ORIF. Functional outcome parameters (Jupiter and Cassebaum Scores) and demographic data were available for all cohorts; additional parameters (range of motion (ROM), QuickDASH Score) were available for series D and E, Mayo Elbow Performance Score (MEPS), Short Form (SF)-36, and strength were assessed specifically in series E (n = 56; follow-up n = 21). Cross-cohort comparisons of categorical variables used chi-square tests; […]

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### Magnetic resonance imaging-based fat infiltration grading improves reparability prediction in massive rotator cuff tears.

*Hu Q, Zhang G, Wang D, Tung TH, Ding J, Zhou X*

PURPOSE: This study aims to investigate the extent of fat infiltration (FI) in the rotator cuff (RC) muscles in more medial slices from the Y-shaped view (Y-view) on shoulder magnetic resonance imaging (MRI) and assess whether these slices provide a more accurate prediction of the reparability of massive RC tears (massive RCTs).

METHODS: The retrospective study included 57 patients with massive RCTs who successfully underwent arthroscopic surgery between 1 January 2023 and 31 December 2023. Patients were categorized into two groups: the irreparable group and the reparable group. All patients underwent shoulder MRI covering the region from the acromion to the medial border of the scapula in oblique coronal and oblique sagittal orientations. The FI stage of the RC was assessed across three different views to determine which view most effectively predicts the repairability of patients with massive RCTs.

RESULTS: The FI stage of the infraspinatus (IS) muscle across three distinct views demonstrated a significant correlation with the repairability of massive RCTs. The FI stage was more severe in the irreparable group. Receiver operating characteristic (ROC) analysis revealed that the area under the curve for the 1/2 scapula view (0.737) and the most severe view (0.745) exceeded that of the traditional Y-view (0.644). Paired-sample ROC curve analysis revealed a significant differe […]

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## Mid-term clinical and functional outcomes after reverse shoulder arthroplasty with latissimus dorsi transfer.

*Colombini AG, Rab P, Macken AA, Soares MN, Kimmeyer M, Shirinskiy IJ, Popescu IA, Lafosse L et al.*

BACKGROUND: Although reverse total shoulder arthroplasty (rTSA) with concomitant latissimus dorsi transfer (LDT) has been shown to effectively treat external rotation (ER) deficits, there are limited data regarding its outcomes with modern implants and its impact on activities of daily living (ADLs) requiring ER. The purpose of this study was to assess the mid-term clinical and radiographic outcomes of rTSA with concomitant isolated LDT in patients with an ER lag sign and posterior rotator cuff deficiency.

METHODS: This retrospective cohort study with prospective follow-up included consecutive patients who underwent rTSA with concomitant isolated LDT between 2010 and 2022 with a minimum follow-up of 2 years. Primary outcomes included the resolution of ER lag sign, active ER, and the Activities of Daily Living and External Rotation (ADLER) score. Secondary outcomes included the Auto-Constant Score (CS), Subjective Shoulder Value (SSV), activities of daily living requiring internal rotation (ADLIR) score, visual analog scale (VAS) for pain, and radiographic analysis of standardized radiographs.

RESULTS: In total, 32 procedures in 32 patients were identified. Of these, 22 procedures in 22 patients (68% female, 72.9 ± 8.4 years at surgery) were available for follow-up at 4.8 ± 2.2 years postoperatively (response rate 73%). The ER lag sign resolved in 95.5% of patients, the active […]

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## Partial versus full revision in cementless single-stage exchange for hip periprosthetic joint infection: a multicenter comparative study with 10-year survivorship analysis.

*Mu W, Jarusriwanna A, Xu B, Brown SA, Zhang X, Dawes A, Parvizi J, Cao L*

BACKGROUND: Periprosthetic joint infection (PJI) is one of the most challenging complications following total hip arthroplasty (THA). Traditional management typically involves full revision to ensure comprehensive infection eradication. However, for patients with well-fixed implants, partial revision in a single-stage cementless approach may provide a viable alternative, potentially preserving bone stock and reducing operative time. The relative efficacy of this approach compared with full revision remains to be fully explored. This multicenter study aims to determine whether partial revision in cementless single-stage exchange surgery offers comparable infection control outcomes to full revision for select patients with well-fixed implants.

METHODS: We conducted a retrospective, multicenter cohort study involving 226 patients who underwent cementless single-stage exchange hip arthroplasty for PJI between 1 October 2013 and 31 July 2022. Patients were divided into partial revision (n = 61) and full revision (n = 165) groups. The primary outcome was treatment success, defined as the absence of clinical symptoms and signs of infection at a minimum follow-up of 2 years.

RESULTS: The success rates were 77.0% for the partial revision group and 80.0% for the full revision group, with no significant difference (p = 0.629). Both groups showed comparable 10-year survival rates for ove […]

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# Journal of Orthopaedic Trauma

Volume 40, Issue 9 · September 2026 · 10 new articles

## Internal Rotation and Transthoracic Lateral Radiographs Frequently Disagree on Sagittal Angulation in Humeral Shaft Fractures (AO/OTA 12).

*Zhang D, Brooks J, Schimizzi G, Erdman MK, Maassen N, Christiano AV*

OBJECTIVES: The aim of this study was to compare humeral shaft fracture sagittal angulation between transthoracic and internal rotation radiographs at the time of injury, and to identify fracture patterns associated with discrepancies between views.

DESIGN: Retrospective cohort study.

SETTING: Single Level 1 academic trauma center.

PATIENT SELECTION CRITERIA: Adult patients presenting with closed humeral shaft fractures (AO/OTA 12A-C) between May 2018 and December 2024 were screened for inclusion. Patients with radiographs that included both internal rotation and transthoracic lateral views at the time of injury before bracing or splinting were included.

OUTCOME MEASURES AND COMPARISON: Demographic variables, injury mechanism, AO/OTA classification, and fracture location were collected. Sagittal angulation was measured on both views, and differences of greater or less than 10 degrees between views were noted. Acceptable humeral shaft fracture alignment was defined as less than 20 degrees of sagittal angulation. A paired sample t test was used to compare angulation between the 2 views. Fisher Exact Test was used to compare fracture characteristics between patients with less than 10 and more than 10 degrees of difference.

RESULTS: A total of 31 patients were included. The cohort was 54.8% female, with a mean age of 42.4 ± 20.4 years. The 2 views disagreed on acceptable align […]

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## Results of a 2025 US Practice Survey of Orthopaedic Traumatologists.

*Puckett HD, Borgida JS, Del Olmo M, Lundy D, Cunningham BP, Cannada LK*

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## Shifts in Medicare Reimbursement for Common Lower Extremity Orthopaedic Trauma Procedures, 2006-2024.

*Hoveidaei AH, Mosalamiaghili S, Feng JE, Anoushiravani AA*

OBJECTIVES: The aim of this study was to analyze Medicare reimbursement trends from 2006 to 2024 for open treatment of proximal femoral fractures [Current Procedural Terminology (CPT) 27236], femoral shaft fractures (CPT 27506), tibial shaft fractures (CPT 27759), and bimalleolar ankle fractures (CPT 27814), using data from the Centers for Medicare and Medicaid Services (CMS) Physician Fee Schedule (PFS).

DESIGN: This study analyzed Medicare reimbursement trends for the 4 most common lower extremity trauma procedures from 2006 to 2024.

SETTING: Reimbursement data were obtained from the Centers for Medicare and Medicaid Services (CMS) Physician Fee Schedule (PFS), and procedure frequency data were obtained from the M165Ortho dataset from PearlDiver Technologies.

PATIENT SELECTION CRITERIA: Procedures were selected based on frequency and included CPT codes 27236, 27506, 27759, and 27814.

OUTCOME MEASURES AND COMPARISONS: Reimbursement data were obtained from the CMS PFS using the corresponding Healthcare Common Procedure Coding System codes. Work reimbursement was calculated by multiplying work relative value units (RVUs) by the CMS conversion factor (CF), and total facility reimbursement was calculated by multiplying total facility RVUs by the CF. Values were adjusted to 2024 US dollars using the Consumer Price Index. Percentage changes from 2006 were calculated. A 5-year pr […]

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## Infection Rate by Gustilo-Anderson Classification in Open Tibial Shaft Fractures Treated With an Intramedullary Nail-A Meta-Analysis.

*Schermerhorn JT, McGlone PJ, Torrie AM, Carlini AR, Castillo RC, Abar B, Hendren S, Wesorick BR et al.*

OBJECTIVES: To quantify deep surgical site infection (SSI) rates after intramedullary nailing of open tibial shaft fractures stratified by Gustilo-Anderson (G-A) classification.

DATA SOURCES: A systematic review of MEDLINE, Embase, and Scopus screened for English language studies published from January 2000 through May 2025 in accordance with PRISMA guidelines.

STUDY SELECTION: Studies included skeletally mature patients with open tibial shaft fractures (OTA/AO 42) treated with locked intramedullary nailing and reporting deep SSI rates by G-A type.

DATA EXTRACTION: Two independent reviewers screened studies and extracted data, with conflicts resolved through consensus or adjudication by a third independent reviewer.

DATA SYNTHESIS: A single-arm meta-analysis was performed to estimate pooled infection rates using fixed-effects and random-effects models, with between-study heterogeneity assessed using Cochran Q and the I2 statistic.

RESULTS: Seventeen studies composed of 2063 total patients met inclusion criteria. The overall pooled deep SSI rate was 13.2% [95% confidence interval (CI), 11.8%-14.8%]. Stratified pooled infection rates were 6.2% (95% CI, 3.7%-10.3%) for G-A type I, 7.6% (95% CI, 5.6%-10.1%) for type II, 11.9% (95% CI, 9.5%-14.6%) for type IIIA, 25.0% (95% CI, 18.6%-32.6%) for type IIIB, and 41.8% (95% CI, 21.3%-65.5%) for type IIIC fractures. Subgroup analysis […]

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## Adverse Mental Health Outcomes After Modern External Ring Fixation Versus Internal Fixation for Severe Open Tibia Fractures: A Secondary Analysis of the FIXIT Study.

*Simske NM, Solarczyk JK, Thompson AR, Reider L, O'Toole RV, Carroll EA, Karunakar MA, Obremskey W et al.*

OBJECTIVES: To report the prevalence of adverse mental health outcomes 1 year after severe open tibial shaft fractures and their association with treatment and complication risk factors.

DESIGN: Randomized controlled trial.

SETTING: Multicenter.

PATIENT SELECTION CRITERIA: Eligible patients were 18-64 years of age with severe open tibial shaft fractures (AO/OTA 41-43) treated with modern external ring fixation or internal fixation. All Gustilo-Anderson type IIIB fractures were included and some type IIIA meeting specific severity criteria.

OUTCOME MEASURES AND COMPARISONS: Outcome measures included Patient Health Questionnaire scores for depression symptoms, posttraumatic stress disorder (PTSD) Checklist for DSM-IV (PCL-S), and the Brief Pain Inventory. Primary comparisons assessed depression, PTSD, and pain between treatment groups (modern external ring fixation vs. internal fixation).

RESULTS: A total of 254 patients were treated with external ring fixation (n = 121) or internal fixation (n = 133). The mean age was 39 years (SD = 13) and 16% were female. At 12 months after injury, 34% of patients had Patient Health Questionnaire-9 scores consistent with moderate to severe depression. Risk factors for depression were female sex (RR: 1.70 [1.01-2.86]), high school educational attainment or less (RR: 1.93 [1.27-3.04]), and any baseline mental health condition (RR: 1.91 [1.2 […]

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## Metabolic Derangements Are Independent Additive Risk Factors for Tibial Diaphyseal Nonunion.

*Kotzur T, Peterson B, McCoy L, Pereira DE, Hosseinzadeh P, Martin C*

OBJECTIVES: The aim of this study was to evaluate whether altered metabolic factors (metabolic comorbidities) increase the risk of postoperative tibial diaphyseal nonunion after fracture fixation, and to determine whether additional metabolic risk factors contribute cumulatively.

DESIGN: Retrospective cohort analysis.

SETTING: The TriNetX Research Network.

PATIENT SELECTION CRITERIA: Patients with operatively treated tibial diaphyseal fractures (AO/OTA 42) between January 1, 2013 and December 31, 2023 were identified. A minimum of 1 year of follow-up was required. Patients with a previous diagnosis of tibial nonunion or underlying metabolic bone disease were excluded.

OUTCOMES MEASURES AND COMPARISONS: The primary outcome was tibial diaphyseal nonunion after the index fracture. Outcomes were compared between patients with and without each metabolic comorbidity of interest (obesity, hypertension, diabetes, and dyslipidemia), and across groups stratified by cumulative metabolic burden to evaluate additive risk.

RESULTS: A total of 36,474 patients (mean age 41.5 years, range 18-90; 62.7% male) were included in the analysis. Metabolic comorbidities were independently associated with a higher risk of nonunion, including diabetes (Hazard Ratio (HR) 2.82, $P < 0.001$), hypertension (HR 1.87, $P < 0.001$), obesity (HR 1.45, $P < 0.001$), and dyslipidemia (HR 1.25, $P < 0.001$). When strat […]

PubMed · DOI

## Comparative Efficacy of Anterior MIPO With PHILOS Versus Posterior ORIF With EA-LCP in Distal-Third Humerus Fractures: A Randomized Controlled Trial.

*Garika SS, Yadav P, Jyoti NJ, Bansal H, Mittal S, Trikha V, Sharma V*

OBJECTIVES: To compare anterior minimally invasive plate osteosynthesis (MIPO) using the proximal humerus internal locking system in an upside-down reverse configuration with conventional posterior open reduction and internal fixation (ORIF) using extra-articular locking compression plate for the treatment of extra-articular distal-third humeral shaft fractures.

DESIGN: Prospective randomized controlled trial.

SETTING: Single Level 1 trauma center (New Delhi, India).

PATIENT SELECTION CRITERIA: Adult patients aged 18-55 years with extra-articular distal-third humeral shaft fractures (AO/OTA types 12A-C) presenting within 2 weeks of injury were randomized to undergo anterior MIPO or posterior ORIF.

OUTCOME MEASURES AND COMPARISONS: Radiological union, functional outcomes (Mayo Elbow Performance Score, University of California-Los Angeles shoulder rating scale), and elbow range-of-motion were assessed and compared at 2 weeks, 6 weeks, 3 months, and 6 months postoperatively. Operative time, intraoperative blood loss, radiation exposure, postoperative pain (visual analog scale scores) and complications were also compared.

RESULTS: Thirty patients were included (anterior MIPO, $n = 14$; posterior ORIF, $n = 16$). The mean age was 27.9 years (range 18-55) in the MIPO group and 29.8 years (range 19-48) in the ORIF group, $P = 0.598$. There were 11 men and 3 women in the MIPO group and […]

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## Blade Versus Screw Complication Risk for Intertrochanteric Fracture Fixation.

*Richey Levine A, Cross JL, Klug T, Salameh M, Riedel M, Leslie M*

OBJECTIVE: To determine whether the use of a screw or blade for fixation in intertrochanteric fracture intramedullary nailing is associated with bony complications, specifically cut-through, cut-out, malunion, nonunion, proximal hardware backout, and reoperation.

DESIGN: Retrospective cohort study.

SETTING: Single Level I Trauma Center.

PATIENT SELECTION CRITERIA: Patients with intertrochanteric hip fractures (OTA/AO 31A) between 2014 and 2023 who underwent fixation with the same type of cephalomedullary nails (CMN) with either a blade or screw for proximal femur stabilization were identified based on Current Procedural Terminology Codes 27245.

OUTCOMES MEASURES AND COMPARISONS: Bony complications, specifically cut-through, cut-out, malunion, nonunion, proximal hardware backout, and/or reoperation, were compared across blade and screw implants for patients. This comparison was completed for the entire cohort, with patients stratified by fracture stability, and with the application of propensity matching across fracture and demographic characteristics.

RESULTS: In total, 1078 patients were included. The 615 patients treated with CMN blade had a mean age of 83.1 (range 65-106) years and 451 were women. The 463 patients treated with CMN screw had a mean age of 83.3 (range 65-103) years and 354 were women. No statistical significance was found between CMN blade versus screw co […]

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## Interpretation Accuracy Between Orthopaedic Surgeons and Qualified Remote Spanish Interpreters in a Trauma Setting.

*Jaeblon T, Monarrez R, Molestina C, DaConceicao L, Babecki MG, Bauer BJ, Talwar S, Demyanovich HK*

OBJECTIVES: To assess the interpretation accuracy of remote qualified Spanish medical interpreters (QMI) in an orthopaedic trauma setting and determine whether interpretation method (telephonic or video), delivery speed, and subject complexity were associated with interpretation accuracy.

METHODS: Three clinical vignettes-simple, moderate, and complex-describing fictitious orthopaedic trauma injuries, treatments, complications, and outcomes were written by the investigators. An investigator read each vignette type to a simulated patient through 20 randomly selected QMIs (10 through telephone and 10 through video), all of whom were trained, proficiency-tested, and blinded to the study design, yielding 60 total interpretations. Data collected included interpretation method, English delivery speed (words per minute), and vignette complexity. One orthopaedic traumatologist and 1 bilingual physician assistant evaluated transcriptions of recorded QMI Spanish interpretations using a nonvalidated, criterion-referenced, expert-derived categorical assessment reflecting the degree to which an interpreter preserved the fidelity of the original vignette for patient ascertainment of diagnosis, treatment, risks, and prognosis, termed overall interpretation designation (OID). OID was classified, "acceptable," "borderline," or "unacceptable." A second orthopaedist, blinded to the first 2 inves […]

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## Intramedullary Reaming Biopsy in Diagnosing Metastatic Long Bone Disease.

*Stevens KN, Pan X, Smith KL, Blackburn CW, Jonard BW, Napora JK*

OBJECTIVES: The aim of this study was to describe the diagnostic accuracy of intramedullary reaming biopsy in cancer patients with long bone lesions or suspected pathologic fractures in determining the presence of metastatic disease.

DESIGN: A retrospective review of electronic medical records.

SETTING: Single academic Level I Trauma Center.

PATIENT SELECTION CRITERIA: Patients with a known cancer diagnosis who underwent intramedullary nail fixation for a long bone lesion or pathologic fracture of the femur, humerus, and tibia for whom intramedullary reamings were sent to pathology between January 2013 and October 2021 were included. Patients for whom open biopsy was performed were excluded.

OUTCOME MEASURES AND COMPARISONS: The primary study measure was intramedullary reaming biopsy sensitivity.

RESULTS: Forty-six reamings from 44 patients were included, with a mean patient age of 69 years (range, 41-85 years) and 35 being female (76%). Approximately half of cases were therapeutic fixation of a pathologic fracture (52%) versus prophylactic (48%) in nature. Most reaming biopsies were from the femur (42/46, 91%), followed by humerus (3/46, 7%), and tibia (1/46, 2%). The most common diagnoses were breast (34%) and lung (20%) carcinoma and multiple myeloma (20%). Thirty-two reamings (70%) were able to establish a diagnosis of metastatic disease. Five samples (11%) were crushe […]

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AOTS

# Archives of Orthopaedic & Trauma Surgery

Volume 146, Issue 1 · August 2026 · 120 new articles

## Correction: Different sagittal cutting plane orientations do not affect posterior tibial slope in open wedge high tibial osteotomy: a Sawbone model study.

*Gurbanov E, Cochard B, Bauer DE, Miozzari HH, Tscholl PM*

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## Resident autonomy in orthopedic trauma: independent femoral IMN by senior residents yields comparable outcomes to fellowship-trained surgeons.

*Balziano S, Nesher G, Ben Nun N, Segev T, Freidlander A, Prat D*

INTRODUCTION: Intramedullary nailing (IMN) of femoral shaft fractures is a fundamental cornerstone procedure in orthopedic trauma and an important benchmark of resident surgical competency. While independent resident operating is essential for skill development, its safety in complex trauma cases remains debated. This study evaluated whether senior residents can safely perform femoral IMN without direct supervision compared with fellowship-trained trauma surgeons.

MATERIALS AND METHODS: A retrospective cohort analysis was performed of adult patients undergoing IMN for isolated femoral shaft fractures between 2015 and 2023 at a level I trauma center. Cases were grouped by primary surgeon: senior residents (postgraduate year ≥ 4) versus attending trauma surgeons. Demographics, fracture characteristics, operative parameters, complications, and revision rates were compared. To minimize selection bias, 1:1 propensity score matching (PSM) was performed based on preoperative covariates including age, comorbidity, fracture type, injury mechanism, and severity.

RESULTS: A total of 207 patients met inclusion criteria; 150 (75 per group) were matched. Operative duration was longer in resident-led cases (median 120 vs. 91 min, $p = 0.003$). Postoperative hospital stay (5.0 vs. 6.0 days, $p = 0.674$) and weight-bearing protocols ($p = 0.218$) were similar. Overall complication rates did not dif […]

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## Racial and ethnic disparities in perceived health and barriers to care among adults with hip osteoarthritis: a national analysis of 16,000 + patients from the all of us program.

*Khan ST, Devadiga SA, Emara AK, Brahmbhatt S, Elmenawi KA, Delaney PG, Piuzzi NS*

OBJECTIVE: To explore racial and ethnic disparities in perceived health, healthcare access, and perceived barriers among adults with hip osteoarthritis (OA) in a large, nationally representative U.S.

METHODS: Cross-sectional analysis of 16,743 adults with hip OA from the All of Us Research Program. Participants self-identified as Black, Hispanic, or White. Outcomes included self-reported general, mental, physical, and social health, provider communication, and structural barriers to care. Between-group comparisons used Chi-square tests.

RESULTS: Black and Hispanic participants were younger, had lower income and education, and relied more on Medicaid than White participants (all $p < 0.001$). They reported higher rates of unaffordable deductibles, difficulty accessing follow-up care, insurance coverage issues, transportation challenges, and medication cost-related nonadherence (all $p < 0.001$) Table 3. Minority groups also more often experienced poor provider respect, trust, and communication. Across all health domains, Black and Hispanic participants reported poorer outcomes and greater emotional distress (all $p < 0.001$).

CONCLUSION: Significant racial and ethnic disparities exist in structural and perceptual barriers to care among adults with hip OA. These findings identify potentially modifiable barriers that warrant further study and may inform future efforts to improve equi […]

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## Postoperative periprosthetic fractures associated with compressive osseointegration: a systematic review of the Zimmer Biomet Compress® device.

*Hashimoto K, Nishimura S, Goto K*

The Zimmer Biomet Compress® Compliant Pre-Stress (CPS) device uses a compressive osseointegration technique for endoprosthetic fixation and provides an alternative to stemed implants. Despite having benefits, such as bone stock retention, periprosthetic fractures continue to remain a concern. Therefore, this systematic review presents the prevalence, risk factors, and outcomes of the CPS device and periprosthetic fractures after treatment. Medical databases (PubMed, Embase, and Google Scholar) were searched to identify studies reporting complications related to the CPS device. Peer-reviewed studies with at least 2 years of follow-up for fracture rates, revision results, and implant survivorship were included. The primary data collected were fracture incidence and timing of complications, and the CPS device was compared with the traditional cemented stem. This review highlights important studies, including a cohort of 221 patients and pediatric data from 36 patients. Adult populations showed a periprosthetic fracture rate of 2.7-3.9%. A distinct temporal pattern emerged: the frequency of fractures was most pronounced within the first 2 years after surgery. In the pediatric context, spindle survivorship was 86.3% at 5-10 years, despite higher rates of mechanical complications. Notably, for most fracture cases, the bone-implant interface was radiographically stable, providing easi […]

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## Hip resurfacing arthroplasty at a minimum of 14 years: revision is concentrated among female patients with dysplasia and small components.

*Lin YC, Chang CH, Hsieh PH, Huang YY, Chen DW, Lau NC*

INTRODUCTION: Hip resurfacing arthroplasty (HRA) is a bone-preserving option for young, active patients. The influence of the underlying diagnosis on long-term outcome remains unclear, particularly the distinction between primary osteoarthritis (OA), developmental dysplasia of the hip (DDH) and non-OA conditions.

MATERIALS AND METHODS: We retrospectively reviewed 104 consecutive metal-on-metal HRAs in 101 patients (three bilateral) performed by one surgeon between 2006 and 2011, stratified into Primary OA ($n = 50$), DDH ($n = 29$) and Non-OA ($n = 25$: avascular necrosis [AVN] $n = 15$, inflammatory arthritis $n = 5$, post-traumatic arthritis $n = 3$, Legg-Calvé-Perthes disease $n = 2$). AVN was analysed additionally as an independent group, and exploratory analyses restricted to female patients and to cups ≤ 50 mm addressed the collinearity of dysplasia, sex and size. Survivorship used Kaplan-Meier estimation with revision as the endpoint.

RESULTS: Implant status was known for all 104 hips; the 91 unrevised assessed hips had a mean follow-up of $16.2 \pm 1.5$ years. Overall 15-year survivorship was 95.2%. Primary OA, Non-OA and AVN each showed 100%; DDH 82.8% (versus Primary OA $p = 0.002$, versus Non-OA $p = 0.031$). All five revisions occurred in female DDH patients with a cup ≤ 50 mm; DDH survivorship remained lower within female patients ($p = 0.025$) and within small cups ($p = 0.006$), with no […]

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## Elective arthroplasty on "high-risk" calendar dates: no evidence for worse outcomes in 43,000 procedures.

*Walter N, Lowenberg DW, Heiss C, Lau EC, Rupp M*

BACKGROUND: Concerns regarding the timing of elective surgery remain common among patients and may influence surgical scheduling. Superstition-associated calendar dates such as Friday the 13th are frequently perceived as "high-risk" time points, yet evidence regarding their impact on arthroplasty outcomes is lacking. This study investigated whether primary total hip arthroplasty (THA) and total knee arthroplasty (TKA) performed on these dates are associated with differences in revision risk, mortality, or procedure frequency.

METHODS: A nationwide retrospective cohort study was conducted using Medicare physician service records (2009-2019). Primary THA and TKA procedures were identified in a 5% sample of Medicare beneficiaries aged ≥ 65 years. Outcomes included aseptic revision, septic revision, and mortality. Competing-risk survival analyses (Fine-Gray) and multivariable Cox regression models adjusted for demographic, clinical, and socioeconomic covariates were applied. Procedure volumes were compared between superstition-associated and control dates.

RESULTS: A total of 945 arthroplasties were performed on Fridays falling on the 13th. No significant differences were observed in aseptic revision, septic revision, or mortality compared with control dates. For Friday the 13th, hazard ratios were 0.80 (95% CI 0.44-1.47) for aseptic revision, 0.83 (95% CI 0.31-2.25) for septic r […]

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### Evaluation of a novel strap fixation method for biomechanical testing of sheep Achilles tendons.

*Basci O, Aydemir S, Husemoglu RB, Çeltik M*

BACKGROUND: Reliable fixation of tendon specimens is a major challenge in biomechanical testing, as slippage and tissue damage can reduce accuracy and reproducibility. Although various clamp- and suture-based fixation methods have been reported, there is still no widely accepted, practical, low-cost technique that preserves tendon integrity during testing. This study aimed to compare commonly used fixation techniques and to evaluate whether a novel strap-based fixation method provides a more stable and tissue-preserving alternative for biomechanical testing of sheep Achilles tendons.

METHODS: Forty fresh-frozen cadaveric sheep Achilles tendons obtained from a local butcher were used, and the specimens were stored at - 20 °C until testing. After controlled thawing at room temperature, all tendons were repaired using the Bunnell suture technique with No. 0 Vicryl. Four different proximal fixation methods were evaluated: (1) sutured fixation with strap, (2) direct clamp fixation, (3) Kocher clamp fixation, and (4) Krackow suture fixation. All specimens were mounted on a universal testing machine and subjected to a load-to-failure protocol at a constant loading rate of 10 mm/s. Maximum force (N), displacement at failure (mm), and mechanical failure mechanisms were recorded for each sample. Data were analyzed using the Kruskal-Wallis test with Bonferroni-corrected Mann-Whitney U te […]

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### Implementation and early results of same-day hip and knee replacement in a public hospital without prior same-day surgical experience, a feasibility study.

*Jørgensen CC, Sveen TM, Johansen B, Sander M, Steen-Hansen C, Sundstrup MS, Rothe C, Lange KH*

BACKGROUND: Hip and knee replacements are increasingly performed as same-day surgery (SDS). However, the feasibility of SDS pathways in departments without prior experience is unknown. We report on the implementation process, feasibility and early results of implementing a SDS protocol in a department without prior experience with same-day hip and knee replacement.

METHODS: The study design was based on Bowens feasibility study framework on implementation processes. Data were collected prospectively through patient reported questionnaires and medical records during, and from the local Department of Digitalisation and Analytics after implementation. Outcome data included discharge on day of surgery, 30-days primary-care contacts, 90-days readmissions and patient satisfaction.

RESULTS: Implementation was initiated as a dedicated fellowship from September 2023 to October 2024. Patients were admitted and discharged through a dedicated day-surgical department with transfer to the regular ward in case of admission. The first SDS patient underwent surgery the 4th of September 2023. By October 2024, 150 of 713 operated patients (21.0% CI:18.2-24.2) had SDS. Of these, same-day discharge was achieved in 125 (83.3% CI:76.6-88.5) patients of which 6 (4.0% CI:0.9-7.1) were readmitted within 90-days. From October 2024 to October 2025, 178 of 800 patients were planned for SDS (22.3% CI:19.5 […]

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### Relationship of foot anthropometry with Modified Pirani score during Ponseti correction of idiopathic congenital talipes equinovarus: a prospective observational study.

*Suresh M, Pinto DJV, Rajasekharan P*

BACKGROUND: The Ponseti method is the gold standard treatment for idiopathic congenital talipes equinovarus (CTEV). Clinical monitoring during correction primarily relies on the Modified Pirani score, which evaluates deformity severity based on clinical findings. However, objective quantitative assessment of morphological foot changes during treatment remains limited. This study aimed to evaluate the relationship between serial foot anthropometric measurements and Modified Pirani scores during Ponseti correction of idiopathic clubfoot.

METHODS: A prospective observational cohort study was conducted in 30 children with idiopathic unilateral CTEV undergoing Ponseti correction at a tertiary care center between June 2024 and February 2026. Five anthropometric parameters were measured weekly for eight weeks: great toe to midheel distance (GT-MH), little toe to midheel distance (LT-MH), second toe to midheel distance (ST-MH), medial malleolus to midheel distance (MM-MH), and lateral malleolus to midheel distance (LM-MH). Modified Pirani scores were assessed simultaneously by two independent observers. Pearson correlation analysis and inter-observer reliability testing were performed.

RESULTS: Significant improvement was observed in all anthropometric parameters and Pirani scores over the study period ($p < 0.001$). Mean GT-MH increased from 80.73 ± 3.36 mm to 87.56 ± 3.41 mm, LT-MH f […]

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### Union outcomes and associated factors in adolescent scaphoid fractures.

*Torun Ö, Girgin AB, Ka■■kç■ M, Acar A, Çevik HB*

PURPOSE: Scaphoid fractures (SF) in adolescents differ from adult fractures in fracture pattern, healing capacity, and skeletal maturity. Limited data exist on factors affecting union and healing time in this population. This study evaluated clinical, radiological, and treatment-related factors associated with union and healing time in adolescent SF.

METHODS: This study included 77 adolescents (aged 10-18 years) treated for SF. Chronological age, bone age, fracture characteristics, treatment modality, union status, and healing time were recorded. Fractures were classified by anatomical location, displacement, and time to diagnosis (< 6 weeks or ≥ 6 weeks). CT-based morphometric measurements, including scaphoid height, lateral intrascaphoid angle, and dorsal cortical angle, were obtained from initial computed tomography images. Functional outcomes were assessed using the QuickDASH and patient-rated wrist evaluation (PRWE) scores.

RESULTS: Union was achieved in 63 patients (81.8%), with a median union time of 8 weeks (range, 6-10 weeks). Union rates were highest in distal pole fractures (100%) and lowest in proximal pole fractures (33.3%). Non-displaced fractures had higher union rates than displaced fractures (92.5% vs. 56.5%), and fractures diagnosed within 6 weeks demonstrated better union outcomes. Surgical treatment, performed in 16.9% of patients, was associated with lowe […]

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### The effect of surgeon volume on outcomes after total hip arthroplasty: a systematic review and meta-analysis.

*Meissner N, Ortwig K, Ramos-Pascual S, Agu C, Stoeve J, Schrednitzki D, Rolvien T, Halder AM*

PURPOSE: To synthesize, critically appraise and systematically review studies comparing outcomes after primary total hip arthroplasty (THA) performed by low- vs. high-volume surgeons, and to propose evidence-based threshold definitions for these categories.

METHODS: This review followed the PRISMA guidelines and was registered in PROSPERO. Medline and Embase were searched on March 24, 2025. Prospective and retrospective clinical studies were included if they reported THA thresholds for low- and high-volume surgeons. Articles not written in English or German were excluded. Outcomes were pooled and presented in forest plots. Methodological quality was assessed.

RESULTS: Of 872 identified articles, 18 were included, reporting on 796,061 patients undergoing THA. Methodological quality was high in all studies (>6 of 7 points). Thresholds used to define high- vs. low-volume were determined using arbitrary values ($n = 11$), tertiles ($n = 1$), quartiles ($n = 3$), percentiles ($n = 2$), or receiver operating characteristic curves ($n = 1$). High-volume surgeons had significantly lower 1-month readmission rates [odds ratio (OR) = 1.6; $p = 0.033$] and 3-month mortality rates (OR = 1.8; $p = 0.008$). In contrast, there were no significant differences between high- and low-volume surgeons for 3-month revision rates (OR = 2.0; $p = 0.228$), 12-month revision rates (OR = 1.8; $p = 0.228$), and 3-month re […]

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### Hexapod-assisted limb salvage after talectomy: outcomes of tibiocalcaneal arthrodesis using the TL-hex system.

*Guerrero Serrano JA, Sánchez Morata E, García López JM, Mencía González S, Vilá Y Rico J*

BACKGROUND: Severe talar destruction due to infection, trauma, avascular necrosis, or neuroarthropathy represents a major reconstructive challenge. Talectomy followed by tibiocalcaneal arthrodesis (TCA) is a limb-salvage option in selected high-risk patients, although optimal fixation remains controversial. Modern hexapod external fixators offer computer-assisted multiplanar control that may improve alignment and fusion after talectomy. The aim of this study was to evaluate the clinical and radiographic outcomes of talectomy followed by TCA using the TL-HEX system in a series of high-risk patients.

METHODS: A retrospective observational study was conducted between January 2020 and January 2024 including all patients who underwent total talectomy followed by TCA using a TL-HEX circular external fixator. Indications included talar destruction secondary to infection, avascular necrosis, post-traumatic collapse, Charcot neuroarthropathy, and aseptic loosening of total ankle arthroplasty. Primary outcomes were radiographic fusion and hindfoot alignment. Secondary outcomes included time to union, functional scores (AOFAS, EFAS), pain (VAS), complications, and patient satisfaction. Preoperative and final follow-up outcomes were compared using paired statistical tests, with significance set at $p < 0.05$.

RESULTS: Thirteen patients (mean age 57 years) were included with a mean follow-u […]

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## Radiographic changes in midfoot alignment after hallux valgus correction with and without concomitant second tarsometatarsal arthrodesis: a comparative study.

*Kim J, Kyung MG, An Y, Kim NY, Lee KM, Lee DY*

INTRODUCTION: Second tarsometatarsal (TMT) joint osteoarthritis occurs in a subset of patients with hallux valgus (HV). While second TMT arthrodesis is performed to address symptomatic arthritis, its association with midfoot alignment changes during HV correction remains unclear. This study aimed to compare changes in radiographic arch parameters between patients undergoing HV correction with and without concomitant second TMT arthrodesis.

MATERIALS AND METHODS: This retrospective comparative study included 73 feet with HV and medial column malalignment (lateral Meary's angle > 10°). The TMT group (n = 37) underwent HV correction combined with second TMT arthrodesis for symptomatic osteoarthritis, while the control group (n = 36) underwent isolated HV correction. These patients were matched to achieve comparable baseline characteristics. Radiographic parameters of the transverse arch (M2-M5 angle, C1-M2 distance) and medial column alignment (lateral Meary's angle) were evaluated preoperatively and at a minimum 1-year follow-up. Functional outcomes were assessed as a secondary measure.

RESULTS: The TMT group demonstrated greater changes in transverse arch parameters compared with the control group. The M2-M5 angle decreased by 2.1° (95% CI, 1.4° to 2.7°) in the TMT group, whereas no significant change was observed in controls (-0.2°; 95% CI, -0.7° to 0.4°). The C1-M2 distance […]

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## Coronal shear fractures of the distal humerus in children and adolescents: proposal for a modified classification.

*Lieber J, Esser M, Fernandez FF, Schmittenbecher PP, Schneidmueller D, Strohm P, Kaiser MM, Kraus R*

PURPOSE: Coronal shear fractures (CSF) of the capitulum and trochlea humeri are rare injuries during growth age. No systematic, prospective analysis of the classification, procedure, and outcome has yet been found in the literature.

MATERIALS AND METHODS: Based on the results of a multicentre study of CSF in growing patients, we propose a redesign of the AO Paediatric Comprehensive Classification of Long-Bone Fractures (PCCF) for children and adolescents. We also assessed inter-rater agreement consistency with the modified classification and calculated the intra-class correlation coefficient (ICC) and its 95% confident interval (CI).

RESULTS: A detailed classification has been proposed that refers to the location of the fragment, the extent to which the trochlea is involved, and the distinction between single and multiple fracture injuries. This modified AO classification distinguishes between five types of injury (13-E/3.1I - 13-E/3.1IV and 13-E/3.2), replacing the current code 13r-E/8.1. Inter-rater reliability using this modified classification was moderate (ICC 0.452 (95% CI 0,284;

0,661). This is more accurate than all previous classifications tested for CSF.

CONCLUSION: A revised classification of coronal shear fractures of the distal humerus in children and adolescents is required. Our proposed classification considers the various types of this injury and their implic […]

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### Correction: Prevalence and risk factors for stem subsidence following reverse shoulder arthroplasty using on-lay press-fit short stems.

*Nishiura R, Nakazawa K, Manaka T, Hirakawa Y, Ito Y, Terai H*

PubMed · DOI

### Reverse shoulder arthroplasty vs. hemiarthroplasty for proximal humerus fracture: in-hospital outcomes in a national trauma registry.

*Raizah A*

BACKGROUND: Reverse shoulder arthroplasty (RSA) has rapidly supplanted hemiarthroplasty for acute proximal humerus fracture, driven by superior functional outcomes and lower revision rates. Whether this shift has altered perioperative morbidity at the population level remains poorly characterized. This study compared in-hospital outcomes between RSA and hemiarthroplasty in a national trauma cohort.

METHODS: This retrospective cohort study used the National Trauma Data Bank (2017-2022). Adults ( > = 18 years) admitted with proximal humerus fracture who underwent RSA (n = 2657) or hemiarthroplasty (n = 428) during the index hospitalization were included. The primary outcome was non-home discharge; secondary outcomes were any in-hospital complication, extended length of stay exceeding 7 days, and intensive care unit (ICU) admission. Multivariable logistic regression and inverse probability of treatment weighting (IPTW) were used to address confounding by indication.

RESULTS: Among 3085 patients (median age, 73 years; 2304 [74.7%] female), RSA patients were older (median, 74 vs. 64 years; P<.001) and more frequently female (77.3% vs. 58.6%; P<.001). After multivariable adjustment, RSA was not significantly associated with non-home discharge (adjusted odds ratio [aOR], 1.20; 95% confidence interval [CI], 0.93-1.55), any complication (aOR, 1.01; 95% CI 0.58-1.77), extended length o […]

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### Impact of cognitive impairment on postoperative delirium and discharge disposition following knee arthroplasty: a Japanese nationwide propensity score-matched study.

*Mori Y, Tarasawa K, Tanaka H, Kanabuchi R, Mori N, Fushimi K, Aizawa T, Fujimori K*

PURPOSE: As populations age, the number of older patients undergoing knee arthroplasty continues to increase. Cognitive impairment is prevalent among older adults and has been associated with adverse perioperative outcomes; however, its impact on postoperative delirium and discharge disposition following knee arthroplasty remains insufficiently defined, particularly in large Asian populations.

METHODS: We conducted a nationwide retrospective cohort study using Japan's Diagnosis Procedure Combination database from April 2016 to March 2023. Older patients who underwent primary total knee arthroplasty (TKA) or unicompartmental knee arthroplasty (UKA) were identified. Cognitive impairment was defined using ICD-10 diagnostic codes at admission. The primary outcome was postoperative delirium, and secondary outcomes included discharge to home, length of hospital stay, and perioperative blood transfusion. Propensity score matching (1:1) was performed to adjust for demographic factors, comorbidities, and surgical characteristics. Multivariable logistic regression and sensitivity analyses were conducted.

RESULTS: Among 259,319 eligible patients, 3,934 matched pairs were identified after propensity score matching. Patients with cognitive impairment had a significantly higher risk of postoperative delirium compared with those without cognitive impairment (absolute risk difference, 3.2%; […]

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### Association of primary tumor surgery with survival outcomes in patients with bone metastases across multiple cancer types.

*Xu PH, Shi B, Li T, Tian M, Fang M, Cheng M, Huang W, Ren F et al.*

BACKGROUND: Synchronous bone metastases (BM) have a notable impact on the survival rates of patients across various cancer types. While primary tumor surgery (PTS) has shown potential benefits, its impact across different cancers and patient subgroups needs further clarification. This study aimed to investigate the survival benefits of PTS across various cancer types.

METHODS: This retrospective cohort study analyzed data from the Surveillance, Epidemiology, and End Results (SEER) database, covering 53,015 patients diagnosed with BM from 2010 to 2019. Propensity score matching (PSM) was performed to reduce baseline imbalance. Overall survival (OS) and cancer-specific survival (CSS) were evaluated using Kaplan-Meier analysis and Cox proportional hazards regression.

RESULTS: PTS substantially improved overall and cancer-specific survival in patients with lung, kidney, breast and bladder cancers. All surgical modalities benefited breast and renal cancer survival. In prostate cancer, PTS produced no overall survival gain; survival varied by surgery type, being better after prostatectomy and worse with local resection. For liver cancer, PTS improved cancer-specific but not overall survival, and the small matched cohort necessitates circumspect interpretation.

CONCLUSION: The findings indicate that PTS can significantly enhance survival in selected patients with BM, particularly i […]

PubMed · DOI

## Liquid nitrogen-treated autograft combined with megaprosthesis versus total femoral replacement for large-volume malignant femoral bone tumors: a comparative study.

*Sang NTQ, Trung ND, Huy LD, Phuc LH, Thang VD, Anh LN, Hung LT, Hien TTT et al.*

BACKGROUND: Limb-salvage surgery for extensive femoral osteosarcoma presents major reconstructive challenges. Conventional options such as total femoral replacement (TFR) and long-segment biological reconstructions are associated with significant complications. We evaluated the clinical outcomes of a novel hybrid approach combining a liquid nitrogen-treated autograft with a megaprosthesis (LN-MP) and compared them with those of TFR.

MATERIALS AND METHODS: This retrospective study included 28 patients (range 8-24 years old) with malignant femoral bone tumors treated between 2020 and 2024. Patients underwent either LN-MP ($n = 10$) or TFR ($n = 18$). Perioperative outcomes, functional results (Musculoskeletal Tumor Society score-MSTS), bone union, oncological control, and survival were analyzed.

RESULTS: LN-MP demonstrated better functional outcomes (MSTS $27.0 \pm 2.1$ vs. $24.4 \pm 2.2$, $p = 0.014$), shorter operative time, and lower blood loss compared with TFR. Length of stay and follow-up duration were comparable. Continuous disease-free survival was higher in the LN-MP group (90% vs. 44%, $p = 0.058$), with lower recurrence (10% vs. 28%). Bone union was achieved in 70% of LN-MP cases at a mean of 9 months, with longer graft length associated with delayed union.

CONCLUSIONS: LN-MP reconstruction may serve as a joint-preserving alternative to total femoral replacement for patients with e […]

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## Tibial bone transport using external fixation in adults: a systematic review.

*Giuli C, Di Pietro G, Farine F, Bocchi MB, Vitiello R, Palmacci O*

INTRODUCTION: Tibial bone transport using external fixation is widely employed for the reconstruction of large and complex tibial bone defects; however, indications and surgical strategies remain highly heterogeneous. Tibial bone transport using external fixation is a demanding but reliable reconstructive technique, characterized by low failure rates and generally favorable outcomes despite frequent complications. Adequate soft-tissue management, selective docking-site grafting, and early functional rehabilitation were commonly reported. Further prospective studies are needed to standardize treatment strategies and improve clinical decision-making.  MATERIALS AND METHODS: A systematic review was conducted in accordance with PRISMA guidelines in April 2025, searching PubMed/MEDLINE and the Cochrane Library. Extracted data included patient demographics, comorbidities, indications, defect length, surgical technique, frame type, external fixation time, healing index, complications, failures, and ASAMI bone and functional outcomes. Methodological quality was assessed using the MINORS score and results were synthesized descriptively.

RESULTS: A total of 3543 patients from 89 studies were included. Post-traumatic infection or osteomyelitis accounted for 58.5% of indications. The mean defect length was 7.04 cm. Bifocal bone transport was performed in 82.4% of cases, and circular exter […]

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### Accuracy and risk factors for poor cup orientation in total hip arthroplasty: a comparison between supine and lateral registration using a portable navigation system.

*Matsumoto S, Tetsunaga T, Yamada K, Okuda R, Masada Y, Yamamoto T, Okazaki Y, Ozaki T*

INTRODUCTION: Accurate acetabular cup orientation is essential in total hip arthroplasty (THA). Portable inertial navigation may improve reproducibility, but accuracy in lateral decubitus THA may depend on the pelvic registration workflow. We compared cup placement accuracy between supine and lateral registration workflows using NAVBIT (Smith & Nephew PLC) and explored factors associated with large navigation errors.

MATERIALS AND METHODS: We retrospectively reviewed 110 consecutive primary cementless THAs performed in the lateral decubitus position using NAVBIT. The supine-registration workflow included 52 hips (supine pelvic registration followed by repositioning), and the lateral-registration workflow included 58 hips (registration and cup implantation in the lateral decubitus position). NAVBIT recorded radiographic inclination (RI) and radiographic anteversion (RA) intraoperatively; postoperative RI and RA were measured via computed tomography. Accuracy was defined as the absolute difference between navigation records and postoperative measurements. We compared navigation accuracy outcomes between propensity score-matched groups. Outliers were defined as an absolute error > 5° in RI or RA; factors associated with outliers were evaluated in the overall cohort.

RESULTS: After propensity score matching, 46 matched pairs (n = 92) were analyzed. Postoperative cup orientation w […]

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### Differences in the coronal plane alignment of the knee (CPAK) measurements using long-leg radiographs and computed tomography.

*Kawaguchi K, Tay ML, Young S*

INTRODUCTION: Coronal plane alignment of the knee (CPAK) parameters are increasingly used to characterize native coronal alignment and guide alignment strategies in total knee arthroplasty (TKA) based on long-leg radiographs (LLR). Computed tomography (CT) based robotic TKA software enables calculation of CPAK parameters based on anatomical reference points. However, potential differences in CPAK measurements between LLR and CT remain incompletely defined. This study compared CPAK coronal alignment parameters and phenotype distributions between LLR and CT, and evaluated factors associated with measurement discrepancies.

METHODS: This was a radiographic subanalysis of a prospective randomised controlled trial including 241 patients undergoing robotic-assisted TKA. Medial proximal tibial angle (MPTA) and lateral distal femoral angle (LDFA) were measured and arithmetic hip-knee-ankle-angle (aHKA), joint line obliquity (JLO) and CPAK phenotype distribution were calculated and classified from preoperative LLR and CT. Discrepancy groups were defined as LLR = CT (≤ 2°), LLR < CT (> 2°) and LLR > CT (> 2°). Associations between discrepancy groups and preoperative deformity and flexion contracture were assessed.

RESULTS: Compared with LLR, CT demonstrated a more varus MPTA (mean 86.7° vs. 87.1°, p = 0.04), a more valgus LDFA (87.1° vs. 87.8°, p < 0.01) and a lower JLO (173.9° vs. 174. […]

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### Conservative and surgical treatment of wrist osteoarthritis: a systematic review.

*Boever J, Unglaub F, Wulf J, Spies CK, Langer MF, Kußmaul AC*

INTRODUCTION: Wrist osteoarthritis leads to pain and functional impairment. Despite conservative treatment options, surgery remains a central therapeutic component. In Germany, the number of patients rose from over 472,000 cases in 2018 to more than 580,000 in 2024. This review examines the most common surgical treatment approaches, focusing on Proximal Row Carpectomy (PRC) and Four-Corner Fusion (4CF), discussing indications, contraindications and potential complications.

METHODS: A systematic and narrative literature review was conducted on treatment options for wrist osteoarthritis. The systematic review focusing on PRC and 4CF followed PRISMA guidelines and included searches in PubMed, Cochrane and Embase. Included were RCTs, systematic reviews and meta-analyses from January 2010 to May 2025 that compared both procedures in terms of pain, range of motion, grip strength, function, complication rates and reoperation rates.

RESULTS: Eight studies were included. Both procedures significantly reduced pain and improved function. PRC showed advantages in range of motion and lower complication rates, while 4CF was associated with higher grip strength. Partial arthrodesis (e.g., scaphotrapeziotrapezoid (STT) arthrodesis or radioscapholunate (RSL)) arthrodesis may be selectively used but are linked to higher nonunion rates. Total wrist arthrodesis or wrist arthroplasty may be consi […]

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## Factors associated with selection of total hip arthroplasty versus hemiarthroplasty for femoral neck fracture in older adults: a nationwide analysis of 99,084 cases.

*Maman D, Steinfield Y, Berkovich Y*

BACKGROUND: Surgical management of displaced femoral neck fractures in older adults typically involves hemiarthroplasty or total hip arthroplasty (THA). Although clinical guidelines suggest that THA may be considered in selected healthier and cognitively intact patients, real-world procedure selection varies widely. This study evaluated patient factors associated with selection of THA versus hemiarthroplasty in a contemporary U.S.

METHODS: A retrospective cohort study was conducted using the 2022 Nationwide Readmissions Database (NRD). Patients ≥ 65 years hospitalized with femoral neck fracture were identified using ICD-10-CM codes. Those treated with internal fixation or non-arthroplasty procedures were excluded. Weighted analyses characterized demographics, comorbidities, and hospital utilization between THA and hemiarthroplasty groups. A multivariable logistic regression model identified factors independently associated with receiving THA. All analyses accounted for NRD survey design.

RESULTS: Among 142,013 operative cases, 99,084 met inclusion criteria (81.8% hemiarthroplasty; 18.2% THA). Patients receiving THA were younger (76.6 vs. 81.9 years), had shorter length of stay (6.06 vs. 7.24 days), and were more frequently discharged home (17.5% vs. 6.2%). Metabolic conditions were associated with increased odds of THA, including obesity (OR 1.17) and sleep apnea (OR 1.12). F […]

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## Effect of distal screw number on axial deformation and fragment-specific stability in AO/OTA 2R3 C2.1 distal radius fractures.

*Dietz L, Spiegel C, Kohler FC, Biedermann U, Zderic I, Feist A, Gueorguiev-Rüegg B, Lenz M et al.*

PURPOSE: To investigate the biomechanical influence of distal screw number in palmar plate fixation of unstable intra-articular distal radius fractures with radiodorsal metaphyseal comminution.

METHODS: Nine matched pairs of cryopreserved human radii were prepared using a standardized AO/OTA 2R3 C2.1 fracture model with a defined radiodorsal metaphyseal defect zone. For each pair, specimens were randomized to palmar fixation using a 2.4-mm variable-angle locking plate with either four or six distal screws. After embedding, constructs were subjected to cyclic axial loading (150 N, 5,000 cycles) simulating early postoperative rehabilitation. Global axial deformation and fragment-specific three-dimensional interfragmentary motion were recorded using an optical motion-tracking system. Statistical analysis was performed using non-parametric tests ($p < 0.05$).

RESULTS: Two specimens from one donor pair were excluded, leaving 16 radii (8 pairs) for analysis. Bone mineral density and initial construct stiffness did not differ between groups. The six-screw configuration showed significantly lower axial deformation after cyclic loading ($p = 0.047$) and lower radial fragment-shaft translation at both time points ($p = 0.028$; $p = 0.045$). Radial fragment-shaft rotation was lower in the six-screw group at baseline ($p = 0.018$), but no rotational difference remained after cyclic loading. In the […]

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## Crowe severity and femoral stem geometry are associated with lower-limb alignment after total hip arthroplasty for unilateral developmental dysplasia of the hip-related secondary osteoarthritis: valgus alignment trajectories.

*Afacan MY, Gurcinar MG, Davulcu CD, Unlu MC, Kaynak G*

BACKGROUND: Total hip arthroplasty (THA) for secondary osteoarthritis related to developmental dysplasia of the hip (DDH) restores hip mechanics. We aimed to quantify pre-to-postoperative changes in ipsilateral knee and global lower-limb alignment after unilateral THA to determine whether alignment changes differ by Crowe classification and femoral stem geometry.

METHODS: This retrospective cohort study included 74 patients who underwent unilateral primary THA for DDH-related secondary osteoarthritis, with a mean follow-up of 45.4 months. The predefined primary radiographic outcome set comprised preoperative-to-postoperative changes and final follow-up values in coronal lower-limb alignment parameters, including mechanical and anatomic lateral distal femoral angles (mLDFA and aLDFA), mechanical medial proximal tibial angle (mMPTA), joint line convergence angle (JLCA), global offset (GO), hip-knee-ankle angle (HKA), knee mechanical axis deviation ratio (KMADR), and femoral mechanical-anatomic axis angle (FMMA). The contralateral reference limb was used for side-to-side radiographic comparison. Patients were stratified by Crowe classification (I-II vs. III-IV), and femoral stem geometry analyses were performed according to conical, wedge, and rectangular stem designs. Clinical outcomes included hip/knee visual analog scale, Harris Hip Score, Knee Society Score, Oxford Knee Score […]

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## Can ChatGPT pass the polish national medical specialization examination in orthopedics and traumatology?

*Maci■g B, Bujak K, Jegierski D, Stolarczyk A*

INTRODUCTION: Artificial intelligence (AI) has evolved rapidly in recent years and is becoming increasingly integrated into many areas of medicine. In the medical field, these systems have attracted considerable attention because of their potential applications in clinical decision support, medical education, scientific communication, and postgraduate training. The aim of the present study was to evaluate whether ChatGPT could achieve a passing score on the Polish National Medical Specialization Examination (Panstwowy Egzamin Specjalizacyjny, PES) in orthopedics and traumatology and to determine how different prompting strategies influenced its performance.

MATERIALS AND METHODS: Authors systematically assessed the performance of ChatGPT-4 across five consecutive official PES exams (from Autumn 2023 to Spring 2025) in orthopedics and traumatology. Each exam was administered using three distinct prompting strategies: Professor prompt, Specialist prompt, Resident prompt. For each exam session (e.g., Spring 2024, Autumn 2023), the prompts were submitted to ChatGPT sequentially. The model's responses were evaluated against the official answer key published by the Polish Center of Medical Exams (Centrum Egzaminów Medycznych, CEM).

RESULTS: The results were averaged across all prompts. The overall accuracy was 81% (range: 68.33%-85.83%). The highest score (85.83%) was recorded mult […]

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## Two-year outcomes following concomitant ACL and extra-articular ligament reconstruction: a nationwide analysis of over 5,000 patients.

*Abid R, J Burkhart R, J Moyal A, M Adelstein J, Kodsy M, G Calcei J, E Voos J, M Apostolakos J*

PURPOSE: The anterior cruciate ligament (ACL) is the most commonly injured and reconstruction ligament in the United States. Associated injuries are common, oftentimes including components of the extra-articular knee. However, complications following concomitant ACL and extra-articular ligaments (EAL) has not been defined in a large-scale, population-based sample. We aimed to analyze outcomes following concomitant ACL and EAL reconstruction using the TriNetX nationwide database.ep METHODS: Deidentified data from the TriNetX Collaborative Network was used to identify all patients who received ACL and EAL reconstruction on the same day. Demographic information including age, sex, race, and geographic data were collected. Rates of medical and orthopaedic complications were analyzed at 30, 90, and 180 days and at 2 years. A propensity matched sub-analysis comparing outcomes patients with concomitant and staged repair was performed. Patients were identified using Current Procedural Terminology (CPT) coding.

RESULTS: We identified 5944 patients who received concomitant ACL and EAL reconstruction with postoperative outcomes available at 30, 90, and 180 days. 5040 patients had data available at 2 years. Serious medical complications like surgical site infection, deep vein thrombosis, acute kidney failure, and sepsis were rare at each point of follow-up. Rates of revision surgery were […]

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### Recovery of gait after ankle fracture surgery following early mobilization and weight-bearing.

*Bauer L, Schlegel AS, Scheuer A, Böcker W, Polzer H, Baumbach SF*

BACKGROUND: Ankle fractures are among the most common fractures in adults. Despite significant advancements in surgical techniques, the postoperative treatment regimens remain conservative. This study aims to objectively evaluate gait recovery following early mobilization and weight-bearing after surgical ankle fracture treatment using instrumented gait analysis.

METHODS: This prospective, single-armed, longitudinal study enrolled adult patients following surgical treatment for isolated ankle fractures with early pain dependent weightbearing. Gait analysis, utilizing a treadmill integrated with a pressure plate and IMU-based motion capture systems, was conducted at 6 weeks, 3-, 6-, and 12-months post-surgery, assessing spatio-temporal and kinematic parameters. Results were compared longitudinally and against a healthy control group.

RESULTS: Longitudinal data were available for 44 patients. Significant ($p<.001$) improvements of self-paced walking velocity (SPV) were observed from 6 weeks (2.2 ± 1.1 km/h), to 3 (2.9 ± 0.9 km/h) and 6 months (3.4 ± 0.9 km/h). The spatio-temporal parameters showed significant changes between 6 weeks and all further time points only at SPV. Kinematic parameters showed significant longitudinal improvements at SPV for hip, knee, and ankle motion (including all three ankle planes). Patient gait parameters at 6 and 12 months were comparable to healthy […]

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### A comparative analysis of three knee arthrodesis techniques to manage failed total knee arthroplasty due to infection: intramedullary nailing, LRS external fixator, and compression plating.

*Rajasekaran RB, Ponniah HS, Palanisami DR, Sundaram VP, Palanivelayutham SS, Jayaramaraju D, Dhanasekaran S, Rajasekaran S*

INTRODUCTION: This study compared external fixation (EF), intramedullary nailing (IMN), and compression plating (CP) for knee arthrodesis (KA) following failed total knee arthroplasty (TKA) due to periprosthetic joint infection, focusing on complications, radiological, and functional outcomes.

METHODS: Patients aged ≥ 18 who underwent knee arthrodesis following failed TKA due to infection from 2008 to 2023 were included. Four surgeons used three techniques: EF, IMN, and CP. Retrospective data on demographics, surgical techniques, complications, and outcomes were collected. Postoperative patient-reported outcome measures (PROMs), pain, and functional status were assessed. Radiographic data was reviewed for limb shortening and union status.

RESULTS: 46 patients (mean age 61.9 years, BMI 28.2), predominantly male (84.8%) were involved in this study. Twenty patients underwent EF, 16 underwent CP, and 10 underwent IMN, all for infected primary TKA. Pseudomonas was the most common infecting organism (15.2%). Average follow-up was 29.1 months. Bony fusion was achieved in 41 knees (89.1%), with five requiring additional surgery; four eventually required amputation due to non-fusion. EF group had the shortest time to union and no major complications or amputations. Time to union was significantly shorter in the CP group compared to the IMN group ($p < 0.001$), and EF group also had a si […]

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## Accuracy of iliac crest reference pin placement in total hip arthroplasty: a CT-based analysis of risk factors for cortical perforation.

*Masada Y, Tetsunaga T, Yamada K, Inoue T, Okuda R, Tetsunaga T, Okazaki Y, Ozaki T*

INTRODUCTION: Accurate iliac crest reference pin placement is essential for safe navigation-assisted total hip arthroplasty (THA). Although complications are uncommon, inaccurate placement may cause neurovascular injury. This study aimed to evaluate pin insertion accuracy using postoperative computed tomography (CT), identify risk factors for cortical perforation, and determine whether CT-detected cortical perforation was associated with postoperative acetabular cup positioning accuracy.

MATERIALS AND METHODS: We retrospectively analyzed 200 primary THAs performed using portable or CT-based navigation between December 2022 and July 2025. Pin position was assessed on CT 1 week postoperatively and classified as complete intraosseous insertion or cortical perforation. Risk factors were examined using univariable and multivariable logistic regression analyses. Perforation direction was evaluated using Firth's penalized logistic regression. Postoperative cup positioning accuracy was assessed using CT-based radiographic inclination (RI), radiographic anteversion (RA), and absolute errors from target angles of 40° for RI and 15° for RA.

RESULTS: Cortical perforation occurred in 70 of 200 hips (35.0%): 32 medial and 38 lateral. On a pin-by-pin basis, 113 of 400 pins (28.3%) demonstrated perforation. In multivariable analysis, smaller ASIS width, lateral decubitus positioning, and 4-m […]

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## Long-term outcomes of arthroscopic single-anchor fixation versus microfracture for osteochondral fractures of the knee weight-bearing area in adolescents and young adults.

*Yang W, Pan Q, Wang H, Peng Y, Meng C, Huang W*

PURPOSE: Osteochondral fractures involving the weight-bearing area of the knee in young patients require stable restoration of the native osteochondral fragment whenever possible. This study aimed to evaluate the long-term clinical and radiological outcomes of arthroscopic single-anchor fixation and to compare them with those of microfracture.

METHODS: This retrospective comparative study included 24 adolescents and young adults with osteochondral fractures of the femoral condyle or trochlear region treated between 2016 and 2017. Twelve patients underwent arthroscopic single-anchor in situ fixation, and 12 underwent microfracture. The fixation technique used one double-loaded suture anchor to obtain multi-point compression of the osteochondral fragment while preserving the subchondral plate. Clinical outcomes were assessed using the Lysholm, International Knee Documentation Committee, and Tegner scores. Magnetic resonance imaging was evaluated using the MOCART 2.0 score. The mean follow-up was 93.7 months.

RESULTS: The single-anchor group showed a significantly higher intraoperative contact area ratio than the microfracture group (96.5% ± 3.2% vs. 85.8% ± 5.6%, $P < 0.001$). At final follow-up, the single-anchor group had superior Lysholm scores (91 ± 5 vs. 83 ± 8, $P = 0.009$), IKDC scores (86 ± 6 vs. 78 ± 7, $P = 0.010$), and Tegner activity levels (5.0 ± 1.1 vs. 3.9 ± 0.9, P = 0 […]

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## Isometric points in cerclage for patellar tendon repair: a 3-dimensional, weightbearing computed tomography kinematic simulation.

*Zindel C, Hodel S, Luo Y, Meisterhans M, Fucentese SF, Vlachopoulos L*

PURPOSE: The gold standard for the management of ruptured patellar tendon is the surgical repair with nonabsorbable sutures and additional reinforcement e.g. cerclage wires as described by McLaughlin in 1947. Despite the early description, the exact placement of such cerclage wires remains unclear. The aim of the study was (1) to analyze how the cerclage elongates during knee flexion and (2) to define the insertion points with the most isometric behavior at the patella and tibial tuberosity (TT).

METHODS: 3D simulations of 10 healthy knees in the range from 0° to 120° of flexion using weightbearing computed tomography scans were generated. 12 insertion points were defined at the patella in equal distances along the lateral and medial border from the proximal to the distal pole. 42 insertion points were defined at the tibial tuberosity (TT) in a 3.0 × 3.0 cm large grid around its center. An automatic string simulation algorithm was used to calculate the length for all possible combinations of the cerclage from 0° to 120° flexion. The most isometric combination of insertion points was found using the isometric score.

RESULTS: The most isometric insertion at the TT is distally ($p < 0.01$) and towards the center of the TT ($p < 0.01$). The most isometric points at the TT is located 61 mm distal to the joint line and 5 mm from the anterior border of the TT in a true lateral view fluo […]

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## Patient mortality exceeds implant revision risk with equivalent survival following unicompartmental versus total knee arthroplasty in octogenarians.

*Innocenti M, Leggieri F, Massenzi M, Stimolo D, Akkaya M, Matassi F, Civinini R*

BACKGROUND: While traditional surgical decision-making emphasizes long-term implant survivorship for octogenarians knee reconstruction, this paradigm may be inappropriate when patient mortality risk substantially exceeds revision risk. This study compared mid-term mortality between unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA) in octogenarians, hypothesizing that implant type would not significantly influence patient survival.

METHODS: A retrospective cohort study was conducted on 238 consecutive octogenarian patients (age ≥ 80 years) who underwent primary knee arthroplasty at a single tertiary orthopaedic center between January 2018 and December 2020. Patients with revision arthroplasty, fracture, tumor, inflammatory arthropathy, or incomplete data were excluded. The primary outcome was all-cause mortality at minimum 5-year follow-up. Secondary outcomes included age-stratified mortality (80-84, 85-89, ≥ 90 years) and identification of independent mortality predictors. Statistical analyses included independent t-tests, chi-square/Fisher's exact tests, standardized mean differences, and multivariable logistic regression adjusting for age, follow-up duration, and registry year.

RESULTS: Of 238 patients, 185 (77.7%) underwent TKA and 53 (22.3%) underwent UKA. Overall mortality was 35.3% (84/238): 33.5% in TKA versus 41.5% in UKA (OR = 1.41, 95% CI: 0 […]

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## Short- to mid-term clinical outcomes and return to sport and work after arthroscopic debridement of the extensor carpi radialis brevis in patients with chronic lateral epicondylopathy.

*Siebenlist S, Doucas A, Hatt FL, Lappen S, Vieider RP, Lacheta L, Kadantsev P*

BACKGROUND: This study aimed to evaluate the short- to mid-term clinical outcomes following arthroscopic extensor carpi radialis brevis (ECRB) debridement in patients with chronic epicondylopathy humeri radialis (cEHR). We hypothesized that arthroscopic ECRB debridement would result in favorable short- to mid-term patient-reported outcomes, improving pain levels, strength, range of motion (ROM) and patient satisfaction.

METHODS: A retrospective case series was conducted on patients who underwent arthroscopic ECRB debridement for cEHR with symptoms > 6 months. Patients were assessed at a minimum of 12 months postoperatively using the Mayo Elbow Performance Score (MEPS), Patient-Rated Tennis Elbow Evaluation (PRTEE), Disabilities of the Arm, Shoulder, and Hand (DASH) score, and Visual Analog Scale (VAS) for pain. Pre- and postoperative scores were compared using DASH and VAS. Maximum isometric wrist extension strength and postoperative ROM were measured bilaterally. Return to sport and work, patient satisfaction, and complications were documented.

RESULTS: Of the 35 eligible patients, 30 completed follow-up (86% follow-up rate; mean age 48.4 ± 6.6 years; range: 36-65 years; 47% male). The mean duration of symptoms before surgery was 12 months (IQR 8-20, range 6-62). The mean follow-up duration was 28.13 ± 10.9 months (range 12-45). At final follow-up, median outcome scores were […]

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## Complications and outcomes after operative treatment of low-energy pelvic ring and acetabular fractures.

*Salonpää KB, Nurkkala J, Lantto I, Kakko S, Kaakinen T, Aitavaara-Anttila M, Liisanantti J*

BACKGROUND: Fragility fracture of the pelvis (FFP) with a low-energy mechanism of injury is relatively common among the growing geriatric population. The incidence of postoperative complications after surgically treated FFP is not well documented. The aim of this study was to report complications after operatively treated FFP and to determine whether these complications would have a detrimental effect on outcomes.

METHODS: Retrospective study of 200 consecutive patients who underwent operative treatment at Oulu University Hospital, Oulu, Finland, between 1.1.2010 and 15.4.2021 for FFP with a low-energy injury mechanism. Postoperative complications were recorded, and associations with outcomes were depicted.

RESULTS: Postoperative complications were recorded in 31 (15.5%) patients. Patients with complications had higher intraoperative bleeding (OR 2.8 (1.0-7.9), p = 0.048) and higher ASA classification 3.4 (1.3-8.4), P = 0.009). Patients with complications had longer hospital stays, but no clear association between complications and mortality or discharge location was observed.

CONCLUSIONS: The incidence of postoperative complications was comparable with that reported in previous literature. Patients who experienced complications had longer hospital stays; however, no definitive association between complications and outcomes was identified.

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## Minced cartilage implantation for cartilage regeneration: a survey of current clinical practices.

*Yilmaz T, Leutheuser S, Niemeyer P, Behrendt P, Roessler P, Gille J, Mumme M, Vogt S*

INTRODUCTION: The objective of this study is to evaluate how minced cartilage implantation (MCI) is currently applied in clinical settings, focusing on the selection criteria, procedural execution, handling of concomitant joint pathologies, and postoperative management.

MATERIAL AND METHOD: An online survey was conducted among all 4,915 members of the German Society for Arthroscopy and Joint Surgery (AGA). Invitations with a survey link were emailed, and the questionnaire was accessible anonymously from January 10 to February 20, 2024. A total of 927 responses were received and analyzed, resulting in a 19% response rate. Demographic data were described descriptively; survey answers were reported as percentages. The 25-item questionnaire, developed by AGA and the Quality Circle for Cartilage Repair & Joint Preservation (QKG e.V.), included single- and multiple-choice questions on experience with cartilage therapy, minced cartilage procedure application and postoperative care, and the overall assessment of this technique within cartilage repair.

RESULTS: MCI has emerged as one of the three most commonly used cartilage repair techniques in the knee joint. Defect sizes up to 4 $cm^2$ represent the largest treated group. The application of the procedure is quite heterogeneous, with some practitioners using a shaver and others manually mincing with a scalpel. Fixation methods also var […]

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## Variability in United States online rehabilitation protocols after open and endoscopic hip abductor repair.

*Anwunah C, Rosenberg S, Hartwell M*

PURPOSE: The purpose of this study was to evaluate readily accessible U.S. based online physical therapy protocols for open and endoscopic hip abductor repairs and to assess the variability in post-operative rehabilitation protocols.

METHODS: Online physical therapy protocols from U.S. based orthopaedic institutions for gluteus medius/minimus repair were collected using a web-based query utilizing search terms "hip abductor repair rehabilitation protocol" and "gluteus repair rehabilitation protocol." Published protocols from both Electronic Residency Application Service (ERAS) academic orthopaedic surgery programs and private groups were collected. All publicly available protocols were included. Protocols were analyzed using a standardized spreadsheet to assess rehabilitation components and timelines. Descriptive statistics (median values, percentages, ranges) were calculated.

RESULTS: 54 rehabilitation protocols were assessed, including 22 (41%) from ERAS-affiliated academic programs or associated physicians. Postoperative weight-bearing was addressed in 53/54 (98%) protocols, with 41 (77%) recommending toe-touch/touchdown/flat-foot/20-lb weightbearing. Median duration of restricted weight-bearing was 6 weeks (range, 2-8 weeks). Regarding range of motion restrictions, 37 (69%) protocols limited hip flexion to 90°, 42 (78%) advised against passive adduction past neutral, and […]

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### The incidence and epidemiology of patellar fractures following medial patellofemoral ligament reconstruction: a systematic review and meta-analysis.

*Abelleyra Lastoria DA, Hunt A, Ross CK, Yemane M, Keynes S, Ghosh A, Hing CB*

INTRODUCTION: Patellar instability can arise from a traumatic event with anatomical predisposing factors increasing the risk of dislocation. Medial patellofemoral ligament reconstruction (MPFLR) is commonly performed to treat patellar instability. The aim of this systematic review was to determine the incidence of and risk factors for patellar fractures after MPFLR.

MATERIALS AND METHODS: The protocol for this review was registered on PROSPERO with registration number CRD420251066107. The PRISMA 2020 checklist was followed. Electronic databases and conference proceedings were searched to the 15th of May 2025. Studies were eligible if they reported incidence of patellar fractures after MPFLR. A random-effects meta-analysis was performed. Included studies were appraised using tools respective of study design.

RESULTS: A total of 4,972 records were screened in the initial search. A total of 110 studies satisfied the inclusion criteria, evaluating 115,496 patients. The evidence was moderate to low in quality. The incidence of patellar fracture ranged from 0 to 9.50%, with a pooled incidence of 1.17% (95% confidence interval [CI] 0.76- 1.68%). There was no difference in incidence of patellar fractures between patients who did and did not undergo bone tunnel fixation (0.94%, 95% CI 0.69-1.24 vs. 1.00%, 95% CI 0.15-2.44%, respectively). There was no difference in fracture incidence […]

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### Predictive accuracy of a general-purpose artificial intelligence model for cut-out following proximal femoral nailing.

*Ben Arie G, Warschawski Y, Amzallag N, Gan-Or H, Ben-Tov T, Khoury A, Horev I, Graif N et al.*

INTRODUCTION: Cut-out is the most consequential mechanical complication after proximal femoral nailing and requires prompt recognition. General-purpose artificial intelligence (AI) models with image-interpretation capability are increasingly accessible, yet their diagnostic performance for this task is unknown. We evaluated ChatGPT's accuracy for predicting cut-out following proximal femoral nailing and compared TFNA and Gamma nail subgroups.

MATERIALS AND METHODS: In this retrospective predictive-accuracy study, 989 patients (683 TFNA, 306 Gamma nail; mean age 83.8 years; cut-out prevalence 2.5%) were analysed. For each case, three baseline radiographs - the injury anteroposterior and lateral views and the immediate postoperative control radiograph, all obtained before any complication could be radiographically present - together with the full clinical data set (excluding complication-related information) were submitted to ChatGPT, which returned a binary prediction (yes/no) and probability estimate (0-100%). The follow-up radiographs on which cut-out becomes apparent were not provided to the model. Ground truth was established on subsequent follow-up radiographs by a departmental review board requiring agreement of two senior surgeons. Accuracy metrics were calculated with Wilson 95% confidence intervals (CI); exact McNemar's and Mann-Whitney U tests assessed directional bia […]

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### Staged management of a compound combined pelvi-acetabular injury with perineal degloving in a young female: a case report and literature review.

*Dhankhar K, Kekatpure AL, Kekatpure AL*

We present the case of a 30-year-old woman, the patient who sustained a compound combined pelvi-acetabular fracture with extensive perineal internal degloving after high-energy trauma. Computed tomography revealed an unstable pelvic ring injury (Tile type C1; anteroposterior compression type III) with right transverse acetabular fracture. The occurrence of extensive internal degloving was confirmed intraoperatively. The injury was treated using a graduated, principle-based approach incorporating early soft-tissue debridement, dead-space management and temporary pelvic stabilization, infection control, and delayed definitive internal fixation. An intercurrent pin-site infection was treated successfully without progression to deep infection. The patient demonstrated progressive radiological union at 4 months with a Harris Hip Score (HHS) of 80 and a Majeed pelvic score of 55. At one-year follow-up, complete union occurred and functional recovery was achieved, characterized by HHS of 90 and a Majeed score of 70. In this case, infection-guided staging as well as extremely aggressive soft tissue treatment may allow for safe delayed fixation even where pelvi-acetabular trauma has been compromised.

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### Concomitant upper extremity fractures do not worsen outcomes in conservatively managed pelvic fractures: a cohort study of 1840 geriatric patients.

*Graif N, Elbaz E, Kazum E, Rachevsky G, Gurel R, Factor S*

INTRODUCTION: Pelvic fractures in elderly patients are associated with significant pain, functional decline, and prolonged rehabilitation, with the majority managed non-operatively. In the conservative management paradigm, intact upper extremity function is critical for mobilization with assistive devices; concurrent upper extremity (UE) fractures may therefore complicate recovery. The aim of this study was to evaluate the clinical outcomes associated with this injury combination in patients with conservatively managed pelvic fractures.

METHODS: A retrospective cohort study was conducted at a Level 1 trauma center, including all patients aged ≥ 65 years admitted with pelvic ring or acetabular fractures managed non-operatively between January 2010 and January 2024. Patients were stratified by the presence of a concomitant UE fracture sustained during the same traumatic event. Primary outcomes were hospital length of stay (LOS) and all-cause mortality at multiple time points. Secondary outcomes included readmission rates and systemic infection rates.

RESULTS: Of 1840 patients, 1693 (92.0%) sustained isolated pelvic fractures and 147 (8.0%) had concomitant UE fractures. Groups were comparable at baseline across all measured covariates. LOS did not differ significantly between groups ($11.0 \pm 14.2$ vs. $11.8 \pm 16.9$ days; $p = 0.796$). Mortality was similar at all evaluated time points […]

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### Single double-loaded anchor repair of isolated Lafosse III subscapularis tears in patients not involved in physically demanding activities provides satisfactory functional and radiologic outcomes at minimum 5-year follow-up.

*Aral F, Oklaz EB, Ahmadov A, Asiret E, Delice C, Guler Oklaz EB, Akdulum I, Kanatli U*

PURPOSE: To evaluate the outcomes of single-anchor repair for isolated Lafosse III subscapularis tears in patients not engaged in physically demanding activities.

METHODS: Patients with isolated subscapularis tears who underwent arthroscopic surgery between 2014 and 2020 were retrospectively evaluated. The inclusion criteria were patients who underwent single double-loaded anchor repair for Lafosse III tears and were not involved in physically demanding activities. Assessments included the American Shoulder and Elbow Surgeons (ASES) score, Constant-Murley Score (CMS), Visual Analog Scale (VAS), range of motion, specific clinical tests, and magnetic resonance imaging encompassing tendon integrity, cross-sectional area (CSA), vertical diameter, and transverse diameters.

RESULTS: 27 patients (mean age $52.4 \pm 10.3$ years; mean follow-up $80.8 \pm 18.6$ months) were included in the study. The final follow-up demonstrated significant improvements in the ASES ($44.0 \pm 12.4$ to $82.4 \pm 14.1$), CMS ($49.1 \pm 15.3$ to $85.3 \pm 11.1$), VAS ($5.7 \pm 2.1$ to $1.3 \pm 2.2$), and internal rotation (L4 to T8) ($p < 0.001$ for all). The positivity of clinical tests decreased substantially: Bear-Hug, from 93% to 18%; Belly-Press, from 89% to 15%; and Lift-Off, from 89% to 18% ($p < 0.001$ for all). Radiological analyses indicated maintained tendon integrity in 93% of patients and increases in CSA ($1.292 \pm 453$ to 1.544 […]

PubMed · DOI

### Shoulder injuries in rugby union: a systematic review and meta-analysis.

*Doyle J, Bellamy M, Fowler K, Keiller H, Gunasekera R, Funk L*

Shoulder injuries are common in Rugby Union, resulting in prolonged absence and considerable burden. Reported incidence, severity, and mechanisms vary widely, reflecting the multiple high-impact actions in the sport. Various management approaches exist to treat a wide range of pathologies. This systematic review and meta-analysis aimed to synthesise evidence on the epidemiology, mechanisms, and treatment of shoulder injuries in rugby. PUBMED, SCOPUS, and WEB OF SCIENCE were searched from database inception until 1st June 2025. From 1,099 abstracts screened, 37 studies were included. The pooled match incidence was 11.02 injuries/1,000 player-hours (95% CI 7.23-16.82), though heterogeneity was very high ($I^2 = 97\%$) and the 95% prediction interval was wide (2.11-57.61 injuries/1,000 player-hours), indicating the pooled rate should be read as an average across diverse populations rather than a single representative value. Higher rates were seen in male elite (13.74/1,000 h) and high school/university players (12.28/1,000 h). Training incidence was considerably lower (0.20/1,000 h). Substantial between-study heterogeneity was observed across all incidence analyses, reflecting genuine variation across playing levels, sex, and surveillance methods. Tackling was the predominant injury mechanism, most often affecting the tackler. Bankart, Latarjet, and Bristow procedures all produced fav […]

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## The pre-operative physical component subscale of the 12-item Short Form Health Survey was the only factor associated with 2-year clinical outcome and satisfaction after hip abductor tendon repair.

*Ebert J, Edwards P, Klinken S, Janes G*

INTRODUCTION: Surgical repair in the context of symptomatic hip abductor tears unresponsive to non-operative management has demonstrated encouraging outcomes. This study sought to investigate demographic, injury/management and post-operative factors associated with clinical outcome after hip abductor tendon repair.

METHODS: A total of 104 patients undergoing repair were included. Patients were assessed pre-surgery and 2-years post-operatively with the modified Harris Hip Score (mHHS), Oxford Hip Score (OHS), physical (PCS) and mental (MCS) component subscales of the 12-item Short Form Health Survey (SF-12), and satisfaction. Regression analysis assessed the contribution of pre-operative demographic (age, sex, body mass index, PCS, MCS), injury/treatment (symptom duration, prior corticosteroid injections, gluteus minimus/medius tear grade and fatty infiltration) and post-operative (hip abduction strength symmetry) variables, to the 2-year mHHS and being 'very satisfied'.

RESULTS: Mean improvement in the mHHS and OHS from baseline to 2 years was 36.0 points (95% CI, 31.9 to 40.2; $p < 0.001$) and 17.5 points (95% CI, 15.8 to 19.3; $p < 0.001$), respectively. Seven (6.7%) patients presented with symptomatic re-tearing. Of the 97 patients with 2-year follow-up, 78 (80.4%) were 'very satisfied'. The pre-operative SF-12 PCS was the only factor associated with the 2-year mHHS ($B = 0.46$; […]

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## Temporising external fixation reduces loss of reduction compared with plaster splinting in ankle fracture-dislocations: a systematic review and meta-analysis of cohort studies.

*Suravaram V, Solomon L, Bhat S, Wittauer M*

PURPOSE: The optimal temporising strategy for ankle fracture-dislocations is a highly debated topic. These unstable injuries are commonly managed with staged fixation to allow for soft tissue recovery, using either plaster splinting or ankle-spanning external fixation. This study aimed to compare the efficacy and safety of these two approaches.

METHODS: A systematic search of MEDLINE, Embase, Scopus, Cochrane CENTRAL, and Google Scholar was conducted. Comparative studies evaluating temporary external fixation versus plaster splinting in skeletally mature patients with ankle fracture-dislocations were included. Meta-analyses were performed to assess differences in loss of reduction prior to definitive fixation, soft tissue complications, and time to definitive fixation. Risk of bias was assessed using ROBINS-I tool, and certainty of evidence was appraised using GRADE.

RESULTS: Nine retrospective cohort studies comprising 1638 patients were included (589 external fixation, 1049 splinting). Loss of reduction occurred in 3% of injuries in the external fixation cohort compared with 14.6% in those managed with plaster splinting (RR 0.22; 95% CI 0.09-0.52). Rates of overall soft-tissue complications were similar between cohorts. The data regarding time to definitive fixation were highly heterogeneous across studies. Overall certainty of evidence was moderate for LOR and very low for […]

PubMed · DOI

### Comprehensive ultrasound assessment of thumb metacarpophalangeal collateral ligament injuries reveals frequent concomitant abnormalities.

*Bain O, Farkash U, Atzmon R, Shemesh S, Ohana N*

INTRODUCTION: Ulnar collateral ligament (UCL) tears of the thumb are well-recognized injuries, while radial collateral ligament (RCL) and dorsal extensor-related abnormalities are less frequently studied. The pattern of concomitant ultrasound abnormalities in patients after thumb trauma is not well documented. The aim of this study was to characterize concomitant UCL, RCL, and dorsal extensor-related abnormalities within a selected cohort referred to thumb ultrasound after trauma, and to explore associations with injury mechanism.

MATERIALS AND METHODS: A retrospective review was performed of 42 consecutive adult patients who underwent high-resolution ultrasound of the thumb metacarpophalangeal joint between April 2014 and February 2025 for suspected ligamentous injury after acute trauma. Patients were included if ultrasound demonstrated pathologic findings in at least one stabilizing structure, UCL or RCL, and the extensor mechanism was systematically assessed. Collateral ligament injuries were graded using standard sonographic criteria, while dorsal extensor-related findings were recorded descriptively. Associations between injury severity and mechanism of injury were tested using chi-square or Fisher's exact analysis, as appropriate.

RESULTS: UCL pathology was present in 35/42 (83.3%) patients, RCL pathology in 23/42 (54.8%), and dorsal extensor-related abnormalities in 17 […]

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### Holding the line: predictors of coronal and sagittal loss of reduction after tibial plateau fracture fixation.

*Hesmerg MK, Benner JL, Joosse P, Keijser LCM*

INTRODUCTION: Tibial plateau fractures (TPFs) are challenging fractures that often suffer from secondary Loss of Reduction (LoR) after fracture fixation. In this study, we aimed to investigate the factors and characteristics associated with LoR after surgical fixation, using measurements of both coronal and sagittal displacement.

METHODS: A retrospective analysis of 213 TPF patients (79.3% female, mean age 57.5 ± 14.4 years) surgically treated between 2017 and 2024 in a level 1 trauma center was performed. Patients were included if they had a minimum of 100-days radiological follow-up. Loss of reduction was defined as late postoperative depression ≥ 3 mm, condylar widening ≥ 5 mm, varus/valgus or tibial slope collapse > 5 degrees, breakage, loosening, or conversion to total knee arthroplasty (TKA). Radiographic images were assessed for preoperative, direct postoperative, and late postoperative (up to 1 year) displacement. Univariable and multivariable logistic regression analyses were employed to study factors associated with LoR..

RESULTS: At a mean follow-up of 325 days (SD ± 161) 77 patients (36.2%) suffered from LoR. The adjusted multivariable logistic regression model revealed that patient age (OR 1.03, $p = 0.028$), involvement of the posterior column (OR 2.87, $p = 0.014$), preoperative step-off (OR 1.08, $p < 0.001$), postoperative step-off (OR 1.61, $p = 0.007$), and postope […]

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### Using an App based algorithm to determine axial rotation and pelvic tilt at the time of an AP radiograph: a sawbone study.

*Boettner F, Premkumar A, Sterneder M, Aboaf C, Haralambiev L*

INTRODUCTION: Axial rotation of the pelvis impacts inclination and anteversion measurements of the acetabular component on plain radiographs. Currently the axial rotation is not taken into account when providing acetabular component measurements on plain radiographs. The current study evaluates a novel App based algorithm to determine pelvic axial rotation at the time of an AP pelvis radiograph.

MATERIALS AND METHODS: AP and lateral radiographs were taken of the sawbone pelvis to calibrate the App. The sawbone was then rotated 11 times to the left (average 5.3°, range 0.2°-10.5°) and 10 times to the right (average 5.2°, range 1.2°-11.3°). The sawbone was then tilted forward 14 times (average 5.6° range 0.1°-11.6°) and 13 times backward (average 5.9°, range 0.1°-11.2°). The amount of axial rotation and tilt of the sawbone pelvis was determined using a digital caliper. AP pelvis radiographs were taken for each position. Using a novel App based calculation algorithm the amount of axial rotation and pelvic tilt was determined from the AP Pelvis radiograph.

RESULTS: The algorithm determined the axial rotation of the pelvis at the time of the AP pelvis radiograph with an average difference of 0.6° (range 0°-1.8°) compared to the digital caliper measurement. For measuring pelvic tilt, the difference was 2.3° (range 0.4°-4.5°) compared to the digital caliper measurement.

CONCLUSIONS […]

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## Risk factors for implant-associated infection after bone allograft reconstruction: a 20-year retrospective cohort study.

*Scheurer F, Smolle M, Hauer G, Ruckenstuhl P, Valentin T, Leithner A*

AIMS: Reconstruction of bone defects is a key challenge in orthopaedic surgery. While autologous bone grafts are considered the gold standard, osseous allografts are increasingly used as an alternative as they allow for treatment of large bone defects. However, they may be associated with an increased risk of postoperative infections. This study aims to analyze the frequency of and risk factors for implant-associated infections after bone allograft transplantation.

METHODS: All patients who underwent reconstruction with bone allografts at a single orthopaedic and trauma unit between 2003 and 2023 were retrospectively included. Data on demographics, disease-associated parameters, risk factors at index surgery, and development of implant-associated infections during follow-up were documented. Potential risk factors for development of implant-associated infections during follow-up were investigated with logistic regression models.

RESULTS: A total of 220 patients were included (46.8% males; median age at surgery 46.5 years [IQR: 22-67]; $n = 23$ with implant-associated infections during follow-up). Factors independently associated with subsequent implant-associated infections were smoking ($p = 0.001$), BMI ($p = 0.015$) and radiotherapy ($p = 0.013$). Gender, age, preoperative chemotherapy, allograft material, preoperative C-reactive protein, hemoglobin and reason for surgery (tumor, n […]

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## Why it's better to believe in a larger definition of the diametaphyseal junction zone in pediatric distal radius fractures.

*von Schrottenberg C, Beck SM, Schwerk P, Fitze G, Schultz J*

INTRODUCTION: Diametaphyseal radius fractures (DMRF) in children are characterized by special biomechanical behavior. They present with a stereotypical loss of reduction (radial translation and ulnar tilt) of the distal fragment when stabilized with elastic stable intramedullary nailing (ESIN), the standard osteosynthesis for pediatric diaphyseal forearm fractures. No consensus has been found on the definition of the diametaphyseal junction zone (DMJZ). Some authors claim that diaphyseal fractures immediately proximal to the metaphysis, as defined by the AO Pediatric Comprehensive Classification of Long Bone Fractures (AO-PCCF), can safely be stabilized with ESIN. They confine the DMJZ to a small part within the metaphysis. Other definitions of the DMJZ extend slightly more proximally, including a small part of the distal diaphysis. The forearm fracture index (FFI) calculates the ratio of the fracture's distance to the radius' growth plate over its width and defines DMRF to have an FFI between 1 and 2. The aim of this case series was to demonstrate that these particular fractures, which can be considered diaphyseal according to the AO-PCCF, still exhibit biomechanical characteristics typical of DMRF.

MATERIALS AND METHODS: Radiologic and clinical outcomes of 11 DMRF, that would be considered diaphyseal by the AO-PCCF classification but fall into the DMJZ when using the FFI, ar […]

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## Optimizing initial biomechanical strength ("Time Zero") of posterior repair in total hip arthroplasty: a biomechanical comparison of suture materials and knot configurations.

*Chen M, Liu X, Chen D, Li X, Du H, Sun Y, Wang H*

INTRODUCTION: Posterior capsule repair (PCR) is commonly performed during total hip arthroplasty (THA) via the posterolateral approach to improve postoperative stability. Because biological healing of the repaired posterior structures requires time, the repair construct must provide sufficient immediate mechanical support after surgery. This study aimed to identify the suture material and knot configuration with favorable initial mechanical performance in a standardized in vitro model.

MATERIALS AND METHODS: This in vitro biomechanical study was conducted in two sequential phases. Phase 1 evaluated the baseline tensile properties of three commonly used suture materials: #2 Ethibond, #0 Vicryl, and 3 - 0 PGLA. Phase 2 used a standardized porcine dermal surrogate model to compare eight suture configurations, including single- and double-strand techniques. The primary biomechanical outcomes included maximum failure load, load at 2 mm gap formation, and construct stiffness during uniaxial quasi-static load-to-failure testing.

RESULTS: In Phase 1, non-absorbable #2 Ethibond showed significantly greater ultimate tensile strength (92.44 ± 8.72 N) and higher stiffness than the absorbable alternatives ($P < 0.001$). In Phase 2, double-strand configurations outperformed all single-strand techniques. Among them, the double-strand Nice knot achieved the highest maximum failure load (104.04 […]

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## Intravenous versus oral tranexamic acid in shoulder arthroplasty.

*Parmar T, Adio A, Daher M, Khan A, Horneff J, Abboud J*

PURPOSE: Tranexamic acid (TXA) is used in shoulder arthroplasty to reduce blood loss. While intravenous (IV) TXA is standard, oral TXA offers advantages in cost and administration. This study compares 90-day complications and postoperative laboratory values between IV and oral TXA in total shoulder arthroplasty (TSA).

METHODS: A retrospective cohort study using TriNetX identified adults undergoing primary TSA (2005-2025) receiving TXA on the day of surgery, stratified by route (IV vs. oral). One-to-one propensity score matching was performed. Ninety-day outcomes included transfusion, thromboembolic events, cardiac and renal complications, neurologic events, infection, wound dehiscence, readmissions, emergency department visits, opioid utilization, and mortality. Postoperative laboratory outcomes included hemoglobin, erythrocyte count, hematocrit, and platelet count. Outcomes were compared using odds ratios (ORs) with 95% confidence intervals (CIs) and t tests where appropriate.

RESULTS: After matching, 5722 patients were included in each cohort. No significant differences were observed between IV and oral TXA in 90-day transfusion rates ($P = 0.873$) or thromboembolic events, including deep vein thrombosis and pulmonary embolism. Rates of myocardial infarction, stroke, cardiac ischemia, acute renal failure, infection, readmission, emergency department visits, opioid use, and mo […]

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## Robotic-arm assistance in revision total hip arthroplasty improves the accuracy of acetabular component positioning: a cadaver-based study.

*Fitzgerald K, Borukhov I, Chandra V, Smitterberg C, Mont MA, LiArno S*

INTRODUCTION: Acetabular component positioning is a key factor for stability, impingement, wear, and biomechanics in revision total hip arthroplasty (THA). However, anatomic distortion and previous implants may complicate positioning. Although evidence on computed-tomography (CT)-based robotic assistance in revision THA is limited, its planning features and precision may improve surgical plan execution. This study aimed to measure the precision and accuracy of acetabular component positioning of revision robotic-arm assisted total hip arthroplasty (RA-THA) compared with a published manual primary total hip arthroplasty (M-THA) benchmark for: (a) acetabular inclination; (b) acetabular version; and (c) center of rotation (COR), with comparison by surgeon case volume.

MATERIALS AND METHODS: There were seven cadavers (nine hips) that previously underwent primary THA revised with RA-THA by independent surgeons with varying primary RA-THA case volumes, surgical approaches, and acetabular component types. Due to surgical protocol deviations, two hips were excluded leaving seven hips from six cadavers. Pre- and postoperative CT scans were compared to calculate absolute error for inclination, version, and COR; normality was established with the Anderson-Darling test and two-variance tests assessed differences in standard deviations ($\alpha = 0.05$). Mean differences between cadaver-based rev […]

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## Correction: Modified rotational wedge distal metatarsal osteotomy versus chevron osteotomy for hallux valgus: long-term radiographic and clinical outcomes.

*Aybar A, Gokkus K, Özden E, Çetin MH, Çetin MÜ*

PubMed · DOI

## Isolated compartment arthrosis of the knee: contemporary concepts, persistent controversies, and future directions in unicompartmental knee arthroplasty.

*Akkaya M, Innocenti M*

PubMed · DOI

## Preoperative lower radiographic osteoarthritis severity in primary total knee arthroplasty increases revision risk by tenfold compared to severe osteoarthritis.

*Stroobant L, Vermue H, Droogsma T, Van Onsem S, Arnout N, van Hellemondt GG, Victor J, Heesterbeek PJC*

BACKGROUND: Dissatisfaction following primary total knee arthroplasty (pTKA) is higher in patients with preoperative low-grade osteoarthritis (OA), but its impact on revision risk remains unclear.

QUESTIONS/PURPOSES: This study aimed to identify predictors for revision surgery within 10 years following pTKA, focusing on radiographic severity of OA using the Kellgren-Lawrence (KL) grading system, along with other demographic and radiographic factors.

PATIENTS AND METHODS: This retrospective case-control study was conducted at two European tertiary referral centers. The case group included 142 patients who underwent aseptic revision total knee arthroplasty (rTKA) between 2007 and 2023, within 10 years following pTKA. A 2:1 control group of 284 patients who had pTKA between 2011 and 2014, with no revision surgery within 10 years, was selected. Collected data included age, sex, body mass index (BMI), side of surgery, and American Society of Anesthesiologists (ASA) classification. Radiographic data were collected, requiring a preoperative radiograph to assess the KL grade. Univariate analyses identified potential predictors for rTKA, and multivariable logistic regression assessed their relationship with revision risk.

RESULTS: Patients requiring rTKA were significantly younger at the time of pTKA. For each additional year of age, the revision risk decreased by 6% (OR 0.94, 95% CI […]

PubMed · DOI

## Correction: Non-tobacco nicotine dependence and postoperative complications following operative fixation of tibial shaft fractures: a retrospective cohort analysis.

*Berk A, Bogart M, Eghtedari C, Good L, Florentino S, Moyal A, Burkhart R, Ochenjele G et al.*

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## Comparison of pediatric proximal femoral locking plates and dynamic hip screws in the fixation of pediatric intertrochanteric fractures, a biomechanical study.

*Wang JH, Shih CA, Wang CH, Lee CC, Wu KW, Wang TM, Kuo KN, Hong CK*

Treatment options for pediatric intertrochanteric fractures range from conservative casting in young children to surgical fixation in older patients, most commonly using dynamic hip screws (DHS) or pediatric proximal femoral locking plates (PF-LCP). As there is limited biomechanical evidence favoring one method over another, this study aimed to compare the biomechanical stability of DHS and PF-LCP for the treatment of pediatric intertrochanteric fractures using a synthetic bone model. Sixteen synthetic composite femurs were allocated into two groups stabilized with either a DHS or a PF-LCP. To simulate Delbet type IV intertrochanteric fractures, a vertical osteotomy was created in each specimen. Biomechanical evaluation was carried out using a universal testing machine, assessing axial stiffness, vertical, horizontal, and total displacement, as well as ultimate load to failure under standardized loading conditions. Comparative statistical analyses were then performed to determine differences in biomechanical performance between the two fixation methods. Group P demonstrated significantly smaller vertical and total displacements (0.15 ± 0.15 mm and 0.41 ± 0.31 mm) compared with the DHS group (0.59 ± 0.42 mm and 0.75 ± 0.33 mm) ($p = 0.018$ and $p = 0.045$), whereas horizontal displacement showed no difference ($p = 0.142$). Axial stiffness and ultimate failure load were comparable bet […]

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### Impact of severe varus deformity on the outcome of mobile-bearing unicompartmental knee arthroplasty: a comparison of patients with varus deformity of less than and more than 15 degrees using matched-pair analysis.

*Mullaji AB, Gupta AK, George GK, Kazi R*

INTRODUCTION: The outcome of patients undergoing UKA with larger preoperative deformity (> 15º varus) and lesser deformity is sparsely compared in the literature. This matched-pair retrospective study attempts to fill this research gap by comparing the effect of preoperative and postoperative residual varus on outcome.

METHODS: Patients undergoing UKA with preoperative varus deformity > 15º (HKA > 15º group) were selected and matched for age and gender with those with < 15º of preoperative varus deformity (HKA < 15º group). There were 72 age and gender-matched pairs knees. HKA was measured and Forgotten Joint Score, OKS, VAS and WOMAC scores were obtained at a minimum follow up of 5 years. Postoperative residual varus was classified into 3 groups based on postoperative HKA: varus < 3º, varus 3 to 7º and varus > 7º.

RESULTS: Mean correction of deformity was significantly more ($p < 0.001$) in the HKA > 15º group (10.5º) when compared to HKA < 15º group (4.5º). There was no significant difference in outcome measures OKS ($p = 0.21$), FJS ($p = 0.35$) and VAS ($p = 0.6$) between patients in the two groups. There was no significant correlation between the preoperative and postoperative HKA of the patient with postoperative OKS, FJS and VAS in the two groups. There was no statistically significant difference between FJS ($p = 0.73$) of the three postoperative varus groups. Excellent to good […]

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### A multimodal imaging approach to predict carpal tunnel syndrome after distal radius fractures.

*Önüt M, Alkol N, Önalo■lu Y, Özer M, Apayd■n E, Talmaç MA*

OBJECTIVE: Distal radius fractures are commonly associated with the development of carpal tunnel syndrome (CTS), yet current evidence primarily focuses on radiographic alignment parameters. The role of carpal tunnel morphology, particularly total carpal tunnel cross-sectional area (CSA) measured on computed tomography (CT), remains insufficiently investigated. This study aimed to evaluate the relationship between CSA, volar tilt, and the development of CTS in conservatively managed distal radius fractures, and to assess the predictive performance of these parameters when used in combination.

METHODS: This retrospective case-control study included 150 patients, comprising 50 patients who developed CTS and 100 controls without CTS. CSA was measured on CT scans obtained at initial presentation, while volar tilt was assessed using standard radiographs. Additional variables, including the presence of ulna fracture and fracture type, were analyzed. Group comparisons were performed using nonparametric tests. Multivariable logistic regression analysis was conducted to identify independent predictors of CTS. Diagnostic performance was evaluated using receiver operating characteristic (ROC) curve analysis.

RESULTS: Patients who developed CTS had significantly lower CSA and higher volar tilt values (both $p < 0.001$). The presence of an ulna fracture was inversely associated with CTS (p = […]

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### Unicompartmental knee arthroplasty and total knee arthroplasty in patients with 15 or more degrees of varus deformity: age- and gender-matched comparative study of radiographic and clinical outcomes.

*Mullaji A, Chopperla S, George GK, Kazi R*

INTRODUCTION: To date, few studies have directly compared matched cohorts of patients with severe varus deformity undergoing UKA versus TKA. The current study seeks to address this gap. The hypothesis was that, in patients with deformity of > 15°, postoperative PROMs following UKA would be inferior to those achieved with TKA.

METHODS: Records of patients operated for UKA and TKA from January 2016 to December 2021 were evaluated, and the data for patients with deformity of ≥ 15° were selected. The UKA patients were then matched for age and gender with TKA patients. 52 matched pairs of patients were obtained. The HKA, mLDFA, MPTA were measured. OKS, SF12, and FJS were collected. The magnitude of change in the mLDFA and MPTA was assessed using the Estimated Marginal Means.

RESULTS: The mean follow-up times for UKA and TKA were 3.7 years and 5.5 years, respectively. The mean deformity correction (ΔHKA) was significantly greater in TKA (16.3° ± 4.5°) than UKA (10.6° ± 3.1°) ($p < 0.001$), with significant changes in mLDFA and MPTA ($p < 0.001$). In UKAs, 68% of patients demonstrated a postoperative HKA between 171.2° and 175.7°, whereas in TKAs, values clustered between 174.5° and 179.8°. Mann-Whitney U test demonstrated no statistically significant differences between the UKA and TKA cohorts across all assessed PROMs, including OKS, SF-12, and FJS ($p > 0.05$ in all). 98% of the UKA an […]

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### Acetabular superior wall fractures: a distinct acetabular fracture entity.

*Gänsslen A, Krappinger D, Liodakis E, Clausen JD, Omar-Pacha T, Lindtner RA, Sehmisch S, Osche DB*

INTRODUCTION: A transitional "grey-zone" exists between typical locations of classical trapezoidal anterior wall and large-fragment posterior wall acetabular fractures. Part of this grey zone is the superior acetabular dome, which is commonly involved in acetabular fracture patterns, especially in geriatric patients. The aim of this study was to characterize isolated acetabular superior wall fractures and to assess their relevance as a distinct fracture pattern.

METHODS: A retrospective analysis of pelvic fracture databases from three tertiary trauma centers was performed. Acetabular fractures with superior dome involvement were screened, and cases with isolated superior articular involvement were included. Overall, seven patients were identified. Clinical and radiological data were analyzed regarding fracture morphology, associated fracture characteristics (including dislocation, comminution, marginal impaction, intra-articular fragments, and femoral head injury), concomitant pelvic ring injuries, treatment, and clinical outcomes.

RESULTS: Isolated superior wall acetabular fractures occurred following both high-energy and low-energy mechanisms, including simple falls. The mean age was 48 years and four of seven patients were male. Hip dislocation or subluxation was observed in four cases (anterior-superior or purely superior). Surgical treatment was performed in six cases, p […]

PubMed · DOI

### A novel biomechanical model for reproducing and analysing the causal mechanisms of PFNA "cut-in" phenomenon.

*Rasappan K, Joshua K, Chou SM, Shaw LKRM, Huang D, Yew A, Kwek EBK*

INTRODUCTION: "Cut-in" is a rare but serious complication associated with cephalomedullary nail fixation for intertrochanteric hip fractures, particularly with Proximal Femoral Nail Antirotation (PFNA). This phenomenon involves paradoxical superomedial migration of the helical blade through the femoral head. Although increasingly recognised, the underlying mechanisms remain incompletely defined. This study aims to describe a new superolateral tensile loading set up in a bidirectional loading model, with the aim of incorporating possible mechanical deviations introduced by abductor weakness.

MATERIALS AND METHODS: This study was conducted in two stages. An initial stage with synthetic femurs (SYNBONE® 2420) was used to calibrate the set up. Within this stage, intramedullary canal diameter was compared (12 mm vs. 18 mm). In the final stage, 5 osteoporotic synthetic femurs (Sawbones® 3503) were used. In both stages, standardised AO/OTA 31-A1.1 intertrochanteric fracture were created in the synthetic femurs and fixed with PFNA-II implants. All femurs were subjected to cyclical bidirectional loading with a universal testing machine. Each cycle consisted of axial compression (720 N) and superolateral tensile loading (120 N) at 2 Hz. This configuration was designed to simulate altered hip biomechanics postulated with the trendelenburg gait. Testing was continued until mechanical fail […]

PubMed · DOI

## Routine intraoperative infection testing in presumed aseptic hip and knee revision: a matched cohort retrospective study.

*Loppini M, Rocchi C, Di Maio M, Chiappetta K, Bulgarelli A, Grappiolo G*

INTRODUCTION: Periprosthetic joint infections (PJIs) complicate 1-2% of primary and up to 4% of revision arthroplasties, and their diagnosis remains challenging due to the lack of a diagnostic gold standard. Differentiating PJI from aseptic failure is particularly difficult in low-grade or biofilm-related infections. Routine intraoperative microbiological screening is not recommended in presumed aseptic revisions. This study aimed to evaluate the association between routine intraoperative microbiological screening and implant survival in presumed aseptic revision arthroplasty. A secondary objective was to identify preoperative predictors of unexpected PJIs.

METHODS: Retrospectively collected data of patients undergoing presumed aseptic THA and TKA revisions between 2013 and 2019 were included. Two matched cohorts were compared: a screened group (2016-2019) undergoing routine intraoperative microbiological sampling, and an unscreened one (2013-2015) without systematic screening. Kaplan-Meier survival analysis with log-rank testing assessed overall and infection-free survival. Univariate conditional logistic regression and receiver operating characteristic (ROC) analyses evaluated potential associations between preoperative serum markers and unexpected PJI.

RESULTS: A total of 559 patients were included, of whom 295 underwent screening and 264 did not. Unexpected infections occ […]

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## Artificial intelligence advancements for orthopaedic clinical reasoning: longitudinal assessment of newer models (ChatGPT-5, Grok-3, Gemini 2.5 Flash) compared to clinicians.

*Agharia S, Soroush S, Ameen D, Zhou Y*

INTRODUCTION: This descriptive study aimed to longitudinally evaluate the performance of contemporary large language models - ChatGPT-5, Gemini 2.5 Flash, and Grok-3 - on orthopaedic clinical multiple-choice tasks, benchmarked against pooled clinician consensus. A secondary aim was to assess whether recent advances in generative AI translated into improved alignment with clinician consensus compared with previous AI models.

MATERIALS AND METHODS: A total of 97 multiple-choice clinical cases spanning eight orthopaedic subspecialties were sourced from OrthoBullets and previously benchmarked against aggregated responses from thousands of practising clinicians. Using identical methodology to our 2023 study of ChatGPT-3.5, ChatGPT-4, and Bard, each model was prompted with standardised case stems and response options. The primary outcome was the proportion of AI responses matching the most popular clinician response; secondary analyses assessed agreement within 10% and 20% of clinician consensus, performance on 'controversial' (< 25% margin) questions, and inter-model concordance using Cohen's kappa coefficients.

RESULTS: Gemini 2.5 Flash achieved the highest alignment with clinician consensus (69.1%), followed by Grok-3 (66.0%) and ChatGPT-5 (58.8%). None of the LLMs refused to respond to any prompts, representing a reduction from 7.2% from our 2023 study. Subspecialty analysis de […]

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## Trajectory of urinary incontinence symptoms following total hip arthroplasty for hip osteoarthritis.

*Kadowaki M, Fukutani S, Ushio K, Uchio Y*

INTRODUCTION: Urinary incontinence (UI), a prevalent quality-of-life-limiting condition, frequently coexists with hip osteoarthritis, potentially through pelvic floor dysfunction related to obturator internus muscle involvement. Several reports suggest improvement after total hip arthroplasty (THA), although postoperative time-course changes remain unclear. This study aimed to clarify the temporal trends in UI after THA and examine association of postoperative UI trajectories with functional recovery.

MATERIALS AND METHODS: This retrospective study included 118 female patients who underwent THA for hip osteoarthritis between October 2021 and March 2024. UI, assessed using the International Consultation on Incontinence Questionnaire-Short Form (ICIQ-SF), was defined as ICIQ-SF ≥ 1 preoperatively and at 3, 6, 9, and 12 months postoperatively, and classified as stress (SUI), urge (UUI), or mixed (MUI). Hip-related patient-reported outcomes were evaluated using the Japanese Orthopaedic Association Hip Disease Evaluation Questionnaire (JHEQ). Associations with BMI, age, surgical approach (direct lateral/Bauer vs. modified Watson-Jones/OCM), and UI type were tested using Mann-Whitney U, chi-squared, and repeated-measures ANOVA tests.

RESULTS: Of 118 patients (mean age at surgery, 67 years; mean BMI, 24.3 kg/m2) preoperatively, 73 (61.9%) reported UI (mean ICIQ-SF 7.88), including S […]

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### The grand piano ratio: an adjunctive intraoperative screening tool for detecting excessive femoral component external rotation in total knee arthroplasty.

*Avci Ö, Öztürk A, Güler BO, Bayrak HÇ, Kaya AÖ, Do■an Y*

BACKGROUND: Accurate femoral component rotation is essential for optimal outcomes in total knee arthroplasty (TKA), yet reliable intraoperative assessment remains challenging. The "grand piano sign" has been described as a qualitative visual cue, but its quantitative clinical value has not been clearly established. This study aimed to evaluate the diagnostic performance of the grand piano ratio as an intraoperative tool for detecting excessive femoral component external rotation using advanced postoperative imaging as a reference standard.

METHODS: A retrospective analysis was conducted on 170 knees undergoing primary TKA. The intraoperative grand piano ratio was measured after completion of all femoral resections and before trial component implantation using the anterior femoral resection surface. Postoperative femoral component rotation was assessed using MAVRIC-sequence magnetic resonance imaging referenced to the surgical transepicondylar axis (sTEA). Excessive femoral component external rotation was defined as postoperative external rotation greater than 3° relative to the sTEA reference. Receiver operating characteristic (ROC) analysis was performed to evaluate diagnostic performance of the grand piano ratio and determine the optimal cutoff value, while multivariable logistic regression analysis was used to identify independent predictors of excessive external rotation. […]

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### The shoulder stiffness scale: a validated tool for monitoring stiffness in arthroscopic rotator cuff reconstructions.

*Bouaicha S, Biner M, Selman F, Wieser K, Müller AM, Audigé L*

BACKGROUND: Postoperative shoulder stiffness after arthroscopic rotator cuff repair is common and clinically heterogeneous. Established shoulder scores assess pain and function, but do not specifically isolate stiffness during follow-up. The aim of this study was to validate the newly developed Shoulder Stiffness Scale (SSS) as a pragmatic monitoring tool for postoperative shoulder stiffness after arthroscopic rotator cuff repair (ARCR).

METHODS: From a cohort of 973 primary arthroscopic rotator cuff reconstructions, patients without concomitant joint pathology of the same or contralateral arm were included. The new SSS was based on three items related to shoulder pain, subjective and objective ROM ranging from 0 to 10 points. It was collected before surgery, as well as 6 and 12 months postoperatively, along with other pain level parameters, Constant Score (CS), Subjective Shoulder Value (SSV), Oxford Shoulder Score (OSS). Changes of the SSS and its subscales over time, Cronbach's alpha for internal consistency, bottom and ceiling effects were reported. Validity was investigated through SSS associations with other parameters of similar dimension, and functional scores.

RESULTS: The selected cohort with 740 patients showed postoperative shoulder stiffness in 12.7 %. The SSS averaged 5.1 points (SD 2.5) preoperatively, and decreased significantly to mean 2.9 (SD 2.4) and 1.6 (S […]

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### Isolated screw fixation of posterior wall fractures.

*Clausen JD, Özbek H, Omar-Pacha T, Sehmisch S, Gänsslen A*

INTRODUCTION: Posterior wall (PW) fractures are common acetabular fractures and are most often treated by open reduction and internal fixation. The gold-standard of surgical stabilization is screw fixation of large wall fragments with additional buttress plating. Some reports focussed on screw fixation alone presenting adequate results.

MATERIAL AND METHODS: From a total of 208 PW-fracture treated between 1972 and 2008, 57 patients were identified with open reduction and internal fixation (ORIF) using isolated screw fixation. These patients were analyzed regarding demographical data (patient age, sex), type of accident, injury mechanism, Injury Severity Score (ISS), concomitant pelvic ring injuries, associated injuries, perioperative parameters and long-term results.

RESULTS: 44 patients were male and 13 were female with a mean age of 36.8 years. More than 90% sustained high-energy trauma with 34 sustaining multiple injuries or polytrauma. The mean Injury Severity Score was 11.5 points. All patients had fracture dislocations, which were reduced within 24 h after admission except one patient with secondary transfer. 11 patients presented with an initial sciatic nerve deficit. 29 patients had a single PW-fragment, 16 presented with two PW-fragments and in 12 patients marginal impactions were present. Surgery was performed in average after 7 days using the Kocher-Langenbeck appr […]

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## Research progress on debridement, antibiotics, and implant retention (DAIR) for the treatment of periprosthetic joint infection after artificial joint replacement.

*Feng W, Zhang Z, Pu R, Wang H, Liu R, Sun Y, Zhang G*

Periprosthetic joint infection (PJI) is one of the most serious complications following artificial joint replacement, posing significant treatment challenges and a heavy clinical burden. Debridement, antibiotics, and implant retention (DAIR) has become the preferred treatment strategy for acute PJI due to its advantages of minimal invasiveness, functional preservation, and cost-effectiveness. This article reviews the latest advances in optimizing the indications, timing of intervention, key technical details, and multimodal anti-infection strategies for DAIR. It focuses on discussing patient selection models (e.g., KLIC score), the impact of modular component exchange on biofilm eradication, the synergistic role of chemical debridement, and the central position of biofilm-active therapy in penetrating biofilms. Furthermore, this article proposes that future efforts should integrate molecular diagnostics, intelligent antimicrobial materials, and artificial intelligence prediction models to achieve individualized precision therapy for PJI, providing a theoretical basis for clinical decision-making.

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## Tranexamic acid in high-risk shoulder arthroplasty patients: safety across thromboembolic, cardiac, renal, and neurologic risk profiles.

*Parmar T, Adio A, Handy A, Daher M, Kellish A, Khan A, Abboud J*

INTRODUCTION: Tranexamic acid (TXA) effectively reduces blood loss and transfusion requirements in shoulder arthroplasty. However, concerns regarding thromboembolic, neurologic, cardiac, and renal complications have limited its use in medically high-risk patients. This study evaluated temporal trends and the safety of TXA in high-risk patients undergoing total shoulder arthroplasty (TSA).

METHODS: A retrospective cohort study was conducted using TriNetX, identifying patients who underwent shoulder arthroplasty (2012-2025). Patients were stratified by preexisting high-risk conditions, including prior thromboembolism, renal failure, atrial fibrillation, seizure disorders, and visual disturbances. Within each subgroup, patients receiving TXA were propensity score-matched to those who did not. Perioperative TXA utilization trends were assessed. Ninety-day postoperative outcomes were compared, including transfusion requirements, thromboembolic events, cardiac complications, renal failure, neurologic events, infections, readmissions, emergency department visits, and mortality. Outcomes were reported as odds ratios (ORs) with 95% confidence intervals (CIs).

RESULTS: Reported perioperative TXA use began in 2012 and increased through 2025, with over 70% of patients in both standard- and high-risk cohorts receiving TXA. TXA use was associated with significantly lower transfusion rates […]

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## Osteoporosis treatment gap prior to femoral fracture and prevalence of pharmacological risk factors: a prospective observational study.

*Böck L, Kerschan-Schindl K, Frossard M, Hajdu S, Crevenna R, Anditsch M, Stemer G, Antoni A*

INTRODUCTION: Hip fractures are a major complication of osteoporosis and are associated with high morbidity, mortality, and costs. Despite clear guideline recommendations, pharmacological osteoporosis treatment remains underutilized. In addition, medications that increase fall and fracture risk further contribute to fracture occurrence. The aim of this study was to assess guideline concordance of osteoporosis therapy and to quantify the prevalence of medications associated with increased fall and fracture risk prior to the fracture in patients admitted with femoral fractures.

MATERIALS AND METHODS: In this prospective observational study, 145 patients aged ≥ 55 years admitted with femoral neck, per- and subtrochanteric, femoral shaft and distal femur fractures to a tertiary academic trauma center between April and June 2025 were included. Pre-admission medication was assessed using structured medication reconciliation. Guideline concordance of pre-fracture osteoporosis therapy was evaluated according to the current Austrian osteoporosis guideline (Dimai et al., 2024). Fall-risk-increasing drugs (FRIDs) were identified using the STOPPFall criteria, and fracture-associated medications were recorded based on Austrian osteoporosis guideline-defined drug classes.

RESULTS: Among 145 included patients (mean age 80.4 years; 71% female), 117 (81%) met criteria for pharmacological oste […]

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## Gait speed recovery after iliofemoral ligament-preserving hip arthroscopy for femoroacetabular impingement: associations with early postoperative gait kinematics.

*Machida S, Tsutsumi M, Utsunomiya H, Kudo S*

INTRODUCTION: Whether iliofemoral ligament-preserving hip arthroscopy facilitates recovery of gait speed-an indicator of functional recovery following hip arthroscopy in patients with femoroacetabular impingement-remains unclear. We investigated whether gait speed improved after iliofemoral ligament-preserving hip arthroscopy and examined whether pelvic jerk, defined as the time derivative of acceleration and assessed using inertial sensors at 1 month postoperatively (1M), was associated with gait speed at 6 months (6M).

MATERIALS AND METHODS: Eighteen patients who underwent primary hip arthroscopy for femoroacetabular impingement were included. Gait speed and pelvic jerk were assessed during a 10-m walk at a self-selected speed, using an inertial sensor positioned at the third lumbar level. Peak pelvic jerks during the 1st and 2nd halves of the stance phase (1st - and 2nd -peak pelvic jerks) were calculated. Measurements were obtained preoperatively and at 1M and 6M. Gait parameters were compared across time points using a linear mixed model. Bayesian regression was used to compare the observed change in gait speed from preoperatively to 6M with previously reported values. Associations between gait variables at 1M and gait speed at 6M were also examined.

RESULTS: Gait speed was significantly higher at 6M than that pre surgery (preoperative, 1.15 [95% confidence interval, 1.0 […]

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## Computed tomography findings in 11,504 adult patients with traumatic brain injury: a large real-world cohort study with a S100B subgroup analysis.

*Clar C, Puchwein P, Moshammer M, Sadoghi P, Kramer D, Leithner A, Reinbacher P*

OBJECTIVE: The aim of this study was to characterize acute cranial CT findings in a large cohort of adult patients with traumatic brain injury and to explore the diagnostic performance of S100B in a selected subgroup. The hypothesis was that most cranial CT scans in adult patients with traumatic brain injury would not demonstrate acute traumatic abnormalities, and that, within a selected subgroup, negative S100B levels would be associated with the absence of such findings on CT.

METHODS: This retrospective cohort study included 11,504 adult patients presenting with traumatic brain injury to a level I trauma center between 04/2016 and 05/2024. All patients underwent cranial CT, while serum S100B measurements were available in a selected subset. CT findings were classified as normal or showing acute traumatic abnormalities. In the S100B subgroup, biomarker levels were analyzed in relation to CT findings to assess their diagnostic performance for excluding acute pathology.

RESULTS: A total of 11,504 adult patients with traumatic brain injury were included; 27.4% were inpatients and 72.6% outpatients. Overall, 81.9% of CT scans showed no acute pathology, while 10.7% demonstrated abnormalities. 7.4% of all cases were excluded. Acute TBI-related interventions were rare (3.7% of inpatients vs. 0.3% of outpatients). In patients with dementia and those receiving anticoagulation, CT sc […]

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## Early post-operative CRP is a better predictor of DAIR failure than pre-operative CRP in total knee PJI.

*Beadel H, Chaffey R, Kim K, Zhu M, Coleman B*

INTRODUCTION: Debridement, antibiotics and implant retention (DAIR) surgery is often performed as a first-line treatment for periprosthetic joint infection (PJI), however, success rates are variable. This study aimed to determine whether post-operative C-reactive protein (CRP) levels predict DAIR failure.

MATERIALS AND METHODS: A multi-centre retrospective cohort study was performed over a 15-year period. All total knee arthroplasty (TKA) patients undergoing DAIR for first episode PJI were included. CRP was measured at admission and for 6 weeks post-operatively. Treatment success was defined as implant retention without the need for revision surgery or long-term suppressive antibiotics. Trends in CRP were compared between two groups: failed and successful DAIR. Receiver operating characteristic (ROC) curves were analysed.

RESULTS: 189 DAIR procedures were included, of which 49% (92) failed and 51% (97) were successful. Mean follow-up was 7.5 years. Overall, CRP trended down following surgery. Mean CRP was significantly higher in the failed DAIR group at all time points. The greatest difference was at week one post-operatively (mean 76 vs. 101, $P < 0.01$). This remained significant when patients experiencing DAIR failure within 90 days were excluded (mean 76 vs. 107, $P < 0.01$). CRP was most accurate as a predictor of failure at week one post-operatively, with an area under the […]

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## Tuberosity refixation in reverse proximal humerus fracture arthroplasty: a prospective cohort study with two-year follow-up.

*Oberrauter CH, Hess F, Welter J, Pape HC, Mayer S, Dullenkopf A, Mazzucchelli R, Marintschev I*

INTRODUCTION: To evaluate the outcomes of a simplified tuberosity refixation technique using a two-suture construct in patients with complex proximal humerus fractures treated with reverse shoulder arthroplasty (RSA). The technique reduces the number of fixation sutures rather than the traditional four to five, aiming to decrease operative complexity while promoting greater tuberosity (GT) healing and improving the healing rate. A secondary exploratory analysis compared clinical and radiological outcomes between cemented and uncemented humeral stem fixation.

MATERIALS AND METHODS: This prospective study included 73 patients who underwent RSA (Global Unite Fracture Reverse System, DePuy Synthes, Warsaw, Indiana, USA). Clinical and radiological outcomes were assessed at two-year follow-up. Bone quality was evaluated using the Deltoid Tuberosity Index (DTI). Functional outcomes and GT healing were analyzed, and stem fixation type was assessed in a secondary subgroup analysis.

RESULTS: The mean patient age was 79 ± 8 years (range 61-95), and 65 (89%) were female. Osteoporotic bone quality (DTI < 1.4) was observed in 51 (70%) of cases. At two years, the median outcome scores were: Subjective Shoulder Value, 90 (IQR 80-95); Constant-Murley score, 76 (IQR 73-81); active forward flexion, 140° (IQR 120-160); abduction, 140° (IQR 120-160); external rotation, 8 (IQR 8-8); and internal r […]

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## Is it time for a systematic and objective assessment of knee laxity? The impact of objective knee laxity after TKA on patient-reported outcomes: a comprehensive review.

*Vermue H, Klincke V, Stroobant L, Tampere T, Victor J, Arnout N*

PURPOSE: Over the past decade, robot assistance has emerged as an innovative and rapidly advancing tool in total knee arthroplasty (TKA). Despite technological advancements, the evaluation of the soft-tissue envelope during TKA often relies on subjective and rudimentary assessments by the surgeon. This literature review explores the relationship between objective intraoperative and postoperative laxity measurements in the coronal and sagittal planes and patient-reported outcome measures (PROMs) following TKA.

METHODS: A literature review was conducted using PubMed/MEDLINE and Embase to identify studies on intra- or postoperative laxity measurements in TKA and their correlation with PROMs. Inclusion criteria required studies on TKA patients with quantifiable coronal and/or sagittal laxity measurements and postoperative PROMs linked to these measurements.

RESULTS: Thirty-four studies met inclusion criteria: ten on sagittal laxity, twenty-seven on coronal laxity, and three on both. Follow-up ranged from 1 month to over 8 years. Sagittal laxity thresholds of 5-10 mm under applied loads of 89-133 N were associated with better functional outcomes, whereas excessive (> 10 mm) or overly tight (< 5 mm) laxity correlated with reduced satisfaction. Medial laxity in the coronal plane was linked to lower PROMs, while the impact of lateral laxity on PROMs was less conclusive.

CONCLUSIONS: […]

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## Silver-impregnated occlusive dressing is associated with a lower rate of acute surgical site infection after direct anterior total hip arthroplasty.

*Tsai YY, Shen PH, Wu CC, Pan RY, Lin LC, Wang SH*

INTRODUCTION: The direct anterior approach (DAA) for total hip arthroplasty (THA) has gained increasing adoption in recent years; however, it has been associated with a higher risk of wound-related complications compared with other surgical approaches. Silver-impregnated occlusive dressings have demonstrated antimicrobial potential, though their role in DAA THA remains underexplored.

MATERIALS AND METHODS: We conducted a retrospective cohort study with a historical control group, reviewing 321 consecutive primary DAA THAs performed by a single surgeon (September 2020-December 2025). Standard sterile gauze was used before August 2021 (n = 76) and a silver-impregnated occlusive dressing (AQUACEL® Ag SURGICAL; ConvaTec, Deeside, UK) thereafter (n = 245). Despite the chronological difference between cohorts, baseline demographic and clinical characteristics did not differ significantly between groups (Table 1). The primary endpoint was acute surgical site infection(SSI) within 30 days (CDC criteria); all patients had ≥ 3 months follow-up to capture early wound-related reoperations as secondary outcomes.

RESULTS: Four acute SSI events occurred (1.25%). SSI incidence was 3.94% (3/76) with gauze versus 0.41% (1/245) with the silver-impregnated dressing (Fisher's exact p = 0.043). This corresponds to an absolute risk reduction (ARR) of 3.54% (95% CI, 0.34-10.57) and a number needed t […]

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## Minimally invasive stabilisation of anterior pelvic ring injuries through anterior infix versus percutaneous anterior retrograde pubic screw: a prospective comparative cohort study.

*Goda El-Hamalawy A, Abdelmoneim M, Al-Sayed M, Youness M, Sadek F, Kassem E*

BACKGROUND: Management of unstable anterior pelvic ring fractures has shifted toward minimally invasive fixation techniques. This prospective randomized cohort study compares anterior subcutaneous internal fixator with percutaneous retrograde pubic ramus screw fixation, evaluating radiological outcomes, functional scores, complications, and peri-operative parameters.

METHODS: Fifty adult patients with unstable anterior pelvic ring injuries (Tile B & C; Young-Burgess lateral compression and vertical shear mechanisms) were enrolled between 2023 and 2025 and randomized into two equal groups: group 1 (internal fixator) and group 2 (retrograde pubic screw). Both groups underwent individualized posterior ring fixation as required. Outcomes assessed included operative time, blood loss, fluoroscopy time, union rate, Matta radiological score at 6 months, Majeed and Pelvic Outcome Scores at 12 months, and post-operative complications.

RESULTS: Management of anterior pelvic injury showed that retrograde screw fixation demonstrated significantly shorter operative time (32.0 ± 9.2 vs. 50.3 ± 8.4 min, $p < 0.001$) and less intraoperative blood loss (68.3 ± 16.2 vs. 122.3 ± 31.8 ml, $p < 0.001$). Radiological outcomes were comparable: excellent Matta score in 68% vs. 64% ($p = 0.834$). Union time showed no significant difference (14.1 ± 2.4 vs. 15.1 ± 1.5 weeks, $p = 0.076$). Functional outcomes at […]

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### Prioritisation of assessments, diagnostic classifications, and outcome measures in Perthes disease: a Delphi survey of international health professionals.

*Ball S, Davies LM, Gray K, Pacey V*

INTRODUCTION: The aim of this study is to reach consensus and prioritise the most clinically important assessments, diagnostic classifications, and outcome measures by an international panel of experts for Legg-Calve-Perthes disease.

MATERIALS AND METHODS: A three-round modified e-Delphi survey was conducted between April and May 2025. Participants rated the clinical importance of 14 clinical assessments, 26 radiological assessments, eight diagnostic classifications, and 26 outcome measures identified by a scoping review. Consensus was defined as ≥ 75% agreement. In the final round, items in each category were prioritised for overall condition and by specific age groups (< 6 years, 6-8 years, > 8 years).

RESULTS: Thirty-eight health professionals, from nine countries participated in round 1. Thirty-four completed all three rounds, a retention rate of 89.5%. Consensus and prioritisation for clinical use were achieved for 10 clinical assessments, three radiological assessments, three diagnostic classifications, and three outcome measures.

CONCLUSION: Paediatric specialists have reached consensus and prioritised methods to evaluate and diagnose Legg-Calve-Perthes disease in clinical practice. These results represent an important step towards improving continuity of care and the foundation for future development of clinical practice guidelines.

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### Compensatory activation of the deltoid and remaining rotator cuff muscles in patients with rotator cuff tears using real-time tissue elastography.

*Hoshikawa K, Oishi R, Yuri T, Uno T, Nagai J, Giambini H, Mura N*

INTRODUCTION: Rotator cuff (RC) tears are a prevalent condition leading to dysfunction and shoulder pain. Spatial changes in the behavior of the deltoid and remaining RC muscles during arm elevation in patients with RC tears have not been investigated. The purpose of this study was twofold: (1) to investigate factors related to the changes in the deltoid muscle behavior during scapular plane abduction (scaption) in patients with RC tears, and (2) to examine the functional relationship between the deltoid and remaining RC muscles, particularly remaining infraspinatus (ISP) and teres minor (TMin) muscles, in these patients using ultrasound elastography.

METHODS: Twenty-four patients (mean age, 66 ± 6 years) with RC tears (mean antero-posterior diameter, 17.4 ± 8.3 mm) were evaluated. Antero-posterior and medio-lateral diameters of the posterosuperior RC and Subscapularis (SSC) tears were measured intra-operatively. Muscle elasticity was evaluated at passive and active states, as a proxy for muscle activation, at 30°, 60°, and 90° during scaption for the deltoid, ISP, and TMin muscles using real-time tissue elastography. Based on the deltoid muscle activity ratio at 30° and 90° scaption, patients were divided into normal and abnormal activation groups.

RESULTS: Activity of the deltoid muscle in the abnormal activation group exhibited consistently high activity across various ele […]

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### Pin orthosis extension block pinning versus conservative treatment for doyle type 4B mallet fractures.

*Col KA, Demirsu O, Ayd■n M, Surucu S, Yilmaz M, Atlihan D*

INTRODUCTION: Although numerous treatment options have been reported for mallet fractures, a universally accepted gold-standard approach has not yet been established. The purpose of this study was to compare the clinical and radiographic outcomes of the pin-orthosis extension-block pinning technique (PO-EBPT) with those of conventional conservative treatment in patients with Doyle type 4B mallet fractures.

MATERIALS AND METHODS: This study included 62 patients with Doyle type 4B mallet fractures involving 20-50% of the distal interphalangeal (DIP) joint articular surface, treated between March 2022 and April 2024. Patients were randomized into two groups: Group 1 (n = 33) underwent PO-EBPT, whereas Group 2 (n = 29) received conservative treatment with splint immobilization. Outcome measures included DIP joint extension lag, range of motion, fracture union, complication rates and functional outcomes according to the Crawford criteria.

Follow-up evaluations were performed at 2, 4 and 6 weeks and at 3, 6 and 12 months.

RESULTS: A total of 62 patients were analyzed (33 PO-EBPT; 29 conservative). No statistically significant differences were observed between the groups with respect to sex, affected side, injured finger, or complication rates ($p = 0.461$, $p = 0.658$, $p = 0.763$ and $p = 0.165$, respectively). However, the PO-EBPT group demonstrated significantly improved DIP joint exten […]

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## Impact of cannabis use and tobacco smoking on outcomes of open carpal tunnel release surgery: a nationwide study in the United States.

*Hoveidaei AH, Esmaeili S, Gilmor R, Chen Z, Conway JD, Ingari JV*

PURPOSE: To evaluate the impact of cannabis use, tobacco smoking, and their concurrent use on postoperative complications following open carpal tunnel release (CTR) using a nationwide database.

METHODS: A retrospective cohort study was conducted using the PearlDiver database (2010-2022) to identify adult patients who underwent open CTR. Patients were categorized into four groups: cannabis-only users (n = 800), tobacco-only users (n = 167,803), concurrent users (n = 3529), and non-users (n = 167,803 matched controls). Postoperative complications, surgical site infection, wound disruption, joint stiffness, and complex regional pain syndrome, were assessed at 6 weeks and 3 months postoperatively.

RESULTS: At 6 weeks and 3 months, surgical site infection was more common among concurrent cannabis and tobacco users, cannabis-only users, and tobacco-only users. Wound disruption occurred more frequently in concurrent users and tobacco-only users, but not in cannabis-only users. Complex regional pain syndrome was associated only with tobacco use. No association was found between any substance use and joint stiffness.

CONCLUSION: Cannabis use, tobacco smoking, and especially concurrent use increase the risk of surgical site infection and wound disruption following open CTR. Tobacco use is independently associated with complex regional pain syndrome, while cannabis may have a protectiv […]

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## Timing and microbiological profile influence long-term outcomes after debridement, antibiotics, and implant retention (DAIR) in acute hip periprosthetic joint infection.

*Muñoz-Mahamud E, Perdomo-Lizarraga JC, Combalia A, Alías A, Serra A, Fernández-Valencia JÁ, Verdejo MÁ, Soriano Á*

INTRODUCTION: The strategy of debridement, antibiotics, and implant retention (DAIR) represents one of the main therapeutic modalities for acute periprosthetic joint infection (PJI). However, reported outcomes remain highly variable, with success rates ranging from 16 to 92%. This study aimed to evaluate long-term treatment failure-free survival and identify risk factors for treatment failure in patients with acute hip PJI treated with DAIR using time-to-event analytical methods.

MATERIAL AND METHODS: This retrospective study evaluated the treatment failure rate in 115 patients treated with the DAIR strategy for acute PJI following primary or aseptic revision hip arthroplasty between 1999 and 2018. Potential predictors of treatment failure-free survival, including patient characteristics, infection microbiology, and surgical timing, were analyzed using univariate tests, Cox proportional hazards regression (hazard ratio [HR]), and Kaplan-Meier survival estimates.

RESULTS: Among the 115 patients included, 56 were women and 59 were men; 48 were aged 75 years or younger and 67 were older than 75 years. In 88 cases, the infection occurred after primary arthroplasty, whereas 27 followed aseptic revision surgery. The mean follow-up was 7.1 years (SD 4.4). At five years, treatment failure occurred in 30.4% of cases. Procedures performed within seven days of diagnosis showed lower fai […]

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## Surgical treatment of varus unicompartmental knee osteoarthritis: indications, trends, and outcomes-a narrative review.

*Burgio C, Bosco F, Cobisi C, Zepeda K, Giustra F, Capella M, Mayman D, Debbi E et al.*

INTRODUCTION: Varus unicompartmental knee osteoarthritis (UC-KOA) is a frequent and heterogeneous clinical condition characterized by compartment-specific cartilage degeneration and frequently associated with lower-limb malalignment. The progressive understanding of knee biomechanics and alignment has led to the development of surgical strategies specifically tailored to address isolated compartment disease. The purpose of this narrative review is to evaluate indications, current trends, and clinical outcomes of the main surgical options for varus UC-KOA, with particular focus on realignment osteotomies and unicompartmental knee arthroplasty (UKA).

MATERIALS AND METHODS: A narrative review of the literature was conducted using major biomedical databases. Evidence related to biomechanics, alignment correction, osteotomy techniques, UKA indications, implant evolution, complication profiles, and comparative outcomes was analyzed and synthesized thematically.

RESULTS: Biomechanical and clinical evidence consistently identifies malalignment as a key driver of isolated compartment degeneration. Corrective osteotomies effectively restore physiological load distribution and provide durable outcomes in selected patients with extra-articular deformity. Advances in implant design and surgical techniques have substantially improved UKA survivorship, allowing broader patient selection wit […]

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## Preoperative NarxCare overdose risk scores greater than 100 are associated with worse PROMs improvements and dissatisfaction after primary THA.

*Khan ST, Emara AK, Patel SV, Elmenawi KA, Ibaseta A, Pasqualini I, Zhang C, Cleveland Clinic Adult Reconstruction Research Group et al.*

INTRODUCTION: The NarxCare Overdose Risk Score (ORS) is a weighted measure of patient-specific prescription drug use. In patients undergoing primary total hip arthroplasty (THA), this study aimed to evaluate (1) the trend of the ORS from preoperative to 1-year postoperatively, and (2) the association of preoperative ORS with achievement of clinically meaningful improvements in patient reported outcome measures (PROMs), and satisfaction at 1-year.

METHODS: All patients who underwent primary unilateral elective THA at a USA tertiary healthcare system from January 2016-December 2022 were eligible. Patients with incomplete PROMs or missing baseline NarxCare ORS were excluded, leading to the inclusion of 5,424 patients. Multivariable regression models were used to assess the association between ORS and 1-year PROMs, including achievement of the minimum clinically important difference (MCID) and Patient Acceptable Symptom State (PASS) for the Hip Disability and Osteoarthritis Outcome Score (HOOS) subdomain scores, as well as satisfaction.

RESULTS: Preoperatively, 46.1% (n = 2501) had a NarxCare ORS of 0, 24.6% (n = 1336) had scores between 100 and 199, and 16.2% (n = 876) in the 200-299 range. After adjusting for confounding variables, a preoperative ORS of 100-199 was associated with failure to achieve MCID in HOOS Pain (OR 1.91;CI 1.32-2.77;p = 0.001), and HOOS JR (OR 1.54;1.10-2 […]

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## Only a partial shield: local vancomycin postpones but does not prevent hip and knee prosthetic infections.

*Darup A, Ettinger M, Savov P, Brand S, Seeber GH, Stauss R*

INTRODUCTION: Periprosthetic joint infection (PJI) represents a severe complication following total hip arthroplasty (THA) and total knee arthroplasty (TKA). The intraarticular application of vancomycin powder is increasingly integrated into perioperative standards as an innovative strategy for PJI prevention. Yet the efficacy of this method remains a topic of debate. The aim of this retrospective cohort study was to evaluate whether topical vancomycin application reduces the incidence of PJI after primary total joint arthroplasty (TJA) and whether it is associated with an increased risk of wound complications.

MATERIALS AND METHODS: In this retrospective monocentric cohort study, all primary THA and TKA procedures performed between January 1, 2022, and December 31, 2023, were analysed. From January 1, 2023, intraarticular vancomycin powder was implemented as part of the standard perioperative protocol for PJI prevention. The PJI rates, postoperative surgery-related complications, and time to infection were compared between the vancomycin and control groups.

RESULTS: A total of 1,499 patients were included in the study. Patients were divided into two groups: a vancomycin group (VG, n = 818) and a control group (CG, n = 681). No statistically significant group differences were observed for wound healing disorders (p = 0.775) or PJI rates (1.0% VG vs. 0.9% CG, p = 0.846). Howev […]

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## Validity and prognostic value of a novel intraoperative assessment tool for reduction in trimalleolar fractures: the SGSC checklist approach.

*Gao H, Xu G, Ma J, Wang Y, Wang J, Zhou R, Liu Z, Wang B*

BACKGROUND: Trimalleolar fractures present substantial surgical challenges, and the quality of fracture reduction significantly influences patient outcomes. Current intraoperative assessment methods, including fluoroscopy and arthroscopy, each possess inherent limitations when used independently. This study aimed to validate the effectiveness of a novel intraoperative evaluation tool, the SGSC (Step, Gap, Syndesmosis, Congruity) "traffic light" checklist, that integrates arthroscopic and fluoroscopic assessment, and to evaluate its predictive value for clinical prognosis.

METHODS: This single-center retrospective cohort study enrolled 36 patients who underwent surgical treatment for trimalleolar fractures. Validation proceeded in two stages: first, we calculated inter-rater reliability (Kappa statistic) and concordance with postoperative CT as the reference standard to establish the reliability and validity of the SGSC checklist (accuracy validation); second, patients were stratified into "All-Green" and "Non-All-Green" groups based on intraoperative SGSC assessment, and we compared AOFAS scores and early radiographic degenerative changes at final follow-up (prognostic validation).

RESULTS: For accuracy validation, inter-rater agreement between the two assessors demonstrated excellent reliability (Kappa = 0.827; 95% CI: 0.644-1.000). The overall concordance between SGSC check […]

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## Radiographic assessment of femoral, acetabular and global offset following hip spacer implantation in staged total hip arthroplasty".

*Cavagnaro L, Ferrari E, Providenti V, Marciante V, Carrega G, Formica M*

BACKGROUND: The number of total hip arthroplasties (THAs) performed worldwide is steadily increasing, and, consequently, the demand for revision procedures is expected to rise as well. two-stage revision procedure continues to be considered the gold standard in many clinical scenarios. During the first step, failure to restore femoral offset has been implicated as a contributor to instability and mechanical failure, although a clear consensus is still lacking.

OBJECTIVES: The primary aim of this study is to evaluate the restoration of biomechanical parameters of the hip-specifically leg length discrepancy (LLD), femoral offset (FO), acetabular offset (AO), and global offset (GO)-following implantation of a specific type of articulating hip spacer during the first stage of a two-stage revision for PJI. A secondary objective is to assess the correlation between variations in offset parameters and the rate of interstage mechanical complications.

MATERIALS AND METHODS: We retrospectively reviewed all patients undergoing staged revision with a specific articulating spacer between 2020 and 2022 at a single institution. Harris Hip Scores, Oxford Hip Scores, and Visual Analogue Scales for pain were obtained at different time points. Radiographic analysis included LLD, FO, AO and GO measurements on the affected (spacer and post-reimplantation) and the contralateral side. Complications […]

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## Simulated tears of antero-posterior rotator cuff force-couple and reduced glenoid concavity decrease anterior glenohumeral stability: a robot-assisted biomechanical analysis of a load and shift sequence.

*Karlsfeld MJ, Wermers J, Tänzler S, Sußiek J, Herbst E, Katthagen JC*

BACKROUND: The glenoid concavity and the compression applied by the rotator cuff (RC) are essential for glenohumeral stability (GHS). This study aimed to determine how different simulated rotator cuff tears (RCT) and the glenoid depth influence GHS.

METHODS: A Load and Shift sequence was performed with eight fresh-frozen cadaveric shoulders in a robotic-assisted setup. Differently configured static loading of the reinforced RC and deltoid muscle (DLT) simulated intact RC and anterior, superior, anterosuperior, posterosuperior, mass, and complete RC plus DLT tears. Anterior dislocation forces and their changes were determined as indicators of stability. The glenoid depth and Bony shoulder stability ratio (BSSR) were defined as indicators of concavity. To assess GHS, the maximal force (Favg), the maximal force increase (dFavg), and their mean deviations (ΔFmax, ΔdFmax) to the intact configuration during the sequence were evaluated.

RESULTS: Simulated tears of the subscapularis tendon (SCP) (ΔFmax = 7.90 N, ΔdFmax = 1.54 N/mm) and simulated tears of the infraspinatus (ISP) + teres minor (TM) tendons (ΔFmax = 7.19 N, ΔdFmax = 1.48 N/mm) resulted in greater differences to the intact shoulder than simulated tears of the supraspinatus tendon (ΔFmax = 3.82 N, ΔdFmax = 1.16 N/mm). High correlations were observed between maximal force concerning glenoid depth (r = 0.81) and BSSR (r = 0 […]

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## Non-tobacco nicotine dependence is associated with increased complications following clavicle open reduction internal fixation.

*Adio AA, Goyal RK, Skiena A, Daher M, Horneff JG, Kassam HF, Hill BW, Abboud JA*

BACKGROUND: Tobacco dependence is well established to impair bone healing and increase postoperative complications. However, the independent effects of nicotine, separate from combustion-related factors, remain unclear. This study aimed to evaluate the impact of non-tobacco nicotine dependence (NTND) following open reduction and internal fixation (ORIF) of clavicle fractures, including midshaft and lateral fracture subtypes.

METHODS: A retrospective cohort study was performed using a large national database. Patients undergoing ORIF of clavicle fractures were identified using procedural and diagnostic codes. Patients with non-tobacco nicotine dependence (e.g., vaping, patches, or gum) were identified and propensity score matched to (1) patients without nicotine exposure and (2) patients with tobacco nicotine dependence. In a secondary analysis, patients were stratified by fracture type (midshaft and lateral), and outcomes were compared between nicotine exposure and no nicotine exposure within each subgroup. Outcomes were assessed at 90 days and 1 year and included medical complications, number of opioid prescriptions, and implant-related outcomes. Risk ratios (RRs) with 95% confidence intervals (CIs) were calculated, with $p < 0.05$ considered significant.

RESULTS: After matching, 936 non-tobacco nicotine users were identified. Non-tobacco nicotine dependence was associated wit […]

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## Association between timing of surgery and postoperative outcomes in older adults with distal femur fractures: a systematic review and meta-analysis.

*Hoshika S, Yamamoto N, Saka N, Tsuge T, Sato T, Isaji Y, Nakao S*

INTRODUCTION: The association between surgical timing and clinical outcomes in older adults with distal femur fractures remains unclear. We conducted a systematic review and meta-analysis to evaluate whether earlier surgery is associated with improved outcomes in this population.

MATERIALS AND METHODS: In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, two independent reviewers searched MEDLINE, Embase, the Cochrane Central Register of Controlled Trials, ClinicalTrials.gov, and the World Health Organization International Clinical Trials Registry Platform from inception to October 8, 2024. The primary outcome was 30-day mortality, and secondary outcomes included longer-term mortality and postoperative complications. Risk of bias was assessed using the Quality in Prognosis Studies tool, and the certainty of evidence was evaluated using the Grading of Recommendations Assessment, Development and Evaluation approach.

RESULTS: Of 6,101 records screened, 16 retrospective cohort studies comprising 31,213 patients met the inclusion criteria. Early surgery was not associated with reduced 30-day mortality (adjusted odds ratio [OR]: 0.77, 95% confidence interval [CI]: 0.56-1.05; low-certainty evidence) or with longer-term mortality at 90 days (crude OR: 0.91, 95% CI: 0.51-1.60), 180 days (crude OR: 0.46, 95% CI: 0.12-1.77), or 1 […]

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## Is a central cavity necessary for bioactive glass-ceramic spacers in plated ACDF? A retrospective comparison of solid versus cavity designs.

*Kang TH, Lee G, Kang B, Han J, Cho M, Lee JH*

PURPOSE: Traditionally, interbody cages feature a central cavity for graft packing. However, for bioactive glass-ceramic (BGS-7) spacers achieving fusion via surface-mediated osseointegration, this cavity may be biologically redundant and structurally disadvantageous. We compared radiological and clinical outcomes of plated anterior cervical discectomy and fusion (ACDF) using BGS-7 spacers with or without a central cavity.

METHODS: We retrospectively reviewed patients undergoing plated ACDF using solid non-cavity (17 patients, 24 levels) or central-cavity (17 patients, 33 levels) BGS-7 spacers. Radiographic fusion was evaluated using dynamic interspinous distance (primary, ≤ 2 mm) and segmental angular motion (secondary, ≤ 4°), alongside CT-based osseointegration (direct contact ≥ 50%). Subsidence (≥ 2 mm) and clinical outcomes were evaluated at 12 months.

RESULTS: Both groups demonstrated comparable 12-month clinical improvements ($p > 0.05$). Radiographic fusion rates were equivalent between non-cavity and central-cavity groups based on the primary distance (91.7% vs. 84.8%, $p = 0.687$) and secondary angular criteria (79.2% vs. 63.6%, $p = 0.331$). The non-cavity group showed a trend toward superior segmental stability, with a smaller dynamic gap distance (0.82 ± 0.64 mm vs. 1.53 ± 1.99 mm, $p = 0.064$). CT-based fusion perfectly matched (90.5% vs. 90.9%, $p = 1.000$). Subsidence ra […]

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## Clinical outcomes of internal iliosciatic fixation using a conventional reconstruction plate via an anterior approach for acetabular fractures involving the posterior column.

*Xiong H, Chen H, Cao S, Chen X, Wang F*

BACKGROUND: Acetabular fractures involving the posterior column remain surgically challenging because of the complex three-dimensional anatomy and restricted operative corridor. As an alternative to traditional posterior approaches, internal iliosciatic fixation through an anterior approach has gained increasing acceptance. Reconstruction plates, which offer the advantages of low cost and convenient intraoperative bending, are extensively used in clinical settings.

METHODS: A retrospective case series was conducted on 36 consecutive patients with acetabular fractures involving the posterior column who underwent internal iliosciatic fixation using an intraoperatively contoured reconstruction plate between July 2019 and November 2022. The modified Stoppa approach (n = 26) or the pararectus approach (n = 10) was employed. Fracture reduction was assessed using the modified Matta criteria, and hip function was evaluated using the modified Merle d'Aubigné-Postel score at one-year follow-up. Operative time, blood loss, fracture union, and complications were recorded.

RESULTS: The mean age of the patients was 42.6 ± 14.5 years (range: 21-71 years), with 25 males and 11 females. Mechanisms of injury included traffic accidents (n = 21), falls from height (n = 10), skiing injuries (n = 3), and others (n = 2). According to the Letournel-Judet classification, there were 13 both-column fra […]

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## Novel UHMWPE sleeve for stabilizing loose suture anchors in rotator cuff repair.

*Lin CK, Jaw FS, Sam SS, Siow TF, Yii SC, Chang YJ, Young TH*

Suture anchors are frequently used for repairing torn rotator cuffs. However, post-operative anchor loosening is a common complication often caused by low bone density. The purpose of this study was to evaluate if a novel UHMWPE cortex locking sleeve could improve the anchor pullout strength after revision in synthetic bone. The initial failed anchor fixation was simulated by inserting a 4.5 mm anchor and pulling it out of the bone, which created an enlarged insertion hole for the revision anchor. After revision, two loosening conditions were simulated; (i) intra-operative anchor loosening where the revision anchor was directly pulled out from the block after insertion, and (ii) post-operative anchor loosening where the revision anchor was first subjected to cyclic loading and then pulled out. For both loosening conditions, the pullout strength and energy absorbed was recorded for the following configurations: (i) Group A, same-diameter anchor (SDA; 4.5 mm × 10 mm); (ii) Group B, larger-diameter anchor (LDA; 5.5 mm × 10 mm; ); and (iii) Group C, same-diameter anchor combined with the novel sleeve (SDA + NS). For intra-operative anchor loosening, the highest pullout strength was recorded in the SDA + NS group, with a mean value of 370.077 ± 32.699 N, which was significantly greater than both SDA and LDA groups ($p < 0.05$).

Similarly with post-operative anchor loosening, the SDAP […]

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### Impact of rheumatoid arthritis on complications and hospital outcomes following reverse shoulder arthroplasty: evidence from 389,135 hospitalizations in the national inpatient sample.

*Mahamid A, Elgabsi M, Khatib M, Murad H, Qawasmi F, Lavon E, Zahalka S, Yassin A et al.*

OBJECTIVE: To evaluate the impact of rheumatoid arthritis (RA) on perioperative complications and hospital outcomes following reverse total shoulder arthroplasty (RTSA) using the National Inpatient Sample (2016-2021).

METHODS: Adult patients undergoing RTSA were identified using ICD-10 procedure codes. Outcomes included prolonged length of stay (LOS), high hospital charges, inpatient complications, and mortality. Multivariable logistic regression adjusted for demographic, clinical, and hospital characteristics. Survey weights were applied to generate nationally representative estimates.

RESULTS: A total of 389,135 patients underwent RTSA, including 18,140 (4.66%) with RA. In unadjusted analyses, RA patients had higher rates of prolonged LOS (17.8% vs. 14.8%) and acute blood loss anemia (10.7% vs. 8.6%). After multivariable adjustment, RA remained an independent predictor of prolonged LOS (adjusted OR 1.18, 95% CI 1.14-1.23, $p < 0.001$) and acute blood loss anemia (adjusted OR 1.25, 95% CI 1.18-1.33, $p < 0.001$), but not urinary tract infection. In-hospital mortality was rare and could not be modeled due to zero events in the RA group. RA patients also had slightly longer hospitalizations and higher hospital charges.

CONCLUSION: RA is independently associated with prolonged hospitalization and increased risk of perioperative blood loss anemia following RTSA. These findings high […]

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### Long-term results of thumb carpometacarpal joint arthrodesis.

*Vogler P, Benedikt S, Bode S, Keiler A, Stock K, Arora R*

INTRODUCTION: Thumb carpometacarpal (CMC 1) arthrodesis remains an established treatment option for patients with high functional demands and strength requirements. Long-term data on mobility, complication rates and clinical outcomes remain limited. This study aimed to analyze long-term clinical and radiological results after CMC 1 arthrodesis.

MATERIALS AND METHODS: A single-center, retrospective study included patients who underwent CMC 1 arthrodesis between 2004 and 2024. Following medical record analysis, eligible patients were invited to a standardized clinical and radiological follow-up examination. Outcomes included DASH, PRWHE and MHQ scores, pain intensity, mobility, strength and patient satisfaction. Radiological assessment covered fusion signs, implant position and adjacent joint arthrosis. The median follow-up was 9 years.

RESULTS: 34 patients were included, 16 participated in follow-up. The median DASH score was 14 points, PRWHE 4 points and MHQ 80 (operated) vs. 84 (contralateral). Grip strength reached 85% of the contralateral side, with preserved pinch strength. Compensatory adaptation of adjacent joints was observed regarding range of motion. Patients reported no relevant resting pain and only minimal load-related discomfort. 15 of 16 patients were satisfied with the outcome. Non-union occurred in 4 of 34 cases (11.8%), associated with absent cancellous bone […]

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### Does the tendon retract in flexor hallucis longus muscle laceration? A cadaveric study.

*Ayd■n M, ■ahin Y, Ya■ar H, Çiçek F, Cerano■lu FG, Mert A, Ç■naro■lu S*

INTRODUCTION: Flexor hallucis longus (FHL) tendon injuries are common, particularly in open injuries caused by sharp objects. However, data regarding FHL tendon retraction following hallux-level FHL tendon lacerations remain limited. This study investigated the retraction and shortening of the FHL tendon using cadaveric models.

MATERIALS AND METHODS: Six fresh-frozen below-knee amputated cadaveric feet (mean age: 65 years, range: 62-88) were utilized. FHL tendons were dissected, and lacerations were simulated at the hallux level. Tendons were subjected to various tensile forces (20-120 N) using a mechanical device. Force was measured in Newtons via a loadcell, and retraction distance was recorded in millimeters using an encoder. Data were summarized as mean ± standard deviation (SD), median, and quartiles.

RESULTS: The FHL tendon retracted gradually as applied force increased. An average retraction of 7 mm was measured at 20 N, reaching 21 mm at 60 N. Beyond 60 N (> 21 mm retraction), the other four toes also flexed. This occurs due to the anatomical connection between the FHL and flexor digitorum longus (FDL) tendons.

CONCLUSIONS: Only slight retraction was observed in hallux-level FHL tendon lacerations. At most, the findings suggest that distal FHL lacerations may exhibit limited proximal retraction under the tested conditions, which may have implications for surgical pla […]

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## Early-onset rice body synovitis of the knee in children under three years: a case series and review of the literature.

*Zhu L, Liu C, Liu Y, Mao B, Huang Y, Wen J, Liu Y, Qian T*

OBJECTIVE: Rice body synovitis reports have been rarely reported in children under three years of age. We aimed to explore the clinical characteristics and treatment experience of rice body-like synovitis in children's knees.

METHODS: We retrospectively analyzed the clinical and imaging data of 4 cases (5 knees) of rice body-like synovitis in children treated at our hospital from January 2014 to December 2021. Among the patients, 1 was male and 3 were female, with an average age of 22.5 months. The average length of symptoms before admission was 1.86 months. Clinical presentations included swelling and limping, with no fever or rash.

RESULTS: Three patients with unilateral onset and one patient with bilateral onset underwent unilateral arthroscopic excision of synovial lesions. All 4 patients had pathological findings consistent with chronic inflammation, leading to a postoperative diagnosis of juvenile idiopathic arthritis (JIA). Postoperatively, all patients received standardized pharmacological treatment and were followed for 24 to 48 months in collaboration with the rheumatology department. None of the 4 knees undergoing surgery experienced recurrence, and the symptoms of the knee treated conservatively resolved spontaneously. In the last follow-up session, MRI and ultrasound showed only mild synovial thickening, with effusion and loose bodies disappearing.

CONCLUSION: R […]

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## Extravertebral temperature exposure during spinal radiofrequency ablation: an experimental surrogate assessment of neural injury risk.

*Weigert A, Hoffmann RT, Büttner A, Zaccaria R, Wegener V, Holzapfel BM, Wegener B*

BACKGROUND: Radiofrequency ablation (RFA) has become an important minimally invasive option for the treatment of painful spinal tumors and benign bone lesions. While its effectiveness within the vertebral body is well established, there is ongoing concern about unintended heat exposure of nearby neural structures. Experimental data describing how heat propagates beyond the vertebra during spinal RFA are still limited. This experimental in vitro study aims to assess the extravertebral temperature exposure during spinal radiofrequency ablation as a surrogate parameter for potential neural injury risk under controlled conditions.

METHODS: This experimental in vitro study was performed on four fresh-frozen human lumbar spines. Radiofrequency ablation was carried out using a commercially available system and standard clinical protocols. Two anatomical configurations were examined: ablation within the vertebral body and ablation within the pedicle. Temperatures were measured at predefined locations outside the vertebra, including the spinal canal and cortical boundaries, using multiple thermocouples. To aid interpretation of radiofrequency-related measurement artifacts, an additional control setup using an electrothermal heating system without radiofrequency energy was included. Temperature data were evaluated descriptively.

RESULTS: In the control experiments, temperature profiles […]

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## Giant cell tumor of bone: exploring HtrA1 as a potential predictor of recurrence.

*Mirioglu A, Dalkir KA, Gokmen MY, Bagir M, Ates KE, Deveci MA, Gönlü■en G, Ozbarlas HS*

INTRODUCTION: Giant cell tumor of bone (GCTB) is a locally aggressive tumor with a considerable risk of recurrence, but reliable predictive markers are limited. High-temperature requirement A serine peptidase 1 (HtrA1) has been implicated in the progression of several cancers and may also play a role in GCTB recurrence.

MATERIALS AND METHODS: This retrospective study included 34 patients with GCTB treated with curettage between 2002 and 2016. HtrA1 expression was evaluated separately in giant cells (GC) and mononuclear cells (MNC) using immunohistochemistry. Based on high or low expression in each cell type, patients were categorized into four expression patterns: High GC / High MNC, High GC / Low MNC, Low GC / High MNC, and Low GC / Low MNC. Clinical data, including recurrence status and time to recurrence, were analyzed.

RESULTS: The High GC / Low MNC group showed the highest recurrence rate at 71.4%, compared with 28.6% in the High GC / High MNC group and 31.6% in the Low GC / Low MNC group. No recurrence was observed in the Low GC / High MNC group, which included only one patient. The High GC / Low MNC group showed a non-significant trend toward higher recurrence compared with all other groups combined, with a similar trend toward shorter recurrence-free survival.

CONCLUSION: HtrA1 expression patterns in GCTB may provide preliminary insight into recurrence risk. Although […]

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### Coronal hamate body fractures with dorsal fourth and fifth carpometacarpal joint instability: fracture patterns, injury mechanisms, surgical strategies, and outcomes.

*Lee SH, Chung HJ, Cha SM*

INTRODUCTION: Coronal hamate body fractures are rare injuries, and studies regarding their morphology and management are limited. This study analyzed fracture patterns, injury mechanisms, associated injuries, surgical strategy, and clinical outcomes in patients with coronal hamate body fractures involving more than one-third of the articular surface and concomitant carpometacarpal (CMC) joint instability.

METHODS: Fifty-eight male patients were retrospectively reviewed. Fractures were classified as dorsal oblique or coronal splitting types based on computed tomography findings. Surgical treatment involved open reduction and multiple screw fixation of the hamate using a K-wire-guided technique, with supplementary Kirschner wire stabilization for associated metacarpal base fractures or CMC joint instability. Radiographic and functional outcomes were evaluated after a minimum follow-up of 6 months.

RESULTS: Dorsal oblique fractures (63.8%) occurred more frequently than coronal splitting fractures (36.2%) and were most commonly associated with punching (OR, 4.9; $p = 0.01$), whereas falls from height were more common in coronal splitting types. Concomitant fractures occurred in 82.8% of cases, most frequently involving the fourth metacarpal base. Among 35 patients with a minimum 6-month follow-up, 34 achieved successful healing. In the 33 patients with functional outcome data, the […]

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### Effect of spinal fusion length on hip and knee osteoarthritis: a comparative cohort study of no fusion, short-segment fusion, and long-segment fusion.

*Sülek Y, Demirkale ■, Erto■rul R, Özdemir HM*

PURPOSE: Lumbar fusion alters spinal-pelvic biomechanics and may influence degenerative changes in the lower extremities. This study investigated whether fusion length is associated with radiographic progression of hip and knee osteoarthritis.

METHODS: In this retrospective, single-center cohort study, patients aged 40-65 years who underwent lumbar surgery between 2010 and 2021 were grouped as no fusion (N), short fusion (S; ≤3 levels), or long fusion (L; ≥4 levels). Hip joint space width was assessed using minimum joint space (MJS) and superointermediate joint space (SJS), including femoral head-standardized measures (sMJS and sSJS). Knee medial joint space width (KMJS) was measured when available. Spinopelvic parameters (LL, PI, PT, SS, PI-LL) and functional outcomes (ODI, HHS, WOMAC) were recorded. Multivariable linear and logistic regression analyses were performed to identify independent predictors of joint space narrowing and osteoarthritis risk.

RESULTS: Hip joint space narrowing differed significantly among groups, with the highest annual narrowing rates in the long-fusion group, followed by the short-fusion group, and the lowest in the no-fusion group ($p < 0.001$). In contrast, knee KL grade change and annual KMJS narrowing did not differ significantly between groups ($p > 0.05$). In logistic regression, each additional fused level increased the odds of postoperative MJ […]

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### Biomechanical comparison of inverted triangle and L-shaped screw configurations with medial buttress and anteromedial support plate in Pauwels type III femoral neck fractures.

*Erem M, Demirta■ U, Y■ld■r■m S, Selçuk E, Çopuro■lu C*

INTRODUCTION: Surgical stabilization of Pauwels Type III femoral neck fractures remains a significant challenge due to high vertical shear forces. While the medial buttress plate is a recognized solution, it requires extensive deep dissection. This study aims to compare the biomechanical performance of various screw configurations combined with either a medial buttress or anteromedial support plate.

MATERIALS AND METHODS: Twenty-five third-generation synthetic femurs were used to create a standardized 70-degree (Pauwels III) fracture model. Specimens were divided into five groups ($n = 5$): (A) Inverted triangle (IT) screws with a Pauwels screw, (B) Inverted triangle (IT) with medial buttress plate (MBP), (C) Inverted triangle with anteromedial support plate (ASP), (D) L-configuration with medial buttress plate (MBP), and (E) L-configuration with anteromedial support plate (ASP). Axial loading was applied at 2 mm/min until construct failure, defined objectively by the real-time force-distance curve.

RESULTS: Although no statistically significant difference was found between groups ($p = 0.102$), a large effect size was observed ($\eta^2 = 0.309$). Group C (IT + ASP) demonstrated the highest mean failure load (1695 ± 494.6 N). Conversely, Group E (L-configuration + ASP) exhibited the lowest stability (977.2 ± 195.4 N) with a remarkably narrow standard deviation. The majority of failures […]

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### Reverse total shoulder arthroplasty is a surgical option for patients over 60 years of age with locked shoulder dislocation.

*Sebastia-Forcada E, Miralles-Muñoz FA, Martinez-Mendez D, Albero-Catala L, Climent-Peris V, Lizaur-Utrilla A*

BACKGROUND: in locked dislocation of the shoulder, instead of reducing back to the glenoid, the humeral head remains incarcerated on the glenoid in a locked fashion. This clinical situation is fairly uncommon. It is essential to conduct an individual evaluation of each patient to determine the appropriate treatment.

OBJECTIVE: the aim of this study was to evaluate the functional outcomes of reverse total shoulder arthroplasty (rTSA) in the treatment of locked shoulder dislocation.

METHODS: patients with locked shoulder dislocation who underwent reverse shoulder prosthesis surgery and were admitted to our center between 2007 and 2023 were reviewed. The primary outcome was the Constant score. Secondary outcomes included the adjusted Constant, UCLA and DASH scores. Additionally, any signs of radiologic loosening were also documented.

RESULTS: the series consisted of 10 patients, six men and four women, with a mean age of 68.0 years. The average time from the traumatic injury to surgery was 7.5 months. All patients showed improved Constant, Adapted Constant, UCLA, and DASH scores compared to their preoperative values. When comparing the outcomes of chronic posterior and anterior dislocations, no differences in functional outcomes or shoulder motion were observed after rTSA implantation. There were no complications during or after surgery.

CONCLUSION: The results of the present […]

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### Does the order matter? Comparing the order of stem placement and fracture reduction in revision total hip arthroplasty for Vancouver B2 and B3 periprosthetic femur fractures.

*Antonioli SS, Khury F, Kennedy MF, Haider MA, Duke A, Schwarzkopf R, Aggarwal VK*

BACKGROUND: Vancouver B2 and B3 periprosthetic femur fractures (PPFF) have posed significant treatment challenges due to stem instability and lack of adequate femoral bone stock. This study investigated subsidence, survivorship, and outcomes of Vancouver B2 and B3 fractures, based on the order in which revision stem placement and fracture reduction occurred during revision total hip arthroplasty (rTHA).

METHODS: This retrospective, cohort study included 46 rTHAs between June 2011 and April 2023. Included patients underwent rTHA for Vancouver B2 or B3 PPFF with minimum one-year radiographic and two-year clinical follow-up. All patients were treated with diaphyseal-engaging tapered fluted titanium stems and stem modularity decisions were based on surgeon preference. Cohorts were separated based on if stem placement (SF, n = 19), or fracture reduction (RF, n = 27) occurred first. Patient demographics, intraoperative information, and clinical and radiographic outcomes were collected.

RESULTS: The SF and RF cohort showed no statistically significant differences in rate of subsidence ≥5 mm [26.3%[SF], 22.2%[RF], P = 0.749), rate of subsidence ≥ 10 mm (15.8%[SF], 14.8%[RF], P = 0.928), nor average subsidence (4.1 mm[SF], 4.4 mm[RF], P = 0.861). We found no statistically significant differences in surgery-related clinical outcomes or all-cause revision rates within a two-year follow- […]

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## Offset reconstruction in stemless and stemmed shoulder hemiarthroplasty: influence on long-term outcomes.

*Trefzer R, Gurda A, Deisenhofer J, Weishorn J, Hariri M, Koch KA, Wolf M, Bülhoff M*

BACKGROUND: Shoulder hemiarthroplasty (HA) is a well-established treatment for various humeral-side pathologies. Overstuffing should be avoided because it is associated with glenoid wear; however, specific radiological reconstruction targets remain unclear. This study assessed the association between radiological joint restoration in stemmed and stemless HA and long-term functional outcomes.

METHODS: Patients who underwent stemmed or stemless HA between 2001 and 2011 were included in a long-term follow-up. Cases with incomplete or non-calibratable digital radiographs or concomitant glenoid procedures were excluded. Clinical outcomes were evaluated using the Constant-Murley Score (CMS) and satisfaction questionnaires. Radiographic parameters-lateral glenohumeral offset (LGHO), lateral humeral offset (LHO), and center of rotation (COR)-were measured on standardized anteroposterior radiographs pre- and postoperatively by two independent observers. Differences from baseline to final follow-up (ΔLHO, ΔLGHO, ΔCOR) were calculated. Interobserver reliability was determined using the intraclass correlation coefficient (ICC). Associations between radiological measures and clinical outcomes were analyzed using linear regression; intergroup comparisons used unpaired t-tests ($p < 0.05$).

RESULTS: Forty-eight patients (12 stemmed, 36 stemless HA) were examined after a mean follow-up of 16.6 […]

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## Emergency access to the subclavian vessels by non-thoracic surgeons: a cadaver-based learning model for orthopedic trauma surgery.

*Grechenig P, Gänsslen A, Dauwe J, Wittig U, Sagmeister M, Koutp A, Puchwein P, Sadoghi P et al.*

PURPOSE: The aim of this study was to evaluate the procedural time and safety of an infraclavicular approach to the subclavian vessels for damage control of major upper extremity vascular injuries. The study specifically focused on the performance of non-thoracic surgeons using a cadaveric model.

METHODS: A sample of 100 orthopedic trauma surgeons (residents, specialists, and attendings) was recruited from an AO cadaveric dissection course. Emergency infraclavicular access was performed on 50 cadavers over two consecutive days. The procedural time, successful vessel clamping, participant self-assessment, and the resulting learning curve were analyzed.

RESULTS: On day one, 27% (9/33) of participants who successfully achieved correct clamping of both subclavian vessels reported feeling confident. By day two, this proportion increased to 71% (55/77). Comparing day 1 to day 2 we found an improvement of 35% (subclavian artery) and 37% (subclavian vein) in the correct identification of subclavian vessels in our sample. The evaluations of this study show that there is no correlation between surgical experience and successful emergency access.

CONCLUSION: Anatomic dissection is of paramount importance for teaching rare and demanding surgical techniques. This study demonstrates that anatomical workshops significantly improve procedural safety and self-assessment when accessing subcla […]

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## Operative timing and surgical complexity in hand trauma: a multicenter analysis of replantation and non-replantation centers in Germany.

*Nicklas A, Will Marks P, Dragu A, Schmidt H, Rotärmel A, HandTraumaRegister DGH*

BACKGROUND: Hand injuries are common traumatic conditions that often require specialized surgical care. Differences in hospital structures and levels of specialization may influence the timeliness and complexity of treatment. This study evaluates quality of care across different hospital types, focusing on time-to-skin-incision and surgical management of finger and hand injuries.

MATERIALS AND METHODS: This retrospective multicenter study is based on data from the HandTraumaRegister of the German Society for Hand Surgery (HTR DGH). A total of 16,726 surgically treated cases with documented finger or hand injuries recorded between 2018 and 2023 were analyzed. Hospitals were categorized as non-replantation centers, replantation centers, or FESSH-accredited replantation centers. Demographic characteristics, injury-patterns, treatment modalities, and time-related parameters were assessed. Statistical analysis included descriptive statistics, group comparisons, and multivariate regression models to evaluate factors associated with time-to-skin-incision.

RESULTS: The study population consisted predominantly of male patients (75.79%), and 91.6% were right-handed. Soft tissue injuries without fractures or amputations accounted for 46.57% of cases. Non-replantation centers demonstrated the longest time-to-skin-incision (median 13 h), whereas replantation centers (median 3.75 h) and FE […]

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## Neglected acromioclavicular joint injury in coracoid process fractures: a retrospective cohort study.

*Lee MS, Lee CH, Jo YH, Cho WT, Seo YW, Choi WS*

INTRODUCTION: Coracoid process fractures are frequently associated with injuries to the superior suspensory shoulder complex, including the acromioclavicular (AC) joint. However, concomitant AC joint injuries may be overlooked in the acute phase, potentially limiting treatment options. This study aimed to investigate the association between coracoid process fractures and SSSC injuries and to identify imaging-based predictors of concomitant AC joint injury.

METHODS: A retrospective single-center cohort study including 60 patients with coracoid process fractures was conducted between February 2016 and April 2023. Coracoid fractures were classified based on anatomical location, and associated SSSC injuries were categorized according to the involved structures. These injuries were evaluated using 3D shoulder CT. Morphological deformity of the SSSC was assessed by measuring the glenoclavicular distance (GCD)-a longitudinal parameter-on final follow-up clavicle AP radiographs. In base fractures, the medial displacement gap was measured on sagittal CT. Statistical analyses included logistic regression and ROC curve analysis to identify predictors of AC joint injury.

RESULTS: Among the 60 patients, fractures were located anteriorly in 27 cases, in the middle region in 5 cases, and at the base in 28 cases. AC joint injury was identified in 15 patients, all of whom had base-type fractu […]

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## Subclinical median nerve alterations after distal radius fractures: a comparative analysis of volar plate fixation and cast immobilization using dynamic ultrasonography and electrophysiology.

*Karapinar SE, Arslan E, Oguzlar FC, Hatip AY, Akgun VO, Dincer R, Asci S, Kizilkaya V*

BACKGROUND: Distal radius fractures may affect the median nerve through trauma-related and treatment-related mechanisms. While overt neurological deficits are uncommon, subclinical median nerve alterations may occur following both conservative and surgical treatment.

PURPOSE: To determine whether treatment modality influences subclinical median nerve behavior using combined dynamic ultrasonographic and electrophysiological assessment.

METHODS: This retrospective observational study included 42 patients with unilateral distal radius fractures. 18 patients were treated conservatively with circular casting, and 24 patients underwent volar plate fixation.

Ultrasonographic assessment of the median nerve was performed at the distal radius level in neutral wrist position, as well as during wrist flexion and extension, using a dynamic ultrasonographic approach, and was combined with electrophysiological evaluation including sensory and motor nerve conduction studies. In the surgically treated group, only patients with Soong type 1 or type 2 plate positioning were included.

RESULTS: Clinical functional outcome scores were comparable between groups, despite a higher frequency of mild clinical signs such as positive Tinel test and night pain in the plate group. However, ultrasonographic evaluation demonstrated significantly greater position-dependent median nerve enlargement and flatte […]

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## Impulsivity and inattention in hand tendon injuries: a case-control study revealing distinct profiles for work-related accidents.

*A■ta■ Ertan E, Demir Karak■l■ç G, Ertan MB*

INTRODUCTION: The impact of attention deficit, impulsivity, and anger on various types of injuries is a subject of ongoing research. Hand tendon injuries are frequently encountered clinical conditions; however, their potential relationships with the aforementioned factors have not been previously investigated. This study aimed to examine the mechanisms of hand tendon injuries from certain psychiatric perspectives.

MATERIALS AND METHODS: 32 patients presenting to the physical therapy and rehabilitation outpatient clinic with hand tendon injuries and 32 healthy controls were evaluated. The assessment included sociodemographic data, causes of injury, Quick Disabilities of the Arm, Shoulder, and Hand (Q-DASH) and Visual Analog Scale (VAS) scores, as well as the physical and functional consequences of the injury. Patients and controls were assessed for attention deficit, impulsivity, and anger using the Adult ADHD Self-Report Scale (ASRS), Barratt Impulsivity Scale-11 (BIS-11), and State-Trait Anger Expression Inventory self-report scales. Statistical analyses of the results were performed.

RESULTS: Patients with tendon injuries were found to have significantly higher scores in ASRS attention and BIS-11 Motor impulsivity compared to the control group ($p = 0.048$, $p = 0.040$). Individuals with hand tendon injuries resulting from work accidents demonstrated significantly lower ASRS to […]

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## Comparative analysis of the improvement in shoulder rotation in complex shoulder fractures treated by reverse shoulder arthroplasty compared to open reduction and internal fixation: a systematic review and meta-analysis.

*Albadran A, Alsugair R, AlHarthi A, Alshangiti H, Alqahtani H, Alsugair G, Ababtain R, Alhajress D et al.*

PURPOSE: To compare the clinical and functional outcomes of open reduction and internal fixation (ORIF) and reverse shoulder arthroplasty (RSA) in patients aged ≥ 65 years with complex proximal humeral fractures (PHFs).

METHODS: A systematic review and meta-analysis was conducted according to the PRISMA guidelines. PubMed, Web of Science, ScienceDirect, EBSCO, and the Cochrane Library were searched for randomized controlled trials (RCTs) and cohort studies published in English without date restrictions. Eligible studies compared RSA and ORIF in elderly patients with PHFs and reported functional, radiographic, or complication outcomes. Pooled data were analyzed using a random-effects model.

RESULTS: Twenty-two studies involving > 34,000 patients were included. RSA was associated with greater forward flexion and a trend toward improved abduction, whereas internal rotation favored ORIF without reaching significance. No significant differences were observed in external rotation. The functional scores (Constant-Murley, Oxford Shoulder Score) were similar, and the complication and reoperation rates did not differ significantly between the groups.

CONCLUSION: RSA offers modest advantages in forward flexion compared with ORIF but does not consistently improve overall functional scores or reduce complications. RSA may be preferred in elderly patients with poor bone stock or rotator c […]

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## Changes in kinematic behavior and collateral ligament strain after medially stabilized TKA using a novel intraoperative navigation platform: a cadaveric study.

*Van de Vyver A, Taylan O, Peersman G, Demeulenaere M, Bialy M, Lauwagie T, Scheys L*

OBJECTIVES: This study aimed to analyze to what extent a novel intraoperative navigation platform (Next-AR, Medacta) for total knee arthroplasty (TKA) allows to restore the native knee joint kinematics and strains in the medial collateral ligament (MCL) and lateral collateral ligament (LCL) throughout a squatting motion.

MATERIALS AND METHODS: Computed tomography (CT) scans of 6 native cadaver legs were used to design patient-specific guides. Bony landmarks and virtual single-line collateral ligaments were identified to acquire real-time intraoperative feedback on bone resection, implant alignment, tibiofemoral kinematics, and collateral ligament elongations using the Next-AR system. The specimens were subjected to squatting (35°-100°) motion using a physiological ex vivo knee simulator while maintaining a constant vertical ankle load of 110 N through active quadriceps and bilateral hamstring controls. Subsequently, each knee underwent a medially-stabilized TKA (Medacta) with mechanical alignment technique and was retested under the same conditions as in their native state. The tibiofemoral and patellofemoral kinematics, along with collateral ligament strains, were computed from 3D marker trajectories using a six-camera optical system (Vicon). MCL and LCL insertions-ant, mid, and post bundles-were identified in relation to bone-pin markers using a wand.

RESULTS: Both native a […]

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## Regional differences in the management of patients with mild traumatic brain injury and antithrombotic therapy-an Austrian survey.

*Sebek B, Dammerer D, Putzer D, Moser L*

BACKGROUND/PURPOSE: The number of patients with mild traumatic brain injury (TBI) receiving oral anticoagulation is increasing. Up-to-date diagnostic and treatment algorithms are missing in Austria. The aim of this survey is to collect and analyse Austria-wide data on the care of this patient group.

METHODS: The survey was carried out using "Jotform" between 27 November 2023 and 31 January 2024, asking questions relating to the diagnosis and treatment of patients with mild TBI receiving antithrombotic therapy. Patient groups comprising individuals with alcohol addiction and hemophilia were incorporated as well with a view to the paucity of available literature on these groups. The survey was sent to all orthopedic-traumatological and neurosurgical departments in Austria.

RESULTS: 39 of 67 (58.2%) orthopedics & traumatology departments and 4 of 10 (40%) neurosurgery departments participated in the survey. The share of departments reporting routine submission of different patient groups to cranial computed tomography (CCT) was 38 (95%) for patients taking antiplatelet agents, 39 (97.5%) departments for patients taking vitamin K antagonists (VKAs), 39 (97.5%) for patients taking direct oral anticoagulants (DOACs), 11 (27.5%) for patients with alcohol addiction, and 28 (70%) for patients with known hemophilia. Considering the separate groups of patients not living in a nursing fa […]

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## Extracorporeal shockwave therapy in bone fractures: a systematic review.

*Santos TIR, Souza E Silva LC, Bonifacio M, Campos Melo CM, Sena BG, Simões MC, Risso Parisi J, da Silva Fernandes AC et al.*

INTRODUCTION: The increasing incidence of fractures and cases of delayed union or nonunion has encouraged the search for non-invasive therapies. Extracorporeal shockwave therapy (ESWT) has been proposed as a strategy to stimulate bone regeneration and improve fracture healing.

MATERIALS AND METHODS: A systematic review was conducted following PRISMA guidelines, with searches performed in PubMed, Scopus, Web of Science, and Embase (2005-2025). Clinical studies evaluating ESWT for fracture treatment with radiographic assessment of bone healing and the presence of a control group were included.

RESULTS: A total of 834 studies were initially identified, of which 7 were included in the final analysis. Most studies demonstrated positive effects of ESWT, including improved bone healing, pain reduction, and functional recovery.

However, there was considerable heterogeneity in treatment protocols and relatively small sample sizes.

CONCLUSION: ESWT shows potential as a non-invasive therapy to stimulate fracture healing, particularly in cases of delayed union or nonunion. However, further controlled clinical trials and standardized protocols are needed to confirm its clinical effectiveness.

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### Clinical outcomes and complication rates of minimally invasive plate osteosynthesis (MIPO) versus open reduction and internal fixation (ORIF) in midshaft clavicular fractures: a comparative study.

*Lin G, Zheng S, Zhang Y, Liu Q, Liu H*

BACKGROUND: Conventional open reduction and internal fixation (ORIF) remains the standard approach for midshaft clavicle fractures, yet its traumatic nature has driven the need for minimally invasive techniques. Locking minimally invasive plate osteosynthesis (MIPO) demonstrates clinical potential through biomechanical superiority and tissue-preserving characteristics.

OBJECTIVE: To compare the clinical efficacy and safety of MIPO versus ORIF for midshaft clavicular fractures.

METHODS: A retrospective cohort study included 203 patients (MIPO group:101 cases; ORIF group:102 cases) treated between 2022 and 2024. Baseline characteristics, functional outcomes (DASH and Constant-Murley scores), and complications were analyzed using Mann-Whitney U and chi-square tests.

RESULTS: No baseline differences were observed ($P > 0.05$). Functional outcomes were equivalent between groups (DASH score: $P = 0.906$; Constant-murley score: $P = 0.646$). The overall complication rate was significantly lower in the MIPO group than in the ORIF group (7.92% vs. 36.27%, $P < 0.001$), particularly in nonunion (0% vs. 5.88%, $P = 0.039$) and sensory disturbance (6.93% vs. 27.45%, $P < 0.001$). This advantage is also evident in the subgroup of patients with AO type 2 C fractures, where MIPO similarly demonstrates a significantly lower incidence of these two complications.

CONCLUSION: MIPO achieves comparable fun […]

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### Isolated scaphoidectomy for type II SLAC and SNAC wrists: retrospective case-series at long-term follow-up.

*Altissimi M, Pataia E, Saracco M, Braghiroli L*

BACKGROUNDS: Stage II scapholunate advanced collapse (SLAC) and Scaphoid non-union advanced collapse (SNAC) wrists are currently treated either by scaphoidectomy and Four Corner Fusion or by Proximal Row Carpectomy. In our retrospective study, isolated scaphoidectomy demonstrated clinical non-inferiority when compared to standard procedures and could be supplemented by a Four Corner Fusion only at a later time if needed.

METHODS: Between 2006 and 2011, 13 patients / 14 wrists (mostly males, manual workers) underwent the surgical procedure. 8 patients were affected by SNAC and 5 by SLAC. The average age was 56,5. All patients were manual workers. The mean follow-up was 15 years. These patients were reviewed a first time in 2012 and once again in 2024. Subjective outcome, DASH score, return to work, ROM and grip strength were evaluated. Radiographic assessment was done in all 14 wrists at the first follow-up and in 6 at the last long-time review. Statistical analysis was conducted using the SPSS software.

RESULTS: 12 patients had resumed the same work at a mean of 3 months after surgery. Only one patient required a Four Corner Fusion after 2 years. 11 of the 13 enrolled patients declared their satisfaction for the procedure at the longest follow-ups. Clinical results remained stable over time. Radiological deterioration was seen in 6/13 wrists at the first follow-up and in all […]

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### Loop versus anchor tenodesis of the long head of the biceps tendon: a prospective randomized controlled pilot study.

*Bischofreiter M, Kölblinger C, Gattringer M, Azesberger A, Gruber MS, Rittenschober F, Kindermann H, Ortmaier R*

BACKGROUND: The long head of the biceps tendon is a frequent source of anterior shoulder pain, often treated with tenodesis. While anchor tenodesis is well established, it carries risks of implant-related complications. This study evaluates an implant-free "loop tenodesis" technique as a potentially safer and cost-effective alternative.

METHODS: In this prospective, double-blinded, randomized exploratory study, 48 patients undergoing arthroscopic biceps tenodesis were randomized to either loop (n = 24) or anchor (n = 24) tenodesis. Clinical outcomes were assessed using ASES, Constant, and DASH scores, as well as range of motion, supination strength, satisfaction, aesthetic appearance, and return-to-sport. Statistical analysis was performed using the Mann-Whitney U and Wilcoxon signed-rank tests to compare non-parametric data.

RESULTS: Both groups showed significant improvements with similar gains in functional scores. Supination strength recovery was greater in the loop group (103% vs. 93%). There were no significant differences in range of motion or complications. Notably, only the anchor group showed cases of Popeye deformity (2 patients). Return to sport was achieved in 95.3% of patients, typically within 3 months. No revision or implant-related issues occurred.

CONCLUSION: Loop tenodesis offers equivalent clinical outcomes to anchor-based techniques, with fewer aesthetic […]

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# I Injury

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## Cumulative traumatic life events and increased risk for emergency department and inpatient utilization after physical injury hospitalization.

*Huynh M, O'Connor S, McManamen M, Buggaveeti AE, Russo J, Wang J, Iles-Shih M, Shoyer J et al.*

INTRODUCTION: Few large-scale investigations have examined the association between the cumulative burden of traumatic life events and population-based assessments of emergency department (ED) and inpatient utilization among physically injured trauma survivors.

METHODS: The investigation was a secondary analysis of a randomized clinical trial that examined the association between cumulative traumatic life events and ED/inpatient utilization after physical injury hospitalization. All 445 physically injured trauma survivors included in the investigation were aged ≥ 18 and underwent screening for elevated posttraumatic stress symptoms. Three types of patient characteristics were examined for potential significant associations with the longitudinal ED/inpatient utilization outcome: pre-injury, baseline at the time of index physical injury admission, and events that occurred in the year post-injury. Regression analyses were used to assess the association between these clinical and demographic characteristics and ED/inpatient utilization in the 12-36 months after the index injury admission.

RESULTS: Approximately one-third of patients reported ≥ 7 serious traumatic life events before the index injury admission. Approximately 62% of patients had one or more ED/inpatient visits over the course of the 12-36 months post-injury. Clinical and demographic characteristics that were signific […]

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## Agreement between ChatGPT and emergency physicians in laceration management: A prospective study.

*Sa■lam Gürmen E, Yorganc■o■lu M, Oral A*

INTRODUCTION: Although large language models (LLMs) like ChatGPT are increasingly used in clinical reasoning, their reliability in procedural decision-making remains questionable. Wound management provides a standardized framework for evaluating their performance in practical scenarios. The primary goal of this study was to assess the level of agreement between ChatGPT's recommendations and emergency physicians' decisions in the management of traumatic lacerations.

METHODS: The Emergency Department of Manisa Celal Bayar University Hafsa Sultan Hospital served as the site of this prospective, comparative study, conducted from February to May 2025. A cohort of 792 patients with traumatic lacerations was included. Five structured questions were presented to ChatGPT (GPT-4 model) to obtain

information on suture material, technique, count, wound classification, and the necessity of antibiotics. Physicians' documented decisions were compared with their recommendations. Concordance was measured using Cohen's kappa (κ), and subgroup analyses were conducted by wound size and location.

RESULTS: The median age of the cohort was 28 years (IQR 16-40, range, 2-80), the proportion of males was 79.9%. The leading cause of injury was metal-related trauma, accounting for 47.7%. Across all decision domains, there was considerable to almost perfect agreement between ChatGPT and physicians (κ = 0 […]

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## Video-assisted thoracoscopic surgery versus open thoracotomy for thoracic trauma in hemodynamically stable patients: An updated systematic review and meta-analysis.

*Barrientos ML, Zapata CAL, López CAD*

BACKGROUND: Video-assisted thoracoscopic surgery (VATS) is increasingly used in thoracic trauma, yet high-certainty comparative evidence against thoracotomy remains limited. We updated the only prior comprehensive meta-analysis (Wu et al., 2015) and re-derived all estimates from source-verified study-level data.

METHODS: Six databases were searched through December 2025. Studies comparing VATS with thoracotomy in hemodynamically stable adult patients with thoracic trauma were eligible. Random-effects models with restricted maximum likelihood estimation and Knapp-Hartung adjustment were used; heterogeneity was assessed with $I^2$ and 95% prediction intervals (PI). PROSPERO CRD420261322930.

RESULTS: Thirty studies were included. VATS was associated with fewer postoperative complications (RR 0.56, 95% CI 0.42-0.74; $I^2$ = 0%), shorter hospital stay (MD -4.30 days, 95% CI -5.12 to -3.49), shorter operative time (MD -37.52 min, 95% CI -47.73 to -27.32), and less intraoperative blood loss (MD -119.09 mL, 95% CI -159.16 to -79.01). Perioperative mortality favoured VATS but was of borderline significance (RR 0.48, 95% CI 0.24-0.97; $I^2$ = 0%) and was not significant when the single dominant study was removed (RR 0.56, 95% CI 0.27-1.19). The pooled VATS-to-thoracotomy conversion rate was 13.0% (95% CI 5.9-26.2%). Prediction intervals excluded the null for complications and hospital stay but […]

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## Immersive virtual reality simulation for technical skills acquisition in pneumothorax management and chest tube insertion: A randomized controlled trial among surgery interns.

*Leon S, Rouhi AD, Baji D, Goldberg DW, Hwang J, Weber D, Fernandez FB, Cannon JW et al.*

BACKGROUND: Immersive virtual reality (IVR) offers a cost-effective, high-fidelity approach to enhance technical skills training. This study assessed the efficacy of IVR simulation delivered on a head-mounted display (HMD) for technical skills acquisition in pneumothorax management and chest tube insertion (CTI) among first-year surgery residents.

METHODS: A novel IVR module on pneumothorax management and CTI was developed using Unity software by a multidisciplinary team of surgeon-educators and software engineers. Forty-five surgical residents were randomly assigned to either IVR instruction or traditional classroom didactics. Participants in both groups then performed CTI on a high-fidelity task trainer, and technical skills acquisition was measured using a validated ten-item Objective Structured Assessment of Technical Skills (OSATS) tool.

RESULTS: The IVR group (n = 21) outperformed the control group (n = 24) in mean OSATS score [39.8 (4.4) vs 36.7 (2.9), p = 0.008]. The IVR group also scored higher in rib blunt dissection [4.14 (0.48) vs 3.79 (0.59), p = 0.035], economy of time and motion [3.71 (0.72) vs 3.21 (0.51), p = 0.009], and required less tutor assistance [4.00 (0.89) vs 3.33 (0.70), p = 0.008]. Five HMDs were utilized at a total cost of $1749.95.

CONCLUSION: In this randomized controlled trial, a novel, cost-effective IVR module demonstrated superior technical […]

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## Re-triage of severely injured patients in no-fault vs. at-fault states.

*Visenio MR, Thivalapill N, Silver CM, Sullivan GA, Thomas AC, Reddy S, Ghneim M, Ryan JL et al.*

IMPORTANCE: State no-fault laws enforce fewer administrative requirements for automotive insurance to pay hospitals for injury care costs than states without no-fault laws (at-fault states).

OBJECTIVE: Determine if no-fault states had lower re-triage (urgent inter-hospital transfer) of patients injured in motor-vehicle collision (MVC) than at-fault states.

DESIGN: This was an observational retrospective cohort study.

SETTING: FL, NY, MD and WI Healthcare Cost and Utilization Project State Emergency Department and Inpatient Databases from 2016 to 2021.

PARTICIPANTS: Severely injured [injury severity score (RISS) > 15] Medicare patients.

EXPOSURE: FL and NY had no-fault laws, while MD and WI were at-fault states.

MAIN OUTCOME: Re-triage was defined as same-day inter-hospital transfer to a high-level trauma center. Mixed effects multivariable logistic regression estimated the association between no-fault states and re-triage controlling for age, sex, race, comorbidity, RISS, diagnosis, trauma designation, hospital for-profit, adjusting for clustering within hospitals and states. A moderation analysis with an interaction term for mechanism * fault laws was conducted to estimate whether no-fault laws moderated adjusted odds of re-triage differently among patients severely injured in MVC versus falls.

RESULTS: In total 6447 Medicare patients in 345 hospitals were severely inju […]

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## Comparable outcomes for non-operative intracranial hemorrhage at Level III vs Level I/II trauma centers.

*Lord S, Arda Y, Karikis I, DeWane MP, Paranjape CN, Ng-Kamstra JS, Maurer L, Kaafarani HMA et al.*

INTRODUCTION: Most patients with isolated, closed blunt intracranial hemorrhage (ICH) do not require neurosurgical intervention. However, outcome data for those managed entirely at Level III trauma centers remain incompletely characterized.

METHODS: Using the 2017-2020 Trauma Quality Improvement Program (TQIP), we compared clinical outcomes for non-operative ICH patients treated at Level I/II versus Level III trauma centers. The primary endpoint was composite in-hospital adverse event counts per patient, modeled using negative binomial regression and reported as adjusted incidence rate ratios (aIRRs). Secondary analyses included patients with ISS > 15, ICU admission by trauma center level, predictors of ICU admission, and unplanned ICU transfers. To address triage selection bias, we also compared time to ED discharge in a cohort of patients who met the same inclusion and exclusion criteria and were transferred from a Level III trauma center.

RESULTS: A total of 24,132 patients were included after complete-case exclusion (Level I/II: 22,846; Level III: 1286). Composite adverse event counts were 0.067 events per patient at Level I/II centers and 0.042 at Level III centers. After adjustment, there was no statistically significant difference in composite adverse event counts between groups (aIRR 0.82, 95% CI 0.56-1.18). Mortality was lower at Level III centers (aOR 0.12, 95% CI 0 […]

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## Drug screening in patients with depressed GCS: An evaluation of current practice and potential bias.

*Serio F, Jenkins PD, Barton C, Loss L, Tinoco-Garcia L, Kiraly L, Brasel KJ*

BACKGROUND: Substance use contributes to trauma morbidity, mortality, and recidivism, yet screening practices among trauma patients are inconsistent. We examined urine drug screening (UDS) utilization among trauma patients presenting with depressed Glasgow Coma Scale (GCS).

METHODS: We retrospectively reviewed adult trauma activations (age ≥18) with admission GCS ≤ 12 at a Level I trauma center (2020-2022). The primary outcome was whether UDS was obtained. Secondary measures included demographic and clinical predictors, UDS results, and referral for screening, brief intervention, and referral to treatment (SBIRT). Logistic regression assessed associations with testing.

RESULTS: Of 661 patients, 39% underwent UDS; 67% of these were positive. Patients aged 31-50 were more likely to be screened, while those ≥ 81 and those discharged from the emergency department were less likely.

Cannabis was the most commonly detected drug. Nearly all patients with positive UDS were referred for SBIRT.

CONCLUSIONS: Trauma patients with depressed GCS are under-screened despite high positivity. Standardized screening may improve detection and intervention.

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### Effectiveness of magnets in removing metallic foreign bodies.

*Aslantürk O, Ergen E, Karakaplan M, Özde■ HU, Al F, Ertem K*

BACKGROUND: Metallic foreign bodies (MFB) in the extremities are a common clinical problem that requires removal to prevent complications and to avoid contraindications to magnetic resonance imaging. While magnetic extraction represents an alternative to conventional surgical techniques, systematic data regarding its efficacy and clinical utility remain limited. This study aimed to evaluate the clinical efficacy of magnetic extraction for MFBs in extremities, its impact on radiation exposure, and success rates across different anatomical locations and foreign body types.

METHODS: We retrospectively reviewed data on patients who underwent MFB removal surgery between 2017 and 2025 at a tertiary trauma center. Of the 364 patients reviewed, 51 met the inclusion criteria after excluding non-metallic foreign bodies, cases in which magnets were not used, and incomplete records. Demographic data, anatomical location, foreign body characteristics, use of image intensifier, and magnet efficiency were recorded.

RESULTS: A magnet was used for 52 MFBs in 51 patients. Magnetic extraction of MFBs was successful in 87% (45/52) of cases. The success rate was 100% for metal splinters but decreased to 63% for the needle-type foreign bodies ($p < 0.001$). Magnetic extraction succeeded in all upper-extremity cases (28/28, 100%) and in 68.2% (15/22) of lower-extremity cases; the single patient with […]

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### The economic value and financial burden of informal caregiving during 3 months after geriatric fracture admission: A longitudinal cohort study.

*Raats JH, Sluijter AJ, Lintmeijer L, van der Velde D, Schuijt HJ*

BACKGROUND: While fragility fractures result in significant healthcare expenditure, the (in)direct costs of fracture-related informal care are rarely acknowledged. This study quantifies the economic value of informal care and the financial burden among working-age caregivers of older adult patients following a fracture-related hospital admission.

METHODS: This longitudinal cohort study included working-age (<67 years) informal caregivers of patients aged ≥ 65 years who were admitted for any fracture. Caregivers reported pre-fracture caregiving details and at 2 weeks, 4 weeks and 3 months post-discharge. The primary outcome measure was the economic value of informal care in Euro (€) per week, estimated using the opportunity cost method. Secondary outcomes included a composite financial burden score ranging from 0 (low) to 6 (high) and dichotomized (low vs. high) financial worry score. Linear mixed models were constructed to assess associations of time and caregiver/patient characteristics with informal caregiving costs.

RESULTS: The 67 included caregivers were on average 54 years old, mostly female (66%), and patients' children (79%). Mean weekly informal care costs were €91.66 (95%CI 54.62-128.71) before admission, €244.80 (95%CI 164.62-342.99) at 2 weeks, €214.57 (95%CI 149.46-279.68) at 4 weeks, and €133.58 (95%CI 92.01-175.16) at 3 months post-discharge. In linear mixed mo […]

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### Epidemiology and outcomes of geriatric patients with severe injury in Korea: Nationwide community-based study, 2016-2023.

*Lee SY, Song KJ, Hong KJ, Park JH, Kim TH, Jeong J, Choi SK, Choi YH*

INTRODUCTION: As the number of geriatric populations increases in the aging era, the injury of the older adults is also increasing. This study aimed to investigate the epidemiologic trends and injury characteristics of geriatric patients with severe injuries by age groups and mechanism, using a nationwide database in Korea.

MATERIALS AND METHODS: A cross-sectional observational study was conducted for all emergency medical service (EMS) treated geriatric (≥65 years) patients with severe injuries in Korea between 2016 and 2023. Patients who met the criteria for a field-based injury triage scheme or had a prehospital physiologic abnormality were defined as severe injury. Exposure was age group: young old (65-74), middle old (75-84), and oldest old (≥85 years) groups. The distribution of age groups was investigated by year. Injury characteristics and outcomes were compared by age group and mechanism. A Multivariable logistic regression analysis was used to investigate the predictors of mortality of geriatric patients with severe injuries.

RESULTS: Among 319,567 patients with severe injuries transferred by EMS over 8 years, 92,085 (28.8%) were over 65 years of age. The proportion of older adults increased from 25.7% in 2016 to 32.0% in 2023. Among older adults, young old was 46.4% (42,742), middle old was 39.3% (36,182), and oldest old was 14.3% (13,161). The proportion of oldest […]

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## Emergency department burden of e-scooter injuries in Victoria, Australia: A statewide analysis of injury patterns and prevention implications, 2022-2025.

*Sharwood LN, Hayman J, Rezaei-Darzi E, Laska C, Eager D, Martiniuk A*

BACKGROUND: E-scooter use has expanded rapidly in Australia alongside evolving legislation and changes in e-scooter availability. However, the population-level impact on injury burden and emergency care services remains incompletely characterised, particularly in the absence of standardised external cause coding for e-scooter injuries.

METHODS: We conducted a retrospective statewide analysis of emergency department (ED) presentations for e-scooter-related injuries in Victoria, Australia, using the Victorian Emergency Minimum Dataset (VEMD). Cases from January 2022 to December 2025 were identified using predefined text-search terms applied to narrative fields, with manual validation. Descriptive analyses examined temporal trends, demographic characteristics, injury patterns, health service use, and seasonal variation. Temporal patterns were interpreted in the context of evolving e-scooter legislation and shared-hire programme changes.

RESULTS: A total of 4694 e-scooter-related ED presentations were identified, including 79 pedestrians struck by an e-scooter. Annual presentations increased from 825 in 2022 to 1,373 in 2024 and remained high in 2025 (1358). Riders were predominantly male (71.4%) with a mean age of 31.1 years; however, 15.4% of presentations involved children aged 2-14 years. Falls accounted for 74.1% of injuries, and 67.2% occurred on roads, streets, or highways […]

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## Dual free flap reconstruction: A configuration-based algorithm using parallel and serial designs.

*Ha Y, Kim JT, Hong SE, Kim YH*

BACKGROUND: In severe trauma, extensive or geometrically complex soft-tissue defects often exceed the capacity of a single free flap. Dual free flaps offer a versatile solution, yet guidance on flap choice and configuration remains limited. We assessed outcomes using a configuration-based algorithm distinguishing parallel and serial designs.

METHODS: From January 2015 to July 2025, 32 patients at two centers underwent dual free-flap reconstruction for wide or elongated defects, predominantly in the setting of trauma. Flap selection was guided by defect geometry and recipient-vessel accessibility: parallel configurations were used for wide defects (first flap to the main recipient vessel, second flap to the serratus anterior branch of the thoracodorsal system), whereas serial configurations were applied for elongated defects (first flap to proximal inflow, second flap to the distal descending branch of the lateral circumflex femoral artery). Patient characteristics, flap types and dimensions, complications, and clinical outcomes were analyzed.

RESULTS: The mean age was 51.1 years, and trauma was the most common etiology (43.8%). Parallel configurations were used in 27 cases (84.4%) and serial configurations in 5 (15.6%). The mean surface areas of the first and second flaps were 281.7 ± 134.9 $cm^2$ and 247.9 ± 179.0 $cm^2$, respectively, with a total reconstructed area of 529.6 ± 24 […]

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## More than meets the eye: the continuing importance of tertiary trauma survey even in moderate (ISS 9-15) trauma.

*Knight C, Burke JR, Hind S, Smith SR*

INTRODUCTION: Tertiary Trauma Surveys (TTS) are designed to identify injuries which are not recognised during the primary and secondary surveys. These injuries occur in up to 40% of patients and are commonly referred to in the literature as "missed", we prefer the term "unrecognised" as this term does not infer the failure of primary and secondary survey, rapid assessments designed to identify life and limb threatening injuries. The incidence and definition of unrecognised injuries varies across studies, and several risk factors have been identified internationally. The aim of our study was to investigate the sensitivity and specificity of TTS and the incidence and risk factors for unrecognised injuries in a large UK Major Trauma Centre (MTC).

METHODS: Data was prospectively collected for all major trauma patients admitted to our MTC over six months. All patients over 16 years with an ISS≥ 9 were included. Patients with a length of stay of less than 24 h or palliated were excluded. All patients were examined using a specific TTS protocol. Patient records were reviewed one year post admission to identify any injuries diagnosed after discharge.

RESULTS: A total of 293 patients were included. Unrecognised injuries were identified at TTS in 18.8% (55/293) patients, most commonly affecting the face or head and eyes. Patients with unrecognised injuries had a higher admission ISS (p […]

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## CTA-detected incidence and descriptive fracture patterns of arterial injury in 539 surgically treated tibial plateau fractures.

*Valderrama J, Narváez F, Cancino J, Raffo JF, Bosselin R, Zamora F, Siegel S, Scheu M et al.*

BACKGROUND: Tibial plateau fractures (TPF) are complex injuries associated with significant soft tissue and periarticular complications. Although vascular injuries are well described in lower limb trauma, their incidence in tibial plateau fractures has not been clearly established.

METHODS: Retrospective cohort study including patients with surgically treated TPF at a level I trauma center between 2016 and 2024. CTA was selectively performed based on clinical judgment. Post hoc descriptive risk scenarios were constructed to organize observed fracture patterns. Analysis was primarily descriptive.

RESULTS: 539 surgically treated TPF were included. Mean age was 43.7 years (18-75), 73.84% were male. Left knee was affected in 54.92%. High-energy trauma accounted for 70.87%. The CTA yield was 3.66% (6/164), corresponding to an overall observed incidence of 1.11% (6/539). Certain fracture patterns and clinical scenarios were more frequently observed among cases with arterial injury; however, these observations are based on very few events and should be interpreted cautiously as descriptive findings only.

CONCLUSION: The CTA-detected incidence of arterial injury in this series of surgically treated tibial plateau fractures was 1.11%. Certain fracture patterns and clinical contexts were observed among the small number of cases with arterial injury; however, these findings are descrip […]

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## Diagnostic value of serum troponin in stable patients at risk for blunt cardiac injury.

*Cleary HL, Bernard MD, Davenport DL, Bernard AC*

INTRODUCTION: There are no gold standard criteria for diagnosing blunt cardiac injury (BCI). While the combination of electrocardiogram (ECG) and serum troponin is considered the most sensitive screening method for those being considered for discharge, little evidence exists regarding the diagnostic value and clinical utility of initial and serial serum troponin levels in stable patients who are being admitted with a possible BCI. Therefore, this study seeks to determine if troponin level is an independent predictor of adverse cardiac events in stable, admitted patients at risk for BCI.

METHODS: This was a five-year retrospective study using the trauma database at a University Level I Trauma Center. The study population included adult trauma patients presenting with a physician diagnosis of BCI or a sternal fracture who met prespecified stability criteria (SBP ≥ 90 mmHg, HR < 110bpm, shock index < 1, GCS ≥ 14). The data collected included all troponin values and ECG interpretations as well as echocardiogram interpretations, when performed. A patient was classified as having an adverse cardiac event if they were diagnosed with a new arrhythmia requiring treatment, had cardiac surgery, or suffered cardiac-related mortality. Sensitivity, specificity, positive and negative likelihood ratios, and diagnostic odds ratios were calculated to analyze the diagnostic performance of the di […]

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## Prevalence and presenting symptoms of intracranial hemorrhage in older adults presenting with subacute head injuries in a level-2 trauma center: A retrospective cohort study.

*Raats JH, Schuijt HJ, Hoogervorst ELJ, Velde DV*

INTRODUCTION: Due to our ageing population, the incidence of subacute head injuries such as chronic subdural hematoma is increasing. As current clinical decision rules regarding CT imaging are in context of acute head injuries, it remains unclear what presenting symptoms of subacute (>48 h after initial trauma) head injuries in older adults are relevant for clinical practice. To aim of this study is to describe the prevalence of traumatic intracranial hemorrhage in older adults presenting to the emergency department with subacute head injuries and identify symptoms associated with traumatic ICH.

METHODS: This retrospective cohort included patients > 65 years who underwent CT imaging for subacute head injuries in the emergency department of a single level-2 trauma center between 2020-2024. The primary outcome was traumatic ICH. Secondary outcomes included hospital length of stay, anticoagulation reversal, neurosurgical intervention, and 90-day mortality.

RESULTS: 374 patients were included, with a mean age of 79.1 (SD 6.7) and 170 (54.5%) were male. 62 (16.6%) patients had ICH, of which 40 (68%) were large lesions with mass effect or epidural hematoma. In patients with ICH, median hospital length of stay was 7 (IQR 4- 11) days, 8/18 (44%) anticoagulated patients received pharmacological reversal, 25 (40%) received neurosurgical intervention, and five (8%) died within 90 days. […]

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## The use of temporary intravascular shunting to improve resource utilization is safe and effective for penetrating peripheral vascular trauma.

*Strobl HV, Coopwood CK, Bolaji T, McKinley WI, Hoefer LE, Polcari AM, Dowzicky PM, Zakrison TL et al.*

BACKGROUND: Temporary Intravascular Shunting (TIVS) is well described in peripheral vascular trauma, typically for orthopedic fixation or damage control. Due to a high volume of penetrating trauma, our urban Level 1 trauma center began utilizing TIVS frequently to minimize resource constraints. Our hypothesis is that TIVS is not inferior to immediate repair.

METHODS: Penetrating peripheral arterial injuries were identified from the institutional trauma registry from 5/2018-8/2024. Patients were categorized as having undergone immediate repair (ImR) vs delayed repair (DR), following TIVS placement. The DR group was further divided into resource utilization (DR-RU) and damage control (DR-DC) subgroups. DR-RU included patients receiving < 6 units of packed red blood cells intraoperatively, with a final intraoperative lactate < 5.0 mmol/L, and no other body cavity explored. DR-DC included shunted patients not meeting these criteria. The DR-RU subgroup was compared to both ImR and DR-DC using univariable tests.

RESULTS: Of 228 patients included, 118 were in the ImR group, 47 in the DR-RU subgroup, and 63 in the DR-DC subgroup. Trauma surgeons performed 76% of the ImR, 99% of the TIVS and 97% of the definitive repairs in the DR group. Median shunt dwell time was 12.9 h (IQR 9.7-17.7) in the DR-RU group. When comparing ImR to DR-RU respectively, there was no statistically significan […]

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## Cranial CT scans rarely result in neurosurgical transfers in older adults over 80 years with acute head injuries: A retrospective cohort study.

*Raats JH, Schuijt HJ, Kroes T, Hoogervorst ELJ, van der Velde D*

PURPOSE: Due to an ageing population and low threshold for imaging in older adults with acute head injuries, the resulting number of cranial CT scans continues to increase. While current clinical decision rules aim to identify all intracranial traumatic injuries, the clinical significance of these findings in older adults remains disputed. This study describes traumatic findings on CT imaging and neurosurgical care transfers of older adults with acute head injuries.

METHODS: This retrospective cohort included patients ≥ 65 years who received cranial CT imaging for an acute head injury in a single level-2 trauma center, without neurosurgical care available, between 2020-2024. Intracranial hemorrhage was classified as a minor or major lesion (i.e., with or without mass effect). The primary outcome measure, transfer rate to neurosurgical care, was compared between age groups 65-80 and > 80. Secondary outcomes included hospital length of stay, anticoagulation reversal, 30-day readmission, and in-hospital and 90-day mortality.

RESULTS: In 4581 included CT reports, minor and major traumatic cranial lesions were found in 315 (6.9%) and 88 (1.9%) patients, respectively. The prevalence of traumatic findings was comparable between age groups. Five out of 45 patients aged > 80 with major lesions were transferred to neurosurgical care, compared to 15 out of 43 patients aged 65-80 (11.1% […]

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## Impact of "Kids' Court" on traffic infraction recidivism in Dar es Salaam: A novel approach to a troublesome problem.

*Hopkins M, Reich J, Kalolo S, Swai N, Rweyendela AG, Stoffan A, Guerrero A*

Road Traffic Injuries are a major problem in Sub-Saharan Africa. Recidivism after road traffic infraction has not previously been studied in this setting. This study evaluated a novel elementary school based program to reduce repeat minor traffic offenses near schools by having children serve as "judges" to rhetorically interrogate offenders. Offenders were randomly allocated into standard penalties or participation in Kids' Court (412 total) and followed via quarterly interrogation of a traffic police database for 1 year. While the number of repeat offenders was similar in both groups (88 vs. 85, $p = 0.84$), the total number of offenses was improved in the Kids' Court group and approached significance but did not achieve it (344 vs. 404, $p = 0.065$). The average time to first repeat offense was significantly decreased in the Kids' Court group (127 vs. 154 days, $p = 0.0019$). This program reduces the number of repeat offenses and lengthens the time to repeat offense. It likely has a yet unquantifiable long-term effect on the children that participate as "judges" in the program.

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## Development and measurement properties of the Safety Threats and Adverse Events in Trauma (STAT) taxonomy: A critical review.

*Nazir A, Shore E, Keown-Stoneman C, Grantcharov T, Nolan B, FIRST60 Investigators*

BACKGROUND: Adverse events are common in trauma care, with most occurring during the initial resuscitation phase. The Safety Threats and Adverse Events in Trauma (STAT) taxonomy is a comprehensive tool that was created to evaluate both technical and non-technical errors in trauma resuscitations. The objective of this study is to critically appraise the development and measurement properties, including sensibility, reliability, and validity of the STAT taxonomy.

METHODS: A literature search was conducted in MEDLINE, Embase, and CINAHL, supplemented by citation searching. Studies describing the development, evaluation, or adaptation of the STAT taxonomy were included. Evidence was synthesized narratively across tool development, sensibility, reliability, and validity.

RESULTS: Seven studies met inclusion criteria, encompassing tool development, pilot testing, reliability assessment, and adaptation. The STAT taxonomy was developed using a systematic review and a RAND-modified Delphi process involving 22 trauma experts, resulting in 67 adverse events across nine domains. Early evidence supports strong content validity, with comprehensive coverage of both technical and non-technical domains. Feasibility was demonstrated across simulation and clinical trauma video review settings. Inter-rater reliability was consistently high, with agreement rates of 90.1% in simulation studies and […]

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## ED discharge after trauma transfer has increased over time: A single tertiary care center analysis.

*Etra JW, Pfaff AC, Kaul S, Seshadri A, Berrigan M, Dombek M, Morris RE, Fleishman A et al.*

OBJECTIVES: Regionalization of care and changes in hospital and provider networks have led to increased transfer of patients to an accredited trauma center. We aimed to identify trends in trauma transfers and injury patterns that are likely to result in tertiary center admission versus discharge from the emergency department (ED).

METHODS: All patients transferred from 2014 to 2021 to a single level I tertiary trauma center were identified. Using demographic, encounter, and diagnosis coding data from our trauma registry, specific injuries were stratified based on body system or type of injury. Numbers of transfers by year were analyzed using Wilcoxon rank sum test and chi-squared or Fisher's exact test. Univariable and multivariable logistic regression was used to identify injury patterns that correlated with tertiary hospital admission/intervention or discharge from the ED.

RESULTS: 9307 trauma transfer patients were included. On multivariable regression, thoracic or orthopedic injuries were less likely to result in discharge from the ED (OR 0.34, 95% CI 0.27-0.42; OR 0.33, 95%CI 0.28-0.39). GCS of < 14 was also associated with a decreased likelihood of discharge (OR 0.13, 95% CI 0.08-0.22). Soft tissue injuries, spinal injuries, and head injuries were more likely to be discharged from the ED without requiring admission (OR1.24, 95%CI 1.09-1.40; OR 1.68, 95%CI 1.43-2.00; OR […]

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### Implementation of a smartphone-based ultrasound program to improve timely diagnosis of life-threatening injuries in Cameroon: Results from a prospective, multisite feasibility study.

*Rosenbloom L, Emeh AN, Tambe J, Haddad K, Marpha M, Maffo CJ, Mbebi EF, Driban M et al.*

BACKGROUND: Undiagnosed hemorrhage is the leading cause of preventable trauma death in Cameroon, yet few injured patients receive imaging to diagnose hemorrhage. A smartphone-based ultrasonography (SBU) curriculum was developed to rapidly train frontline healthcare providers in Cameroon to perform and interpret eFAST. SBU has demonstrated promising educational efficacy, but its clinical feasibility in this technology-constrained setting is unknown. We evaluate the feasibility and acceptability of a SBU pilot at three trauma centers in Cameroon.

METHODS: We implemented a six-month feasibility pilot at three Cameroonian hospitals participating in the Cameroon Trauma Registry (CTR). Trauma providers were trained to perform SBU eFAST using a novel 5-hour curriculum, then asked to perform SBU on all injured patients as part of the trauma evaluation. Feasibility was assessed as the proportion of CTR patients with completed eFAST. Acceptability was assessed as the proportion of users rating SBU ≥ 68 on the previously validated System Usability Scale (SUS).

RESULTS: Trauma care providers completed SBU eFAST on 87% (608 of 701) of eligible patients, compared to a pre-study diagnostic imaging rate of 5.5% (38 of 648). Completion was highest at the rural and regional low-volume centers (100%, n = 42 and n = 75, respectively) and lower at the high-volume urban tertiary referral center (8 […]

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### Insurance-based disparities in traumatic brain injury length of stay and discharge dispositions: A retrospective cohort analysis of the national trauma data bank.

*Strelko O, Spiro E, Lui A, Tarapore PE, Huang MC, Manley GT, Yue JK, DiGiorgio AM*

BACKGROUND: Insurance-based disparities associated with outcomes in traumatic brain injury (TBI), including acute care discharge dispositions, remain understudied. This study aims to evaluate the associations between Medicaid versus private insurance (PI) on hospital length of stay (LOS) and discharge disposition in patients with TBI across severities.

METHODS: This is a retrospective multicenter cohort study utilizing data from United States trauma centers from the National Trauma Data Bank (NTDB) years 2014-2016. Inclusion criteria were: identified to have TBI based on ICD-9 codes. Exclusion criteria were: missing gender, race, Glasgow Coma Scale (GCS) score, or LOS; not admitted as inpatient to hospital; did not have Medicaid or PI; age < 14 or > 89 years; extracranial AIS= 6. Patients were grouped into those with Medicaid and those with PI.

RESULTS: Of 808,120 NTDB patients with TBI, 188,756 met inclusion criteria (Medicaid: N = 58,867, PI: N = 129,889). Overall, PI patients were older and more frequently female, with differences in racial distribution and TBI severity compared to Medicaid patients. On multivariable analysis, Medicaid patients demonstrated longer inpatient LOS across all TBI severities, corresponding to a 7.3% increase after mild TBI (95% CI: 6.4%-8.2%, p < 0.001), a 7.4% increase after moderate TBI (95% CI: 2.8%-12.2%, p = 0.001), and an 8.7% increase af […]

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## Early operative management in trauma: A nationwide comparison of time to surgery and survival at trauma centers and non-trauma centers.

*Eriksson J, Mattsson A, Larsson E, Oldner A, Oelreich EV*

BACKGROUND: Trauma is a leading cause of death and disability, both in Sweden and globally. Timely surgical intervention and level of care have been identified as important determinants of outcome, yet it is not known whether time to surgery differs between trauma centers and non-trauma centers in Sweden, or whether previously observed survival differences also apply to patients undergoing early operative management.

METHODS: This retrospective national cohort study included adult trauma patients (≥18 years) who underwent surgery within six hours of hospital arrival between 1 January 2019 and 21 March 2023. Data were obtained from Swedish national quality registries and administrative registers held by the National Board of Health and Welfare (Socialstyrelsen). The primary outcome was time from hospital arrival to initiation of surgery. Secondary outcomes were time to urgent surgery and 30-day mortality. Associations between trauma center status and outcomes were examined using regression models adjusted for demographic and injury-related factors.

RESULTS: A total of 1129 adult trauma patients who underwent surgery within six hours of hospital arrival were included, the majority presenting with moderate to severe injuries; of whom 659 (58.4%) were treated at trauma centers and 470 (41.6%) at non-trauma centers. Patients treated at trauma centers had higher injury severity and […]

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## Turret truck-related head and neck injuries in a Japanese wholesale market: Clinical characteristics, risk factors, and the need for preventive measures.

*Kushi K, Hifumi T, Kamijo E, Ito K, Shirasaki K, Ro B, Mochizuki T, Nonaka T et al.*

BACKGROUND: Turret trucks are widely used to handle cargo in Japanese wholesale markets. Under Japanese law, helmet use is not mandatory for turret truck operators, which potentially increases the risk of head and neck injuries. This study aimed to describe the clinical characteristics, risk factors, and outcomes of turret truck-related head and neck injuries as the first reported analysis of this injury mechanism.

METHODS: We conducted a retrospective observational study of patients with turret truck-related head and neck injuries who were transported to St. Luke's International Hospital between January 2011 and June 2024. Demographic data, injury characteristics, and outcomes were collected. The primary outcome was the necessity for hospital admission, and the secondary outcome was functional status at discharge, measured using the modified Rankin Scale (mRS).

RESULTS: Of the 67 patients analyzed, 48 (72%) were turret drivers and 19 (28%) were non-drivers. The cohort was predominantly male (97%) with a median age of 64 years. Twenty-four patients (36%) required hospital admission, with a higher proportion among turret drivers (44%) than among non-drivers (16%). Among turret drivers, all patients with severe injuries (Injury Severity Score ≥16) and those with significant head and neck injuries (Abbreviated Injury Scale ≥3) required hospitalization. Despite injury severity, m […]

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## Emergency medical services response times to motor vehicle crashes increased in the USA over the period 1987-2020.

*Dubois S, Gravelle P, Savage DW, Maxwell H, Bédard M*

INTRODUCTION: The timing of Emergency Medical Services (EMS) notification, crash scene arrival, and hospital arrival may impact motor vehicle fatalities. We examined EMS response time intervals over the past three decades, considering the effects of weather, vehicles involved, time of day, and location.

METHODS: We used the Fatal Accident Reporting System to compute and describe annual (1987-2020) EMS response time intervals. This included total time (i.e., crash-to-hospital), as well as the intervals between four key timepoints: crash, crash notification, crash scene arrival, and hospital arrival. We examined the proportion of fatal crashes with total intervals under 60 min (i.e., the "golden hour"), and where the crash arrival-to-hospital interval was under 30 min (the "beneficial timeframe"). Additionally, analyses were stratified by crash factors including weather (poor/clear) number of vehicles involved (single/multiple), time of day (early morning/rest of the day), and location (urban/rural).

RESULTS: A total of 310,001 fatal crashes were analyzed. Between 1987-1994 total median response times ranged between 40 and 42 min. By 1999, intervals had increased to 45 min; elevated intervals were evident through 2009. By 2020, observed intervals had returned to 41 min. Paralleling this pattern, crashes with "golden hour" intervals decreased from 77.0% in 1987 to 72.4% in 2009 […]

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### All-terrain vehicle related urethral injuries: An evaluation of the National Trauma Database.

*Hertz A, VanDyke M, Hudak S*

BACKGROUND: U.S. Consumer Product Safety Commission reported that from 2018 to 2020 there were 94,700 off-highway vehicle accidents with 92% of those injuries related to all-terrain vehicles (ATV). An estimated 300 deaths were in children less than age 16. There has not been any previous evaluation of urethral injuries resulting from ATV accidents. We sought to define the volume and demographics of ATV-related urethral injuries.

METHODS: Data was obtained from the National Trauma Database (NTDB) Trauma Quality Program over a 7-year period (2016-2022). ICD-10 codes were used to identify patients who sustained urethral injury in ATV-related accidents. Injury and patient related demographic data was reviewed. R: A Language and Environment for Statistical Computing was used for data review and statistical analysis. Data were presented and analyzed using standard statistical methods RESULTS: During this time frame, there was a total of 7679,101 injuries recorded in the NTDB. There was a total of 7398 urethral injuries (0.1%), with 181 (2.5%) of those being related to ATV-accidents. 90.6% of patients were male, and there was a median age of 30 (range 5-76, IQR 29). The median injury severity score (ISS) was 17 (range 4-50, IQR 15). There was no association between age and higher ISS (r = 0.07, p = 0.36). Passengers were more likely to be less than age 16 (p = 0.001). Interestingly, […]

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### Challenges and learning curve in adopting resuscitative endovascular balloon occlusion of the aorta for trauma patients: A retrospective multicenter study.

*Kim K, Kang BH, Kim DH, Yu B, Chang SW, Jung PY, Heo Y, Kang WS*

BACKGROUND: This study aimed to assess the learning curve of REBOA from the first case in trauma centers using cumulative sum (CUSUM) analysis.

MATERIALS AND METHODS: This study enrolled consecutive trauma patients who visited five trauma centers from December 2015 to December 2021. To monitor the effectiveness of REBOA, we performed risk-adjusted cumulative sum (RA-CUSUM) analysis for mortality due to exsanguination. For individual risk adjustment, we implemented the least absolute shrinkage and selection operator (LASSO) logistic regression model. We then calculated the CUSUM for: (1) the time from common femoral artery (CFA) access to confirmation of REBOA placement (RP-CUSUM), (2) the total occlusion time (OT-CUSUM), (3) door-to-balloon time (DB-CUSUM), and (4) the time from injury to admission (IA-CUSUM). To determine whether observed deviations were statistically significant, a V-mask was superimposed on the CUSUM curve.

RESULTS: A total of 251 patients were enrolled. The overall mortality rate was 67.7% (170/251), and the mortality rate due to exsanguination was 49.0% (123/251). The RA-CUSUM model was developed using a LASSO logistic regression approach. In three hospitals, the RA-CUSUM showed an improvement after 5-34 procedures. However, the RA-CUSUM showed fluctuations with deterioration in the other two hospitals. The RA-CUSUM fluctuations exceeded the V-mask contr […]

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### Venous thromboembolism burden in IVC injuries: Is there an optimal strategy in prevention?

*Himmler A, Preston JD, Guindi A, Wagner V, De Leon Castro L, Fransman RB, Nguyen J, Smith RN et al.*

INTRODUCTION: Venous thromboembolism (VTE) is a common complication in trauma patients; however, there have been limited studies on VTEs and the prevention thereof in patients with inferior vena cava (IVC) injuries. This study aimed to examine VTE incidence and the preventative efficacy of VTE prophylaxis regimens after operative IVC repair.

METHODS: A 12-year retrospective review was performed of all patients who presented with IVC injuries to an urban level 1 trauma center. A subgroup analysis of VTE incidence, prophylaxis regimen (i.e., prophylactic anticoagulation [PAC] and/or antiplatelet [AP]) efficacy, and other clinical variables was performed on patients who underwent a primary or patch IVC repair and survived > 72 h.

RESULTS: A total of 132 patients presented with IVC injuries requiring operative management, with 56% overall survival. Among the 66 patients who received primary or patch repair and survived > 72 h, 27% had a VTE during index hospitalization and/or readmission. VTEs occurred a median of 9 days post-injury, and the most common type of VTE was deep vein thrombosis, followed by IVC thrombosis and pulmonary embolism. There were no significant differences in any clinical variables compared between patients who developed VTEs during index hospitalization and those who did not, including VTE prophylaxis timing, hospital or ICU length-of-stay, and injury sever […]

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## Association between social vulnerability, urbanicity, and post-injury outcomes following nonfatal motor vehicle crashes.

*Grossman MK, Rink AJ, Foley LD, Ott RL*

BACKGROUND: Motor vehicle crashes (MVCs) are a leading cause of injury and death in the United States. Community-level factors, such as social vulnerability and urbanicity, have been associated with risk of death; less is known about how these factors impact nonfatal, post-injury outcomes. This study examined the association between social vulnerability and urbanicity with hospital length of stay (LOS) and hospital discharge disposition among MVC patients.

METHODS: Patients aged 18 years and older who were admitted to a Montana regional trauma center with a non-fatal injury following an MVC from 2016 to 2024 were included in the study. The CDC Social Vulnerability Index (SVI) was used to quantify social vulnerability at the census tract level and scores were divided into tertiles representing low, medium, and high vulnerability. Urbanicity was defined using RUCA codes based on patient residence. Generalized estimating equations with a binomial distribution were used to estimate the joint association between SVI and urbanicity with discharge disposition (home vs. facility) and prolonged LOS (≥7 days), controlling for injury severity, patient demographics, and comorbidities.

RESULTS: Of the 668 patients, 529 (79%) were discharged home and 179 (27%) had a prolonged LOS. Among metropolitan patients, higher SVI rankings were associated with increased odds of discharge home; patien […]

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## Integrated surgical management of forequarter lateral implosion injury: Technical considerations and early outcomes.

*Chua SKK, Balasubramaniam S, Cole PA, Tan BY*

Concomitant ipsilateral fractures of the clavicle, scapula, and ribs, termed forequarter lateral implosion injury, represent a severe but underrecognized injury pattern resulting from high-energy lateral shoulder trauma. While the surgical indications for isolated chest wall and shoulder girdle injuries are well described, guidance on the integrated management of this combined injury complex remains limited. We describe a reproducible multidisciplinary approach for the concurrent surgical fixation of clavicle, scapula, and rib fractures, illustrated through two cases of forequarter lateral implosion injury resulting from high-energy road traffic accidents. Preoperative planning incorporated CT three-dimensional (3D) reconstructions and patient-specific 3D printed models to facilitate pre-operative planning, with a multidisciplinary team involved for incision planning and fixation sequencing. Surgery was performed in a single setting with the patient in the lateral position, utilizing muscle-sparing approaches and a staged fixation strategy to address the clavicle, scapula, and ribs through coordinated exposures. Simultaneous osseous stabilization allowed restoration of the superior shoulder suspensory complex integrity and chest wall mechanics, enabling immediate postoperative shoulder mobilization and aggressive pulmonary rehabilitation. Both patients demonstrated early pain r […]

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## Palliative and end of life care in older major trauma - A point prevalence evaluation in England, Wales and Scotland.

*Cole E, Jarman H, PECOT collaborators*

INTRODUCTION: Traumatic injury in older people is a significant health burden with higher mortality rates than younger cohorts. Survival following older trauma may be complicated by the patients pre-injury state and clinical uncertainty. Timely identification of palliative and end-of-life care needs may be challenging for acute clinical teams, and treatment escalation planning is not routinely embedded in trauma care. This point prevalence snap-shot aimed to evaluate treatment escalation discussions and palliative/end of life care (EoLC) practice in older major trauma patients at a national level.

METHODS: A one-day point prevalence "flash-mob" audit was conducted across Major Trauma Centres (MTCs) and Trauma Units (TUs) in England, Wales and Scotland. All trauma patients aged ≥ 65 years in hospital were eligible for inclusion. Patients with and without treatment escalation plans (TEPs) and those on care pathways were analysed.

RESULTS: Data from 957 patients in 49 hospitals were included and median time from injury was 11 days (interquartile range 4-24). A TEP or equivalent was documented in 393 patients (41.0%). Among patients with a TEP, there were more aged > 85 years (165/393 (41.9%), than in those without a TEP (167/564 (29.6%), $p < 0.001$). Clinical frailty scoring was performed in 657 patients (68.6%), and where recorded, TEPs were associated with increased frailty (CF […]

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## Interhospital variation in highest-level trauma activation and its association with mortality: A 37-center cohort study of level I and II trauma centers in the US.

*Orlando A, Mazumder H, Leichtle S, West MA, Joseph B, Kaafarani H, Hunt JP, Shen Y et al.*

BACKGROUND: Trauma team activation protocols are critical for mobilizing resources in the care of severely injured patients. In the US, the American College of Surgeons (ACS) specifies minimum criteria for the highest-level (full) trauma activation (fTA), but hospitals retain discretion to add criteria, potentially leading to variability in activation practices and resource utilization. The extent of this variation and its impact on patient outcomes is unknown. The aim of this study was to quantify inter-hospital variability in fTA use and its relationship to mortality.

METHODS: We conducted a multicenter, retrospective cohort study of adult trauma patients treated at 37 Level I and II trauma centers across the United States from 2017 to 2019; transfers were excluded. Mixed-effects logistic regression models were used to quantify inter-hospital variability in fTA utilization and total mortality (death+hospice), adjusting for 12 patient and hospital-level characteristics. Correlation analyses assessed the relationship between adjusted hospital-specific fTA rates and adjusted total mortality.

RESULTS: Overall, 158,696 patients were included, with 34,374 (21.7%) receiving a fTA. The median age was 53 yrs, with 59% male, 71% White, 88% blunt, and a median Injury Severity Score of 9. Use of fTA varied widely (3.3% to 54.1%, median [IQR]=19.3% [13.6-27.3%]) and the adjusted odds of […]

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## Alternative trials of recipient site vessel in maxillomandibular reconstruction with multisegment fibular flaps including only one pedicle anastomosis.

*Akbay E, Gülmez M■*

BACKGROUNDS: The objective of this study is to closely examine the preferred anastomoses of the superior thyroid artery, facial artery, lingual artery, maxillary artery, and superficial temporal artery anastomoses for the reconstruction of maxillomandibular defects caused by firearm injuries. The study will also examine the reasons for choosing these arteries, their advantages and disadvantages, surgical techniques, complications, and their postoperatively clinically and scintigraphically assessable viability.

METHODS: The present study encompasses a cohort of ten patients who sustained maxillomandibular injuries and underwent surgical intervention employing a multisegment fibular flap. In all cases, following bone fixation, microvascular anastomoses were sutured under a microscope using 9-0 nylon sutures. Flap viability was initially assessed clinically; in addition, all patients underwent a bone scan on the fifth day after surgery using an intravenous infusion of Tc-99m methylene diphosphonate (MDP).

RESULTS: In the study, 10 patients who underwent maxillomandibular multisegment flap surgery received the most appropriate microvascular anastomosis for their pathology. Of these patients, the six most demonstrative cases, which best represent the relevant artery used for anastomosis, have been detailed.

CONCLUSIONS: Although the aim of this study was not to establish a defini […]

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### An epidemiological analysis of extracorporeal membrane oxygenation use in trauma.

*Wallis MC, Rizzo JA, O'Neil ER, Sams VG, Schauer SG*

BACKGROUND: Trauma is the leading cause of death in younger adults and children. Severe polytrauma predisposes patients to the failure of multiple organ systems, including the cardiovascular and respiratory systems, at times necessitating extracorporeal membrane oxygenation (ECMO). ECMO use in the setting of trauma is increasing, yet little data exists describing current practice patterns. We performed an epidemiological analysis of ECMO use and outcomes in trauma patients.

STUDY DESIGN AND METHODS: We analyzed data from the Trauma Quality Improvement Program Registry from 2017 to 2023. Procedure codes were used to identify the application of ECMO. Pediatric patients were those < 18 years of age. We analyzed two groups from this data set: one including all patients who required ECMO and a separate group including only pediatric cases. We used descriptive and inferential statistical methods.

RESULTS: There were 8,014,737 encounters of which 1919 had documented ECMO use. Within that group, 224 were < 18 years of age. The median time from hospital arrival to the first initiation of ECMO was 44 h (interquartile range [IQR] 5-147). The incidence per year ranged from 1.9 to 2.9 events per 10,000 encounters. Survival ranged from 59% to 68% per year. The number of facilities with documented ECMO use annually ranged from 103 to 158 and overall increased during the time of the study. I […]

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### The role of palliative care in older persons trauma.

*Baxter M*

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### Shock and Trauma Risk Index Nomogram (STRIN): A bedside physiology-based tool for early-high-acuity care triage in major trauma.

*Saronni S, Mecca G, Rampin C, Agarossi A, Ripoll-Gallardo A, Colzani G, Cimbanassi S, Chiara O*

BACKGROUND: Major trauma is a leading cause of mortality and permanent disability, especially in younger adults. Early identification of high-risk patients is critical for appropriate care. Tools like the Shock Index (SI) or New Trauma Score (NTS) enable rapid bedside assessment but show variable performance across populations.

OBJECTIVE: To develop and internally test the Shock Trauma Risk Index Nomogram (STRIN), integrating Shock Index (SI), Glasgow Coma Scale (GCS), oxygen saturation (SpO■), and mean arterial pressure (MAP) for early risk stratification of time-critical resource use in major trauma.

METHODS: Data of a retrospective cohort (March 2022-December 2024) from a single level-1 centre retrieved from Lombardy registry was considered. Adult patients (≥16 years) with major trauma and complete vital signs were included. Composite outcome was emergency surgery (ES) within 6 h, massive transfusion (MT) (≥10 units of packed red blood cells within 24 h or ≥4 units within first hour), Intensive Care Unit (ICU) admission and interventional radiology procedures (IR). The model was developed using multivariable logistic regression with Restricted Cubic Splines. Performance was assessed through discrimination (AUROC), calibration (Hosmer-Lemeshow test, Brier score), and internal testing via bootstrap resampling. Comparative performance against SI and NTS included AUROC, Net Re […]

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## Predictors of traumatic brain injury amongst secondary school students in England: A retrospective cohort study using electronic health records.

*Benavente MT, Wilson R, Jackson J, Birnie K, Mytton J, Ijaz S, Booker M, Burrell A et al.*

BACKGROUND: Traumatic brain injury (TBI) in children and adolescents is a leading cause of disability and mortality, with long-term health-related consequences. There is little evidence describing the children and adolescents most at risk of TBI.

OBJECTIVE: To identify the demographic and clinical predictors of TBI in secondary school-aged children in England.

PARTICIPANTS AND SETTING: Linked Clinical Practice Research Datalink and Hospital Episode Statistics Admitted Patient Care data were used to identify patients aged 11-18 years registered with a GP surgery in England between 2013 and 2021.

METHODS: Multivariable Cox regression was used to assess the association between demographic and clinical risk factors and time to medically attended TBI.

RESULTS: The analytical sample included 402,249 children, 2.3% of which had a TBI presenting to primary or hospital care when aged 11-18 years. In the fully adjusted model, increased risk of TBI was associated with being male; less socioeconomically deprived; having a history of fracture, abuse, depression, or previous TBI; having two or more previous GP visits, having more previous Emergency Department visits; and having fewer hospital admissions.

CONCLUSION: Using a nationally representative dataset we were able to identify which children were most at risk of TBI in their secondary school years. TBI is often preventable and targ […]

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## Delivering psychosocial care and support in traumatic injury follow-up: A qualitative study of clinician's experiences.

*Wake E, Ranse J, Marshall AP*

INTRODUCTION: The purpose of trauma follow-up care after hospital discharge is to support patients with serious injuries recover to their fullest extent. Whilst the physical health of patients is a central focus of follow-up care, psychosocial support is described as challenging and inadequate, with few trauma follow-up services able to provide integrated psychosocial support inclusive of routine access to psychosocial experts. As trauma follow-up care within Australia and New Zealand (Australia and New Zealand) is predominantly provided by nursing and medical trauma clinicians, exploring how psychosocial care and support is provided by these clinicians is warranted. The broad aim of this qualitative study was to explore trauma clinicians' experiences in providing trauma follow-up care with this paper focusing on the provision of psychosocial care and support.

METHODS: Semi-structured interviews were used to explore the experiences of trauma clinicians in providing follow-up care after hospital discharge to patients who have experienced serious injury. Twenty participants were purposively selected from public hospitals throughout Australia and New Zealand who provided trauma follow-up care after hospital discharge. Qualitative data were transcribed verbatim and analysed using inductive content analysis.

RESULTS: Psychosocial care and support provided by trauma service clinici […]

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## Diagnostic criteria and clinical course of inappropriate antidiuresis and cerebral salt wasting syndrome following traumatic brain injury: A retrospective cohort of 351 severe trauma patients.

*Laurent C, Deffontis L, Bouchdoug K, Dagod G, Girard M, Festoc M, Capdevila X, Charbit J*

BACKGROUND: Hyponatremia is common in traumatic brain injury (TBI) population, and is associated with poor outcomes. The main mechanisms are the syndrome of inappropriate antidiuresis (SIAD) and cerebral salt wasting syndrome (CSW). This study aimed to assess hyponatremia prevalence and time of onset in patients with TBI and assess differences between SIAD and CSW.

METHOD: This retrospective cohort study was conducted between 2015 and 2018 in our level 1 trauma center. Patients admitted to the intensive care unit with TBI were included. Three subgroups were determined using urinary clinical and biological criteria: SIAD, CSW, and Undetermined. Predictive factors were assessed for each subgroup, especially the influence of free water, sodium, or fluid intake.

RESULTS: Among 351 trauma patients with TBI, 57 (16 % [95 %CI 12 %20 %]) developed hyponatremia within 30 days. 30 (9 % [95 %CI 6 %12 %]) developed a SIAD, 13 (4 % [95 %CI 2 %6 %]) a CSW, 14 (4 % [95 %CI 2 %6 %]) an undetermined form. CSW subgroup had higher Simplified Acute Physiology Score II (SAPS II), Injury severity score ISS, more cerebral injuries, intracranial pressure monitoring, mechanical ventilation than the SIAD subgroup. Fluid, sodium and free water intake in the 48 h preceding hyponatremia did not influence CSW or Undetermined hyponatremia development, whereas excessive free water intake in the 48 h precedi […]

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### Video review analysis of early common femoral arterial access in trauma: Can we identify occult shock?

*Murali S, Cannon J, Joergensen S, Zone AI, Forsythe L, Brinson M, Chreiman K, Abella BS et al.*

BACKGROUND: Early common femoral artery (CFA) access in trauma resuscitation has the potential to improve hemodynamic monitoring and facilitate interventions. However, data on its utilization and impact are limited. This study aimed to collect objective data on early CFA access.

METHODS: We conducted a prospective observational study using trauma video review at a Level 1 trauma center. Critically injured trauma patients were included based on predefined criteria. Video and chart review were used to collect data on patient demographics, injury characteristics, procedural details of CFA access (including time of procedural milestones), and hemodynamic measurements. Trauma team leaders were also surveyed to determine their impressions on early arterial access.

RESULTS: Among 72 patients, 34.7% underwent early CFA access. Early CFA access was associated with blunt mechanism ($p < 0.001$), lower presenting GCS ($p = 0.006$), and worse functional status at discharge ($p = 0.028$). The median time from arrival to visualizing an arterial waveform was 16.5 min (IQR 13.2, 26.2). Ultrasound improved first-pass success rates (median attempt number: 1 vs 2, $p = 0.030$). Noninvasive systolic blood pressure measurements were, on average, 13.5 mmHg higher than invasive measurements; the discrepancy was more pronounced in patients with arterial SBP < 90 mmHg, and 9 paired measurements revealed occu […]

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### Outcomes and safety of selective non-operative management of penetrating liver injuries: A prospective study.

*Kapnias K, Kivilo M, Saar S, Talving P, Navsaria P*

BACKGROUND: Highly selective nonoperative management (SNOM) of penetrating liver injuries is increasingly implemented, however, still underutilized. This ambivalence is derived from the anxiety that hollow viscus injury may be missed, especially after gunshot wounds (GSWs). As a result, exploratory laparotomy is routinely performed in patients with penetrating abdominal trauma (PAT), frequently resulting in both negative and non-therapeutic laparotomies. The aim of this study was to assess the safety of nonoperative management (NOM) of penetrating liver injuries.

PATIENTS AND METHODS: A prospective study including patients with penetrating liver injuries admitted to a high-volume level I trauma centre was conducted over a 52 - month period (April 2015 - July 2019). Patients suitable for SNOM included: 1) penetrating injury to the right upper quadrant or right thoracoabdominal area 2) absence of peritonism 3) hemodynamic stability 4) GCS > 13 and no spinal cord injury 5) evidence of liver injury and absence of hollow viscus organ injury on CT scanning. Patients presenting with peritonism, hemodynamic instability, evisceration, or unreliable physical examination underwent immediate laparotomy. Primary outcome was failure of NOM and mortality.

RESULTS: Overall, 243 patients had liver injury and met inclusion criteria. Immediate laparotomy (IL) and SNOM groups included 168 (68 %) […]

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## Epidemiology of ranger injuries among members of the Game Rangers Association of Africa: Service-related risk and the call for targeted prevention strategies.

*Viljoen C, van Mechelen W, Sykes L, Clark J, de Bruin L, Mushonga T, Rensburg DCJV*

OBJECTIVES: To determine the epidemiology of ranger injuries, risk factors for injury, and health support needs among members of the Game Rangers Association of Africa (GRAA).

METHODS: We employed a retrospective cross-sectional design to collect data online using a questionnaire developed specifically for this study. We used convenience sampling to include 120 rangers aged 18 years or older who were actively working in Africa.

RESULTS: Our findings showed a career injury prevalence of 52 % and period prevalence (past 12 months) of 33 %. The lower limb was the most frequently injured anatomical region (38 %), followed by the upper limb (31 %) and trunk (19 %), specifically affecting the hand (19 %), ankle (14 %), and foot (13 %). The most common injury types included skin lacerations (29 %), followed by muscle injuries (23 %), and fractures (10 %). Working for more than 15 years as a ranger is associated with an increased risk for career injury (OR=2.63) compared to working for 10 years or less. Injuries resulted in a median of 7 days (IQR=0-20 days) lost from work, with most injuries requiring either emergency medical care (31 %) or consultation with a medical doctor (26 %). Rangers need better equipment, cardiovascular fitness training programs, safety training, and mental health support.

CONCLUSION: One in every two rangers associated with the GRAA sustains an injury acro […]

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## The effectiveness of an early management guideline for older patients following blunt thoracic injury.

*O'Donohoe RB, Mathew J, Winter N, Fitzgerald M, Hendel S, Dockrell L, Lodge M, Hooper A et al.*

BACKGROUND: Thoracic injury is a leading cause of trauma related morbidity and mortality worldwide. Older patients (aged ≥65 years) who have sustained rib fractures following blunt chest wall trauma are at greater risk of serious pulmonary complications, prolonged hospital stay, and death when compared with the general population. The Blunt Chest-Wall Injury and Rib Fracture Pathway (BCWP) was introduced at our institution in July 2021 and advocates for early, aggressive analgesia, specialist support from the Acute Pain Service as well as consideration of regional techniques and Intensive Care Unit admission.

METHODS: A retrospective review of the prospectively maintained Alfred Hospital Trauma Registry identified all patients aged ≥ 65 years admitted to our Level 1 Adult Trauma Centre following blunt chest wall injury between the 1st of January 2019-1st of January 2024. The primary outcome measure was proportion of patients receiving an adequate analgesia (AA) regimen. Secondary outcomes included in-hospital mortality and hospital LOS.

RESULTS: 2488 patients were admitted during the 5-year study period, with 1168 admitted prior and 1320 post BCWP introduction. Mean age was 78.3 (SD 8.4) and 78.8 (SD 8.3) while most patients [57.0 % (n = 666) & 56.3 % (n = 743)] were male. There was a significant difference in the proportion of patients who received AA, with 87.7 % (n = 1158) […]

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## Posterior interosseous artery flap for severe hand injuries: Outcomes of reconstruction combined with local and regional flaps.

*Ertem H, Ad■güzel ■F, Özde■ HU, Aslantürk O, Ertem K*

BACKGROUND: Severe hand injuries with extensive soft-tissue loss present a significant reconstructive challenge. Achieving stable coverage while preserving hand function often necessitates the combined use of regional and local flaps. The posterior interosseous artery (PIA) flap, in combination with regional flaps harvested from non-salvageable digits, may offer an effective solution.

METHODS: A retrospective analysis was performed on patients with severe hand injuries who underwent reconstruction using a PIA flap between 2022 and 2025. Patients were treated with either an isolated PIA flap or a PIA flap combined with local flaps, including fillet flaps harvested from non-salvageable digits or rotational flaps, depending on the extent and location of the defect. Demographic characteristics, injury mechanisms, defect locations, and surgical details were recorded. Postoperative complications and functional outcomes were assessed using fingertip-to-palm distance, Quick DASH score, and VAS for pain.

RESULTS: Fourteen patients with severe hand injuries were included in the study. The mean age was 39.4 years, and the mean follow-up period was 19.7 months. Five patients with complex, multi-site defects underwent combined reconstruction using a PIA flap with local flaps (four with fillet flaps, one with a rotational flap), while nine patients were treated with an isolated PIA flap. S […]

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### Trends and outcomes following diagnostic laparoscopy for blunt abdominal trauma in the United States.

*Atkins KK, Gallaher JR, Schneider AB, Charles AG*

BACKGROUND: The role of diagnostic laparoscopy in adults with blunt abdominal trauma and the effect of negative laparoscopy on mortality is not well delineated.

METHODS: We reviewed the National Trauma Data Bank (2007-2019) for adults sustaining blunt abdominal trauma who underwent operative intervention. We performed a doubly robust, augmented inverse propensity weighted multivariable logistic regression to estimate the effect of a negative diagnostic laparoscopy on mortality in adults with operative blunt abdominal trauma.

RESULTS: 87,864 patients met the inclusion criteria. Diagnostic laparoscopy occurred in 6.6% (n = 5816) of patients, with a 21.1% (n = 1226) conversion to laparotomy rate. The rate of negative diagnostic laparoscopy was 28.6% (n = 1665). Negative laparoscopy patients had a 49% reduction in odds of mortality (OR 0.51, 95%CI 0.47 - 0.56, p < 0.001) compared to negative laparotomy patients. Patient's that underwent laparoscopy, found to have intra-abdominal injury, had a similar reduction in odds of mortality compared to negative laparotomy patients (OR 0.54, 95% CI 0.51 - 0.57, p < 0.001).

CONCLUSION: Diagnostic laparoscopy may be safe for adults with blunt abdominal trauma and prevent significant morbidity and mortality from a negative laparotomy.

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## European Journal of Trauma & Emergency Surgery

Volume 52, Issue 1 · June 2026 · 120 new articles

### Orbital floor fractures: clinical characteristics, mechanisms of injury, and surgical outcomes in a 25-year single-center study.

*Kasap ■, Ayd■n MS, Özler G, Ni■anc■ M, Özvurmaz O*

PURPOSE: Orbital floor fractures are common injuries following blunt midfacial trauma, presenting as isolated defects or as part of complex craniofacial fractures. This study aimed to evaluate clinical characteristics, fracture patterns, management strategies, and postoperative outcomes in surgically treated patients.

METHODS: This retrospective single-center study included 56 patients who underwent surgical treatment for orbital floor fractures between 2000 and 2025. Demographic data, injury mechanisms, imaging findings, associated fractures, clinical presentation, surgical techniques, and outcomes were analyzed. Minimum follow-up was 3 months. Statistical analysis was performed using IBM SPSS Statistics version 25.0, with p < 0.05 considered significant.

RESULTS: Among 56 patients, 80.4% were male, with a mean age of 38.6 years (10-70). Assault (44.6%) was the most common cause, followed by in-vehicle accidents (21.4%) and falls (17.9%). Inferior wall fractures were most frequent (55.4%), followed by medial (28.6%) and lateral (16.1%) involvement; no superior fractures were observed. Common findings included periorbital edema/ecchymosis (75.0%), diplopia (26.8%), and gaze limitation (19.6%); infraorbital hypoesthesia occurred in 12.5%. Associated fractures involved the maxilla (60.7%) and zygoma (46.4%).Surgical treatment included exploration/elevation (46.4%), titanium mes […]

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## Epidemiology and healthcare burden of implant-associated complications in Germany: a nationwide population-based analysis.

*Anees H, Heiss C, El Khassawna T*

PURPOSE: Implant-associated complications remain a major challenge in orthopaedic and trauma surgery, particularly among elderly patients requiring revision procedures or infection management. This study analysed nationwide trends and healthcare burden of implant-associated complications in Germany.

METHODS: Nationwide German hospital discharge data from 2000 to 2023 were analysed and complemented by Diagnosis-Related Group (DRG) data from 2024. Cases were identified using ICD-10-GM code T84 and its subcategories. Outcomes included temporal trends, age-standardised incidence, demographic characteristics, in-hospital mortality, length of stay (LOS), and DRG-based healthcare resource utilisation.

RESULTS: Hospitalisations for implant-associated complications increased from 45,121 in 2000 to a peak of 95,154 in 2011, followed by a decline to 72,846 in 2023. Age-standardised incidence decreased by approximately 31% between 2010 and 2023. Segmented regression identified a temporal breakpoint in 2010.6, with annual percentage changes of + 6.21% before and - 3.55% after the breakpoint. Annual hospitalisations for periprosthetic joint infection increased from 5,367 to 11,215, while osteosynthesis-associated infections increased from 2,469 to 7,059. Mean LOS decreased from 20.8 to 11.8 days, whereas in-hospital mortality remained stable. Compared with low-complexity cases, infection-r […]

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## Effect of periprosthetic greater trochanter fractures (Vancouver AG) on primary stability of the femoral stem in total hip arthroplasty: a biomechanical micromotion analysis.

*Bruder J, Goronzi M, Cavalcanti Kußmaul A, Wulf J, Pachmann F, Holzapfel BM, Böcker W, Kistler M*

PURPOSE: Primary stability is a crucial factor for the long-term success of cementless total hip arthroplasty (THA). Periprosthetic fractures represent an important cause of loss of primary implant stability. While Vancouver AG fractures rarely lead to stem loosening, their effect on the metaphyseal anchoring of the stems has not yet been investigated. The present study aimed to evaluate the influence of a Vancouver AG fracture on the primary stability of the CLS Spotorno stem.

METHODS: Six fourth-generation composite femora were implanted with cementless CLS Spotorno stems (Zimmer, Warsaw, USA). Primary stability was determined by three-dimensional digital image correlation under dynamic loading conditions (300-1700 N, 1 Hz) simulating physiological gait. Micromotion was recorded at six standardized measurement points (proximal, mid, distal, medial and ventral) before and after a simulated Vancouver AG fracture. Data was statistically analyzed using paired one-sided t-tests ($\alpha = 0.05$).

RESULTS: The results revealed no significant increase in micromotion at the primary anchorage zone of the stem. However, a non-significant increase in micromotion was observed at the prosthesis tip (DGventral and DGmedial), where the micromotion values increased from 164.4 µm (± 59.6 µm) and 140.4 µm (± 75.6 µm) to 193.1 µm (± 48.8 µm) and 186.9 µm (± 60.5 µm), respectively.

CONCLUSION: Vanco […]

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## Injury patterns, management and outcomes of hemodynamically unstable pelvic fractures at an Australian level 1 trauma centre: a 10-year registry analysis.

*Louw L, Raubenheimer K, Blakeney WG, Young S, Balogh ZJ, Weber DG*

INTRODUCTION/BACKGROUND: Pelvic ring injuries (PRI) in traumatic shock are associated with over 30% mortality even in high income countries' mature trauma systems, but recent Australian data showed no haemorrhage related mortality in this group. We aimed to describe the epidemiology, management and outcomes of PRI with haemodynamic instability managed in a state's only Level 1 adult trauma centre.

DESIGN: We conducted a retrospective cohort study of all trauma patients with major PRI (Abbreviated Injury Scale (AIS) > 3) and hemodynamic instability (defined as systolic blood pressure (SBP) <90 mmHg or a shock index (heart rate (HR)/SBP) > 0.7) presenting to the state adult trauma centre between 2013 and 2022. Data on demographics, injury patterns, management strategies, and outcomes were collected from the State Trauma

Registry and hospital records.

RESULTS: Of 1369 patients with pelvic or acetabular fractures during the 10-year study period, 152 (11%) had major PRI with hemodynamic instability at presentation. Most were male (72%) with a median age of 35 years. Motor vehicle and motorbike crashes accounted for 52% of injuries. The median Injury Severity Score was 33. Massive transfusion protocols were activated in 45% of cases, while angioembolisation was utilised in 5.9%. Overall in-hospital mortality was 7.2% and mortality due to haemorrhage was 2%.

CONCLUSIONS: In our mat […]

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## Time to surgery and its impact on length of stay and mortality for neck of femur fractures at a major trauma centre.

*Tang LY, Ho SYS, Zhou AK, Geetala R, Krkovic M*

PURPOSE: Neck of femur (NOF) fractures are a major public health concern. Timely surgical intervention (current guidelines is within 36 h) is associated with improved survival, shorter hospital stays, and better functional outcomes. However, surgical delays remain common in hospitals, especially in major trauma centres, and their impact on length of stay (LOS) and mortality is not well defined across different NOF fracture subtypes.

METHODS: We conducted a retrospective cohort study of 4415 patients with NOF fractures, classified into intracapsular (n = 2507) and extracapsular (n = 1908). The primary outcome data gathered were LOS (from surgery to discharge) and in-hospital mortality. Regression analyses were performed to assess associations between LOS, and mortality, adjusting for age, sex, Charlson Comorbidity Index (CCI), American Society of Anesthesiologists (ASA) score, intra-operative blood transfusion, and surgical methods.

RESULTS: For intracapsular fractures, delays in time to surgery were significantly associated with increased LOS in patients expected to have short to median lengths of stay, after adjusting for other variables ($p < 0.05$). Delayed surgery was associated with increased mortality for both intracapsular and extracapsular fractures ($p < 0.0001$). Survival curves showed significant differences for the surgical delayed group. After controlling for time to […]

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## Pattern of blast injuries. A Systematic Review: Part 1-Improvised, multiple and unspecified explosive devices.

*Neubert A, Seidelmann M, Gaeth C, Bieler D*

PURPOSE: This systematic review aims to comprehensively identify and categorize the specific injury patterns associated with IEDs and other explosive devices across military and civilian populations to improve medical preparedness, resource planning strategies, and treatment approaches.

METHODS: A search on PubMed und Scopus was conducted on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion type - Part 1 IED and multiple/unspecified explosives. A narrative data synthesis was performed. Reporting was assessed with the JBI tool for case series.

PROSPERO: CRD42023391736.

RESULTS: 33 studies with 4,373 victims (mainly military personnel) from 2002 - 2022 reported injuries caused by IEDs. The average age was 25.1 years. 97.2% were male. Injuries to the upper and lower extremities were predominant in all study populations followed by head injuries among military personnel, and thorax and abdomen injuries among civilians. 37 studies with 24,600 casualties (mean age 25.7, 85.8% males) were included in multiple/unspecified explosives. Extremities were predominant followed by facial injuries among military personnel, and thorax injuries among civilians.

CONCLUSION: The first part of this systematic review on the pattern of blast injuries shows that injuries to the extremities, includ […]

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## Clinical and laboratory predictors of mortality in Fournier's gangrene: prognostic role of the HALP score.

*Yavuz A, Ersöz V, Çamöz MS, Yurdakul ZN, ■■deci Tut G, Demirci BF, Koçsoy NS, Dönmez M*

OBJECTIVE: Fournier's gangrene (FG) is a rapidly progressive necrotizing soft-tissue infection associated with substantial mortality. This study evaluated the prognostic performance of the Hemoglobin-Albumin-Lymphocyte-Platelet (HALP) score and routinely available clinical and laboratory parameters in patients with FG.

METHODS: This retrospective cohort study included 236 consecutive patients who underwent surgical treatment for FG between February 2019 and May 2025. The primary outcome was in-hospital mortality. Prognostic factors were evaluated using univariate and multivariable logistic regression analyses, and discriminative performance was assessed by receiver operating characteristic (ROC) curve analysis.

RESULTS: The overall in-hospital mortality rate was 12.3%. Non-survivors had significantly lower HALP scores, hemoglobin, hematocrit, albumin, lymphocyte, and monocyte levels (all $p < 0.05$). In univariate analysis, younger age, disease confined to the perianal region, and higher albumin, monocyte, hemoglobin, hematocrit, and HALP values were associated with survival. In the primary multivariable model, advanced age (OR = 0.966, 95% CI: 0.936-0.997; $p = 0.030$) and monocyte count (OR = 5.243, 95% CI: 1.055-26.049; $p = 0.043$) remained independent predictors of survival. Hemoglobin showed the highest discriminative performance (AUC = 0.781), followed by albumin (0.767), he […]

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### Surgical stabilization of rib fractures is associated with lower adjusted odds of adverse pulmonary outcomes across RibScore-defined injury severity: a single-center cohort study.

*Muldiiarov V, Liu JL, Campbell A, Bouldoukian S, Daro A, Kemp SE, Matos M, Cantrell E et al.*

PURPOSE: Surgical stabilization of rib fractures (SSRF) is increasingly used for selected patients with severe chest wall injury, but its relationship with pulmonary outcomes after accounting for RibScore-defined injury severity remains unclear. This study evaluated adverse pulmonary outcomes after SSRF versus nonoperative management (NONOP) among adults with traumatic rib fractures.

METHODS: Single-center retrospective cohort study at an ACS-verified Level I trauma center from January 2016 to April 2023. Adults with CT-confirmed blunt traumatic rib fractures were categorized as SSRF or nonoperative management. RibScore was calculated from initial CT imaging. The primary outcome was a composite of pneumonia, invasive mechanical ventilation for more than 48 h, or tracheostomy. Multivariable logistic regression adjusted for demographics, RibScore, thoracic and extra-thoracic injury severity, calendar year, COPD, and current smoking status.

RESULTS: Among 3,066 patients, 444 underwent SSRF and 2,622 were managed nonoperatively. SSRF patients had greater radiographic chest wall injury severity, including a higher median RibScore than NONOP patients (2 vs. 0; $p < 0.001$). Composite adverse pulmonary outcome occurred in 14.6% of SSRF patients and 16.6% of NONOP patients. In adjusted analysis, SSRF was associated with lower odds of composite adverse pulmonary outcome (aOR 0.57, 95% C […]

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### Physician-staffed prehospital units for severely injured patients in the Stockholm trauma system - is there a survival benefit?

*Helmersson K, Lapidus O, Alm T, Bäckström D, Rubensson Wahlin R*

BACKGROUND: The role of prehospital physicians for patients with major trauma has previously been examined with mixed results. No studies have assessed the survival benefit of physician-staffed prehospital units in the context of the Stockholm trauma system.

AIM: To investigate the association between exposure to prehospital physicians and mortality for severely injured trauma patients in the Stockholm trauma system.

METHODS: A total of 11,063 trauma patients were obtained from the Swedish Trauma Registry and stratified according to physician exposure in the prehospital setting. Logistic regression was used to evaluate the association between prehospital physician exposure and 30-day mortality and Glasgow Outcome Score (GOS).

RESULTS: No significant reduction in 30-day mortality was observed in the P-EMS group compared to the EMS group (OR 1.06, p = 0.71). However, transport with helicopter emergency medical services (HEMS) was independently associated with greater odds of survival. The P-EMS group had significantly lower GOS at discharge compared to EMS (OR 1.72, p < 0.001).

CONCLUSION: The potential benefit of prehospital physician exposure for major trauma patients remains uncertain. However, HEMS transport was associated with improved survival. Further research is needed to determine which patients benefit from prehospital physicians and to evaluate the optimal allocati […]

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## Clinical impact and economic burden of electric scooter-related facial fractures in Barcelona.

*Lugilde-Guerbek A, López-Masramon B, García Sánchez V, Andreu-Sola V, Collado-Delfa JM, Pueyo Morillo C, Arteaga A, Grabosky-Elbaile A et al.*

PURPOSE: The landscape of facial trauma has evolved with the integration of micromobility devices into urban environments. This study aimed to characterize the clinical profile of electric scooter (e-scooter) accidents compared with traditional etiologies (traffic, assaults, falls) and evaluate the economic burden imposed on the public health system.

METHODS: A cohort of 119 patients who underwent surgery for facial fractures at a Spanish tertiary hospital between 2020 and 2024 was analyzed. To accurately assess economic impact, a specific subgroup of 51 "pure facial fractures"-those with isolated injuries-was utilized for cost analysis.

RESULTS: E-scooter users (10.1% of the total cohort) were significantly younger (mean age 27.9 years, p = 0.044), had shorter hospital stays (7.7 days, p = 0.007), and lower ICU requirements (16.7%) than the traditional group. Adverse outcomes accounted for 29.4% of cases overall, with a significantly higher rate in the e-scooter group compared to other etiologies (58.3% vs. 26.2%; p = 0.039). The median cost for an e-scooter injury was €8,092; though higher than other causes, this was not statistically significant (p = 0.33).

CONCLUSION: E-scooters generate complex facial injuries with clinical severity and complication rates comparable to high-energy trauma. The high economic cost and reconstructive complexity, suggest a need for stricter […]

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## Significant other reported outcome measures within European cohorts of severely injured patients: A systematic review.

*Van Ditshuizen JC, Berg LVD, Verhofstad MHJ, Leliefeld PHC, Jongh MAC, Hartog DD*

PURPOSE: Major trauma can have a long-term impact, leading to diminished health-related quality of life (HRQoL) for patients and their significant others (SOs). Patient reported outcome measures (PROMs) make this impact visible, providing expectation- and self-management. European studies assessing Significant Other Reported Outcome Measures (SOROMs) in cohorts of severely injured patients were reviewed.

METHODS: A systematic literature search was conducted. European studies from 2000 onwards assessing SOROMs in severely injured patient cohorts were included. Two reviewers independently screened all records, consensus was reached by a third reviewer.

RESULTS: 2,909 studies were screened, resulting in 30 inclusions. Studies originated from Western and Northern Europe, overrepresented by traumatic brain injury (n = 27, 90%). Forty-three different SOROMs were identified. Twenty (67%) studies reported subacute and long-term reduction in HRQoL, and high prevalence (28-56%) of depression, anxiety, and stress within SOs. Outcomes of SOs and patients run parallel and are associated with coping strategies. Sixteen (53%) studies reported around 50% of unmet family needs (information and emotional support), and a disproportionate burden of care associated with loneliness and life satisfaction.

CONCLUSION: SOs of severely injured patients reported reduced HRQoL, with high levels of depr […]

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## Early treatment goal uncertainty as a barrier to regional analgesia and respiratory deterioration in traumatic rib fractures: a propensity-weighted analysis.

*Yi SB, Wu JH, Zhang XN, Lin DC, Zheng YL*

PURPOSE: Patients with multiple rib fractures (≥ 3) admitted to the intensive care unit (ICU) are at high risk of early respiratory deterioration, for which timely and effective analgesia is essential. Although multimodal analgesic strategies are widely recommended, their real-world implementation may be influenced by variability in early clinical decision-making, including treatment goal uncertainty (TGU). We aimed to investigate the association between early TGU, analgesic strategy selection, and respiratory outcomes in critically injured patients with rib fractures.

METHODS: This single-center observational cohort study included 236 adult ICU patients with ≥ 3 radiologically confirmed rib fractures who did not require invasive mechanical ventilation at admission (2020-2024). TGU within the first 48 h was assessed using blinded electronic medical record review as a process-level marker of variability in early care prioritization. Analgesic strategies were categorized as regional-based analgesia (RA), systemic-dominant analgesia (SA), or multimodal conservative management. The primary outcome was respiratory deterioration, defined as a composite of escalation of respiratory support, endotracheal intubation, or pulmonary complications requiring intervention. Associations were evaluated using propensity score overlap weighting and adjusted regression models.

RESULTS: High TGU […]

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## Multifocal necrotizing fasciitis: what makes it different? A German trauma center experience.

*Keßler F, Frank J, Störmann P, Wagner N, Saman AE, Marzi I, Leiblein M*

BACKGROUND: Multifocal necrotizing fasciitis (MulNF) is a rare presentation of necrotizing fasciitis (NF) characterized by involvement of more than one non-contiguous anatomical site during the same disease episode. Comparative data between monofocal and multifocal NF remain limited.

METHODS: We performed a retrospective single-center cohort study of patients treated for NF between 2015 and 2026. Patients were classified as MNF (single site) or MulNF (multiple non-contiguous sites) based on chart review. Demographics, comorbidities, microbiology, laboratory values, organ failure, ICU course, surgical burden, and outcomes were compared.

RESULTS: Forty-seven patients were included, of whom five (10.6%) had multifocal NF. Four of these showed clearly synchronous non-contiguous involvement, whereas one developed a delayed second non-contiguous focus within the first four days of the same disease episode. Compared with MNF, MulNF was associated with higher ICU burden, more frequent mechanical ventilation, and a greater number of debridements and total operations. Higher rates of septic shock, hemodynamic failure, and renal replacement therapy were observed in MulNF. No deaths occurred in the small multifocal cohort, whereas 11/42 patients with monofocal NF died during hospitalization. Given the small multifocal sample size and multiple comparisons performed, all findings should be […]

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## Trauma-related retrobulbar hematoma: predictive risk factors and clinical outcome in 52 cases.

*Bicsák Á, ■ahin H, Brandt L, Hassfeld S, Bonitz L*

PURPOSE: Retrobulbar hematoma (RBH) is a vision-threatening emergency. This large-scale study integrates clinical data and 3D analysis to identify prognostic factors, refine diagnosis, and support guideline-based emergency management.

METHODS: This study standardizes terminology, applies AO fracture classification, and uses CT-based 3D segmentation to analyze anatomical patterns, treatment protocols, providing evidence-based guidance for diagnosis, surgical management, and postoperative care.

RESULTS: The Dortmund Maxillofacial Trauma Registry (2018-2025) recorded 52 RBHs among 7,671 trauma patients (0.7%). RBH occurred predominantly in middle-aged males and elderly females, with overproportionate incidence of anticoagulant therapy. Most fractures clustered around the orbital floor, medial wall, zygomatic surface, often near the Frankfurt horizontal plane. RBHs were mainly extraconal; poor outcome was predicted with intraconal location, optic nerve contact, orbital floor fracture, high RBH/orbital volume ratio, and aspirin use. Surgical decompression was performed in 15 cases (28.9%); 65,4% (34 of 52) retained vision. Neural network analysis confirmed fracture site and hematoma position as key prognostic factors over demographic or comorbidity variables.

CONCLUSION: In 7,671 midface fracture cases, RBHs occurred in 0.7% (n = 52), mainly after trauma. Falls predominated in ae […]

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## Patterns of negative trauma laparotomy worldwide: a sub-analysis of a global dataset.

*Laird WLA, Bath MF, Amoako J, Wohlgemut J, Nuño-Guzmán CM, Khajanchi M, Smith BG, Hobbs L et al.*

PURPOSE: Trauma remains a major contributor to global disability. While the trauma laparotomy can be a life-saving procedure in abdominal injury, a select proportion are performed with no pathology identified intra-operatively, potentially exposing patients to unnecessary morbidity. We aimed to assess the global prevalence of negative laparotomy and identify any influencing factors at both the patient- and system-level.

METHODS: This was a secondary analysis of the GOAL-Trauma study, a multicentre prospective international observational study on trauma laparotomy patients, conducted from April to December 2024. We defined a negative laparotomy as where no intra-abdominal injuries were identified during the index operation. A multivariable logistic regression was performed to identify predictive factors for a negative laparotomy.

RESULTS: Of the 1769 patients included, 128 patients (7.2%) underwent a negative laparotomy. Regression analysis demonstrated that a penetrating mechanism of injury (OR 2.37, CI:1.53-3.69, $p < 0.001$) and a lower grade of surgeon (surgical registrar relative to consultant - OR 2.91, CI:1.90-4.47, $p < 0.001$) were independent predictors for negative laparotomy. Neither human development index nor hospital resource level impacted negative laparotomy rates.

CONCLUSION: One in every fourteen trauma laparotomies performed in our global cohort was negative, […]

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## Preoperative stay under 48 h after admission for proximal femur fractures is linked to fewer complications and better outcomes.

*Moskal M, Feulner S, Grimm A, Müller-Heck R, Heep H, Godolias P*

PURPOSE: To evaluate the association between admission-to-surgery time (with a focus on < 48 h) and postoperative complications and discharge mobility in patients with proximal femur fractures (PFFs), using a large administrative dataset from routine inpatient care in Germany.

METHODS: In this retrospective database analysis, 128,092 patient records from external inpatient quality assurance (North Rhine-Westphalia, Germany, 2016-2020) were included. All patients underwent surgical treatment for PFFs and were stratified by time to surgery as early surgery (Group A; within 48 h of admission) or delayed surgery (Group B; more than 48 h after admission). Two cohorts were analysed separately: hip arthroplasty (HA: early 57,635; delayed 7,504) and osteosynthesis (OST: early 59,466; delayed 3,487). Associations between admission-to-surgery time, postoperative complications, pre-admission anticoagulant therapy and inability to walk at discharge were assessed using descriptive statistics and binary logistic regression. Patients unable to walk before the fracture were excluded from the mobility analysis. Reporting was guided by the STROBE statement.

RESULTS: In-hospital mortality was 4.7% after early versus 9.7% after delayed surgery (OST) and 5.8% versus 9.3% (HA). Delayed surgery was associated with higher odds of postoperative complications (OST: OR 1.21, 95% CI 1.11-1.34; HA: OR 1. […]

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## Impact of routine use of the GFAP and UCH-L1 TBI blood test for management of mild TBI on patient care in a European emergency department.

*Brüning-Wolter F, Wolff N, Schrader M, Musaelyan K, Chandran R, Hoffmann M, Marino JA, McQuiston B et al.*

OBJECTIVES: Blood biomarkers glial fibrillary acidic protein (GFAP) and ubiquitin C-terminal hydrolase L1 (UCH-L1) were recently introduced into clinical practice for mild traumatic brain injury (mTBI) assessment. However, real world data demonstrating TBI biomarkers' impact on the emergency department (ED) practice is lacking. This study aimed to describe the experience of a single tertiary hospital ED that introduced TBI biomarkers GFAP and UCH-L1 measurements using Alinity i® TBI test into routine mTBI assessment.

METHODS: Retrospective data were extracted from hospital records of patients who had TBI test ordered in the ED. Test results and subsequent clinical care pathway data (CT order, CT scan result, admission or discharge) were analysed.

RESULTS: Data was available for 460 mTBI patients (median age 43, 54% male). Of the 216 patients (47%) with a negative TBI test, 198 did not have a CT scan, and 18 had a CT scan with no abnormalities detected. Among 244 patients with a positive TBI test, 162 had CT scans, with 12 (6.6%) showing intracranial lesions. Overall, among all 180 scanned patients every patient with intracranial lesion findings had a positive TBI test result. The test could rule out CT scans in 47% of mild TBI patients. In practice, 42% of patients avoided a CT scan based on the TBI test results with additional 12% of patients ruled out based on clinical asse […]

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## Electric bicycle versus conventional bicycle crashes: comparative analysis of injury patterns, severity, and the modifying effect of body mass index.

*Bauer D, Maier JP, Holzfurtner L, Pietsch C, Wagner FC, Schmal H, Erdle B, Mühlenfeld N*

PURPOSE: E-bike trauma is often presumed to result in greater injury severity than conventional cycling, yet evidence remains inconsistent. Furthermore, the impact of factors such as Body Mass Index (BMI) on injury patterns is poorly understood. This study therefore evaluated whether e-bike trauma is associated with increased overall severity or distinct regional injury patterns.

METHODS: This retrospective single-center cohort study analyzed 261 adult trauma-bay patients (e-bike n = 37; conventional bicycle n = 224). Primary endpoints included Injury Severity Score (ISS) and regional maximum Abbreviated Injury Scale (AIS) scores. Secondary endpoints included physiological burden (SAPS II) and resource utilization (LOS, TISS-10). Multivariate logistic regression was used to identify independent predictors of intracranial hemorrhage, e-bike riding and extremity fractures. Event counts and events-per-variable (EPV) ratios are reported for each model.

RESULTS: E-bike riders had a significantly higher BMI than conventional cyclists (26.1 vs. 24.6 kg/m²; p = 0.029). However, the absolute difference of approximately 1.5 kg/m² is modest, and both group medians fall within the normal-to-overweight range. No statistically significant differences were observed in ISS, physiological burden, or resource utilization. Conventional cyclists had a higher proportion of patients with AIS Pelvi […]

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## Effect of management in intermediate care unit versus conventional hospitalization on long-term mortality and autonomy in patients with rib fractures aged 75 years and older: a retrospective study.

*Vallod E, Debord-Peguet S, Mikail N, Termoz A, Tripoz S, Abrard S, Gueret-du-Manoir M*

PURPOSE: Rib fractures in older adults are associated with substantial morbidity and mortality. Management in an intermediate care unit (IMCU) may improve outcomes, but its long-term benefit remains uncertain. We compared IMCU versus conventional management regarding 12-month mortality and autonomy in patients aged ≥ 75 years with rib fractures.

METHODS: We conducted a retrospective observational cohort study in two Lyon teaching hospitals (Hôpital Edouard Herriot and Hôpital Lyon Sud, Hospices Civils de Lyon, France) between January 2015 and December 2019. The primary outcome was a composite of death or institutionalization within 12 months of hospital admission. Secondary outcomes included discharge modality, opioid prescription at discharge, hospital length of stay, unplanned readmission within 6 months, and ADL/IADL scores. Logistic regression was adjusted for hospitalization unit, age, sex, pre-existing institutionalization, and associated injuries.

RESULTS: A total of 196 patients were included: 126 (64%) in the conventional group and 70 (36%) in the IMCU group. Death or institutionalization within 12 months occurred in 31 patients (26%) in the conventional group and 8 (12%) in the IMCU group (p = 0.026). However, after adjustment, conventional hospitalization was not significantly associated with the primary outcome (OR 2.25, 95% CI 0.90-6.09; p = 0.092). Institutional […]

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### The road to the Pearl-String Technique: lessons learned from early IMT modifications using RIA autograft and cancellous allograft granules.

*Hückstädt M, Scholl C, Langwald S, Klauke F, Mendel T, Kobbe P, Fischer C*

BACKGROUND: The induced membrane technique (IMT) is widely used for reconstruction of large bone defects; however, reported outcomes vary considerably among technical modifications. One potential mechanism impairing consolidation is graft sedimentation, particularly when low-density cancellous autograft harvested using the reamer-irrigator-aspirator (RIA) is applied. This study evaluates a standardized modification of the IMT and explores mechanical factors potentially affecting graft consolidation.

METHODS: In this retrospective single-center study, 28 patients were treated following eradication of osteomyelitis using a defined modification of the IMT. Mean follow-up was 7.4 years. Defect length ranged from 3.2 to 15.7 cm. During the second stage, RIA-harvested autologous cancellous bone was combined with lyophilized cancellous bone allograft granules. Definitive fixation was achieved using intramedullary nailing or locking plate fixation; one patient was treated with a ring fixator.

RESULTS: Primary consolidation was achieved in 57.1% of cases. Aseptic nonunion occurred in 10.7%, with secondary consolidation achieved after revision in all affected patients. Recurrent infection was observed in 35.7%; infection control was achieved in 90% of these cases following additional procedures, and reconstruction was ultimately completed using alternative strategies. Median time to fu […]

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### Non-tourniquet strategy with high-position low-perfusion in distal humeral fracture surgery: reduces swelling without increasing blood loss.

*Wang Z, Ai X, Ren T, Wu B, Zhang K, Li Z, Xue H, Li M et al.*

INTRODUCTION: To evaluate the feasibility of non-tourniquet strategy combined with "High-position Low-Perfusion" positioning in distal humeral fractures ORIF (DHF-ORIF), and to investigate its effects on perioperative blood loss, postoperative rehabilitation, and functional outcomes.

METHODS: Prospective inclusion of patients aged 18-45 with distal humeral fractures, randomly divided into an experimental group without tourniquets (Group A) and a control group with traditional tourniquets (Group B). The Group A adopted a lateral position with high and low perfusion of the affected limb, and received intravenous injection of TXA (15 mg/kg) combined with continuous flushing with 4 ■ physiological saline during the operation; The control group received routine use of tourniquets.

PRIMARY OUTCOMES: perioperative blood loss, elbow ROM and degree of limb swelling.

SECONDARY OUTCOMES: Postoperative VAS score, Mayo elbow joint score and DASH score, and incidence of complications.

RESULTS: No significant difference was observed in total perioperative blood loss between the Group A and the Group B (240.78 mL vs. 228.53 mL, p > 0.05). However, the Group A demonstrated significantly reduced postoperative drainage volume and hidden blood loss (p < 0.05). The Group A exhibited significantly superior outcomes in terms of VAS scores and limb swelling, compared with the control group. Additi […]

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### Impacts of intracranial pressure monitoring on mortality and length of stay in severe traumatic brain injury: a systematic review and meta-analysis.

*Merakis M, Kim N, Velli G, Balogh ZJ*

PURPOSE: This systematic review and meta-analysis examined the association between intracranial pressure monitoring (ICPm) and mortality in patients with severe traumatic brain injury (sTBI), with additional focus on functional neurological outcomes and intensive care unit (ICU) and hospital length of stay (LOS).

METHODS: This review followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was registered with PROSPERO (CRD42025643607). MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials were searched through August 1, 2025, for studies comparing ICP-monitored and non-ICP-monitored patients with traumatic brain injury. Eligible study designs included randomized controlled trials, prospective observational cohort studies, retrospective observational cohort studies, registry-based cohort studies, and case-control studies. Study quality was assessed using the Strengthening the Reporting of Observational Studies in Epidemiology checklist, Risk of Bias 2.0 for randomized trials, and the Oxford Centre for Evidence-Based Medicine Levels of Evidence framework.

RESULTS: Thirty-one studies met inclusion criteria. The mortality meta-analysis included 121,799 patients. Pooled crude analysis showed lower mortality in ICP-monitored patients than in non-monitored patients (OR 0.79, 95% CI 0.68-0.91, p = 0.002), with high heteroge […]

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## Prehospital point-of-care lactate for identification of severe injury in trauma team-activated patients: a prospective multicenter cohort study.

*Faul P, Hagebusch P, Bertram E, Naujoks F, Störmann P, Schweigkofler U, Koch D*

BACKGROUND: Accurate prehospital trauma triage remains challenging despite structured activation criteria, particularly in patients without overt physiologic instability. Prehospital point-of-care lactate has been proposed as a physiologic adjunct to support early identification of severely injured patients. This prospective multicenter cohort study evaluated the association between prehospital lactate and injury severity in trauma team-activated patients within a structured guideline-based trauma system.

METHODS: This prospective observational multicenter cohort study included adult trauma patients transported by emergency medical services to three Level I trauma centers and managed in the trauma room following trauma team activation (TTA). Prehospital lactate was measured by capillary earlobe sampling before or during transport and was not used for clinical decision-making. The primary outcome was severe injury, defined as Injury Severity Score (ISS) ≥16. Secondary outcomes were Abbreviated Injury Scale (AIS) ≥4 in at least one body region and intensive care unit (ICU) admission > 24 hours. Diagnostic performance was assessed using receiver operating characteristic (ROC) analysis and predefined lactate thresholds of ≥2 mmol/L and ≥4 mmol/L.

RESULTS: A total of 108 patients were included, of whom 26 (24.1%) had ISS ≥ 16. Prehospital lactate levels were higher in severely inj […]

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## Surgical timing and mortality in acute traumatic subdural hematoma: is the four-hour threshold confounded by injury severity?

*Rakip U, Y■ld■zhan S, Canbek ■, Boyac■ MG, Korkmaz S, Öztürk KH, Aslan MZ, Aslan A*

PURPOSE: Early surgical evacuation within four hours has long been advocated for acute traumatic subdural hematoma (ASDH), yet the supporting evidence is largely observational and unadjusted for injury severity. We examined whether surgical timing is independently associated with in-hospital mortality after accounting for baseline severity.

METHODS: We retrospectively analyzed 91 consecutive patients who underwent surgical evacuation for ASDH between January 2014 and December 2024. The primary outcome was in-hospital mortality; the secondary outcome was functional status at discharge measured by the Glasgow Outcome Scale (GOS). Surgical timing (≤ 4 vs. >4 h) was assessed through unadjusted comparison, stratification by Glasgow Coma Scale (GCS) score, multivariable Firth penalized logistic regression with 1000-iteration bootstrap internal validation, and sensitivity analyses including alternative timing thresholds (1-12 h), restricted cubic spline modeling, and timing-by-severity interaction testing.

RESULTS: In-hospital mortality was 33.0% (30/91; 95% Wilson CI 24.2-43.1%). Crude analysis showed higher mortality with early surgery (50.0% vs. 15.6%, p = 0.001), but these patients had lower admission GCS and higher rates of CT-defined traumatic axonal injury (TAI) and cerebral contusion. After stratification, no statistically detectable mortality difference was observed within […]

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### Association between time to transfusion and outcomes in patients with traumatic haemorrhagic shock: an analysis of a nationwide trauma registry in Japan.

*Shibahashi K, Inoue K, Kato T, Sugiyama K*

PURPOSE: To investigate the association of in-hospital time to transfusion on outcomes of patients with traumatic haemorrhagic shock.

METHODS: Using data from the Japan Trauma Data Bank (January 2019-December 2023), we identified adult patients (aged ≥ 18 years) with haemorrhagic shock following trauma (systolic blood pressure < 90 mmHg; shock index ≥ 1.0 upon hospital arrival) and received ≥ 10 units red blood cells within 24 h of injury. We excluded patients transferred from or to other hospitals after emergency treatment, with burn injuries or cardiac arrest, without transfusion time records, and with first transfusion > 60 min after hospital arrival. Mixed-effect multivariable logistic regression estimated the association between time to transfusion for in-hospital mortality, adjusting for patient characteristics, comorbidities, and injury severity. Heterogeneity in the association between transfusion timing and mortality was assessed through nine subgroup analyses.

RESULTS: Overall, 698 patients (median age 59 years, 66.2% male) from 132 hospitals were analysed, of whom 265 (38.0%) died during hospitalisation. Mean time to transfusion varied from 4 to 58 min across hospitals. A positive association was observed between time to transfusion and in-hospital mortality. Each 1-min delay in transfusion initiation was associated with higher in-hospital mortality (adjusted odds […]

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### Neutrophil phenotyping reflects injury severity when base excess and lactate remain normal.

*van Eerten FJC, Duebel LV, de Fraiture EJ, Brands SR, van Wessem KJP, Vrisekoop N, Venhuizen JH, de Groot MCH et al.*

INTRODUCTION: Accurate assessment of injury severity is essential for optimal trauma care, as it dictates timing and intensity of interventional strategy. Conventional markers, such as base excess and lactate levels are usually borderline deranged in patients with midrange injury severity. This study explored whether determining circulating neutrophil phenotypes in these patients (ISS 9-35) might aid in the assessment of injury severity.

METHODS: A descriptive single-center cohort study of trauma patients ≥ 16 years between 2013 and 2023 was conducted. Data was collected from the Dutch National Trauma Registry and the Utrecht Patient-Oriented Database (< 10% missing values). Base excess, lactate levels and neutrophil phenotypes were routinely measured in emergency department blood samples. The dynamics of these parameters was analyzed in the context of increasing Injury Severity Scores (ISS).

RESULTS: 807 patients were included, of whom 591 were classified as midrange severely injured. In these patients, banded neutrophils showed a greater fold increase between ISS 9 and 35 compared with base excess and lactate (6.7, 2.7, and 1.5, respectively). Furthermore, the number of banded neutrophils exceeded the upper interquartile range of controls at lower trauma severities compared to base excess or lactate (ISS 17 vs. 30 and 39, respectively).

DISCUSSION: Although base excess and […]

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### Limb occlusion pressure-based individualized tourniquet pressure management in patients undergoing internal fixation for extremity fractures: a prospective comparative feasibility study.

*Yang J, Hu P, Wang L*

OBJECTIVE: This study aimed to evaluate the feasibility of limb occlusion pressure (LOP)-based individualized tourniquet pressure management and to quantify the reduction in applied tourniquet pressure compared with a traditional fixed-pressure protocol in patients undergoing internal fixation for extremity fractures.

METHODS: A prospective comparative study was conducted at Civil Aviation Shanghai Hospital from January to September 2025. Patients undergoing internal fixation for extremity fractures were enrolled and allocated to either the LOP-based group (n = 171) or the traditional fixed-pressure group (n = 235) according to the attending surgical team and operative scheduling, without randomization. The LOP-based group received individualized tourniquet pressure based on preoperative LOP measurement plus a safety margin, while the control group received standardized pressures (40-50 kPa; 300-375 mmHg). The primary outcome was applied tourniquet inflation pressure. Secondary outcomes included demographic characteristics, extremity location, operative duration, and subgroup analyses stratified by extremity location, age, and sex; multivariable linear regression was performed to adjust for potential confounders.

RESULTS: The LOP-based group demonstrated significantly lower mean tourniquet pressure compared with the traditional group (32.82 +/- 3.63 kPa [246 +/- 27 mmHg] vers […]

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## Impairment in activities of daily living and textbook outcome after major emergency abdominal surgery.

*Mima K, Nitta H, Shikiji Y, Nakashima R, Uemura S, Morinaga T, Itoyama R, Umezaki N et al.*

PURPOSE: Textbook Outcome (TO) has been a useful tool to assess benchmark patient-centred surgical outcomes in elective abdominal surgery. This study aimed to evaluate the association of preoperative activities of daily living (ADL) status with TOs following major emergency abdominal surgery.

METHODS: This retrospective study included 301 consecutive patients who underwent major emergency abdominal surgery for high-risk acute abdominal conditions, including perforated peptic ulcer, intestinal ischemia, bowel obstruction, bowel perforation, and severe intraabdominal bleeding or infection, at Kumamoto Regional Medical Center between April 2019 and March 2025. Preoperative ADL was evaluated using the Barthel Index (range, 0 to 100), with higher scores indicating greater independence. Barthel Index ≤ 85 was defined as ADL impairment. According to a recent international consensus, TO was defined as discharge to "home" without major postoperative complications (Clavien-Dindo grade ≥ III). Multivariable logistic regression analysis was conducted to assess the association of preoperative ADL with TO achievement, adjusting for clinical features and surgery-related factors.

RESULTS: Among the 301 patients, 90 (30%) had preoperative ADL impairment (Barthel Index ≤ 85) and 211 (70%) achieved TO. Multivariable analysis revealed that preoperative ADL impairment was independently associated […]

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## Early calcium replacement after minimal blood transfusion is associated with adverse outcomes.

*Hewgley WP, Griffin RL, Baird E, Hashmi ZG, Baker SJ, Jansen JO, Kerby J, Holcomb JB et al.*

PURPOSE: Calcium disturbances are common in patients with traumatic injuries, exacerbated by shock and blood product resuscitation. Recently, empiric calcium supplementation has been recommended with the first unit of blood transfusion, despite a lack of data supporting this practice. We hypothesize that there are risks associated with calcium administration in patients receiving small volume transfusion.

METHODS: This is a retrospective review of patients admitted to our Level 1 trauma center from 2018 to 2024 who received 1-3 units of red blood cells (RBCs). Patients were stratified by calcium levels and whether they received early calcium supplementation. A logistic regression was used to estimate odds ratios for the association between calcium levels, calcium supplementation, and outcomes. The primary outcome was in-hospital mortality. Secondary outcomes included acute kidney injury, myocardial infarction, acute respiratory distress syndrome, and venous thromboembolism. The model was adjusted for age, sex, ISS and shock index.

RESULTS: A total of 2,680 patients were included. When stratified by calcium levels and supplementation status, patients were similar in terms of age, sex, mechanism and injury severity score, but shock index was higher among patients receiving calcium supplementation. After adjustment, hypocalcemic patients who received calcium supplementation had […]

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### Peripheral perfusion index as a predictor of severe shock in trauma patients: a prospective observational study.

*Sundar H, Purushothaman V, Mathuram SR, Chandy GM, Sharma SL, Nayak S, James JD*

BACKGROUND: Early recognition of haemorrhagic shock is critical in trauma care, particularly in low- and middle-income countries where delayed referral is common. Conventional vital signs may remain normal despite ongoing hypoperfusion. The Perfusion Index (PI), derived from pulse oximetry, is a non-invasive and continuously available marker of peripheral perfusion. This study evaluated the ability of PI to identify severe haemorrhagic shock in trauma patients and compared its performance with the Shock Index (SI) and Modified Shock Index (mSI).

METHODS: This prospective, single-centre study was conducted at a Level 1 trauma centre between January and May 2024. Adult Priority 1 trauma patients were enrolled and classified as having no/mild shock (ATLS grades 1-2) or severe shock (grades 3-4). PI was recorded during initial assessment. SI, mSI, lactate, injury severity, transfusion requirement, ICU admission, and 24-hour mortality were analysed. Receiver operating characteristic (ROC) analysis and multivariable logistic regression were performed.

RESULTS: A total of 120 patients were included (mean age 40.4 ± 17.2 years; 90% male), predominantly with blunt trauma (85% road traffic injuries). Severe shock was present in 35 patients (29.2%). Median PI was significantly lower in severe shock compared with no/mild shock (0.54 [IQR 0.28-1.31] vs. 1.33 [0.71-2.16], p = 0.008). Mean […]

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### A prospective study on abdominal trauma: patterns of injury, clinical presentation, organ involvement, associated injuries, and outcomes at a tertiary hospital in Mogadishu, Somalia.

*Mohamud AA, Hassan AI, Ali AK, Jubur AA, Ahmed MR, Mohamed SS, Mohamed MM, Bashir AM*

BACKGROUND: Trauma is a major global public health problem and one of the leading causes of death and disability worldwide, particularly among individuals younger than 44 years of age. Approximately 6 million people die each year as a result of traumatic injuries, with nearly 90% of these deaths occurring in low- and middle-income countries. Previous epidemiological studies have reported postoperative complication rates ranging from 18% to 42% and mortality rates between 8.5% and 13%.

OBJECTIVE: To evaluate the patterns of injury, clinical presentation, organ involvement, associated injuries, management strategies, and outcomes among adult patients with abdominal trauma in Mogadishu, Somalia.

METHODS: A prospective observational study was conducted from August 2023 to July 2024 at Mogadishu Somali-Türkiye Recep Tayyip Erdo■an Training and Research Hospital. A total of 146 adult patients with abdominal trauma were enrolled. Data were collected using a pretested structured questionnaire and analyzed using Stata version 16. Univariable and multivariable logistic regression analyses were performed to identify factors independently associated with in-hospital mortality.

RESULTS: The study population consisted predominantly of young adult males, with a mean age of 30.18 ± 9.2 years, and males accounted for 97.3% of all patients. Penetrating trauma was the predominant mechanism of […]

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### Comparison of fluoroscopic versus electromagnetic distal locking in humeral shaft osteosynthesis using intramedullary nail: a retrospective study.

*Madeja R, Szeliga J, Voves J, Pometlova J, Konde A, Janosek J, Dousa P*

PURPOSE: Intramedullary nailing is a standard method for osteosynthesis of diaphyseal humeral fractures. Distal locking, however, typically requires fluoroscopic guidance, which increases radiation exposure. This study aimed to compare standard fluoroscopy-guided distal locking with the Ezy-Aim system (Austofix, Australia) based on electromagnetic pulse detection from the perspectives of surgical times, radiation exposure, and complications.

METHODS: A retrospective study included 58 patients treated for diaphyseal humeral fractures (AO/OTA 12A, 12B) between 2019 and 2023. Group A (n = 28) received osteosynthesis using the Ezy-Aim system; Group B (n =

30) underwent fluoroscopy-guided distal locking. Data on operative time, fluoroscopy time, radiation dose, and complications were analysed.

RESULTS: In all, distal locking was not successful at the first attempt in 14.3% (4/28) of screws placed with Ezy-Aim and 20.0% (6/30) when fluoroscopy was used. Time and dose analyses were performed on the remaining 48 patients. The total surgical time was significantly longer with the Ezy-Aim system (median 58 min) compared to fluoroscopy (median 52 min, p = 0.034), primarily due to the time required for the device assembly. However, distal screw placement was faster with Ezy-Aim (median 12 min vs. 14 min, p = 0.019). Radiation exposure was significantly lower in the Ezy-Aim group (median […]

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### Patterns of patient worry following major emergency abdominal surgery: a 180-day follow-up study.

*Parner MK, Jensen LR, Gamst-Jensen H, Burcharth J, Kokotovic D*

PURPOSE: Major emergency abdominal surgery is associated with high postoperative morbidity and mortality. Beyond physical complications, patients experience psychological distress. This study aimed to describe the distribution of patient-reported worry across independent cross-sectional samples and its association with Days Alive and out of Hospital (DAOH) over 180 days postoperatively.

METHODS: Prospective cohort at Copenhagen University Hospital Herlev (March 30, 2023-March 31, 2024). Degree of Worry (DOW) was surveyed at discharge and postoperative day (POD) 30, 90, and 180, categorized as physical, mental, and physical-generic domains of worry. DAOH was computed cumulatively and incrementally for 0-30, 30-90, and 90-180 days. Associations between high DOW (hDOW) and subsequent DAOH were tested with Mann-Whitney U.

RESULTS: hDOW was reported by 36.1% at discharge, 22.4% at POD 30, 45.0% at POD 90, and 32.0% at POD 180. Across 180 days, the most frequent worries were relapse (21.3%), disease (14.1%), functional level (12.0%) and daily living (9.1%). The proportion of physical worries was significantly higher at POD 30 than at discharge and significantly lower at POD 180; mental worries varied between time points without statistically significant differences. hDOW at POD 30 was associated with fewer cumulative DAOH by POD 90 (p = 0.008).

CONCLUSION: Patient-reported worry f […]

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### Outcomes and complications of metallic vs. bioabsorbable screw fixation for displaced medial epicondylar fractures in pediatric and adolescent patients.

*Jhala T, Scherer S, Esser M, Lieber J, Fuchs J, Dietzel M*

PURPOSE: The use of bioabsorbable screws in orthopedics and trauma surgery is increasing. The major advantage of not having to remove the implant is of particular interest in pediatric surgery. Due to drawbacks, most notably implant fracture, bioabsorbable screw osteosynthesis has not yet surpassed the gold standard of metallic screw osteosynthesis. This single-centre study compares metallic screw and bioabsorbable screw osteosynthesis of medial humeral epicondyle fractures in children and adolescents over a two-year period.

MATERIALS AND METHODS: Retrospective analysis of surgically treated (screw fixation with either bioabsorbable magnesium-based or metallic screw) patients with a medial humeral epicondyle fracture between 2020 and 2022 at the author's institution.

RESULTS: Seventeen patients with medial epicondyle fractures underwent surgical treatment using either bioabsorbable screws (n = 8) or metallic screws (n = 9). Baseline characteristics, surgical approach, and postoperative care were similar between groups. Two complications (Clavien-Dindo I) occurred in the bioabsorbable group due to implant failure (screw breakage), though without clinical sequelae. These implant failures led to a significantly increased number of postoperative radiographs in the bioabsorbable group (median 4.8 vs. 2.5; p = 0.014). Functional outcomes (LOM, MEPI) and follow-up durations were com […]

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### Characteristics and temporal trends of mobility scooter accidents in the Netherlands: a multicenter cohort study.

*Stensen Z, Peters A, Keereweer D, Krul I, van Osch F, de Groot B, Barten D*

PURPOSE: Mobility scooters are increasingly used by older individuals. The aim of this study was to assess the characteristics and temporal trends of mobility scooter accidents (MSAs) in the Netherlands over a 10-year period using a representative dataset derived from fourteen Emergency Departments (EDs).

METHODS: In this retrospective multicenter cohort study, we analyzed MSA-related ED visits between January 2014 and December 2023, using the Dutch Injury Surveillance System (DISS). Information about age, sex, hospital admission, type and location of injury, mechanism of injury, and operational errors was analyzed. Injury severity was classified using the maximum abbreviated injury score (MAIS). Trend analyses were only performed for ED visits of patients with MAIS2+ injuries.

RESULTS: In total, 2,702 MSA-related ED visits were identified. Forty-nine percent of patients was male and the median age was 75 years (IQR 64.0-83.0). The most common mechanism of injury was a mobility scooter rollover (34.2%). From 2014 to 2023, the number of MSA-related ED visits of patients with MAIS2+ injuries increased by 38.0%. In the 70-79 age group, this rise was 91%. The total number of traumatic brain injuries (TBI) rose steeply bij 103.0%, which includes a modest (+34.0%) increase in moderate-severe TBI. The hospital admission rate was 30.1%, with a median length of stay of 7 days (IQR 3.0 […]

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## "Is the effect of intraoperative 3D imaging beneficial in plate fixation of distal radius fractures: a matched-pair analysis".

*Zumsteg JS, Pieringer A, Hinkelmann MA, Isenmann K, Pape HC, Allemann F, Barthel C*

BACKGROUND: Distal radius fractures (DRFs) are among the most common adult fractures and are frequently treated with volar plate fixation. While intraoperative 3D imaging has been proposed to improve detection of malreduction and implant malposition, its impact on clinical and radiological outcomes remains unclear. This study aimed to evaluate whether intraoperative 3D imaging provides advantages over conventional 2D fluoroscopy. Outcome criteria were accuracy of fracture reduction, plate positioning relative to the watershed line, complication rates, operative time, and functional outcomes following volar plate fixation of distal radius fractures.

METHODS: This retrospective single-centre study included patients aged ≥ 16 years who underwent volar plate fixation for distal radius fractures between 2016 and 2024. Patients were assigned to a 3D imaging group or a conventional 2D fluoroscopy group. To ensure comparability, patients were matched 1:1 using exact AO/OTA subgroup matching and propensity score matching for age and sex. Outcomes included operative duration, radiographic reduction quality, plate positioning, complications, and wrist range of motion. Matched-pair comparisons were performed using appropriate paired statistical tests.

RESULTS: Of 1,259 patients screened, 418 patients (209 matched pairs) were included after exact AO/OTA subgroup matching and propensity sc […]

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## High-velocity gunshot chest trauma: tailoring patient care through insights from a mass casualty experience.

*Azulay AA, Stein M, Tabo LY, Lebenthal A, Ruderman L, Refaely Y*

OBJECTIVES: This study describes the management strategies and clinical outcomes of a cohort of patients sustaining high-velocity military rifle gunshot wounds to the thorax during a mass casualty incident (MCI). The analysis focuses on damage control surgical techniques, the incidence and timeline of venous thromboembolism (VTE), regional pain management, and early psychiatric protocols.

METHODS: A descriptive retrospective analysis was conducted on patients presenting with a single penetrating high-velocity ballistic wound to the thorax on October 7, 2023. Inclusion criteria were limited to patients with a projectile trajectory traversing the lung parenchyma, managed by the cardiothoracic surgery department. Injuries to immediately adjacent structures (subclavian vessels, brachial plexus, clavicle, and scapula) were recorded as associated injuries. Patients presenting with multi-system trauma involving entirely separate, distant anatomical regions were excluded.

RESULTS: The cohort comprised 11 male patients (age range: 19-49 years; 9 soldiers, 2 civilians) injured by 7.62 mm assault rifle rounds. Six patients (54.5%) required emergency thoracotomy, sternotomy, or VATS within the first 24 h due to ongoing hemorrhage; primary hemorrhage control consisted of deep parenchymal lung suturing (n = 5) and temporary chest packing (n = 1). Four patients (36.4%) underwent delayed, el […]

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## Elevated risk of wound and systemic complications in MRSA-colonized patients undergoing lower extremity orthopedic trauma surgery.

*Areti AS, Hafeez H, Sangineni PV, Argao L, Deveza L, Dawson J*

PURPOSE: Preoperative colonization with methicillin-resistant Staphylococcus aureus (MRSA) is a known risk factor for adverse outcomes in arthroplasty but remains under-investigated in orthopedic trauma. This study evaluates the impact of preoperative MRSA colonization on postoperative complication rates in patients undergoing lower extremity orthopedic trauma surgery.

METHODS: We conducted a retrospective cohort study using the TriNetX Research Network, identifying adult patients who underwent lower extremity orthopedic trauma procedures over a 20-year period. Patients with a documented MRSA diagnosis within three months before surgery were matched 1:1 to MRSA-negative controls based on demographics and comorbidities. Postoperative outcomes were assessed within three months of surgery, including surgical site infections (SSIs), wound dehiscence, organ dysfunction, thromboembolic events, and healthcare utilization. Odds ratios (ORs) and p-values were calculated, with statistical significance set at $p < 0.05$.

RESULTS: A total of 1,868 MRSA+ and 629,342 control patients were identified. After matching, MRSA+ patients remained at significantly increased risk for wound dehiscence (OR = 1.80, p = 0.005), pneumonia (OR = 2.33, $p < 0.001$), sepsis (OR = 3.71, $p < 0.001$), acute renal failure (OR = 1.45, $p < 0.001$), blood transfusion (OR = 1.74, $p < 0.001$), deep vein thrombosis (OR = 2 […]

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## A global contrast shortage: a descriptive study exploring IV contrast utilization and outcomes in trauma patients at a level I trauma center.

*Zakaria H, Johansen C, Waldrop C, Wan B, Graham C, Fetzer D, Minei J, Bruns B et al.*

PURPOSE: Computed Tomography (CT) with intravenous (IV) contrast is critical to the evaluation of the trauma patient. A pandemic-related lockdown occurred in Spring 2022 in the city where iohexol iodinated contrast media) is manufactured, resulting in a global shortage. We explored the potential impact of this shortage on contrast utilization and patient outcomes.

METHODS: A retrospective study was performed of all trauma patients who underwent CT between January and December 2022 at a level I trauma center. Pre-contrast conservation (PRE) period was defined as January to April 2022, and post-conservation (POST) as May to December 2022. Demographics, utilization rates (bolus/patient) and outcomes were evaluated. Chi-square tests were performed for categorical variables and Mann-Whitney U tests for continuous variables. Statistical significance was set at $p < 0.05$.

RESULTS: 1,097 patients were included; 509 in the pre-shortage period, 588 post-shortage. 857 CTs with contrast (1.68 contrast bolus/patient) were performed pre-conservation and 979 post-conservation (1.66 contrast bolus/patient) (p = 0.04). Maximum Abbreviated Injury Scale (AIS) head, head/neck, and extremity were higher post-shortage ($p < 0.05$); however, mortality was higher in the pre-conservation group ($p < 0.0012$). There were 9 cases of Acute Kidney Injury (AKI, 8 PRE, 1 POST), 22 delayed diagnoses (14 PRE, 8 P […]

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## Trauma stabilization points in armed conflict: operational evolution and strategic reflections from mosul to Gaza.

*Marta C, Luca P, Kai-Hsun H, Areej AF, Athanasios G, Wessam EG, Jonas Z, Johan VS et al.*

BACKGROUND: Forward trauma care remains a critical gap in emergency responses to armed conflict, particularly where conventional medical evacuation is disrupted. Trauma Stabilization Points (TSPs) were introduced in previous conflicts to bring life-saving care closer to the point of injury. To improve standards and adaptability, the World Health Organization convened a Technical Working Group to develop operational guidance for TSPs, drawing on diverse field experiences, including the ongoing Gaza deployments.

CASE PRESENTATION: Between February and July 2024, three TSPs were established in Gaza, managing over 4,000 consultations. Initially focused on trauma stabilization and referral, these sites quickly adapted to minor injuries and non-traumatic conditions, reflecting population needs and access challenges. Referral rates varied across sites due to hospital proximity, ambulance availability, and shifting frontlines. Security threats limited forward deployment and safe patient access, requiring high mobility and rapid relocation. Experience from Gaza highlighted key operational principles: locating TSPs near the point of injury; integrating within a functioning trauma referral pathway supported by evacuation capacity and hospital readiness; maintaining clear clinical functions and staffing standards; and using standardized documentation for quality assurance and continuity. […]

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## Acute-on-chronic versus isolated acute subdural hematoma in complicated mild traumatic brain injury: association with radiological mass effect markers in a multicenter cohort.

*Crombé A, Benhamed A, Seux M, Frassin L, Matichard R, L'Huillier R, Millon D, Gorincour G*

PURPOSE: The clinical impact of acute-on-chronic subdural hematoma (acSDH) in mild traumatic brain injury (mTBI) remains incompletely characterized. We assessed whether acSDH, compared with isolated acute SDH (aSDH), is associated with worse neurological presentation and increased radiological markers of mass effect.

METHODS: We conducted a retrospective multicenter cohort study using the IMADIS teleradiology head trauma workflow (103 emergency departments in France, January 2020 to December 2022). Adult patients with complicated mTBI (i.e., GCS 13-15 plus intracranial hemorrhage or skull fracture) and acute SDH on non-contrast head CT were eligible. Exposure was acSDH versus aSDH based on structured radiology reports. Outcomes were (1) GCS < 15 at presentation, (2) SDH maximal thickness ≥ 7 mm, (3) radiological brain herniation, and (4) intermediate or high QueBIC risk category. Univariable and multivariable logistic regression models were used to estimate associations.

RESULTS: Among 935 included patients, 107 (11.4%) had acSDH. Patients with acSDH were older (≥ 75 years: 57.0% vs. 39.7%, p < .0001) and more frequently had dementia (8.4% vs. 3.7%, p = .0465). Compared with aSDH, acSDH was associated with GCS < 15 (37.4% vs. 21.7%, P = .0010) and confusion (41.0% vs. 25.8%, p = .0016). On CT, acSDH was associated with SDH thickness ≥ 7 mm (54.2% vs. 20.4%, OR 4.50, 95%CI 2.9 […]

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## Persistent neuropathic pain symptoms after FPV drone-related thoracic blast injury without rib fractures: a pilot observational study from the Ukraine War.

*Battle C, Baker E, Giamello JD, Melchio R, Dmytriiev D*

PURPOSE: The aim of this exploratory pilot study was to describe First Person View drone (FPV) related, blast-mechanism thoracic trauma with no rib fractures in Ukrainian military patients sustained during the Russia-Ukraine Conflict, and report the prevalence of persistent neuropathic pain symptoms at three months among patients admitted to a rehabilitation and pain medicine service.

METHODS: A retrospective, single-centre observational, exploratory study design was used. Military patients (≥ 16 years old) with blast-mechanism thoracic injuries with no rib fractures, admitted to a hospital's rehabilitation and pain medicine service in Ukraine for ≥ 24 h were included. The PainDETECT survey was completed at three months post-injury. Participant characteristics were summarised using descriptive statistics. Exploratory univariate and multivariate analyses were performed to explore possible associations between variables and the development of persistent neuropathic pain symptoms.

RESULTS: 217 patients were included in the final analysis. Median age was 39 years (IQR: 32-47.5). 131 (60.4%) patients were admitted to critical care with 91 (41.9%) patients requiring mechanical ventilation. The median hospital length of stay was 22 days (IQR: 16-25) and critical care length of stay was 4 days (IQR: 2-5). Persistent neuropathic pain symptoms were reported in 173 out of 217 patients ( […]

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## The association between pancreatic enzyme serum concentrations and blunt pancreatic trauma: a systematic review.

*Sykes B, Gracias D, Bendinelli C, Balogh ZJ*

PURPOSE: Blunt pancreatic trauma is uncommon. When suspected, serum amylase and lipase are measured, but their diagnostic value remains uncertain. This systematic review evaluated the diagnostic performance of serum amylase and serum lipase in detecting blunt pancreatic trauma in adults.

METHODS: MEDLINE, Embase, Scopus and Cochrane Central were searched for studies evaluating serum amylase and/or lipase in adults with blunt pancreatic trauma. Observational studies reporting diagnostic accuracy or clinical outcomes were included. Two reviewers independently screened studies and assessed risk of bias using the Newcastle-Ottawa Scale.

RESULTS: Four observational studies met inclusion criteria. A total of 4937 patients with blunt abdominal trauma, including 150 with pancreatic injury were investigated. Two studies evaluated serum lipase alone and two compared amylase and lipase. Pancreatic injuries were graded according to the American or Japanese Association for the Surgery of Trauma. Early serum amylase measurement (< 90 min post-injury) demonstrated low sensitivity. In contrast, testing performed more than 3 h after injury has shown improved diagnostic accuracy with one study suggesting combined serum amylase and lipase testing can achieve a sensitivity of 85% and specificity of 100%.

CONCLUSION: Early serum pancreatic enzymes measurement does not appear to aid initial manag […]

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### Extension of the fluidics NTF-Biobank of polytraumatized patients with extracellular vesicles: improved vesicle preservation through local processing.

*Weber B, Leppik L, Han J, Becker N, Wang T, Groven RVM, Balmayor ER, Zhao Q et al.*

PURPOSE: Extracellular vesicles (EVs) show diagnostic relevance in trauma, prompting the development of the NTF■EV■Biobank as an extension of the established Network Trauma Research (NTF) plasma/serum biobank. After optimizing the EV isolation and characterization standard operating procedure (SOP) at a single center, this multicenter study evaluates its transferability and examines how sample transport affects EV quality before large■scale implementation in a nationwide biobank setting.

METHODS: Plasma and serum from polytrauma patients and healthy controls (n=10) were processed under the standardized NTF■Biobank protocol and distributed to three centers. EVs were isolated using the unified NTF■EV■Biobank workflow, exchanged between centers, and analyzed for protein markers, particle size and concentration (NTA), total protein content, and morphology (TEM) to compare EV quality across sites.

RESULTS: Implementation of the EV biobank protocol across participating centers, as well as inter-center transport on dry ice, was successfully performed without complications. All samples arrived at the analyzing centers in a frozen state. EVs exhibited typical vesicular morphology as confirmed by transmission electron microscopy and remained positive for CD9 and CD81, indicating that transportation did not affect morphology or surface marker expression. However, transportation did redu […]

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### Whole blood first resuscitation and association of blood product utilization based on mechanism of injury.

*Hadley J, Conner J, Thomas C, Llorente K, Gilani A, White B, Miller Iii P, Nunn A et al.*

INTRODUCTION: Studies evaluating the effect of whole blood (WB) vs. component-based resuscitation accounting for mechanism of injury are lacking. We sought to evaluate the effects of a WB first resuscitation strategy regarding mechanism of injury.

METHODS: This is a retrospective cohort study of trauma patients who received any blood product during resuscitation from January 2016 to November 2021. The primary outcomes were total blood volume utilized within 24 h and 30-day mortality. Multivariable gamma regression with a log link was used for blood volume and logistic regression for mortality. A likelihood ratio test evaluated whether injury mechanism modified the treatment effect.

RESULTS: Between January 2016 and November 2021, 1,013 injured patients received blood products (785 WB-first). We did not detect statistically significant effect modification by injury mechanism for either outcome (interaction $p = 0.44$ for blood volume, $p = 0.25$ for mortality). In multivariable gamma regression, WB-first resuscitation was associated with a ratio of mean total blood volume of 0.71 (95% CI 0.55 to 0.90, $p = 0.005$) relative to components, representing approximately 29% lower utilization. WB-first was not associated with 30-day mortality (OR 0.90, 95% CI 0.48 to 1.71, p = 0.73).

CONCLUSIONS: A WB-first resuscitation strategy is associated with lower blood volume utilized within the f […]

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## Reported complications after spinopelvic fixation for U-type sacral fractures in older adults: a systematic review with narrative synthesis.

*Giordano V, Pinheiro HRS, Pires RE, Varga P, Freitas A, Giannoudis PV*

BACKGROUND: Spinopelvic fixation (SPF), alone or combined with iliosacral or transiliac-transsacral fixation, is often used for highly unstable posterior pelvic ring injuries and some fragility fracture patterns in older adults. However, the evidence specifically describing complications after SPF in older patients with U-type sacral fractures is limited.

METHODS: This systematic review was conducted according to the Cochrane Handbook and PRISMA recommendations. PubMed, Embase, and grey-literature sources were searched from January 1, 2000 to July 31, 2025. Eligible studies included adults aged 60 years or older with U-type sacral fractures or closely related spinopelvic dissociation/FFP IVb patterns treated with SPF, and reporting at least one complication. Because of the small number of eligible studies, probable cohort overlap, heterogeneity in fracture definitions, treatment indications, fixation constructs, and outcome reporting, findings were synthesized narratively rather than pooled quantitatively.

RESULTS: Three studies met the eligibility criteria, comprising 157 patients overall; however, only a subset underwent SPF, and two studies were produced by the same group with probable cohort overlap. The available cohorts mixed bilateral sacral fragility fractures, FFP IVb patterns, and comparative SPF-versus-trans-sacral constructs. Reported adverse events included wound […]

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## Influence of allograft augmentation on surgical outcomes and complications in proximal humerus fractures: a matched-pair analysis.

*Barthel C, Rudolph F, Kalbas Y, Russek R, Pape HC, Allemann F*

BACKGROUND: Proximal humerus fractures predominantly affect elderly patients and pose a growing clinical burden. This study evaluates the impact of allograft augmentation on operative time, fracture reduction quality, and complication rates compared with standard fixation across different fracture morphologies.

METHODS: This retrospective matched-pair study included patients surgically treated for proximal humerus fractures with locking plate fixation at a single tertiary trauma center between 01.01.2014 and 31.12.2024. Patients treated with allograft augmentation (Group A) were matched 1:1 to patients without allograft (Group NA) using exact matching for Mayo FJD fracture morphology and sex, and propensity score matching for age. Radiological outcomes were assessed using pre- and postoperative neck-shaft angles, with normalization defined as correction into the physiological range (120°-150°). Complications and operative parameters were recorded. Paired statistical analyses were applied.

RESULTS: A total of 1,012 patients were treated during the study period; after selection and matching, 102 matched patient pairs (204 patients) were analyzed. Operative time was longer with allograft augmentation (141.3 ± 40.3 vs. 126.9 ± 50.0 min; p = 0.08), reaching statistical significance only in varus posteromedial fractures (144.4 ± 37.8 vs. 122.2 ± 39.6 min; p = 0.007). Although overa […]

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## Impact of a comprehensive trauma pathway on workflow and clinical processes in a level III trauma center.

*Neitzert L, Druetto C, Forno D, Molino P, Vitale D, Grio M*

PURPOSE: Organized trauma systems improve coordination of care and may enhance the efficiency of trauma management, potentially improving outcomes and reducing hospital length of stay and costs. The aim of the study was to evaluate key process-of-care indicators and clinical outcomes before and after the introduction of a comprehensive trauma pathway.

METHODS: We conducted a single-center retrospective-prospective cohort study at Rivoli Hospital, Italy. Trauma patients admitted to the Emergency Department (ED) between 2024 and 2025 who met the criteria for trauma team activation were prospectively included. Outcomes were compared with a retrospective cohort from the period prior to the implementation of the dedicated trauma pathway (2023-2024).

RESULTS: A total of 465 patients were included. Implementation of a comprehensive trauma pathway was associated with significant improvements in several time-dependent quality indicators, including ED boarding time (888 vs. 1021 min; $p = 0.02$), time to imaging (48 vs. 78 min; $p < 0.01$), time to surgery (183 vs. 237 min; $p = 0.03$), time to ICU admission (226 vs. 343 min; $p = 0.02$), and time to interhospital transfer (238 vs. 305 min; $p = 0.04$). We observed a trend toward a reduction in early mortality (2% vs. 3.6%; $p = 0.41$) and 30-day mortality (3.7% vs. 5.9%; $p = 0.62$).

CONCLUSIONS: The adoption of a structured multidisciplinary trau […]

PubMed · DOI

## The DRoPH score: development of a tool for predicting reduction difficulty in proximal humerus fracture osteosynthesis.

*Barthel C, Rudolph F, Kraus M, Pape HC, Allemann F*

BACKGROUND: Proximal humerus fractures (PHFs) are among the most common and challenging fractures due to complex anatomy and variable fracture patterns. While anatomical reduction is critical for successful outcomes, no tool currently predicts the intraoperative difficulty of achieving it. Despite advances in surgical techniques, complications persist. Existing classification systems, such as Codman's, Neer's, AO/OTA, and Mayo-FJD, focus on fracture morphology and displacement but do not address intraoperative complexity. This study aims to develop a new score for the early prediction of intraoperative reduction difficulty in proximal humerus fracture surgery, the DRoPH score (Difficulty of Reduction of Proximal Humerus Fractures).

METHODS: An online survey was conducted among experienced upper extremity surgeons in Switzerland to identify factors influencing the difficulty of reducing and retaining proximal humerus fractures. Inclusion criteria for surgeon grading comprised experience with more than 100 surgically treated proximal humerus fractures. Eligible surgeons rated the relevance of factors influencing reduction difficulty using a five-point Likert scale. A rank-based weighting system derived from expert ratings was applied, assigning two points to the three highest-ranked factors. The resulting DRoPH-Score ranged from 0 to 10, with higher values indicating greater sur […]

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## Interface stability of intramedullary nail locking configurations under combined axial and torsional loading.

*Marianne H, Martin W, Claus G, Anja B, Peter A*

The biomechanical performance of intramedullary nails in metaphyseal regions strongly depends on distal locking configuration. This study aimed to evaluate screw-bone interface stability under combined axial and torsional cyclic loading for different distal locking configurations of intramedullary nails, focusing on the effects of screw number, spatial distribution, interscrew distance, and multiplanar versus coplanar fixation using a standardized synthetic bone model. Six distal locking configurations (n = 5 per group) were tested under stepwise cyclic axial compression, superimposed with a constant torsional moment (± 2 Nm, 2 Hz). Primary endpoints were cycles to failure, cycles to 1 mm axial displacement, and cycles to 0.5° rotation. All constructs failed by progressive screw cut-through within the synthetic bone. The three-screw multiplanar configuration exhibited the highest endurance (≈ 36,000 cycles to failure) and lowest micromotion (≈ 34,000 cycles to 1 mm displacement), significantly outperforming the two-screw configurations overall ($p = 0.002$). Increasing interscrew distance from 10 mm to 34 mm improved migration resistance by approximately 25%, whereas the short 10 mm spacing led to premature failure. Blocked and free-hand configurations showed comparable behavior to the standard setup, indicating no added benefit from toggle elimination alone. Locking configuratio […]

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## Three-dimensional finite element analysis for internal fixation of mandibular condylar fractures: a scoping review.

*Kakei Y, Sader R, Ebener P, Tanaka A, Akashi M, Kuroda R, Thoenissen P*

PURPOSE: Three-dimensional finite element analysis (3D-FEA) has emerged as an essential computational tool for evaluating the biomechanical behaviors of internal fixation systems for mandibular condylar fractures. This scoping review maps and synthesizes the existing literature on 3D-FEA studies investigating the internal fixation of mandibular condylar fractures and explores, in a descriptive manner, differences in methodological focus between European and non-European research centers.

METHODS: PubMed and Cochrane Library databases were systematically searched following PRISMA-ScR guidelines. Original English-language research articles using 3D-FEA to evaluate internal fixation of mandibular condylar fractures were included. Data were extracted using a standardized charting form.

RESULTS: Thirty-six studies met the inclusion criteria. European studies (n = 12) predominantly focused on development of novel plate designs and mechanical testing protocols, whereas Asian and other nationality studies (n = 24) emphasized material comparisons between titanium and bioabsorbable systems, patient-specific modeling, and loading condition analysis. Consensus findings include: (1) double-plate fixation provides superior stability; (2) among 3D plate designs, trapezoid plates demonstrate the best performance for subcondylar and condylar base fractures, whereas alpha and lambda plates may […]

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## Initial surgical management of injuries to the upper extremities for patients with multiple and/or severe injuries - a systematic review and clinical practice guideline update.

*Berk T, Könsgen N, Lechler P, Imach S, Schär M, Bieler D, Horst K, Hildebrand F et al.*

PURPOSE: Our objective was to update the evidence-based and consensus-based recommendations for the initial surgical management of upper extremity injuries in patients with suspected multiple and/or severe injuries based on current evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to September 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared early versus delayed surgical treatment for fractures, vascular injuries, or nerve injuries affecting the upper extremities in patients with multiple and/or severe injuries. Studies comparing limb salvage versus amputation or comparing amputation criteria for the upper extremities were also included. We considered patient-relevant outcomes such as mortality and limb salvage, as well as the sensitivity and specificity of scores for predicting upper extremity amputation. Risk of bias was assessed at the outcome level using ROBINS-I for observational studies and AMSTAR-2 for systematic reviews. We used available meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expe […]

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## Transosseous sutures versus vertical cerclage wire for distal pole patellar fractures: a retrospective comparative cohort study.

*Laso JI, Muñoz JT, Bianchi S, Martínez D, Rojas T, Franulic N, Olivieri R*

PURPOSE: To compare transosseous sutures (TOS) and vertical wire cerclage (VCW) for treating distal pole patellar fractures, evaluating clinical outcomes, patient-reported outcome measures (PROMs), radiographic outcomes, and complications.

METHODS: We conducted a retrospective comparative cohort study of patients surgically treated for patellar distal pole fractures (AO/OTA 34A1b) with either TOS or VCW between 2015 and 2023. Patient characteristics, surgical technique, postoperative data, patellar height, and complications were recorded. Functional outcomes were evaluated at a minimum of 12 months postoperatively using PROMs (Kujala, Lysholm, and Bostman scores). The primary outcome of this study was functional recovery, assessed via the Bostman score at 12 months.

RESULTS: A total of 46 patellar distal pole fractures were included (30 TOS and 16 VCW). Baseline characteristics showed no statistically significant differences between groups. No significant differences were found regarding clinical or functional outcomes. Regarding radiographic outcomes, patellar height remained stable; however, both groups showed significant patellar length variation, with elongation being significantly more pronounced in the TOS group (mean 3.77 mm) compared to the VCW group (mean 0.44 mm; $p = 0.001$). While overall complication rates showed no statistically significant difference, all five ca […]

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### Re■fracture risk following implant removal in consolidated hip fractures: a multicenter retrospective cohort study of 575 patients.

*Mohammed J, Mili-Schmidt V, Wadsten M, Juto H, Wolf O, Crnalic S, Sköldenberg O, Mukka S et al.*

PURPOSE: The risk of re-fracture after implant removal in healed hip fractures remains a clinical concern. This study aimed to determine the incidence of re-fracture following implant removal after osteosynthesis of a hip fracture.

METHODS: We conducted a retrospective multicenter cohort study including patients aged ≥ 50 years who underwent implant removal between 2003 and 2023 after radiologically confirmed consolidation of a hip fracture. Patients were identified using procedural codes. Baseline variables included age, sex, ASA classification, fracture type (femoral neck or trochanteric), and implant type (pins/screws, sliding hip device [SHD], or short/long cephalomedullary nail [CMN]). Patients were followed from implant removal until re-fracture, conversion to arthroplasty, death, or end of follow-up, with a minimum follow-up of 1 year. Cox proportional hazards regression was used to assess associations between implant type and re-fracture risk, adjusting for age and sex. Because the proportional hazards assumption was violated, a time-stratified Cox regression and restricted mean survival time analyses were applied.

RESULTS: A total of 575 patients (median age 73 years, IQR 65-81) were included, with a median follow-up of 53 months (IQR 18-100). Lateral hip pain was the most common indication for implant removal (72.5%). The overall re-fracture incidence was 10.4%. Ris […]

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### Physical functionality two years after major trauma: predictors from a single center cohort study.

*Fetz K, Koetsenruijter J, Rodrigues Recchia D, Rutetzki J, Lefering R, Kaske S*

PURPOSE: Traumatic injuries are a major global public health concern, often leading to impaired physical function and disability. Long-term physical function impairments can greatly affect a patient's health related quality of life, influencing their emotional state and social interactions. This study examines patient-reported physical functionality before and after major trauma, identifying key predictors of poor functional recovery.

METHODS: This retrospective single-center cohort study investigates physical functionality 23 months after a traumatic injury using standardized questionnaires combined with clinical data. Inclusion criteria were Injury Severity Score (ISS) ≥ 9, requiring ICU treatment, and age ≥ 18 resulting in an eligible sample of 515 patients. Health outcomes were assessed using the Trauma Outcome Profile (TOP), the Abbreviated Injury Scale (AIS), and the ISS.

RESULTS: Physical functionality significantly decreased from 94.8 (SD 11.5) to 72.9 (SD 26.7) 23 months post-trauma with almost half of participants still scoring below the critical cutoff of 80 points. The strongest predictor for functional physical impairment was pre-trauma physical functionality (OR 8.21, $p < 0.001$). Other determinants included age (age 50-59 versus 18-29 OR 2.75, $p = 0.001$), as well as an impacted lower extremity including pelvic (OR 1.60, $p = 0.018$).

CONCLUSION: The most importan […]

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### Association between fresh frozen plasma transfusion and mortality stratified by Glasgow Coma Scale scores in isolated traumatic brain injury: a nationwide cohort study in Japan.

*Nagamura T, Aoki M, Yamada K, Terayama T, Seno S, Tomura S, Kiyozumi T*

PURPOSE: Fresh frozen plasma (FFP) is widely used to manage trauma-induced coagulopathy. However, its effectiveness in isolated traumatic brain injury (TBI) management remains controversial, particularly when stratified by severity based on the Glasgow Coma Scale (GCS) scores. This study aimed to examine the association between FFP administration within 24 h and mortality in patients with isolated TBI, overall and stratified by GCS on arrival.

METHODS: We conducted a multicenter retrospective cohort study using the Japan Trauma Data Bank (2019-2023). Adults with isolated TBI (intracranial AIS ≥ 3 and extracranial AIS < 3) were included. Patients were classified into FFP and non-FFP groups based on transfusion within 24 h of admission. Subgroup analyses were performed by TBI severity (severe: GCS ≤ 8; mild-to-moderate: GCS > 8), craniotomy status, and FFP dose. The primary outcome was 28-day mortality. Propensity score matching was applied to adjust for confounding.

RESULTS: Among 12,480 patients (median age 72 years [IQR: 55, 82]; 66.6% male), 513 received FFP. After matching, FFP was not significantly associated with 28-day mortality (OR 0.80; 95% CI 0.59-1.08). In subgroup analyses, FFP was associated with higher mortality in mild-to-moderate TBI (OR 2.38; 95% CI 1.07-5.31) and lower mortality in severe TBI (OR 0.61; 95% CI 0.42-0.88). No significant differences were observ […]

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### Percutaneous fixation of posterior pelvic ring lesions in the elderly: current evidence and recent advances.

*Liodakis E, Schreiber S, Giannoudis VP, Fritz T, Giannoudis PV*

Transsacral percutaneous fixation is increasingly seen as a safe, effective minimally invasive treatment for geriatric posterior ring fractures The use of Navigation appears a safe and efficient method for percutaneous screw placement Multiple new implants have been described for management of these injuries. Level I evidence is still lacking for treatment options in the management of posterior pelvic ring lesions with percutaneous fixation

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### Infrared thermography and dual-layer spectral CT for the detection of femoral vascular compromise: an experimental large animal study.

*Hübner CT, Ricklin J, Müller AL, Klingebiel FK, Stoeck CT, Weisskopf M, Carneiro Gomes A, Cinelli P et al.*

BACKGROUND: Rapid identification of vascular compromise remains a clinical challenge in the acute trauma setting. Conventional diagnostic tools such as CT angiography (CTA) and Doppler-ultrasound might provide decent anatomical and functional assessment. However, they are limited by ionizing radiation, contrast exposure or time requirements. Although not validated, infrared thermography (IRT) might offer a non-invasive alternative capable of detecting surface temperature asymmetries that may reflect underlying perfusion deficits. This study aimed to evaluate the feasibility of thermographic imaging in detecting vascular lesions.

METHODS: As part of a porcine polytrauma model (n = 28), vascular lesion of the right femoral artery was defined as a ≥ 50% diameter reduction on CTA. Temperature differences between both extremities were measured using infrared thermography. SDCT based iodine concentrations distal to the stenosis were quantified to assess perfusion changes. Correlation and regression analyses were performed between stenosis degree, temperature difference, and tissue iodine uptake.

RESULTS: Six animals fulfilled the criteria for relevant vascular compromise. This resulted in a mean vessel diameter reduction behind the lesion of 68% ±18%. The vascular lesion group demonstrated a mean inter-limb temperature difference of 6.93 °C ± 1.78 °C compared to 0.71 °C ± 1.27 °C i […]

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### Fixation strategies for periprosthetic femur fractures around knee replacements: a comprehensive single-center analysis of 102 consecutive cases.

*Nomdedéu J, González-Morgado D, Minguell-Monyart J, Joshi-Jubert N, Teixidor-Serra J, Tomàs-Hernández J, Andrés-Peiró JV*

PURPOSE: To evaluate 1 year mortality, complication rates, and functional outcomes after internal fixation of femoral periprosthetic knee fractures following total knee arthroplasty, and to identify predictors of adverse outcomes and increased healthcare resource utilization.

METHODS: This retrospective cohort study included 102 consecutive patients with femoral periprosthetic knee fractures classified as UCPF VB3 and VC3 treated with internal fixation at a single tertiary center between 2010 and 2023. Interprosthetic fractures and combined fixation techniques were excluded. Primary outcomes were 1 year mortality and postoperative complications. Secondary outcomes included length of stay, discharge destination, and ambulatory status at follow up.

RESULTS: Patients were markedly frail, with a mean age of 82.4 ± 10.8 years and mean Charlson Comorbidity Index of 5.6. One year mortality was 16.7% and was associated with older age, higher ASA class, higher Charlson index, impaired pre fracture ambulation, and fracture related infection. Overall complication rate was 29.4%, including 15.7% infections and 13.7% mechanical failures. Open reduction was more frequent in plate fixation and resulted in longer operative times, without increasing complication rates. Median hospital stay was 18 days, 66.3% required discharge to nursing facilities, and only 7.8% regained independent ambulati […]

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### Epidemiology and management of pediatric femoral fractures at a level I trauma center: a 10-year retrospective monocentric study.

*Dietz SO, Traub F, Schmidt E, Schippers P, Wegner E, Schierjott L, Gercek E, Nowak TE*

PURPOSE: Pediatric femoral fractures are relatively uncommon but represent severe injuries frequently requiring inpatient trauma care. Epidemiological characteristics, fracture patterns, and treatment strategies vary between institutions and regions. This study aimed to analyze the epidemiology, fracture distribution, treatment modalities, complications, and fracture-predisposing comorbidities of pediatric femoral fractures treated at a German level I pediatric trauma center over a 10-year period.

METHODS: All patients aged 0-14 years treated for femoral fractures between January 2012 and December 2021 were retrospectively reviewed. Fractures were categorized using the AO Pediatric Comprehensive Classification of Long-Bone Fractures (PCCF). Demographic variables, fracture characteristics, management approaches, time to consolidation, complications, and relevant comorbidities predisposing to fracture were analysed descriptively in accordance with STROBE recommendations.

RESULTS: A total of 183 patients met the inclusion criteria (67.2% male; mean age 5.2 years). Children aged 2-5 years constituted the largest group (42.1%). Femoral shaft fractures were the most frequent femoral fracture type (70%). Surgical management, most frequently elastic stable intramedullary nailing (ESIN), was increasingly utilized with advancing age. The overall complication rate was 14.2%, with the ma […]

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### Blood-based biomarkers GFAP/UCH-L1 for the diagnosis of mild traumatic brain injury (mTBI): a single-center implementation experience.

*Maegele M, Breidenbach L, Kaufmann J, Keles Y, Uhl E, Schroeer P*

INTRODUCTION: Mild traumatic brain injuries (mTBI) affect millions of people worldwide every year as one of the most common clinical presentations in the emergency department. Diagnosis is mainly based on clinical criteria and computed tomography scans. The use of computed tomography causes high costs, long waiting times in daily clinical practice and radiation exposure. GFAP (glial fibrillary acidic protein) and UCH-L1 (ubiquitin carboxyl-terminal hydrolase-L1) turned out to be potential biomarkers for the diagnosis of mTBI. This study retrospectively evaluates the possible use of these biomarkers combined as negative predictors for excluding brain injuries in patients with suspected mTBI in the emergency department.

METHODS: Adult patients (n = 320) registered in the emergency department at a level 1 trauma emergency center in Germany (Cologne Merheim Medical Center/CMMC) between 11/2023 and 04/2024, with suspected mTBI, Glasgow Coma Scale (GCS) score 13-15 and within 12 h after trauma were considered. All evaluable patients underwent cranial CT (cCT) scans and blood tests for GFAP and UCH-L1 serum concentrations.

RESULTS: Biomarkers GFAP and UCH-L1 were tested positive in 261 patients (82%) while CT detected intracranial injuries in only 29 patients (9%). Biomarkers combined had a sensitivity of 97% and a negative predictive value (NPV) of 98% in mTBI diagnosis with a nega […]

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### Resuscitative endovascular occlusion of the aorta restores cerebral metabolic markers of ischaemia induced by haemorrhagic shock.

*Bader SE, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.*

BACKGROUND: Resuscitative endovascular balloon occlusion of the aorta (REBOA) is used as an adjunct in haemorrhagic shock (HS) to restore proximal perfusion. Its cerebral metabolic effects during haemorrhagic shock, particularly in the presence of elevated intracranial pressure (ICP), remain incompletely characterised.

METHODS: In a controlled porcine model, HS was induced by controlled bleeding to a mean arterial pressure of approximately 40 mmHg. Animals were allocated to a normal ICP group or an experimentally elevated ICP group. Total REBOA (zone 1) was applied for 90 min. Cerebral metabolism was assessed using intracerebral microdialysis with measurements of lactate, pyruvate, and lactate-pyruvate ratio (LPR). Cerebral haemodynamics and ICP were continuously monitored.

RESULTS: HS was associated with a statistically significant increase in LPR in both groups, indicating cerebral metabolic disturbance. Following aortic occlusion, LPR gradually decreased toward baseline in both groups. Animals with elevated ICP demonstrated a transient delay in metabolic normalisation during the early post-occlusion phase. Statistically significant differences between groups were limited to the first 10 min following occlusion. Overall metabolic trajectories were similar thereafter.

CONCLUSIONS: Total REBOA restored cerebral metabolic markers of ischaemia during haemorrhagic shock, as ref […]

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## Delayed single-stage surgical management using a peroneus brevis muscle flap for fracture-related infection after lateral malleolus fixation: a retrospective case series.

*Dyreborg K, Jørgensen AI, Peterlin AAN, Gottlieb H*

BACKGROUND: Fracture-related infection (FRI) following lateral malleolus fracture fixation complicated by soft tissue breakdown represents a challenging clinical problem with no established standard of care. This study evaluates a treatment strategy consisting of prolonged antibiotic suppression until fracture union, followed by delayed single-stage surgical management.

METHODS: This retrospective consecutive case series included 11 patients treated between 2020 and 2023 at a multidisciplinary orthopaedic infection unit. All patients were managed according to a standardized protocol involving oral antibiotic suppression until radiologically confirmed fracture consolidation. This was followed by a single-stage surgical procedure including implant removal, radical debridement, deep tissue sampling, local antibiotic therapy, and soft tissue reconstruction using a distally based peroneus brevis muscle flap with split-thickness skin grafting.

RESULTS: Eleven consecutive patients (median age 58 years) were treated according to the protocol. Six patients underwent antibiotic suppression for a median duration of 147 days prior to surgery. Among patients available for outcome assessment, infection control was achieved in 10 of 10 surviving patients. One patient died of unrelated causes within 14 days postoperatively and was not classified as treatment failure. Soft tissue complication […]

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## Mixed reality improves agreement on surgical approach selection and patient positioning in tibial plateau fracture planning compared to CT, 3DCT and 3D printing.

*Dust T, Henneberg JE, Hartel M, Reiter A, Kylies J, Korthaus A, Streckenbach A, Keller J et al.*

PURPOSE: Tibial plateau fractures require precise preoperative planning, particularly regarding treatment concept, patient positioning, and surgical approach. Mixed reality (MR) may enhance three-dimensional understanding of fracture morphology beyond conventional imaging and physical 3D models. This study investigated whether MR improves agreement on key preoperative planning decisions compared to computed tomography (CT), 3D computed tomography (3DCT), and 3D-printed fracture models.

METHODS: Twelve orthopaedic trauma surgeons (6 junior, 6 senior) evaluated 22 surgically treated tibial plateau fractures (AO/OTA type B or C). Each case was assessed in four steps using (1) CT, (2) 3DCT reconstructions, (3) physical 3D-printed models, and (4) MR visualization using Microsoft Hololens 2. Raters selected (i) treatment concept, (ii) patient positioning, and (iii) surgical approach. Interobserver agreement was analysed using Fleiss' kappa ($\kappa$) and percentage match (PM).

RESULTS: Mixed Reality (MR) yielded the highest interobserver agreement for both surgical approach (PM 32%, $\kappa$ = 0.30) and patient positioning (PM 57%, $\kappa$ = 0.35), particularly among junior surgeons (approach PM 39%, $\kappa$ =

0.28; positioning PM 57%, κ = 0.39). Compared to CT, 3DCT, and 3D printing, MR demonstrated consistent improvements, while agreement on treatment concepts remained high across all modalities (PM > 82% […]

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## Timing of fracture fixation for femur and pelvis fractures in patients with severe traumatic brain injury - an analysis of the TraumaRegister DGU®.

*Bayer TP, Teuben MPJ, Halvachizadeh S, Shehu A, Lefering R, Pfeifer R, Pape HC, Jensen KO*

PURPOSE: Fracture fixation timing and strategy in polytrauma patients with traumatic brain injury (TBI) remain controversial. This study investigates treatment patterns and outcomes for femoral and/or pelvic fractures stratified by TBI severity.

METHODS: Patients in the TraumaRegister DGU® (2016-2022) with pelvic and/or femoral fractures (AIS ≥3) and TBI (head AIS ≥3) were included. Strategies were non-operative management (NOM), early total care (ETC), and damage-control orthopedics (DCO). Outcomes included treatment allocation, fixation timing, and in-hospital mortality.

RESULTS: 985 patients were included (mean age 52.5, SD 26.3 years; ISS 27.8, SD 8.1). Allocation was NOM in 320 (32.5%), ETC in 336 (34.1%), and DCO in 329 (33.4%) patients. Head AIS was 3 in 48.5%, 4 in 31.1%, and 5 in 20.3%. NOM patients were older, had the highest ISS and estimated mortality, and showed the largest proportion of critical TBI (AIS 5: NOM 30.9%, ETC 14.3%, DCO 16.1%). Femoral ETC was mainly performed within the first day (median 0, IQR 0-1 days), whereas pelvic ETC was delayed with increasing TBI severity (median 3, IQR 0-5 days for head AIS 3; 5, IQR 0-7 days for AIS 4). Observed mortality was 37.2% after NOM, 9.2% after ETC, and 10.3% after DCO.

CONCLUSION: ETC in patients with moderate TBI (AIS 3) was associated with reduced observed mortality relative to NOM and matching DCO. Increasi […]

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## From single to dual platelet inhibition FIBTEM assay: fibrinogen replacement thresholds for critical bleeding after trauma.

*Latona A, Winearls J, Ho A, Mitra B*

PURPOSE: The FIBTEM assay in Rotational Thromboelastometry (ROTEM®) measures fibrin-based clot formation. The reagent transitioned from single to dual-platelet inhibition with addition of tirofiban to cytochalasin D, offering more complete platelet suppression. The aim of this study was to assess the impact of this change on transfusion thresholds in trauma.

METHODS: In this retrospective cohort study, we included adult patients with major trauma across Queensland who presented between 2022 and 2025 and had results available for paired ROTEM and conventional coagulation tests on arrival. Patients were sub-grouped into single and dual-platelet inhibition assay groups. Correlations between FIBTEM-A5 and Clauss fibrinogen were analysed, comparing correlation strength and diagnostic accuracy for fibrinogen levels of < 2.0 g/L.

RESULTS: The relationship between FIBTEM-A5 and Clauss fibrinogen was similar using single platelet inhibition assays ($R^2$ 0.61;95%CI: 0.51-0.70) and dual-platelet inhibition assays ($R^2$ 0.70;95%CI: 0.62-0.77; p = 0.94). Predicted FIBTEM-A5 amplitudes at 2 g/L fibrinogen were 7.83 mm (95%CI: 7.42-8.23) and 7.54 mm (95%CI: 7.22-7.87). Diagnostic performance for detecting fibrinogen < 2.0 g/L was comparable between assays, for both FIBTEM-A5 ≤ 10 mm and FIBTEM-A5 ≤ 8 mm (all p > 0.05). For the dual-platelet inhibition assay, sensitivity and specificity were 97. […]

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## Pattern of blast injuries. A systematic review: Part 2 - Landmines, unexploded ordnance and terrorism.

*Neubert A, Gaeth C, Seidelmann M, Bieler D*

PURPOSE: This systematic review analyzes the patterns and distribution of blast injuries from landmine explosions, UXOs and terror attacks including suicide bombing, in civilians and military personnel.

METHODS: A search was conducted in two databases on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion typ. A narrative data synthesis was performed. The JBI tool for case series was used.

PROSPERO: CRD42023391736.

RESULTS: 126 studies were included in the overall systematic review. 45 studies on terror & suicide bombings and 12 studies on landmines & UXOs are analyzed in the present paper. Studies on landmines from 1980 to 2006 included 13,270 individuals (age: 24.0 years (Range 8.2 years - 32.2 years); mainly male (88.9%)) in post-conflict low- and middle-income countries. Lower extremity injuries including amputations were most frequent, followed by eye injuries. Studies on terrorist attacks with a total of 12,518 mainly civilian victims (average age: 33,1 years; 61.6% males) reported on incidents in post-conflict regions and high-income cities mainly from the early 2000s. The injury pattern was multidimensional, with extremity injuries most frequent (28%-30%), including fractures, lacerations, and traumatic amputations (18%-20%).

CONCLUSION: This Systematic Review Part 2 […]

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## Is nail plate fixation the way forward for distal femur fractures with long segmental metaphyseal comminution? - A study of 56 patients.

*Mahesan H, Sundaram VP, Devendra A, Ramesh P, Dheenadhayalan J, Rajasekaran S*

BACKGROUND: Mechanical instability due to long metaphyseal comminution and medial bone gap after lateral locking plate results in varus collapse and implant failure. We believe that fixation in the form of Nail and plate combination provides better stability and prevents complications.

MATERIALS: We performed a retrospective analysis of 56 distal femur fractures with long metaphyseal comminution treated in our institution with nail and plate construct from 2017 to 2023. The mean age was 43.1 years and the mean follow up was 24 Months (4-36 months). Of these 56 patients, 16 were closed injuries underwent primary fixation, and 40 were open injuries requiring staged reconstruction. At the final follow up, patients were assessed for shortening, knee range of movements, Sander et al. functional score, radiological union and loss in alignment.

RESULTS: The mean length of metaphyseal comminution was 10.15 cm(5-18 cm). Primary Iliac crest bone grafting was done in 25 patients. Of the 56 patients, 51 achieved primary union and the mean time to union was 6 months. Of the remaining, 3 patients united after secondary bone grafting and 2 patients had nonunion. The mean loss in aLDFA at the final follow up was 1.75° in 7 patients. Shortening of the limb was noted in 16 patients, and 4 patients had significant shortening of 2 cm. Three patients developed deep SSI, of which one patient requi […]

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## Impact of modified ERAS protocol implementation on length of stay and complications in elderly hip fracture patients: a before-and-after study.

*Liu Y, Ou G, Xie M, Wu F, Wang X, Yang B*

AIM: Hip fractures in elderly patients pose significant healthcare challenges with high complication rates and prolonged hospital stays. Enhanced Recovery After Surgery (ERAS) protocols have shown promise in various surgical specialties, but their application in elderly hip fracture patients, particularly in Traditional Chinese Medicine (TCM) hospital settings, remains limited.

METHODS: This retrospective before-and-after study was conducted at a TCM hospital from July 2022 to July 2025. A total of 300 elderly hip fracture patients (≥ 65 years) were included: 146 in the control group (July 2022 - October 2023) receiving conventional care and 154 in the ERAS group (February 2024 - July 2025) receiving modified ERAS protocol incorporating TCM elements. Primary outcomes were length of hospital stay and perioperative complications within 30 days. Secondary outcomes included functional recovery [Harris Hip Score (HHS), Modified Barthel Index (MBI)], time to first mobilization, pain scores, and healthcare costs.

RESULTS: The ERAS group demonstrated significantly shorter length of stay compared with controls (12.38 ± 3.75 vs. 16.24 ± 5.10 days; mean difference 3.86 days, 95% CI: 2.84-4.88; P < 0.001). Total perioperative complications were reduced (7.79% vs. 15.07%; risk difference: -7.28%, 95% CI: -14.5% to - 0.1%; P = 0.047). The ERAS group showed superior functional outcomes with […]

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### Prehospital endotracheal intubation and 30-day survival in severe trauma: a matched cohort study from Navarra, Spain.

*Zulet Murillo D, Ferraz Torres M, Fortún Moral M, Belzunegui Otano T, Tamayo Rodríguez I*

PURPOSE: Prehospital endotracheal intubation (ETI) is frequently used in major trauma, but its association with 30-day survival remains debated. We evaluated the association between prehospital ETI and 30-day survival and assessed whether sex modifies the age-related increase in mortality risk.

METHODS: We conducted a retrospective cohort study using the Navarra Major Trauma Registry (2010-2019). Patients aged ≥15 years with New Injury Severity Score (NISS) ≥15 were included; patients who died before hospital admission or had incomplete records were excluded. To mitigate confounding by indication, intubated and non-intubated patients were matched 1:1 using prehospital Glasgow Coma Scale (GCS), respiratory distress, and prehospital cardiac arrest as matching variables. Because the proportional hazards assumption for Cox regression was not met, parametric Weibull survival models were fitted, including a sex-by-age interaction term.

RESULTS: Overall, 1,909 patients were included; 212 (11.1%) underwent prehospital ETI. Matching yielded 190 pairs (n=380). In the matched cohort, prehospital ETI was not independently associated with 30-day survival (adjusted HR 0.91, 95% CI 0.65-1.27). Mortality risk was independently associated with lower GCS (per 1-point increase: HR 0.86, 95% CI 0.82-0.91) and respiratory distress (HR 7.63, 95% CI 4.34-13.4). A significant sex-by-age interaction […]

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### Mechanical and manual cardiopulmonary resuscitation in traumatic cardiac arrest: a retrospective observational study.

*El-Menyar A, Ahmed K, Howland IR, Mollazehi M, Mekkodathil A, Rizoli S, Cameron P, de Oliveira Manoel AL et al.*

BACKGROUND: Prehospital Traumatic cardiac arrest has a high fatality rate despite advances in trauma systems and resuscitation strategies. Mechanical chest compression devices have been increasingly adopted to optimize cardiopulmonary resuscitation (CPR) during Emergency Medical Services (EMS) transportation. We aimed to evaluate the impact of CPR (mechanical during transportation versus manual CPR only at the scene) on survival among trauma patients.

METHODS: A retrospective analysis was conducted for patients who received CPR at the scene and during transportation to the hospital between 2016 and 2024.

RESULTS: A total of 610 patients sustaining blunt traumatic cardiac arrest were included. The mean age was 34 ± 12.6 years; 94.6% were male, and 5.4% were female patients. Two hundred twenty (36.1%) received mechanical CPR during transportation to the hospital, and 390 (63.9%) received manual CPR only at the scene. Compared with the manual CPR group, the mechanical CPR group had higher rates of primary chest injury (30.5% vs. 20.5%; p = 0.006), bystander CPR (15.9% vs. 6.7%; p = 0.001), adrenaline administrations (95% vs. 26.4%; p = 0.001), and initial non-shockable rhythm (80% vs. 27.2%; p = 0.001). The two groups were comparable in median Injury Severity Score (ISS), head Abbreviated Injury Score (AIS), and total transport time. Mechanical CPR was associated with a markedly […]

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### Cervical spine clearance in pediatric trauma patients - Consensus algorithm of the Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU).

*Bolte J, Rüther H, Brecht P, Wiersbicki DW, Youssef Y, Jarvers JS, Disch AC*

PURPOSE: Although cervical spine injuries in pediatric patients are rare, clinical and radiological examinations are clinical routine. Currently, no standardized decision algorithm does exist for detection of these injuries in the European region. The proposed algorithm is aiming to safely identify significant injuries through consistent diagnostics while minimizing immobilization and radiation exposure at the same time.

METHODS: The Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU) developed an algorithm based on own scientific investigations and considering primary and secondary literature. The algorithm was developed through a formal consensus process and thereby aligns with international recommendations.

RESULTS: The algorithm uses the initial pediatric Glasgow Coma Scale (pGCS) to group patients to three treatment pathways. Resulting recommendations for imaging, clinical re-assessment, and further management include the child's age, trauma mechanism, and clinical evaluations.

CONCLUSION: Despite their rarity, potential cervical spine injury in children requires accurate diagnosis. Clinical examinations can be challenging, especially in children with compromised consciousness. Increased radiation exposure from X-rays and computer tomography (CT) scans must be considered, particularly as magnetic resonance im […]

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### Development of a low-cost external fixation system: an off-label solution for resource-limited settings.

*Seiser M, Deininger C, Fierlbeck J, Hollensteiner M, Coles P, Reuter C, Tempfer H, Traweger A et al.*

PURPOSE: External fixators are essential for the surgical management of long-bone fractures, particularly in resource-limited settings where access to commercial systems is often restricted by cost and logistics. This study aimed to develop a low-cost external fixation concept using widely available construction materials and evaluate its biomechanical performance under axial loading.

METHODS: An external fixator composed of M10 threaded rods, washers, and nuts was developed. Five configurations were tested: SP (Steinmann pins), KT (threaded K-wires), KS (smooth K-wires), and BKS (bilaterally applied, smooth K-wires). A commercial external fixator served as control (CG). Juvenile bovine ulnae with a standardized fracture gap were subjected to cyclic axial loading. Construct stiffness, elastic deformation, and plastic deformation were assessed.

RESULTS: CG showed the highest stiffness (107.80 ± 14.12 N/mm), followed by SP (87.73 ± 17.92 N/mm), BKS (83.81 ± 10.01 N/mm), KS (49.84 ± 3.80 N/mm), and KT (46.66 ± 2.39 N/mm). Post-hoc analyses showed no significant differences between CG vs. SP and CG vs. BKS, whereas KT and KS were significantly less stiff. Elastic deformation followed a similar pattern. Plastic deformation was greatest in CG, while SP and BKS showed significantly lower values.

CONCLUSIONS: The threaded-rod fixator provides relevant mechanical stability at extreme […]

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### Supplementary fixation in multifragmentary midshaft clavicle fractures: do interfragmentary screws or suture cerclage improve outcomes compared with plate-only fixation?

*Barthel C, Hinkelmann MA, Schibler D, Frei F, Pape HC, Allemann F*

BACKGROUND: Clavicle fractures are common injuries. As displaced or multifragmentary midshaft patterns carry a substantial risk of malunion and nonunion with nonoperative treatment, they have contributed to the increasing use of surgical fixation. Modern operative techniques include plate fixation, intramedullary devices, and hybrid strategies to address comminution and improve fragment stability. Supplementary interfragmentary screws and cerclage constructs have been proposed to enhance fragment control and optimize load transfer in multifragmentary fracture patterns. This study aimed to compare three fixation strategies for operatively treated midshaft clavicle fractures managed with plate osteosynthesis: Group IS (interfragmentary screw), Group CA (cerclage augmentation), and Group POF (plate-only fixation), with respect to operative time, complication rates, and potential biological and mechanical differences.

METHODS: This retrospective study included patients aged ≥ 16 years who underwent plate fixation for multifragmentary clavicle fractures between 2016 and 2024. Patients were categorized into three groups: Group IS (supplementary interfragmentary screw), Group CA (suture cerclage augmentation), and Group POF (plate-only fixation). Demographic, surgical, and radiographic data were collected. Callus formation was assessed by three independent observers. Statistical anal […]

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## Cerebral glycerol during haemorrhagic shock in normal and raised intracranial pressure resuscitated with total REBOA: an experimental porcine study.

*Bader S, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.*

BACKGROUND: Haemorrhagic shock (HS) is frequently associated with secondary cerebral injury due to compromised cerebral perfusion. Resuscitative endovascular balloon occlusion of the aorta (REBOA) is an effective haemorrhage control strategy; however, concerns persist regarding its cerebral effects, particularly in the presence of elevated intracranial pressure (ICP). Cerebral microdialysis (CMD) derived glycerol (CGly) is a recognised marker of cellular membrane stress and injury.

OBJECTIVE: To investigate CGly dynamics during prolonged total REBOA (tREBOA) in HS and to compare cerebral cellular responses between animals with normal and elevated ICP.

METHODS: In this experimental porcine study, eighteen pigs were subjected to controlled HS and resuscitated with tREBOA for 90 min. Animals were allocated to either a normal ICP group (NICPG, n = 9) or an elevated ICP group (EICPG, n = 9). Continuous monitoring of proximal arterial pressure, ICP, and cerebral perfusion pressure was performed. CGly concentrations were measured using CMD throughout the experimental protocol.

RESULTS: tREBOA effectively restored proximal arterial pressure and cerebral perfusion pressure in both groups. CGly concentrations remained relatively stable during the haemorrhage phase and increased progressively during prolonged aortic occlusion. Importantly, CGly responses did not differ significantly be […]

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## Transport and destination hospital for patients with suspected multiple and/or severe injuries - a systematic review and clinical practice guideline update.

*Pavlu F, Könsgen N, Bieler D, Goossen K, Gliwitzky B, Häske D, Faul P, Wölfl C et al.*

PURPOSE: Our aim was to update the evidence-based and consensus-based recommendations on criteria for transport and destination hospital for patients with suspected multiple and/or severe injuries based on available evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to January/February 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared different transport strategies or destination hospitals for patients with (suspected) multiple and/or severe injuries. We considered patient-relevant outcomes such as mortality and neurological outcomes. Risk of bias was assessed at outcome level using ROBINS-I for observational studies and RoB 2 for RCTs. We conducted meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expert consensus was used to develop recommendations and determine their strength.

RESULTS: A total of 127 studies were identified. They addressed helicopter vs. ground emergency medical services (EMS) transport, ground EMS with vs. without a physician, on-scene time, and handover on arrival at the hospital […]

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## Does the addition of peripheral nerve blocks improve analgesia during distal radius fracture reduction? A prospective, randomized, multicenter clinical trial.

*Guerrero-Serrano JA, Santamaría-López A, García-Lamas L, Jiménez-Díaz V, González-Fernández Á, Cecilia D*

PURPOSE: This study aimed to evaluate whether adding a median nerve and a superficial branch of the radial nerve block to hematoma block improves analgesia during distal radius fracture reduction and to identify factors associated with increased pain perception.

METHODS: A prospective, randomized, controlled study was conducted in two tertiary hospitals, including 180 adult patients with displaced distal radius fractures requiring closed reduction. Patients were allocated either to an isolated hematoma block group or to a combined block group receiving hematoma block with median nerve and superficial branch of the radial nerve blocks. Within the combined group, half of the median nerve blocks were performed under ultrasound guidance and half using anatomical landmarks. The primary endpoint was digital pain during traction and fracture reduction, assessed using a visual analog scale (VAS, 0-10). Secondary outcomes included wrist pain at different procedural stages and the influence of demographic and clinical variables including age, sex, laterality, participating hospital, body mass index (BMI), fracture type (AO/OTA), and baseline psychiatric or chronic analgesic medication use.

RESULTS: Digital pain was significantly lower in combined group compared with isolated hematoma block during traction (4.54 ± 2.53 vs. 7.39 ± 1.76; p < 0.001) and reduction (5.36 ± 2.56 vs. 8.33 ± 1. […]

PubMed · DOI

## Under pressure - a sensor-based analysis of simulated prehospital pressure dressing application for hemorrhage management.

*Brauckmann V, Menzel J, Welke B, Decker S, Macke C*

BACKGROUND: The concept of "x-ABCDE", with "X" representing exsanguinating hemorrhage, highlights the priority of immediate bleeding control in trauma care. Despite being a fundamental intervention, pressure dressing techniques lack standardization. Studies show high variability in achieved pressures, often below therapeutic targets or exceeding harmful levels, with no correlation between subjective estimates and objective measurements.

OBJECTIVES: To objectively assess current practices of pressure bandage application and identify influencing factors including material type, technique, and provider experience.

METHODS: This prospective, single-center, simulation-based observational study was conducted from August 2024 to January 2025 at a Level I Trauma Center. Emergency medical providers applied pressure dressings according to personal preference using available materials. Pressure distribution was measured via two calibrated capacitive pressure sensors (61 measurement fields). Analysis included correlation tests, group comparisons, and multiple regression.

RESULTS: A total of 124 emergency medical providers (75% male; 36% paramedics, 44% emergency medical technicians, 13% trainees, 7% emergency physicians) completed pressure dressing applications, with 116 datasets included in final analysis. Mean maximum pressure was 169.35 ± 84.33 mmHg (range 50-412 mmHg) with applicati […]

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## Clinical impact of early postoperative routine radiology in isolated pelvic injuries: a retrospective study.

*Bolierakis E, Ioannou P, Schubert SL, Groven RVM, Bläsius FM, Bode M, Alabdulrahman H, Hildebrand F et al.*

PURPOSE: The clinical value of routine immediate postoperative conventional radiography following surgical fixation of pelvic fractures remains debated. This study assessed the impact of early postoperative X-rays on management decisions and revision surgery in patients with isolated pelvic ring or acetabular fractures.

METHODS: We conducted a retrospective cohort study at a Level 1 trauma center, including adults undergoing surgical treatment for isolated pelvic ring or acetabular fractures between 2020 and 2023. All patients received high-quality intraoperative fluoroscopy and postoperative conventional radiographs. The primary outcome was any change in standard postoperative care based on radiographic findings. The secondary outcome was revision surgery prompted by conventional postoperative imaging.

RESULTS: A total of 216 patients were included (median age 63 years), with 156 pelvic ring injuries and 60 acetabular fractures. Postoperative radiographs were obtained at a median of 3 days post-surgery. Changes in postoperative care based on radiographs occurred in 2 patients (3.3%) within the acetabular fracture group. Revision surgery triggered by conventional imaging was required in 1 patient (0.6%) among pelvic ring injuries. In all cases, additional CT imaging preceded definitive management changes.

CONCLUSION: Routine immediate postoperative conventional radiography a […]

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## Older age and post-traumatic organ failure a TraumaRegister DGU® analysis of 34,469 elderly major trauma patients.

*Bewersdorf TN, Lefering R, Stein S, Streblow J, Findeisen S, Schmidmaier G, Grossner T*

PURPOSE: As the number of elderly major trauma patients (EMTP) increases continuously, this study sought to determine the incidence of organ failure (OF) and multiple organ failure (MOF) and their impact on mortality and recovery status in this cohort.

METHODS: Based on the German TraumaRegister DGU® database this study included EMTP aged between 65 and 99 years with an Injury Severity Score (ISS) of ≥ 16 between 2014 and 2023. OF was defined as Sequential Organ Failure Assessment (SOFA) score of ≥ 3 and MOF as failure of ≥ 2 organs.

RESULTS: The database revealed 34,469 EMTP, here 37% suffered from OF (16.0%) or MOF (21.0%). Mortality was significantly higher in OF and MOF patients compared to non-OF patients (non-OF: 7.4%; OF: 45.0%; MOF: 59.0%; $p < 0.001$). 80.9% of all deaths were associated with failure of at least one organ and 51.1% of non-survivors suffered from MOF. Highest odds ratios (OR) for mortality were found for liver (OR: 5.13, $p < 0.001$), central nervous system (OR: 3.63, $p < 0.001$) and renal failure (OR: 3.25, $p < 0.001$), followed by cardiovascular (OR: 1.95, $p < 0.001$) and lung failure (OR: 1.16, $p = 0.010$). Haemostatic failure had no impact on mortality.

CONCLUSION: OR and mortality differ significantly between distinct OFs due to varying degrees of success in treatment options and systemic impact of these OFs. Clinicians should be aware that OF and MOF o […]

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## Denis pattern-dependent biomechanical behavior of posterior trans-iliac plate fixation compared with bilateral triangular osteosynthesis in unilateral sacral fractures: a finite element study.

*Yang J, Böker KO, Li C, Schilling AF, Ritter Z, Acharya M, Lehmann W*

OBJECTIVE: Posterior trans-iliac plate osteosynthesis (TPO) is commonly used for the stabilization of unilateral vertically unstable sacral fractures. However, its biomechanical performance across different Denis fracture patterns remains unclear. This study aimed to investigate the fracture pattern-dependent biomechanical behavior of posterior trans-iliac plate osteosynthesis (TPO) in unilateral vertically unstable sacral fractures, using bilateral triangular osteosynthesis (BTO) as a high-stability reference construct.

METHODS: A validated three-dimensional finite element model of the lumbopelvic complex was developed to simulate unilateral vertically unstable sacral fractures corresponding to Denis type I, II, and III patterns. Two posterior fixation strategies, TPO and BTO, were constructed for each fracture type. Seven physiological loading conditions were applied, including standing, flexion, extension, axial rotation, and lateral bending. Von Mises stress distribution and fracture displacement were evaluated. Fracture micromotion was quantified by calculating relative displacement changes between paired points along the fracture gap.

RESULTS: When interpreted relative to the high-stability BTO reference construct, TPO showed fracture pattern-dependent variation in fracture displacement. The difference was most pronounced in Denis type I fractures and progressively decr […]

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## Factors associated with in-hospital mortality in necrotising soft tissue infections. a multicentre retrospective cohort study.

*Podda M, Ceresoli M, Virdis F, Rosa F, Pilia T, Murzi V, Dessì A, Tedesco S et al.*

PURPOSE: Necrotising soft tissue infections (NSTIs) are rare but life-threatening conditions associated with high mortality rates. This multicentre study aimed to identify admission variables associated with in-hospital mortality.

METHODS: This retrospective study included adult patients with surgically confirmed NSTIs treated at four high-volume academic referral centres in Italy between 2010 and 2024. Demographic, clinical, physiological, and laboratory variables available at hospital admission were analysed. Categorical variables were compared between survivors and non-survivors using the chi-square test or Fisher's exact test, while quantitative variables were compared using the Student's t test or Mann-Whitney U test. Multivariable logistic regression analyses were performed to identify independent predictors of in-hospital mortality. Results were reported as ORs with 95%CI. A prognostic nomogram was developed from the final multivariable model, including variables independently associated with mortality.

RESULTS: A total of 379 patients were included. In-hospital mortality was 16.7%. In subgroup comparisons, mortality was 17.0% in necrotising fasciitis and 22.4% in Fournier's gangrene (P = 0.275). In the overall NSTI population, age (aOR 1.061, 95%CI 1.037-1.087) and serum creatinine at admission (aOR 1.301, 95%CI 1.040-1.629) were independently associated with mortalit […]

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## Influence of frailty on short-term mortality in older patients with multiple trauma in the emergency department.

*Çelik ■, Toker A, Biçer S, Cebecio■lu ■K*

PURPOSE: To investigate the association between frailty and 28-day mortality in older patients presenting to the emergency department with multiple trauma and to assess the discriminative performance of PRISMA-7 and the Trauma-Specific Frailty Index (TSFI) in frailty assessment in relation to commonly used trauma severity scores.

METHODS: This prospective observational study included patients aged ≥ 65 years with multiple trauma admitted to the emergency department between July 1 and August 31, 2024. Frailty was assessed using PRISMA-7 and TSFI. Patients were stratified into frail and non-frail groups and further categorized according to age groups (65-74, 75-84, and ≥ 85 years). Clinical characteristics, trauma severity scores, and outcomes, including 28-day mortality, were compared. Receiver operating characteristic (ROC) analysis was used to evaluate discriminative performance, and logistic regression analysis was performed to assess the association between scores and mortality.

RESULTS: A total of 144 patients were included. Frailty was significantly associated with increased 28-day mortality, higher ICU admission rates, and longer hospital stays ($p < 0.001$). Mortality increased with age, reaching 30.0% in patients aged ≥ 85 years ($p = 0.006$). Frailty scores (PRISMA-7 and TSFI) increased significantly with age, whereas ISS did not differ significantly across age groups. R […]

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## Mortality prediction in older adults with earthquake-related trauma: a model combining MEWS and laboratory biomarkers.

*Ozen A, Acehan S, Satar S, Bulut EA, Gulen M, N■g■z K, Sevd■mbas S, Sonmez GO et al.*

PURPOSE: This study aimed to identify laboratory parameters and clinical scores that can predict in-hospital mortality among patients aged 65 years and older who presented to the emergency department (ED) with earthquake-related trauma following the February 6 and 20, 2023 earthquakes.

METHODS: In this retrospective observational study, 362 patients with earthquake-related trauma between February 6-21, 2023, were included. Demographic characteristics, laboratory results, and clinical scores such as the Modified Early Warning Score (MEWS), Injury Severity Score (ISS), and Shock Index (SI) were analyzed.

RESULTS: Of the included patients, 53.9% (n = 195) were female, and the median age was 73 years (IQR: 69-80). The overall mortality rate was 9.4%. Multivariable logistic regression analysis identified MEWS (OR: 3.535; 95% CI: 2.397-5.212; $p < 0.001$), urea (OR: 1.009; 95% CI: 1.002-1.015; $p = 0.008$), and hs-Troponin I (OR: 1.001; 95% CI: 1.000-1.002; $p = 0.016$) as independent predictors of mortality. Receiver operating characteristic (ROC) analysis demonstrated an AUC of 0.936 for the combined model including MEWS, urea, and hs-Troponin I (95% CI: 0.875-0.996; $p < 0.001$). Among the variables, MEWS showed the highest AUC (0.903; 95% CI: 0.829-0.976). The optimal cutoff value for MEWS was determined as 1.5, with a sensitivity of 70.6% and a specificity of 97.6%.

CONCLUSIONS: In o […]

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## The economic burden in terms of cost of illness and generic health-related quality of life of posttraumatic long bone non-unions among the adult population of the Netherlands from a societal perspective.

*van der Broeck LCA, Shchurov DG, Gabrio A, Lodewijks AJL, Poeze M, Blokhuis TJ, Evers S*

PURPOSE: This study assessed the societal economic burden in terms of cost of illness and health-related quality of life (HRQoL) of posttraumatic long bone non-unions in the Netherlands.

METHODS: An incidence-based bottom-up approach was used, focusing on adult patients with posttraumatic long bone non-unions. The analysis included healthcare costs, patient and family costs, and productivity losses, measured using the iMCQ and iPCQ questionnaires. Cost evaluations followed Dutch costing guidelines, and productivity losses were calculated using the friction cost method. HRQoL was assessed with the EQ-5D-5L. A deterministic one-way sensitivity analysis varied baseline characteristics by ± 10%. Scenario analyses were conducted from healthcare and patient perspectives, as well as for patient subgroups.

RESULTS: Average costs per patient during the three months before the initial visit at a non-union clinic were €8,928 (healthcare, N = 78), €1,360 (patient and family, N = 58), and €4,313 (productivity losses, N = 59), the average of the observed data calculates to 10,831 (N = 44). The mean EQ-5D-5L utility score was 0.390 (± 0.29 SD). Subgroup analysis showed no significant cost increase for patients with infections or open fractures. However, a higher Non-Union Severity Score and a lower HRQoL were significantly associated with higher total costs.

CONCLUSION: Posttraumatic long […]

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### Nonoperative treatment of Maisonneuve fractures.

*Frühauf J, Bartoní■ek J, Fojtík P, Hor■áková A, Tu■ek M*

PURPOSE: The aim of this study is to describe the basic pathoanatomical characteristics of a stable Maisonneuve fracture and mid-term results of its nonoperative treatment.

METHODS: The study included 17 prospectively collected patients with a mean age of 59 years. The postinjury ankle CT had to meet the following criteria: nondisplaced or minimally displaced (up to 1 mm) fracture of medial malleolus, medial clear space less than 3 mm, nondisplaced or minimally displaced (up to 2 mm) fracture of posterior malleolus, anatomical position or minimal malposition of the distal fibula in the fibular notch (widening of the tibiofibular space up to 2 mm or external rotation of the distal fibula up to 10°). The average follow-up period was 34 months, the final follow-up included CT examination and functional evaluation based on AOFAS and FAOS scores.

RESULTS: A medial malleolus fracture was recorded in 12% cases, a posterior malleolus fracture in 29% patients and a Tillaux-Chaput tubercle fracture in 18% cases. All fractures of the proximal fibula, medial and posterior malleolus healed within 3 months. The position of the distal fibula in the fibular notch did not change in 11 cases compared to the post-injury CT scan, improved slightly in 5 cases, and worsened slightly in 1 case. The average final AOFAS hindfoot score was 96.9 points and the average final FAOS score was 98.7%.

CONCL […]

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### Age-related differences in prehospital triage and critical interventions: a multicenter TraumaRegister DGU® analysis.

*Koch DA, Hagebusch P, Hackenberg L, Pavlu F, Jakobi T, Münzberg M, Lefering R, Schweigkofler U et al.*

PURPOSE: Older trauma patients are known to be undertriaged in the prehospital setting. However, it remains unclear whether these disparities translate into differences in the delivery of critical prehospital interventions. This study aimed to evaluate age-related differences in prehospital triage, allocation, and treatment within a physician-staffed emergency medical service system.

METHODS: A retrospective multicenter analysis of the TraumaRegister DGU® was conducted, including trauma patients aged ≥ 20 years between 2020 and 2023 with trauma team activation and subsequent intensive care unit admission. Patients were stratified into two age groups, 20-59 vs. ≥60 years, with additional subgroups up to 90-99 years. Prehospital interventions, transport modality, and trauma center allocation were analyzed. Furthermore, predefined high-risk scenarios (severe motor vehicle trauma, traumatic brain injury with GCS ≤ 8, and pneumothorax) were evaluated.

RESULTS: A total of 67,069 patients were included, with comparable injury severity across age groups (mean ISS 20.2). Older patients were less frequently transported by air and less often allocated to supra-regional trauma centers. While overall intervention rates were similar, differences emerged in high-risk scenarios. Older patients received endotracheal intubation and tranexamic acid less frequently despite comparable injury seve […]

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### Prediction of risk of death in severely injured patients: the revised injury severity classification score, version 3 (RISC III).

*Lefering R, Imach S, Bieler D*

PURPOSE: Trauma Registries with a focus on severely injured patients use survival as their primary outcome. In order to compare hospitals, interventions, and changes over time, a precise risk of death prediction is mandatory. The German TraumaRegister DGU® uses the RISC II model (Revised Injury Severity Classification, version II) since 2013. The aging trauma population and an improved handling of missing data required the present revision.

METHODS: A total of 53,738 seriously injured trauma patients documented in 2022-2023 served as basis for development and validation (3:1 ratio). Missing values should now be considered to be within normal physiological range except in patients with specific findings indicative for an altered physiology. These findings were suggested by clinical experts and validated by registry data. The increasing age of trauma patients was addressed by additional categories and higher point weights. A logistic regression analysis provided new point weights for all predictors. Precision (observed versus predicted mortality) and discrimination (area under the receiver operating characteristic curve, AUROC) were calculated in the development and validation dataset.

RESULTS: Patients in both datasets were well comparable, with a mean age of 55 years, 69% males, and an average Injury Severity Score (ISS) of 18 points. The rate of missing values ranged from 0% […]

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### Dislodgement forces and cost effectiveness of prehospital securement of peripheral intravenous catheters on moist skin: a randomized controlled clinical trial in healthy volunteers.

*Vogeler J, Schumann S, Heinrich S, Hell J, Schmutz A*

OBJECTIVES: Peripheral intravenous catheters (PIVC) are widely used clinical devices for administering fluids or essential drugs to patients. The failure rate of PIVC remains high and Emergency Medicine has been shown to be a risk factor for dislodgement. When the skin is moist from sweat or fluids, standard intra-hospital dressings and securements fail. In emergency situations, a failed catheter can then critically delay intravenous therapies. The most effective dressing to prevent accidental removal of a prehospital PIVC remains unclear. It was the aim of this study to compare the force required to dislodge a PIVC with four different methods of securing PIVCs used in emergency medicine. In addition, the costs were calculated.

METHODS: Artificial sweat was applied to the skin of 180 volunteers. PIVCs were attached onto the forearm using four different securements (elastic gauze, cohesive gauze, clingfilm and a velcro securing device). Continuously increasing traction force was applied until dislodgement of the respective securement. For statistical tests, either Friedman's test or repeated measures ANOVA was used.

RESULTS: Clingfilm showed the greatest resistance to increasing pulling force with cohesive bandages as close second. The velcro securing device was strongest at resisting low level of forces but fell off sharply at higher force. Elastic bandages were the weakest i […]

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### Injury to the thorax is a dominant contributor to post-traumatic neutrophil mediated inflammatory response.

*van Eerten FJC, Duebel LV, de Fraiture EJ, Hanlo MF, Breuking EA, Brands SR, van den Bosch TD, van Wessem KJP et al.*

PURPOSE: Traumatic injury induces a neutrophil associated inflammatory response mediated by the release of damage-associated molecular patterns (DAMPs). This response can be maladaptive, increasing risk of infection. Since blunt thoracic injuries are associated with high infection rates, we hypothesized that these injuries elicit a stronger neutrophil associated inflammatory response than injuries to other body regions.

METHODS: A cohort study was conducted in adult patients presenting at the trauma bay of a level 1 trauma center between March 2020 and November 2022. These patients underwent routine analysis of their neutrophil associated inflammatory response by phenotyping blood neutrophils using automated flow cytometry. Injury per body region was counted when the Abbreviated Injury Severity (AIS) scores were ≥3 to focus on substantial injuries. The primary outcome was an extensive post-traumatic inflammatory neutrophil response.

RESULTS: 861 patients were included. Blunt internal thoracic injuries showed the strongest association with severe neutrophil associated inflammation (OR 5.7), followed by abdominal (OR 2.6), pelvis (OR 2.5) and lower extremities (OR 2.3), even when corrected for ISS, age and hemodynamic instability. In multivariable analysis, only thoracic internal injuries remained a statistically significant contributor to excessive neutrophil associated inflam […]

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## Major improvement using peroperative 3D compared to 2D imaging for SI screw fixation in traumatic pelvic fractures.

*van Eerten FJC, Benders KEM, van Baal MCPM, Hietbrink F*

INTRODUCTION: Pelvic fractures, though accounting for only 3% of skeletal injuries, carry a high mortality rate, primarily due to the associated risk of hemodynamic instability. Sacroiliac (SI) screw fixation is a common surgical intervention, which can be technically challenging due to the complex pelvic anatomy. Imaging techniques have evolved from 2D to 3D imaging for its improved accuracy in screw placement. The impact of this transition for SI screw placement was evaluated.

METHOD: A single-center observational retrospective study was performed between 2013 and 2023. Patients ≥ 18 years who underwent SI screw placement following pelvic injuries were analyzed. 2D imaging was used until 2021, after which 3D imaging was implemented. Outcomes included the need for revision surgery during initial admission and < 18 months after trauma.

RESULTS: 2D imaging was used in 42 patients and 3D imaging in 44 patients. Baseline characteristics and injury severity of both groups were comparable. In the 2D group, 18 patients (42.9%) had suboptimal placement of the SI screw. Of these, 3 (7.1%) required revision surgery due to malpositioned screws. In contrast, no patients (0.0%) in the 3D group had suboptimally positioned screws or underwent revision surgery, (p < 0.001, p = 0.22).

DISCUSSION: Intraoperative 3D imaging enhanced the accuracy of SI screw placement, leading to an encouragin […]

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## Minimally invasive versus open fixation for flail chest in elderly patients: a prospective cohort study on perioperative safety and long-term outcomes.

*Tian Y, Zheng Z, Ji Z, Lu W, Song N, Wang J*

OBJECTIVE: To systematically evaluate the perioperative safety and short- and long-term efficacy of minimally Invasive fixation using a reversed shape-memory alloy plate versus conventional open fixation for the treatment of flail chest in elderly patients.

METHODS: This prospective cohort study enrolled 130 elderly patients (≥ 60 years) with flail chest between January 2022 and January 2026. After 1:1 propensity score matching, 100 patients were included (minimally invasive group, n = 50; open group, n = 50). Perioperative parameters (operative time, blood loss, C-reactive protein(CRP) at 24 h/48 h/72 h, complications), short-term outcomes (resting/cough-evoked Numerical Rating Scale(NRS) pain scores, non-steroidal analgesic drug duration, rescue analgesic frequency, Patient-Controlled Analgesia Pump(PCA) consumption, time to ambulation, hospital stay), and long-term outcomes (implant dislodgement, 6-month pulmonary function, Douleur Neuropathique 4 Questions Naire(DN4), Medical Outcomes Study 36- Item Short Form Health Survey(SF-36)) were compared.

RESULTS: The minimally invasive group had significantly fewer incision-related complications (P = 0.046), lower resting NRS scores on postoperative days 1 and 3 (P < 0.01; P = 0.041), lower cough-evoked NRS score on day 1 (P = 0.011), shorter non-steroidal analgesic duration (P < 0.01), and lower PCA consumption (P = 0.027). At 6 […]

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## Epidemiology and outcomes of traumatic sternal fractures and associated blunt cardiac injury: a nationwide cohort study in the Netherlands.

*Klei DS, Benders KEM, Leenen LPH, van Wessem KJP*

PURPOSE: Comprehensive data on epidemiology, trauma mechanisms, associated injuries, and outcomes of traumatic sternal fractures are scarce. This study analysed nationwide data to improve diagnosis and management within the Dutch healthcare system.

METHODS: This nationwide retrospective cohort study using the Dutch National Trauma Registry included adult patients admitted with traumatic sternal fractures between 2015 and 2023. Patients with prehospital cardiopulmonary resuscitation or penetrating trauma were excluded. Incidence, patient characteristics, trauma mechanisms, associated injuries, and in-hospital outcomes were analysed. Subgroup analyses evaluated patients with concomitant blunt cardiac injury (BCI).

RESULTS: Of 568,399 adult trauma admissions, 4,765 patients (0.84%) sustained traumatic sternal fractures. Median age was 62 years; 60% were male. Motor vehicle accidents (48%) and falls (28%) were the leading mechanisms. 35% were severely injured (ISS ≥ 16). Associated injuries included rib fractures (51%), spinal fractures (36%), and lung contusions (18%). Critical care unit admission was 40%, with median mechanical ventilation duration of 4 days; median hospital stay was 5 days. In-hospital mortality was 5.7%, and 30-day mortality 6.0%. BCI occurred in 9.5% of patients and was associated with a higher number of injuries and increased injury severity, emergency inte […]

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## Trauma patients ≥ 65 years old constitute 40% of severely injured adult trauma patients in Sweden - a 10-year analysis of geriatric trauma prevalence.

*Bredberg S, Modalen E, Jöneby C, Nilsson D, Lapidus O*

BACKGROUND: The elderly population in Sweden is steadily increasing. Age is an independent predictor of mortality in trauma patients. There is growing evidence that geriatric patients constitute an increasing proportion of the Swedish trauma population. No studies have analyzed temporal trends in the proportion of geriatric trauma patients in the context of the national trauma population in Sweden.

AIM: To determine the proportion of geriatric patients in the Swedish trauma population and analyze temporal trends in geriatric trauma prevalence.

METHODS: The study cohort was 11,969 severely injured trauma patients (NISS > 15) treated in Sweden during 2013-2022, obtained from the Swedish Trauma Registry. The proportion of geriatric patients was compiled annually. Temporal trends were analyzed using weighted linear regression models. Two subgroups were defined based on injury severity (NISS ≥ 25 and NISS ≥ 40).

RESULTS: The proportion of patients ≥ 65 years in the adult trauma population increased from 30% to 40% during 2013-2022 (p = 0.003), while the proportion of patient ≥ 80 years also increased, from 8.6% to 17% (p = 0.009). Similar increases were also seen in the NISS ≥ 25 and NISS ≥ 40 subgroups.

CONCLUSION: There is likely an increase in geriatric trauma prevalence in Sweden, but the full extent of this temporal trend remains uncertain. Trauma patients ≥ 65 years old co […]

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## A totally laparoscopic strategy combining microwave ablation and autologous blood transfusion for patients with initial WSES grade IV traumatic splenic rupture in selected transient responders: a proof-of-concept study.

*Feng Z, Xie D, Hong W, Zheng C*

PURPOSE: Splenic artery embolization (SAE) is the standard intervention for transient responders with high-grade splenic trauma. When SAE is unavailable, contraindicated, or declined, alternative minimally invasive approaches may be warranted. This study evaluates a totally laparoscopic strategy combining microwave ablation (MWA) for hemostasis with intraoperative autologous blood transfusion (ABT) in selected transient responders with initial WSES Grade IV splenic rupture.

METHODS: This retrospective study included 10 consecutive transient responders (October 2021-June 2024) with initial WSES Grade IV splenic rupture who underwent totally laparoscopic MWA combined with intraoperative ABT. All patients exhibited transient hemodynamic response to limited resuscitation (≤ 1500 mL fluids) and preoperative shock index > 0.9. Intraoperative injury was graded by the AAST system. Outcomes included technical success, transfusion requirements, complications, and follow-up.

RESULTS: All 10 patients underwent successful spleen-preserving laparoscopy without conversion. AAST grades: II (n = 2), III (n = 4), IV (n = 4). Mean operative time was 191.3 ± 101.7 min. Mean blood salvage was 1587.5 ± 663.9 mL, yielding 490.8 ± 80.2 mL autologous red cells. No patient required postoperative allogeneic transfusion. Hemoglobin, platelets, and lactate improved significantly by postoperative day 3 (P […]

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## Pediatric pancreatic trauma: a 10-year retrospective analysis from two high-complexity centers in Medellín, Colombia.

*Barrientos ML, Calderon SP, Zapata CAL, Lopez CAD*

PURPOSE: Pediatric pancreatic trauma is uncommon, but it is associated with significant morbidity and an ongoing debate regarding its management. Hemodynamic stability and pancreatic duct involvement are key factors in treatment decisions.

METHODS: A retrospective study analyzed 25 pediatric patients with pancreatic trauma treated between 2010 and 2020 at two high-complexity hospitals in Medellin. Clinical variables, trauma mechanism, imaging findings, management, and complications were analyzed.

RESULTS: Blunt trauma was the predominant mechanism (64%). Preoperative computed tomography was performed in 12 patients (48%); radiological AAST grading was not retrievable in 13 (52%). Immediate surgery was performed in 13 patients (52%), predominantly driven by associated intra-abdominal injuries rather than the pancreatic injury itself. Non-operative management was initially attempted in 9 patients (36%), with a 56% success rate; failures (44%) were related to clinical deterioration and progression of peripancreatic collections. Clinically relevant postoperative pancreatic fistula (ISGPS Grade B) occurred in 3 of 13 operated patients (23%), with a mean closure time of 55.6 days (range 18-110); no Grade A or Grade C fistulas were recorded. Other complications included post-traumatic pancreatitis (24%), pancreatic pseudocyst (8%), and organ/space surgical site infection (8%); most […]

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## Influence of Antithrombotics on Severely Injured Patients Over 50 With Blunt Abdominal Trauma.

*Eibinger N, Limberger B, Hörlesberger N, Puchwein P, Lefering R, Bayer TP, Zumsteg JS, Pape HC et al.*

INTRODUCTION: Severe abdominal trauma represents a critical subset of injuries, particularly in the aging European population, where the prevalence of preexisting anticoagulant or antiplatelet therapy is increasing. While the impact of such medications on outcomes in traumatic brain injury is well studied, limited data exist regarding their influence in patients with abdominal trauma.

MATERIALS AND METHODS: This retrospective cohort study used data from the TraumaRegister DGU® between January 2015 and December 2023. Inclusion criteria were patients aged ≥ 50 years with severe blunt abdominal trauma (AIS Abdomen ≥ 3) without relevant head injury (AIS Head ≤ 3) from Austria, Germany, or Switzerland. Patients were grouped according to pre-injury antithrombotic medication (no medication [NM] vs. antithrombotic medication [AM], with subgroups). Statistical analysis included descriptive comparisons, standardized mortality ratios (SMR) using RISC III, and multivariate logistic regression to identify independent predictors of hospital mortality.

RESULTS: Of 328,281 patients, 4,069 met inclusion criteria (2,831 NM; 1,238 ATM). Patients in the AM group were older (74.1 vs. 62.3 years) and had higher mortality (25.0% vs. 10.4%), particularly in the DOAC subgroup (30.3%). SMR analysis demonstrated increased mortality in the AM group compared to expected rates. In multivariate logistic re […]

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## Predicting limb amputation in trauma-related necrotizing fasciitis of the extremities: insights from a 10-year cohort.

*Yang S, Ren Z, Li Y, Zhao Z, Sun C, Liu L, Jin L, Hou Z*

BACKGROUND: Necrotizing fasciitis (NF) is a rapidly progressive, life-threatening soft tissue infection that in most cases requires aggressive surgical management. Major limb amputation is a devastating limb-related outcome. We retrospectively analyzed NF cases to quantify the rate of amputation and identify associated risk factors.

METHODS: We retrospectively reviewed the medical records of all patients with trauma-related extremity NF treated at Hebei Medical University Third Hospital from Jan 2014 to Dec 2024. Demographics, comorbidities, admission laboratories and key time intervals were compared between patients who did and did not undergo major limb amputation. Multivariable logistic regression identified independent predictors, and receiver operating characteristic (ROC) analysis assessed the discriminative power of continuous variables.

RESULTS: Of 134 patients, 22 (16.4%) required amputation. Compared with the non-amputation group, amputees had a markedly longer pre-hospital delay (median 242 h vs. 65 h), more multiple injuries (63.6% vs. 13.4%) and a higher prevalence of diabetes (59.1% vs. 21.4%). Admission lactate dehydrogenase (LDH) was substantially higher (median 589 vs. 182 U/L). In multivariable analysis, elevated LDH, prolonged time from traumatic injury to hospital admission, multiple injuries and diabetes were independently associated with amputation. LDH […]

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## Is 18 F-FDG PET-CT-guided geographic debridement superior to conventional debridement in appendicular chronic osteomyelitis? a retrospective comparative cohort study.

*Elsheikh A, Elseidy H, Saad H*

BACKGROUND: Determining the surgical extent of chronic osteomyelitis and fracture-related infection (FRI) remains challenging. Adequate resection margins prevent recurrence; however, no standardised preoperative tool objectively defines resection borders.

OBJECTIVE: To compare PET■CT-guided surgical planning versus the local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI in a minority of patients) for chronic limb osteomyelitis and FRI outcomes.

METHODS: Retrospective comparative cohort study of 126 patients treated 2019-2024. Group 1: 84 patients underwent 18 F-FDG PET-CT imaging and map-based geographic debridement. Group 2: 42 patients received surgery after local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI).

PRIMARY OUTCOME: infection recurrence at 12 months. Multivariable logistic regression controlled for confounders.

RESULTS: PET-CT group achieved 3.6% recurrence versus 16.7% in the conventional group ($p = 0.011$), representing 78.5% relative risk reduction and a number-needed-to-treat of 8. Critically, the PET-CT cohort had significantly worse baseline complexity: 3.3-fold longer infection duration (43.1 vs. 13.1 months, $p < 0.001$) and 3.6-fold more prior surgeries (2.48 vs. 0.69, $p < 0.001$). Multivariable analysis confirmed PET-CT-guided surgery as an independent protective […]

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## Digital coordination in mass casualty incidents: a retrospective analysis of prehospital distribution times before and after IVENA-MANV implementation.

*Brauckmann V, Ringe B, Caviglia M*

BACKGROUND: Digital coordination tools have been proposed to improve management during mass casualty incidents (MCIs). In 2021, the IVENA-MANV system was introduced in the Hannover region as an extension of the established IVENA eHealth platform to enable digital patient tracking and hospital capacity management. However, no real-world evaluations have yet quantified their operational impact.

OBJECTIVES: This study aimed to provide the first real-world evaluation of the IVENA-MANV digital coordination system and to determine how its implementation affected prehospital distribution processes and time intervals during mass casualty incidents.

METHODS: A retrospective observational study to assess the impact of the IVENA-MANV digital coordination platform on prehospital times during MCIs in Germany was conducted using dispatch records and IVENA-MANV logs from all MCIs between January 2018 and July 2025 in the Hanover-Region in Germany. Primary outcomes

were the triage-to-evacuation (TtE) and total prehospital time (TPHT) per patient. Mann-Whitney U tests compared pre- and post-implementation groups, and multiple linear regression models examined associations between IVENA-MANV use, MCI-level, and time intervals.

RESULTS: Seventy-three MCIs met inclusion criteria, including 188 documented individual casualty characteristics. 41.1% of incidents used IVENA-MANV. Most MCIs were tra […]

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## The real-world management of acute mesenteric ischemia in Spain: results from a multicenter national survey.

*González-Castillo AM, Calsina Juscafresa L, Gelabert A, Velescu A, Marrano E, Vikatmaa P, Tolonen M*

BACKGROUND: Acute mesenteric ischemia (AMI) remains one of the most lethal vascular emergencies, with in-hospital mortality rates frequently exceeding 50%. Although early diagnosis and timely revascularization are critical, clinical practice remains highly variable. This study aimed to evaluate the real-world management of AMI in Spain, identifying gaps in resources, protocols, and interhospital coordination.

METHODS: A national cross-sectional survey was conducted between March and August 2024 using the Survio® platform. The questionnaire was distributed to general surgeons through national surgical societies and included 27 items covering hospital infrastructure, clinical protocols, diagnostic and therapeutic availability, personal experience, and perceived system-level barriers.

RESULTS: A total of 291 surgeons responded. The median age was 40 years (IQR: 17). Most were consultants (76.3%) working in tertiary (42.6%), secondary (33%), or community hospitals (24.4%). While 97.6% reported access to 24/7 multiphasic CT and 90.4% to round-the-clock radiology, only 51.9% had 24/7 interventional radiology and 60.1% vascular surgery. Just 26.8% had institutional AMI protocols. The median distance to a referral center was 25 km (range: 2-250 km), and 68.4% reported difficulty in patient transfers. While 96.9% felt competent managing AMI, only 36.4% were familiar with the term "int […]

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## Changes in operative and in-hospital outcome of proximal femoral fractures before and after the implementation of work hour restrictions in Switzerland.

*Tewes A, Canal C, Schöb O, Neuhaus V*

PURPOSE: Surgeon fatigue is a recognized risk factor for patient safety; however, reduced working hours may limit surgical training exposure. In 2005, Switzerland introduced a maximum 50-hour workweek for resident surgeons. In view of current discussions on further reductions (42 + 4 h), concerns persist regarding training quality and workforce capacity. This study evaluates the impact of the 2005 Swiss labor law on patient outcomes, operative characteristics, and intraoperative teaching utilizing a standardized trauma procedure.

METHODS: A retrospective analysis of anonymized data from a national database was performed. All patients undergoing operative fixation of trochanteric femur fractures (ICD-10 S72.1) were included. Two four-year periods were compared: pre-regulation (2001-2004) and post-regulation (2016-2019). Primary endpoints were in-hospital complications, mortality, and length of stay. Secondary endpoints included patient characteristics, surgeon seniority, teaching status, operative duration, and time to surgery.

RESULTS: Post-regulation patients were older and exhibited higher American Society of Anesthesiologists classifications despite fewer comorbidities. Complications increased from 9.1% to 17.7%, and mortality from 1.6% to 3.5%. Length of stay decreased from 14 to 9 days. Operative duration decreased by 10 min across all surgeon levels. Resident surgeons i […]

PubMed · DOI

## Early discharges in severely injured patients by ISS score: exploring injury patterns and coding practices.

*Parouei F, Krüsemann EJZ, Poeze M, de Jongh MAC, Hietbrink F, Landelijke Beraadsgroep Traumachirurgen (LBTC)*

PURPOSE: This study examines the prevalence of early discharge (ED) among patients classified as severely injured in the Dutch National Trauma Registry (DNTR) and evaluates whether identifiable patient-, injury-, and system-related factors are associated with the occurrence of ED in this population.

METHODS: This retrospective cohort study analyzed DNTR data from 2015 to 2022, focusing on severely injured patients (Injury Severity Score [ISS] ≥ 16). Patients were grouped as early discharge (ED; discharged within 48 h without in-hospital mortality) or non-ED (NED). Patients' characteristics were compared using descriptive statistics, and a mixed-effect model identified factors associated with ED.

RESULTS: From 2015 to 2022, a total of 37,626 severely injured patients were registered including 1,454 (3.9%) ED patients. This proportion increased from 3.2% in 2015 to 4.8% in 2022. ED was more common in younger patients, males, general practitioner (GP) or self-referred cases, those with severe head injury (Abbreviated Injury Scale [AIS] ≥ 3) and fewer registered injury codes. In the mixed-effects model, younger age (OR up to 2.0), increasing head injury severity (OR: 1.16 per AIS point), less than 3 registered AIS codes (OR: 0.60), referral by a general practitioner (OR 1.68) or self-referral (OR 1.92), and admission year (OR: 1.08 per year) independently predicted early discharg […]

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## Trauma team activation for pediatric patients in Denmark: a multicenter study of criteria, organization, and injury severity.

*Nordestgaard CH, Gude MF, Leth SV, Raaber N*

BACKGROUND: Pediatric trauma team activation (TTA) is intended to ensure timely management of severely injured children, yet both prehospital visitation practices and in-hospital TTA criteria vary across trauma systems. The aim of this study was to describe and compare pediatric TTA criteria, organizational models, and injury severity across all Danish level I trauma centers.

METHODS: We conducted a retrospective multicenter cohort study including all pediatric trauma patients (< 18 years) admitted via TTA at the four Danish level I trauma centers between 2014 and 2024. Center-specific prehospital and in-hospital TTA protocols were obtained through structured inquiries. Injury severity was assessed using Injury Severity Score (ISS) and Abbreviated Injury Scale (AIS). Overtriage was defined as TTA in patients with ISS < 15.

RESULTS: A total of 3,452 pediatric trauma patients were included. Two distinct TTA models (single-criterion and point-based) and four organizational structures were identified. Overall, 14% of patients had ISS ≥ 15, corresponding to high overtriage across all centers. Overtriage rates ranged from 81% to 95% and were highest at centers using point-based TTA models. Mortality was low and did not differ significantly between centers. Among the most severely injured patients (ISS ≥ 25), head and thoracic injuries predominated.

CONCLUSION: Pediatric trauma tri […]

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## Does the operative technique (MIPO vs. ORIF) influence surgical outcomes and complications in proximal humerus fractures? A matched-pair analysis.

*Barthel C, Rudolph F, Kraus M, Kalbas Y, Russek R, Pape HC, Allemann F*

INTRODUCTION: Proximal humerus fractures account for approximately 4-6% of all fractures and are among the most common fractures in the elderly. Surgical treatment is increasingly applied for displaced and unstable fracture patterns. The two most commonly used techniques are open reduction and internal fixation (ORIF) via a deltopectoral approach and minimally invasive plate osteosynthesis (MIPO) using a deltoid-split or anterolateral approach. ORIF allows direct fracture visualization and anatomical reduction but is associated with more extensive surgical exposure. MIPO aims to preserve soft tissues and vascular supply through indirect reduction techniques. This study compares ORIF and MIPO in proximal humerus fractures with regard to surgical parameters, radiographic outcomes, and complications.

METHODS: A retrospective single-center cohort of 392 proximal humerus fractures was analyzed. Seventy-nine patients treated with MIPO were matched to 79 patients treated with ORIF using exact matching for fracture type (Mayo FJD classification) and propensity score matching for age and sex. Primary outcomes included operative time, reduction quality, and complication rates.

RESULTS: The MIPO group showed significantly shorter operative times (97.3 ± 30.8 min vs. 112.2 ± 34.0 min, p = 0.004). No significant differences were observed in reduction quality or overall complication rates […]

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### Risk factors and predictors of in-hospital mortality in necrotizing fasciitis: a five-year retrospective cohort study from a tertiary care center in Colombia.

*Barrientos ML, Cordoba ASR, Valencia MDML, Valencia MMM, Toro DAM, Zapata CAL, Delgado CA*

PURPOSE: Necrotizing fasciitis (NF) is a life-threatening soft tissue infection with limited epidemiological data from Latin America. This study aimed to characterize clinical outcomes, identify factors associated with in-hospital mortality, and develop a clinical prediction model in patients with NF at a tertiary care center in Medellín, Colombia.

METHODS: A retrospective cohort study was conducted including all adult patients (≥ 18 years) with surgically or histopathologically confirmed NF between January 2019 and December 2024. Demographic, clinical, microbiological, and therapeutic variables were collected. Bivariable analysis and multivariable logistic regression were performed to identify independent mortality predictors. A clinical risk score was developed and internally validated using bootstrap resampling and 10-fold cross-validation.

RESULTS: Of 112 identified patients, 111 met inclusion criteria. Overall in-hospital mortality was 23.4% (26/111). Deceased patients were significantly older (median 69.5 vs. 52 years; $p < 0.001$). Serum lactate demonstrated strong prognostic value (AUC 0.719; 95% CI 0.596-0.842; 2.9 vs. 1.8 mmol/L; $p = 0.003$). Septic shock was present in 34.2% and doubled mortality risk (34.2% vs. 17.8%; $p = 0.060$). Polymicrobial infections predominated (74.8%), with a predominantly gram-negative profile and high antimicrobial resistance rates (carbapen […]

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### ESTES recommendations regarding distal radius fractures.

*Lameijer CM, Teuben MPJ, van Delft EAK, Tomazevic M, Kastelec M, Nau C*

Distal radius fractures (DRFs) account for 15% of all fractures in adults. Optimal treatment of DRFs depend on patient and fracture characteristics. Non- or minimally displaced DRFs can be treated nonoperatively with a cast. Although surgical treatment is becoming more popular, the majority of patients with a DRF are still treated nonoperatively. In addition, these considerations may be different for elderly patients. Conflicting evidence exists on the different aspects of treatment of DRFs. These recommendations regarding distal radius fractures describe considerations regarding diagnostic modalities, associated soft tissue injuries, different treatment options, rehabilitation protocol, considerations for treatment of elderly patients, follow-up decisions and possible complications. The aim is to aid the orthopedic traumasurgeon in decision making and treatment of distal radius fractures.

PubMed · DOI

### Small traumatic intracranial hemorrhages identified in routine radiology reports are associated with a low risk of adverse events: a retrospective cohort study.

*Gerdås K, Wigermo A, Offenbartl M, Vestlund S, Rezk F, Vedin T*

PURPOSE: To determine whether routinely available radiology reports, together with basic clinical data, can identify patients with traumatic intracranial hemorrhage (TICH) who are at low risk of adverse events.

METHODS: This retrospective cohort study of adults with TICH in Region Jönköping, Sweden (2019-2021). Clinical data, findings from radiology reports and outcomes were extracted from medical records. Hemorrhage size was classified as small (≤ 4 mm or described as minimal/very small/discrete) or larger. Adverse events were defined as neurosurgical intervention or death directly attributable to the TICH. Risk difference (RD), relative risk (RR), and Firth's penalized logistic regression were used to assess associations with adverse events.

RESULTS: Among 527 included patients, 195 (37%) had small TICHs. None of these patients experienced adverse events, compared with 13.6% neurosurgical interventions and 13.0% trauma-related deaths in the group with larger TICHs (RD 24.5% points, 95% CI 18.4-29.6; RR 97.1, 95% CI 6.1-1557.1; $p < 0.001$). Small TICH size had the strongest association with absence of adverse events. Normal neurological status and GCS 14-15 were also associated with a low risk of adverse events. Anticoagulant or antiplatelet therapy showed no significant association with adverse events.

CONCLUSION: Routinely available radiology reports, combined with basic c […]

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## From fracture to joint injury: association between CT-based fracture characteristics and surgically treated non-bony injuries in tibial plateau fractures.

*Neidlein C, Colcuc S, Ullerich F, Schöllmann H, Steinhausen ES, Kröpil P, Brinkmann N, Dudda M et al.*

PURPOSE: Tibial plateau fractures (TPF) are often associated with non-bony injuries that may impair knee stability and long-term outcomes. However, it remains unclear in which patients preoperative MRI is indicated to detect these injuries, and clear recommendations for surgical decision making are lacking. This study aimed to develop a CT-based predictor of surgically relevant non-bony injuries.

METHODS: This retrospective single-center study included all intra-articular TPF treated between January 2022 and May 2025 at a Level I trauma center. Fractures were classified according to the 10-segment and three-column classifications. Operative reports were reviewed to identify non-bony injuries requiring surgical treatment. Descriptive statistics and multivariate logistic regression were performed to determine predictors of surgically relevant non-bony injuries.

RESULTS: Among 243 patients (mean age 49,3 ± 14,9 years), 34,6% had at least one surgically treated non-bony injury. Posterior column involvement was significantly associated with a higher injury rate (41,4%), particularly affecting meniscus (20,7%), Anterior cruciate ligament (ACL) (17,2%), and medial collateral ligament (MCL) (12,3%). The posterolateral-central (PLC) segment showed the highest segment-specific injury rate (37,6%) and was an independent predictor of ACL injury. Increasing fracture complexity was associa […]

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## Transferrals and clinical pathways of unstable pelvic fractures over the last 10 years - a retrospective analysis of the Trauma Register DGU®.

*Weuster M, Pfeifer R, Seekamp A, Lippross S, Lefering R, Dgu T, Menzdorf L*

PURPOSE: Pelvic fractures are classified as stable or unstable. They correlate with a severity of trauma and the initial medical treatment is decisive. This study evaluated the transferrals of such fractures and described the initial treatment as well as the clinical course.

METHODS: We analysed retrospective data from a large cohort of the TraumaRegister DGU® (TR-DGU), covering the period from 2014 to 2023, comprising a total of n = 397,910 patients. All patients aged ≥ 16 years were included. Injury patterns were described according to the Abbreviated Injury Scale (AIS), the mechanically unstable fractures were classified with an AIS ≥ 3. We considered all participating hospitals within Germany. The patients were subdivided in three groups: Group 1 = primary admitted patients with outcome, group 2 = pre-treated patients transferred in from other hospitals, and group 3 = primary admitted and early (< 48 h) transferred out.

RESULTS: The majority of the patients was male and about 53 years old. Blunt trauma was the leading trauma mechanism. Concomitant injuries (AIS 2+) affected thorax (56%), spinal cord (41%), lower extremities (38%), head (31%) and abdomen (24%). Among primary admitted cases with pelvic fractures (n = 36,398), 21,091 cases (57.9%) had an unstable pelvic fracture (AIS pelvis 3-5). Level 1 trauma centers not only treated 12,836 primary admitted cases with unst […]

PubMed · DOI

## Surgical treatment of coronal shear fractures: short- to mid-term results and risk factor analysis.

*Bauer A, Cornel L, Sauter M, Jakobi T, Münzberg M, Klug A*

BACKGROUND: Coronal shear fractures of the distal humerus are rare but severe injuries. Reconstruction is often challenging, especially in comminuted cases, which is why many surgeons opt for an elbow arthroplasty in those cases. However, total elbow arthroplasty is associated with a variety of potential problems itself. Therefore, the aim of this study was to present the functional and clinical outcome of coronal shear fractures treated by osteosynthetic reconstruction in a short- to mid-term follow-up, and to identify possible risk factors for an inferior outcome.

METHODS: We performed a retrospective follow-up assessment of 51 consecutive patients (30 women; median age 56 years, (IQR 39-62)) who underwent osteosynthetic reconstruction for coronal shear fractures between 2012 and 2022 after a minimum follow-up period of two years. The Mayo Elbow Performance Score, Oxford Elbow

Score, and Disabilities of the Arm, Shoulder and Hand score were evaluated, and all available radiographs were analyzed. All complications and revision procedures were assessed. Univariable and multivariate regression analyses were performed to identify potential risk factors for a poor outcome following osteosynthetic reconstruction.

RESULTS: After a median follow-up period of 43 (IQR 28-78) months, the median Mayo Elbow Performance Score was 100 (IQR 85-100), the median Oxford Elbow Score was 42 (IQ […]

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### Publisher Correction: Experience with patient-specific guides, instead of general surgical experience, improves the accuracy of 3D-guided corrective osteotomies.

*Buist ML, Meesters AML, Colaris JW, Doornberg JN, Ten Duis K, Tabernée Heijtmeijer SJC, Jutte PC, Pijpker PAJ et al.*

PubMed · DOI

### Assessment of a new technique for biological augmentation in sternoclavicular joint dislocations: a cadaveric feasibility and biomechanical evaluation.

*Somberg O, Förster E, Rosteius T, Bernstorff M, Schildhauer T, Königshausen M*

AIMS: Biological augmentation is necessary in chronic sternoclavicular dislocations. However, the placement of drill holes for a graft can be difficult at that region. The aim of this study was to evaluate, as an early proof-of-concept, the feasibility of using a locally available clavicular periosteal flap (CPF) or sternal periosteal flap (SPF) for sternoclavicular joint (SCJ) stabilization and to biomechanically test the periosteal flaps.

METHODS: A cadaveric anatomic feasibility and biomechanical study (level of evidence V) was performed on nine SCJs. The CPF was mobilized from three sides by lifting the flap off the bone, preserving its attachment near the SCJ. Two 1.6 mm drill holes were created in the sternum, and the CPF was folded over the SCJ and fixed with a non-resorbable wire. Additionally, the non-resorbable wire was used in terms of a figure-of-8-bracing through a 1.6 mm clavicular drill hole to secure the joint position. Flap dimensions and sternal overlap were measured. Vice versa SPFs were prepared in selected specimens and fixed in a similar fashion. The CPFs were biomechanically tested using a uniaxial tensile testing machine to obtain values for stress at 50% strain (σ (50%)), maximum force (Fmax), maximum stress (σmax), strain at maximum force (ε(Fmax)), and strain at failure (ε at failure).

RESULTS: CPF mobilization and fixation were successfully achieve […]

PubMed · DOI

### Differentiating injury patterns and outcomes in accidental, suicidal and occupational falls from heights.

*Beyersdorf C, Hussmann B, Wergen N, Lefering R, Schiffner E, Maus U, Jaekel C, Dgu T*

BACKGROUND: Falls from significant heights are a leading cause of severe injuries and can result from accidents, suicide attempts, or occupational incidents. While previous studies have explored general injury patterns associated with falls, little is known about the specific differences between accidental, suicidal, and occupational falls. This study aims to identify distinct injury patterns and outcomes among these fall types, thereby enhancing primary care and decision-making algorithms.

METHODS: We conducted a retrospective analysis using data from the German TraumaRegister DGU® (2014-2023), focusing on patients who experienced falls from heights. Only patients admitted to trauma centers with trauma team activation and a Maximum Abbreviated Injury Scale (MAIS) severity ≥ 3 were included. All falls were categorized in accidental, suicidal, or occupational and distinguished between high (≥ 3 m) and low (< 3 m) falls. Statistical analyses included chi-squared tests for categorical variables and Mann-Whitney U tests for continuous variables, with significance set at $p < 0.05$.

RESULTS: Among 188,708 patients, 29,991 (15.9%) experienced high falls, while 50,674 (26.9%) fell from low height. High falls were predominantly accidental (82.3%), with suicidal falls accounting for 17.7%. Accidental high falls mainly caused thoracic (58.9%) and head injuries (46.2%), whereas suicidal f […]

PubMed · DOI

## Predictive performance of ISS and NISS for clinical outcomes in severely injured trauma patients: a retrospective registry study.

*Säkkinen A, Partio N, Mattila V, Mäenpää H, Riuttanen A*

BACKGROUND: The Injury Severity Score (ISS) and New Injury Severity Score (NISS) are widely used in evaluation of injury severity in trauma patients. The aim of this study is to assess the predictive accuracy of these scoring systems on clinical outcomes.

METHODS: We conducted a retrospective registry study of 1,112 severely injured trauma patients (NISS ≥ 16) treated at Tampere University Hospital (TAUH) between 2015 and 2024. Outcomes included in-hospital mortality, prolonged hospital and ICU stay (≥ 75th percentile), in-hospital intubation, and blood transfusions. Predictive performance was assessed using the area under the receiver operating characteristic curve (AUC) and the Hosmer-Lemeshow (H-L) test for calibration. Subgroup analyses were performed for patients with significant (AIS ≥ 3) head, thorax, and extremity injuries.

RESULTS: A total of 848 (76%) patients had a higher NISS than ISS with a median increase of 9 points. Both scoring systems generally demonstrated fair to considerable (0.7 < AUC < 0.9) discrimination for mortality and intubation, and poor to fair (0.6 < AUC < 0.8) discrimination for blood transfusions and prolonged hospital/ICU stay. ISS had superior discrimination for blood transfusions in the total cohort (AUC: 0.669 vs. 0.630, p = 0.019), head (AUC: 0.737 vs. 0.672, p = 0.004), and thorax (AUC: 0.656 vs. 0.613, p = 0.006) subgroups, and for intu […]

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## Surgical and anesthetic activity in a french emergency field hospital responding to devastating floods.

*Gaudray E, Brac L, Agopian P, Arnaud I, Ribelles P, Buland S, Couret A, Simonet J et al.*

PURPOSE: Floods are the most frequent natural disasters, causing widespread mortality and devastation. In 2023, Storm Daniel triggered catastrophic flooding in Derna, Libya, leading to over 11,300 deaths and leaving more than 10,000 people missing. In response, the French Emergency Medical Team (EMT-FRA) ESCRIM, deployed via the European Union Civil Protection Mechanism, arrived to provide critical medical aid. METHODS: The team set up a field hospital equipped with adult and pediatric care, ICU, radiology, and laboratory services, adhering to strict aseptic conditions. Data collection was performed daily by designated physicians, with records transcribed for analysis. Over the course of the 22-day mission, the EMT-FRA provided care to over 1,600 patients, with 7.6% undergoing surgery. RESULTS: The anesthesia team performed 64 procedures, including 11 critical care assessments and 65 pre-anesthetic consultations. Surgical cases were primarily related to infections (49%), visceral issues (27%), and orthopaedic injuries (25%), including wound debridement, VAC therapy, amputations, and laparotomy. The surgical team carried out 64 surgeries and 126 consultations, with 83% of patients receiving emergency care for the first time. A significant number of procedures (78%) were dirty or infected, reflecting the local health challenges. CONCLUSIONS: The deployment of EMT-FRA demonstrated […]

PubMed · DOI

## Hook plate versus locking plate fixation in lateral clavicle fractures: operative efficiency and cumulative surgical burden in a propensity score-matched cohort study.

*Schibler D, Hinkelmann MA, Frei F, Pape HC, Allemann F, Barthel C*

INTRODUCTION: Hook plates remain controversial in the management of lateral clavicle fractures despite their biomechanical advantages. Concerns include implant-related complications, subacromial irritation, and the need for routine implant removal. The aim of this study was to compare operative efficiency and surgical burden of hook plate fixation with locking plate fixation in lateral clavicle fractures. METHODS: A retrospective cohort analysis was conducted at a single tertiary trauma center between January 1, 2016, and December 31, 2024. Surgically treated lateral clavicle fractures were identified. Propensity score matching was performed based on age, sex, and AO fracture classification to reduce confounding, and matched pairs were compared between hook plate fixation (Group HP) and locking plate fixation (Group LCP). Outcome measures included operative time, complications, implant removal, radiographic callus formation, and refracture rates. RESULTS: A total of 101 lateral clavicle fractures were identified. After propensity score matching, 44 matched pairs (Group HP vs. Group LCP) were analyzed, with comparable baseline characteristics between groups. Operative time was significantly shorter in Group HP (77.8 ± 28.4 vs. 93.5 ± 30.2 min, p = 0.018). Complication rates were higher in Group HP but did not reach statistical significance (18.2% vs. 4.5%, p = 0.093). Implant re […]

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### Diagnostic value of hypoxia-inducible factor 1 alpha and adrenomedullin in acute mesenteric ischemia.

*Bayindir E, Küçükceran K, Özer MR, Ayranci MK, Kilinç ■, Kökbudak N, Bayindir GK*

BACKGROUND: This study aims to evaluate the diagnostic significance of adrenomedullin and hypoxia-inducible factor 1 alpha (HIF1-alpha) levels in an animal model of acute mesenteric ischemia.

MATERIALS AND METHODS: In our study, a total of 21 male New Zealand rabbits were used, and the animals were divided into three groups for the study. Blood samples were taken for adrenomedullin and HIF1-alpha levels at 0, 1, 3, and 6 h from the control, sham, ischemia groups.

RESULTS: HIF levels of the ischemia group at the 3rd and 6th hours were statistically significantly higher than those in the sham (p = 0.002, p < 0.001, respectively) and control groups (p = 0.002, p < 0.001, respectively). Adrenomedullin levels of the ischemia group at the 1st, 3rd, and 6th hours were statistically significantly higher than those in the sham (p = 0.004, p < 0.001, p < 0.001, respectively) and control groups (p < 0.001, p < 0.001, p < 0.001 respectively).

CONCLUSION: In this experimental model of acute mesenteric ischemia, hypoxia-inducible factor-1 alpha and adrenomedullin levels were significantly higher in the ischemia group compared with the control and sham groups. Adrenomedullin demonstrated an earlier increase, while hypoxia-inducible factor-1 alpha increased significantly at later time points. These biomarkers may provide supportive diagnostic value when interpreted alongside clinical and im […]

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### Videolaryngoscope-assisted fibreoptic intubation in anticipated difficult airways of maxillofacial trauma: a prospective randomised comparison with videolaryngoscope-guided nasotracheal intubation.

*Jain K, Nair R, Goyal A, Makkar JK, Singla K, Arora S, Bhatia N*

BACKGROUND: Maxillofacial trauma poses airway management challenges due to edema, distorted facial anatomy and associated bleeding. Further, its fixation requires unhindered access to facial fractures, often necessitating securing the airway naso-tracheally during the intraoperative period. Traditional technique of nasotracheal intubation is associated with 17%-77% incidence of epistaxis. Fibreoptic bronchoscope helps guide nasotracheal tube under vision, decreasing incidence and severity of epistaxis. Moreover, as patients with maxillofacial trauma have difficult airways, every effort should be made to secure these challenging airways with best possible technique that increases first-attempt success rate. Use of fibreoptic bronchoscope as a part of hybrid videolaryngoscope-assisted fibreoptic intubation(VAFI) technique may provide these benefits. Hence, we conceptualized this study comparing the VAFI technique with traditional videolaryngoscope-assisted technique for securing nasotracheal tube in patients with anticipated difficult airways following maxillofacial trauma. METHODS: Following a written, informed consent, we included adult patients with maxillofacial trauma, scheduled for fracture fixation under general anaesthesia, requiring nasotracheal intubation. All patients were randomised into two groups of 30 patients each, with patients in group-VLS undergoing videolaryng […]

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### Malignancy risk and management outcomes in adult intussusception: a single-institution retrospective study.

*Chang HC, Kang JC, Pu TW, Chen CY*

PURPOSE: Adult intussusception is rare and frequently caused by a pathological lead point, including malignancy. We assessed malignancy rates by CT-based anatomical subtype/location, identified predictors of malignancy, and summarized management outcomes to support risk-stratified decision-making in the emergency setting. METHODS: We conducted a single-institution retrospective study at Tri-Service General Hospital (Taipei, Taiwan), including consecutive adults with CT-diagnosed intussusception from January 2015 to August 2024. We collected demographics, symptoms, hemoglobin, CT-based type/location, management, and histopathology. Malignancy was defined by index-admission pathology (malignant vs. benign). Anemia was defined as hemoglobin < 13 g/dL in men or < 12 g/dL in women. Predictors of malignancy were evaluated using univariable and multivariable logistic regression. Primary analyses included pathology-confirmed cases; sensitivity analyses additionally included presumed benign cases. RESULTS: Forty-seven patients were identified; 38 had pathology-confirmed outcomes. In the primary cohort (n = 38), 18 (47.4%) were malignant. Lower hemoglobin was associated with malignancy (OR per 1 g/dL increase, 0.709; 95% CI, 0.529–0.950; p = 0.021), and anemia strongly predicted malignancy (OR, 11.667; 95% CI, 2.438–55.830; p = 0.002). In multivariable analysis adjusting for age and colo […]

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